

ETAS RTA-BSW v12.7.0



RTA-Com Stack Reference Guide

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Contents

1	Communication	7
1.1	Introduction	7
1.2	Supported Features	8
1.3	Extensions	12
1.4	Limitations	30
1.5	Exclusions	32
1.6	Deviations	32
1.7	API Definition	46
1.8	Configuration Advice	49
1.9	Integration Advice	63
1.10	Hardware Requirements	66
2	Efficient COM for Large Data	67
2.1	Introduction	67
2.2	Supported Features	68
2.3	Extensions	69
2.4	Limitations	71
2.5	Exclusions	71
2.6	Deviations	71
2.7	API Definition	72
2.8	Configuration Advice	73
2.9	Integration Advice	74
2.10	Hardware Requirements	77
3	Communication Manager	78
3.1	Introduction	78
3.2	Supported Features	78
3.3	Extensions	81
3.4	Limitations	84
3.5	Exclusions	85
3.6	Deviations	85
3.7	API Definition	88
3.8	Configuration Advice	90
3.9	Integration Advice	95
3.10	Hardware Requirements	98

4	I-Pdu Multiplexer	99
4.1	Introduction.....	99
4.2	Supported Features	100
4.3	Extensions	102
4.4	Limitations.....	103
4.5	Exclusions	103
4.6	Deviations	104
4.7	API Definition	108
4.8	Configuration Advice	108
4.9	Integration Advice	110
4.10	Hardware Requirements	112
5	Mirror.....	113
5.1	Introduction.....	113
5.2	Supported Features.....	113
5.3	Extensions.....	114
5.4	Limitations	114
5.5	Exclusions	115
5.6	Deviations.....	117
5.7	API Definition.....	118
5.8	Configuration Advice	119
5.9	Integration Advice	124
5.10	Hardware Requirements.....	125
6	Network Management Interface	126
6.1	Introduction	126
6.2	Supported Features	127
6.3	Extensions.....	127
6.4	Limitations	131
6.5	Exclusions	131
6.6	Deviations	132
6.7	API Definition	136
6.8	Configuration Advice	138
6.9	Integration Advice	140
6.10	Hardware Requirements	141
7	PDU Router	142

7.1	Introduction	142
7.2	Supported Features	143
7.3	Extensions	146
7.4	Limitations	157
7.5	Exclusions	160
7.6	Deviations	160
7.7	API Definition	161
7.8	Configuration Advice	163
7.9	Integration Advice	166
7.10	Hardware Requirements	167
8	ComBased Transformer	168
8.1	Introduction	168
8.2	Supported Features	169
8.3	Extensions	170
8.4	Limitations	171
8.5	Exclusions	171
8.6	Deviations	172
8.7	API Definition	172
8.8	Configuration Advice	172
8.9	Integration Advice	173
8.10	Hardware Requirements	173
9	rba_SysElem	174
9.1	Introduction	174
9.2	Supported Features	174
9.3	Extensions	174
9.4	Limitations	261
9.5	Exclusions	261
9.6	Deviations	261
9.7	API Definition	261
9.8	Configuration Advice	262
9.9	Integration Advice	263
9.10	Hardware Requirements	263
10	rba_Xfrm	264
10.1	Introduction	264

10.2	Supported Features	264
10.3	Extensions	264
10.4	Limitations	265
10.5	Exclusions	265
10.6	Deviations	265
10.7	API Definition	265
10.8	Configuration Advice	266
10.9	Integration Advice	266
10.10	Hardware Requirements	266
11	rba Comstack Complex Device Driver	267
11.1	Introduction	267
11.2	Supported Features	267
11.3	Extensions	268
11.4	Limitations	268
11.5	Exclusions	269
11.6	Deviations	270
11.7	API Definition	270
11.8	Configuration Advice	270
11.9	Integration Advice	271
11.10	Hardware Requirements	271
12	rba_XCoreCdd	272
12.1	Introduction	272
12.2	Supported Features	272
12.3	Extensions	273
12.4	Limitations	280
12.5	Exclusions	281
12.6	Deviations	281
12.7	API Definition	281
12.8	Configuration Advice	282
12.9	Integration Advice	284
12.10	Hardware Requirements	285
13	Glossary	286

1 Communication

Module name	Com
Package	RTA-COM
Module ID	50
AUTOSAR Version	22-11

1.1 Introduction

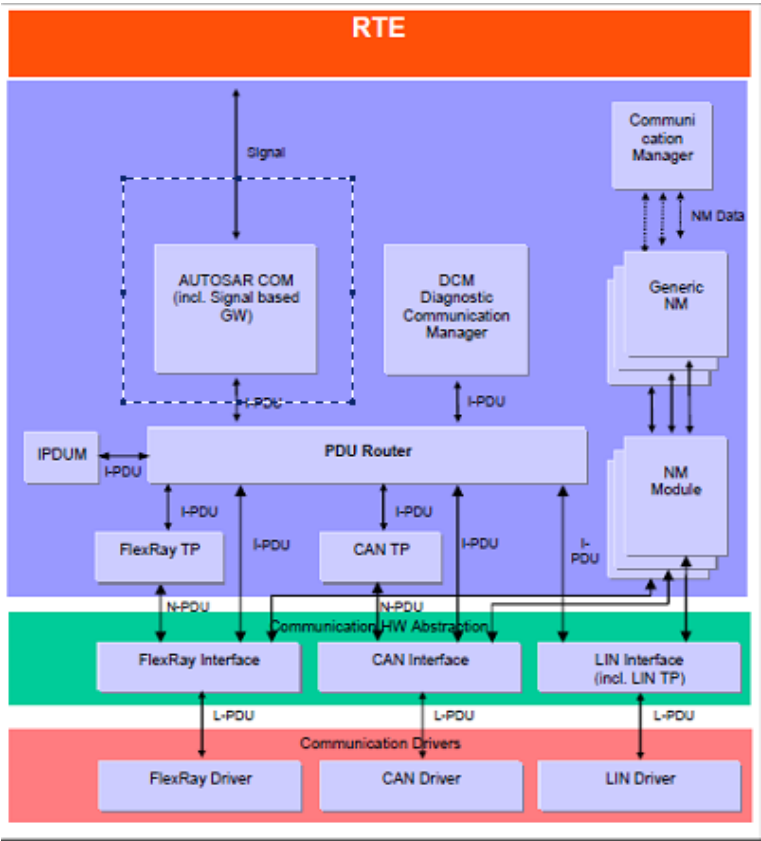


Figure 1.1.Communication stack

The Communication module (Com) is part of the RTA-COM stack, and it is located between the RTE and the PDU Router (PduR).

Com creates PDUs using signals or a signal group received from the RTE, and it sends them to the PduR.

In the other direction, Com extracts signals and signal group from PDUs received from the PduR, and it sends them to the RTE.

1.1.1 Interaction with other Modules

Com directly interacts with various other modules.

RTE

Com is used by RTE to send and receive signals.

PduR

As Com transports either un-fragmented I-PDUs (via simple L-PDUs) or fragmented I-PDUs (via TP), it uses both sets of the PduR module's upper layer APIs - i.e. those that use TP and those that do not. The functionality that Com requires from the PduR can be summarised as follows:

- An indication of incoming I-PDUs.
- A sending interface for outgoing I-PDUs, including the confirmation if an I-PDU has been sent by the communication controller.
- A trigger interface to enable the PDU Router to cause a transmission from Com.
- Buffer handling for TP communication.

BswM

BswM controls the starting and stopping of the sending and receiving I-PDUs via Com.

Trigger Transmit Functionality

The following modules use Com_TriggerTransmit to get the latest data from Com.

- CanNm
- IPduM
- FrIf
- LinIf

Note that the API calls do not directly call a Com API but involve a call to the PduR, which in turn routes the call to Com by calling Com_TriggerTransmit with the appropriate Com handle ID for the corresponding I-PDU.

Module Configuration

This module can be generated by the RTA-BSW Configuration Generator.

For any unsupported automatic configuration, the module can be manually configured by the user.

1.2 Supported Features

Com supports the following Autosar-defined features:

- Communication modes
- Signal Invalidation
- Handling of I-PDU
- Deadline Monitoring
- Signal Groups - Complex Data Types
- Large Data Types

- Dynamic Length Signals
- Update-Bits
- Signal Gateway
- Provision of signal oriented data interface for the RTE
- Packing of AUTOSAR signals to I PDUs to be transmitted
- Unpacking of received I-PDUs and provision of received signals to RTE
- Routing of signals from received I-PDUs into I PDUs to become transmitted
- Routing of signal groups from received I-PDUs into I-PDUs to become transmitted
- Communication transmission control (start/ stop of I-PDU groups)
- Replications of send requests
- Guarantee of minimum distances between transmit I-PDUs
- Monitoring of receive signals (signals timeout)
- Filter mechanisms for incoming signals
- Different notification mechanisms
- Provision of init values and update indications
- Byte order conversion
- Endianness Conversion and Sign extension
- Support of two different transmission modes per I-PDU
- Support of MetaData, of up to 64 bytes, as per AUTOSAR 4.2.2 specification
- Support of SignalGroup Access as an uint8[n] Array
- Support of 1:N Signal Gateway
- Support of ComEnableMDTForCyclicTransmission feature
- Post-Build Selectable

See the AUTOSAR_SWS_COM.pdf document for more details on these functions.

1.2.1 Handling data lengths > 8 bytes

If the data length is larger than 8 bytes, the reception and transmission processes are performed by calling the following APIs:

Com_RxIndication	Reception
Com_StartOfReception	Reception
Com_CopyRxData	Reception
Com_TpRxIndication	Reception
Com_CopyTxData	Transmission
Com_TpTxConfirmation	Transmission

1.2.2 Handling data lengths ≤ 8 bytes

If the data length is less than or equal to 8 bytes, the reception and transmission process are as described below.

Reception (Immediate and Deferred modes)

If Com_RxIndication is called with an I-PDU which is not configured for the Gateway, the Com module performs the following actions depending on how the I-PDU is configured.

- IMMEDIATE mode: Com unpacks data from PduInfoPtr and calls a callback function provided by RTE to immediately update the corresponding signals.
- DEFERRED mode: Com copies data from PduInfoPtr to a buffer, then the scheduled function Com_MainFunctionRx processes this buffer, unpacks the data, and calls a callback function provided by RTE to update the corresponding signals.

Transmission

Com supports packing signals to an I-PDU and transmitting to lower layer upon transmission modes. In the case of transmission from SWC by CAN, LIN and FLEXRAY, the following APIs are called:

Com_SendSignal	CAN / LIN / FLEXRAY
Com_SendSignalGroup	CAN / LIN / FLEXRAY
Com_SendDynSignal	CAN / LIN / FLEXRAY
Com_MainFunctionTx	CAN / LIN / FLEXRAY
Com_SendIpdu	CAN / LIN / FLEXRAY
PduR_ComTransmit	CAN / LIN / FLEXRAY
Com_TxConfirmation	CAN / LIN / FLEXRAY
Com_TriggerTransmit	LIN / FLEXRAY

1.2.2.1 Updating Group Signals

Com supports the updating of group signals using Com_ReceiveSignal, Com_InvalidateSignal and Com_SendSignal.

1.2.3 Multiple MainFunctions

Instead of processing all Ipdis in one Com_MainFunction, you can split the processing for some ComIpdis. See the Configuration Advice section for more details.

1.2.4 Support Signal Gateway Timeout

The destination signal shall use the last value before the source signal times out and shall use the signal timeout value (substitution or initial value) after the source signal times out.

Use Case	Update Bit Position Configured Rx Signal (Group)?	Update Bit Position Configured Tx Signal (Group)?	Requirements	Signal Gatewayed on Timeout (replace with either initial or substitution value)
1	No	No	[Customer Requirement] The destination signal shall use the last value before the source signal times out and will then use the signal timeout value after the source signal times out. Note: An invalid value or initial value can be used if there is no timeout value.	Yes
2	No	Yes	[SWS_Com_00706] If the received signal/signal group has no update bit (ComUpdateBitPosition) configured and the destination signal/signal group has an update bit configured then the AUTOSAR COM module shall set the update bit of the destination signal when a new signal/signal group is received and will be cleared after sending the destination signal/signal group.	No

3	Yes	No	[SWS_Com_00705] If the received signal/signal group has an update bit (ComUpdateBitPosition) configured but the destination signal has no update bit configured and the update bit is not set then the AUTOSAR COM module shall not route the signal/signal group.	No
4	Yes	Yes	[SWS_Com_00703] If the received signal/signal group, and the destination signal/signal group have an update bit (ComUpdateBitPosition) configured with the update bit of the received signal/signal group not set then the AUTOSAR COM module shall not route this signal/signal group.	No

1.3 Extensions

1.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

1.3.1.1 Com/ComConfig/ComIPdu/ComCalibrationNameForIpduControl

Short Name	ComCalibrationNameForIpduControl
------------	----------------------------------

Description	User is provided with this option for each IPDU (To Control IPDU through Calibration) to give an desired calibration name instead of Com_st"PduShort-Name"PduCtrl_C name. Calibration name provided here will be generated in Com generated files and henceforth will be used to access the calibration value to Control IPDU communication. Otherwise, if this parameter is left blank, default calibration name will be generated as Com_st"PduShortName"PduCtrl_C. Note: ComIPduControlViaCalibration parameter should be enabled in ComGeneral for the this parameter field to be used. No validation will be checked on the string provided here, calibration name will be generated as it is.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

1.3.1.2 Com/ComConfig/ComIPdu/ComCalibrationNameForNumberOfRepetitionOfTxPdu

Short Name	ComCalibrationNameForNumberOfRepetitionOfTxPdu
Description	User is provided with this option (For Tx IPDUs configured with number of repetition value) to give a desired calibration name instead of Com_nrOfRepn"ComlpduShort-Name"_C name. Calibration name provided here will be generated in Com generated files and henceforth will be used to access the calibration value to calibrate the number of repetition of Tx IPDUs. Otherwise, if this parameter is left blank, default calibration name will be generated as Com_nrOfRepn"ComlpduShortName"_C. Note: ComTxIpduNumberOfRepetitionCalibration parameter should be enabled in ComGeneral for this parameter field to be used. No validation will be checked on the string provided here, calibration name will be generated as it is.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

1.3.1.3 Com/ComConfig/ComIPdu/ComCalibrationNameForRxIpduFirst-Timeout

Short Name	ComCalibrationNameForRxIpduFirstTimeout
------------	---

Description	User is provided with this option (For Receive IPDUs configured with first timeout value) to give an desired calibration name instead of Com_ti"PduShort-Name"RxFirstTout_C name. Calibration name provided here will be generated in Com generated files and henceforth will be used to access the calibration value to calibrate the first timeout of receive IPDUs. Otherwise, if this parameter is left blank, default calibration name will be generated as Com_ti"PduShortName"Rx-FirstTout_C. Note: ComRxlpduFirstTimeoutCalibration parameter should be enabled in ComGeneral for the this parameter field to be used. No validation will be checked on the string provided here, calibration name will be generated as it is.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

1.3.1.4 Com/ComConfig/ComIPdu/ComCalibrationNameForRxlpduTimeout

Short Name	ComCalibrationNameForRxlpduTimeout
Description	User is provided with this option (For Receive IPDUs configured with Timeout value) to give an desired calibration name instead of Com_ti"PduShort-Name"RxTout_C name. Calibration name provided here will be generated in Com generated files and henceforth will be used to access the calibration value to calibrate the timeout of receive IPDUs. Otherwise, if this parameter is left blank, default calibration name will be generated as Com_ti"PduShortName"RxTout_C. Note: ComRxlpduTimeoutCalibration parameter should be enabled in ComGeneral for the this parameter field to be used. No validation will be checked on the string provided here, calibration name will be generated as it is.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

1.3.1.5 Com/ComConfig/ComIPdu/ComCalibrationNameForTxModeCyclePeriod

Short Name	ComCalibrationNameForTxModeCyclePeriod
------------	--

Description	User is provided with this option (For Transmit IPDUs configured with PERIODIC/MIXED mode) to give an desired calibration name instead of Com_ti"PduShortName"TxPerd_C name. Calibration name provided here will be generated in Com generated files and henceforth will be used to access the calibration value to change the peridiocity of transmit IPDUs (configured with PERIODIC/MIXED mode). Otherwise, if this parameter is left blank, default calibration name will be generated as Com_ti"PduShortName"TxPerd_C. Note: ComTxModeCyclePeriodCalibration parameter should be enabled in ComGeneral for the this parameter field to be used. No validation will be checked on the string provided here, calibration name will be generated as it is.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

1.3.1.6 Com/ComConfig/ComIPdu/ComCalibrationNameForTxPduTime-Offset

Short Name	ComCalibrationNameForTxPduTimeOffset
Description	User is provided with this option (For Cyclic Tx IPDUs configured with ComTxModeTimeOffset value) to give a desired calibration name instead of Com_ti"PduShortName"TxTOffset_C name. Calibration name provided here will be generated in Com generated files and henceforth will be used to access the calibration value to calibrate the TxTimeOffset of cyclic Tx IPDUs. Otherwise, if this parameter is left blank, default calibration name will be generated as Com_ti"PduShortName"TxTOffset_C. Note: ComTxModeTimeOffsetCalibration parameter should be enabled in ComGeneral for this parameter field to be used. No validation will be checked on the string provided here, calibration name will be generated as it is.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

1.3.1.7 Com/ComConfig/ComIPdu/ComIPduCounter

Short Name	ComIPduCounter
------------	----------------

Description	This optional container contains the configuration parameters of PDU Counter.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

1.3.1.8 Com/ComConfig/ComIPdu/ComIPduCounter/ComIPduCounter-ErrorNotification

Short Name	ComIPduCounterErrorNotification
Description	Name of Com_CbkCounterErr callback function to be called. If this
Introduction	parameter is omitted no I-PDU counter mismatch notification shall take place.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

1.3.1.9 Com/ComConfig/ComIPdu/ComIPduCounter/ComIPduCounter-Size

Short Name	ComIPduCounterSize
Description	Size of PDU Counter expressed in bits
Multiplicity	1..1
Value Range	1..8
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.10 Com/ComConfig/ComIPdu/ComIPduCounter/ComIPduCounter-StartPosition

Short Name	ComIPduCounterStartPosition
Description	Position of PDU counter expressed in bits from start position of data content
Introduction	of I-PDU (SDU). Note that PDU counter is not allowed to cross a byte border. The parameter ComIPduCounterStartPosition shall define the bit0 of the first byte like in little endian byte order.
Multiplicity	1..1
Value Range	0..4294967295
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.11 Com/ComConfig/ComIPdu/ComIPduCounter/ComIPduCounterThreshold

Short Name	ComIPduCounterThreshold
Description	Threshold value of I-PDU counter algorithm, see ECUC_Com_00590.
Multiplicity	0..1
Value Range	0..255
Default Value	0
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.12 Com/ComConfig/ComIPdu/ComIPduReplication

Short Name	ComIPduReplication
Description	This optional container contains the information needed for each I-PDU replicated.
Multiplicity	0..0
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

1.3.1.13 Com/ComConfig/ComIPdu/ComIPduReplication/ComIPduReplicaRef

Short Name	ComIPduReplicaRef
Description	Reference to replicas PduR PDUs of this IPDU.
Multiplicity	0..0
Default Value	-
Type	ECUC-REFERENCE-DEF

1.3.1.14 Com/ComConfig/ComIPdu/ComIPduReplication/ComIPduReplicationQuorum

Short Name	ComIPduReplicationQuorum
Description	The number of identical I-PDUs needed for successful voting.
Multiplicity	0..0
Value Range	1..3
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.15 Com/ComConfig/ComIPdu/ComIPduTimeoutNotification

Short Name	ComIPduTimeoutNotification
------------	----------------------------

Description	Name of the callback to provide a Tx/Rx timeout notification on Ipdu level to runtime environment
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

1.3.1.16 Com/ComConfig/ComIPdu/ComIpduNotificationCallback

Short Name	ComIpduNotificationCallback
Description	Notification callback function for IPDU. This function is similar to ComNotification at ComSignal or ComSignalGroup.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

1.3.1.17 Com/ComConfig/ComIPdu/ComTxIPdu/ComMetaDataDefault

Short Name	ComMetaDataDefault
Description	In case an I-PDU refers to a globally configured MetaDataType and no explicit meta data is given for a send request (e.g. by using Com_TriggerIPDUSendWithMetaData), the AUTOSAR COM module uses this configured default meta data for sending.
Introduction	The configured string shall be interpreted as a decimal representation of the characters separated by blanks, e.g. "97 98 100" means a string "abd", where the char "a" is in byte 0 (lowest address), "b" is in byte 1, and "d" is in byte 2 and (highest address).
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

1.3.1.18 Com/ComConfig/ComSignal/ComErrorNotification

Short Name	ComErrorNotification
Description	Only valid on sender side: Name of Com_CbkTxErr callback function to be called.
Introduction	If this parameter is omitted no error notification shall take place.
Multiplicity	0..1
Default Value	-

Type	ECUC-FUNCTION-NAME-DEF
------	------------------------

1.3.1.19 Com/ComConfig/ComSignal/ComInvalidNotification

Short Name	ComInvalidNotification
Description	Only valid on receiver side: Name of Com_CbkInv callback function to be called. Name of the function which notifies the RTE about the reception of an invalidated signal/ signal group. Only applicable if ComDataInvalidAction is configured to NOTIFY.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

1.3.1.20 Com/ComConfig/ComSignal/ComNotification

Short Name	ComNotification
Description	On sender side: Name of Com_CbkTxAck callback function to be called.
Introduction	On receiver side: Name of Com_CbkRxAck callback function to be called. If this parameter is omitted no notification shall take place.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

1.3.1.21 Com/ComConfig/ComSignal/ComTimeoutNotification

Short Name	ComTimeoutNotification
Description	On sender side: Name of Com_CbkTxTOut callback function to be called.
Introduction	On receiver side: Name of Com_CbkRxTOut callback function to be called.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

1.3.1.22 Com/ComConfig/ComSignalGroup/ComErrorNotification

Short Name	ComErrorNotification
Description	Only valid on sender side: Name of Com_CbkTxErr callback function to be called.
Introduction	If this parameter is omitted no error notification shall take place.
Multiplicity	0..1
Default Value	-

Type	ECUC-FUNCTION-NAME-DEF
------	------------------------

1.3.1.23 Com/ComConfig/ComSignalGroup/ComInvalidNotification

Short Name	ComInvalidNotification
Description	Only valid on receiver side: Name of Com_CbkInv call-back function to be called. Name of the function which notifies the RTE about the reception of an invalidated signal/ signal group. Only applicable if ComDataInvalidAction is configured to NOTIFY.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

1.3.1.24 Com/ComConfig/ComSignalGroup/ComNotification

Short Name	ComNotification
Description	On sender side: Name of Com_CbkTxAck callback function to be called.
Introduction	On receiver side: Name of Com_CbkRxAck callback function to be called. If this parameter is omitted no notification shall take place.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

1.3.1.25 Com/ComConfig/ComSignalGroup/ComTimeoutNotification

Short Name	ComTimeoutNotification
Description	On sender side: Name of Com_CbkTxTOut callback function to be called.
Introduction	On receiver side: Name of Com_CbkRxTOut callback function to be called.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

1.3.1.26 Com/ComGeneral/ComCommonIPduTxPeriodFeature

Short Name	ComCommonIPduTxPeriodFeature
------------	------------------------------

Description	Feature to reduce busload during flashing of other ECUs. This can be activated by calling function Com_ReduceBusload(CycleTime), CycleTime represents new cycle time in ticks (multiples of ComTxTime-Base). If this functionality is activated by calling function Com_ReduceBusload(), period of all TX IPDUs (periodic mode IPDUs and periodic part of mixed mode IPDUs) will be changed to new period.
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.27 Com/ComGeneral/ComEnableBufferForSignalProcessingImmediate

Short Name	ComEnableBufferForSignalProcessingImmediate
Description	If enabled, local buffer will be allocated to store the received I-PDU for ComIPduSignalProcessing configured as IMMEDIATE. If disabled, local buffer will not be allocated to store the received I-PDU for ComIPduSignalProcessing configured as IMMEDIATE.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.28 Com/ComGeneral/ComEnableCalibrationPhysicalRepresentation

Short Name	ComEnableCalibrationPhysicalRepresentation
Description	Enable to view Com module calibrations in its physical representation
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.29 Com/ComGeneral/ComEnableConfigInterfaces

Short Name	ComEnableConfigInterfaces
Description	If enabled then the required global interfaces for the IPDU, Signals and SignalGroups will be accessible on the reception side.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.30 Com/ComGeneral/ComEnableJitterFilter

Short Name	ComEnableJitterFilter
Description	If ComEnableJitterFilter is enabled, then ComRxTime-Base is added with the configured time out for the Rx IPDU, after the IPDU is received successfully. The feature is valid in the reception deadline monitoring for the signals and signal groups with update bit and also for the Rx IPDU.
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.31 Com/ComGeneral/ComEnablePduBasedLocks

Short Name	ComEnablePduBasedLocks
Description	If enabled, Pdu based locks shall be enabled in below API's Com_InternalMainFunctionRx, Com_RxIndication, Com_Priv_SendIpdu which avoid multiple short locks. Purpose: To optimise runtime consumption by many short locks. If disabled, existing locks shall still be present in the code which can be further enabled/ disabled through Com_Cfg_SchM.h_tpl file.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.32 Com/ComGeneral/ComEnablePeriodicFramesSync

Short Name	ComEnablePeriodicFramesSync
Description	If enabled, Com support the periodic ComTxIPdu frames synchronization. If disabled, Com will not support periodic ComTxIPdu frames synchronization.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.33 Com/ComGeneral/ComIPduControlViaCalibration

Short Name	ComIPduControlViaCalibration
------------	------------------------------

Description	If switch is enabled, a calibration variable is generated for each IPdu, with the help of which, it is possible to control operation of that IPdu. If the value of the calibration variable is configured to 0, then the IPdu is disabled from communication activity(transmission / reception). In case the value of the calibration variable for an Ipdu is configured greater than 0 OR if ComIPduControlViaCalibration switch is disabled, then the software behaves normally and a call to Com_IpduGroupStart or Com_IpduGroupStop API will result in normal operation of the IPDU.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.34 Com/ComGeneral/ComIPduControlViaRbaNdsEcuVariant

Short Name	ComIPduControlViaRbaNdsEcuVariant
Description	If switch is enabled, it is possible to control communication of any IPdu. If the IPdu status is set to disabled via Com_SetTxIPduControlViaRbaNdsEcuVariant and Com_SetRxIPduControlViaRbaNdsEcuVariant API for transmission and reception respectively, then IPDU will be disabled from communication activity. If the IPdu status is set to enabled or if ComIPduControlViaRbaNdsEcuVariant switch is disabled, then the software behaves normally and a call to Com_IPduGroupStart or Com_IPduGroupStart API will result in normal operation of the IPdu.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.35 Com/ComGeneral/ComIPduGroupControlCoreSpecificMemmap

Short Name	ComIPduGroupControlCoreSpecificMemmap
Description	If the switch is enabled, core specific memmap would be generated for the IpduGroupControl variables, and that as well require the ComIPdu's that belong to specific ComIPduGroup shall have the same EcucPartition
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.36 Com/ComGeneral/ComInitvalOptimization

Short Name	ComInitvalOptimization
Description	If set to TRUE the optimization of the InitialValues of ComSignals and ComGroupSignals will be enabled.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.37 Com/ComGeneral/ComInvalidationForDiscardedTpRxIpdu

Short Name	ComInvalidationForDiscardedTpRxIpdu
Description	Configurable(Backward compatible) switch to select whether the AUTOSAR COM module shall treat all signals of the received large I-PDU exactly like invalidated signals, whenever it is incorrectly received due to any of below scenarios: In case Com_TpRxIndication is called with value E_NOT_OK for the parameter result, When the I-PDU callout in context of Com_TpRxIndication returns false, When the large data(I-PDU) ComIPduCounter mismatch occurs.
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.38 Com/ComGeneral/ComMultiCoreSwitch

Short Name	ComMultiCoreSwitch
Description	If enabled, then some multicore related locks will be activated
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.39 Com/ComGeneral/ComProvideIpduStatusAPI

Short Name	ComProvideIpduStatusAPI
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Description	If enabled, it is possible to check the IPdu status with APIs, each for Tx and Rx IpduS COM_ProvideRxIpduStatus : Returns boolean value of the Rx Ipdu started/Stopped status; Started :True, Stopped :False COM_ProvideTxIpduStatus : Returns boolean value of the Tx Ipdu started/Stopped status; Started :True, Stopped :False The parameter for each of this API is Rx/Tx Ipdu handle Id repectively.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.40 Com/ComGeneral/ComRTESpecificSymbolicNames

Short Name	ComRTESpecificSymbolicNames
Description	If enabled, RTE specific symbolic names will be generated for Com Signals, SignalGroups and GroupSignals along with AUTOSAR specific symbolic names otherwise only AUTOSAR specific symbolic names will be generated.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.41 Com/ComGeneral/ComReadRxIPduLength

Short Name	ComReadRxIPduLength
Description	If enabled then it is possible to read the length of the received IPdu by a call to the API Com_ReadRxIPduLength The API returns COM_SERVICE_NOT_AVAILABLE in case the ComRxPduld is not in range or the corresponding IPdu Group is not started and updates the RxIPdu length with zero In normal case it returns E_OK by updating the RxIPdu length.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.42 Com/ComGeneral/ComRxDualCycleSupport

Short Name	ComRxDualCycleSupport
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Description	This parameter enables/disables to update new IPDU based reception deadline value via Com_SetRxIPduTimeoutTicks service, true: enabled, false: disabled.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.43 Com/ComGeneral/ComRxIpduFirstTimeoutCalibration

Short Name	ComRxIpduFirstTimeoutCalibration
Description	If enabled, a calibration variable Com_ti"PduShort-Name"RxFirstTout_C is generated for each reception IPDU with first ComTimeOut configured. The value of first ComTimeOut can be calibrated and this will be considered as the Rx deadline monitoring parameter for the corresponding IPDU instead of the configured ComFirstTimeout. If disabled, Rx deadline monitoring will be performed based on the value of IPDU ComFirstTimeout configured.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.44 Com/ComGeneral/ComRxIpduTimeoutCalibration

Short Name	ComRxIpduTimeoutCalibration
Description	If enabled, a calibration variable Com_ti"PduShort-Name"RxTtout_C is generated for each reception IPDU with ComTimeOut configured. The value of ComTimeOut can be calibrated and this will be considered as the Rx deadline monitoring parameter for the corresponding IPdu instead of the configured ComTimeout. If disabled, Rx deadline monitoring will be performed based on the value of IPDU ComTimeout configured.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.45 Com/ComGeneral/ComSignalGwOnTimeout

Short Name	ComSignalGwOnTimeout
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Description	Enable/disable routing of timed-out source signals that are replaced based on the ComRxDataTimeoutAction.
Introduction	true: enabled, false: disabled. Supported only for source and destination signal/signal group without ComUpdateBitPosition configured.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.46 Com/ComGeneral/ComTxIPduDeferredProcessing

Short Name	ComTxIPduDeferredProcessing
Description	If enabled, all the event pdu's will be consolidated and transmitted through Com_MainFunctionTx.
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.47 Com/ComGeneral/ComTxIPduSortByCanId

Short Name	ComTxIPduSortByCanId
Description	If enabled, order of transmission of ComTxIPdu with same periodicity will be decided on the configured CanIfTxPduCanId, lower the value higher the priority. Otherwise it is ignored.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.48 Com/ComGeneral/ComTxIpduNumberOfRepetitionCalibration

Short Name	ComTxIpduNumberOfRepetitionCalibration
Description	If enabled, a calibration variable Com_nrOfRepn"ComIPduShortName"_C is generated for each transmission IPdu with default repetition value. The value of repetition can be calibrated and this will be considered as the number of repetition for IPdu. If disabled, repetition will be performed based on the value of IPDU repetition configured. Please Note: This is applicable only for the Tx Ipdu's with Direct/Mixed transmission mode E.g. Calibration to 0 → Message is transmitted one time after a request Calibration to 5 → message is transmitted 5 + 1 = 6 times after request.
Multiplicity	0..1

Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.49 Com/ComGeneral/ComTxModeCyclePeriodCalibration

Short Name	ComTxModeCyclePeriodCalibration
Description	If enabled, a calibration variable is generated for each Transmission IPDU with name Com_ti"PduShort-Name"TxPerd_C. The value in this variable will be considered as the transmission period factor for the corresponding IPdu instead of the the configured ComTxModeTimePeriod. If disabled, then the Tx PDU transmission period behavior will be normal. Please Note: 1. This is applicable only for the Tx Ipdus with Periodic/Mixed transmission mode 2. The value assigned to this should be a multiple of ComTxTime-Base.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.50 Com/ComGeneral/ComTxModeTimeOffsetCalibration

Short Name	ComTxModeTimeOffsetCalibration
Description	If enabled, a calibration variable Com_ti"PduShort-Name"TxTOffset_C is generated for each Cyclic Tx IPDU's with ComTxModeTimeOffset configured. The value of ComTxModeTimeOffset can be calibrated and this will be considered as the Offset for the cyclic type Pdu's(PERIODIC or MIXED) for the corresponding IPdu instead of the configured ComTxModeTimeOffset. If disabled, Offset for the cyclic type Pdu's will be performed based on the value of IPDU ComTxMode-TimeOffset configured.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.51 Com/ComGeneral/ComTxScheduledInDiffTimebase

Short Name	ComTxScheduledInDiffTimebase
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Description	If enabled, then Com_IsTxScheduled(PduIdType ComTxPduId, uint16 ComCallerTaskCycle) function can be called in different task cycle as Com_MainfunctionTx. If disabled, then Com_IsTxScheduled has to be called in same task cycle as Com_MainfunctionTx. When this feature is enabled, the Com_IsTxScheduled API shall use "ComCallerTaskCycle" parameter. Where ComCallerTaskCycle is in milli-seconds, i.e., for 1ms, ComCallerTaskCycle has to be passed as 1.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.2 Additional Services

The following API are available in version v12.7.0 in addition to those defined in AR v22-11

1.3.2.1 Com_ReduceBusload

Syntax	void Com_ReduceBusload(uint16 CommonPeriod)
Parameters	CommonPeriod New period (interms of multiples of Com_mainFunctionTx() raster time)
Description	Changes the Periodicity of all the Tx frames.

1.3.2.2 Com_RestoreBusload

Syntax	void Com_RestoreBusload(void)
Parameters	None
Description	Changes the period of all the Tx frames to a old period from common TX period, set by function Com_ReduceBusload().

1.3.2.3 Com_IsTxScheduled

Syntax	boolean Com_IsTxScheduled(PduIdType ComTxPduId, uint16 ComCallerTaskCycle)
Parameters	ComTxPduId Tx ipdu ID ComCallerTaskCycle Value of caller's task cycle in ms
Description	Checks whether the Tx frame is scheduled to be sent in the coming cycle.

1.3.2.4 Com_ReadRxIPduLength

Syntax	uint8 Com_ReadRxIPduLength(PduIdType ComRxPduId, PduLengthType RxIPduLengthPtr)
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Parameters	ComRxPduId Id of the I-PDU of which the transmission mode shall be changed RxIPduLengthPtr The transmission mode that shall be set
Description	Reads the length of the received IPdu.

1.3.2.5 Com_ProvideRxIpduStatus

Syntax	boolean Com_ProvideRxIpduStatus(PduIdType PduId)
Parameters	PduId Id of the I-PDU of which the IPDU status is required
Description	Returns the status of Rx I-PDU referenced by PduId.

1.3.2.6 Com_ProvideTxIpduStatus

Syntax	boolean Com_ProvideTxIpduStatus(PduIdType PduId)
Parameters	PduId Id of the I-PDU of which the IPDU status is required
Description	Returns the status of Tx I-PDU referenced by PduId.

1.3.2.7 Com_SetRxIPduControlViaRbaNdsEcuVariant

Syntax	void Com_SetRxIPduControlViaRbaNdsEcuVariant(PduIdType IpduId, boolean IpduStatus)
Description	Invoked by Rba_ComScl to set/reset the status of Rx Ipdu.

1.3.2.8 Com_SetTxIPduControlViaRbaNdsEcuVariant

Syntax	void Com_SetTxIPduControlViaRbaNdsEcuVariant(PduIdType IpduId, boolean IpduStatus)
Description	Invoked by Rba_ComScl to set/reset the status of Tx Ipdu.

1.3.2.9 Com_SetRxIPduTimeoutTicks

Syntax	Com_SetRxIPduTimeoutTicks(PduIdType rxPduId_uo, uint16 timeoutTicks_u16)
Parameters	rxPduId_uo Id of the Rx IPDU timeoutTicks_u16 Rx IPDU deadline value in 'ticks', i.e., multiple of referred Main-FunctionRx
Description	Updates new dual-cycle-time IPDU based reception deadline value.

1.4 Limitations

The following features are not supported for version v12.7.0

1.4.1 General Limitations

- Post build Loadable

- Communication Protection
- Transmission mode selection support on the routed Tx Signals or Signal Groups
- Routing of type 1:n for Com Signals and Signal groups
- Read the source and destination description for source and destination Signals

1.4.2 Limitations on Group Signals and Signal Groups

- Filters are not supported for signed Group Signals.
- Filters are not supported for Rx Group Signals.
- If the upper layer reads a Group Signal with an associated cleared update-bit, the last received value is returned.
- A Signal group is considered as BigEndian if the endianness of the Group Signals of the Signal Group is BigEndian.
- All Group Signals within the Signal Group should have the same signal endianness.
- Group Signals are not allowed to have a zero bit size.

1.4.3 Limitations on configuration of Signal Gateway

- The Signal Gateway cannot be used for DIRECT-N and MIXED Transmission Mode I-PDUs, only PERIODIC is supported.
- 1:n routing is not supported.
- A TMS switch is not possible for a Gateway I-PDU (i.e. an I-PDU with signals to be routed).
- If a routed source Signal or source Signal Group has update-bit configured, and if the update-bit is not set, then the value used for TMS for this signal will be the old value.

1.4.4 Miscellaneous Limitations

If a TxIPdu with an application metadata transmission request fails due to the return of E_NOT_OK from PduR_ComTransmit, or discarded from Tx-ComIPdu-Callout, the caller provided metadata will not be stored for further transmissions, irrespective of the configuration of ComRetryFailedTransmitRequests.

- Notification is called without checking update-bit in Com_RxIndication.
- Error Notifications for I-PDUs do not have any parameters passed.
- Data Invalid for negative values is not supported.
- The Group Signal Data Invalid notification function will not be called, only the Data Invalid notification for Signal Groups is supported.
- The ComTxDoubleBuffer switch is ignored, so the TriggerTransmit clearing

of update-bits is done only for the second buffer of an I-PDU.

1.4.5 ComIPduGroup (In ComGroupSignal/ComGwSource)

Support of unreferred ComIPduGroup(s) either in ComIPdu/ComIPduGroup is not currently provided.

1.4.6 ComFilterMask/ComFilterX/ComFilterMin/ComFilterMax (In ComCongif/ComFilter)

Currently negative values are not supported for these parameters.

1.5 Exclusions

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- Com/ComConfig/ComDataMemSize
- Com/ComConfig/ComGwMapping/ComGwDestination/ComGwDestinationDescription/ComGwIPduRef
- Com/ComConfig/ComGwMapping/ComGwSource/ComGwSourceDescription/ComGwIPduRef
- Com/ComConfig/ComIPdu/ComMainFunctionRouteSignalsRef
- Com/ComConfig/ComIPdu/ComTxIPdu/ComMetaDataDefaultItem
- Com/ComConfig/ComIPdu/ComTxIPdu/ComMetaDataDefaultItem/ComMetaDataDefaultValue
- Com/ComConfig/ComIPdu/ComTxIPdu/ComMetaDataDefaultItem/ComMetaDataItemRef
- Com/ComConfig/ComMainFunctionRouteSignals
- Com/ComConfig/ComMainFunctionRouteSignals/ComMainRouteSignalsPartitionRef
- Com/ComConfig/ComMainFunctionRouteSignals/ComMainRouteSignalsTimeBase
- Com/ComConfig/ComMaxIPduCnt
- Com/ComConfig/ComSignal/ComInitialValueOnly
- Com/ComConfig/ComSignalGroup/ComInitialValueOnly
- Com/ComGeneral/ComSupportedIPduGroups

1.6 Deviations

1.6.1 Configuration Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

1.6.1.1 Com/ComConfig/ComGwMapping/ComGwDestination/ComG-

wDestinationDescription/ComBitPosition

[RTA-BSW] Description	Starting position within the I-PDU. This parameter refers to the position in the I-PDU and not in the shadow buffer. If the endianness conversion is configured to Opaque the parameter ComBitPosition shall define the bit0 of the first byte like in little endian byte order. If signal endianness is BigEndian, then MSB(bit) shall be configured, otherwise LSB(bit) shall be configured.
[AUTOSAR] Description	Starting position within the I-PDU. This parameter refers to the position in the I-PDU and not in the shadow buffer. If the endianness conversion is configured to Opaque the parameter ComBitPosition shall define the bit0 of the first byte like in little endian byte order

1.6.1.2 Com/ComConfig/ComGwMapping/ComGwDestination/ComGwDestinationDescription/ComFilter/ComFilterAlgorithm

[RTA-BSW] Description	The range of values is specified below: Always: Filter evaluates to true. MaskedNewDiffersMaskedOld: Filter evaluates to true, if the masked signal value has changed. MaskedNewDiffersX: Filter evaluates to true, if masked signal value is not equal to a specific(FilterX) value. MaskedNewEqualsX: Filter evaluates to true, if masked signal value is equal to a specific(FilterX) value. Never: Filter evaluates to false. NewIsOutside: Filter evaluates to true, if signal value is outside a predefined(FilterMax, FilterMin) boundary. NewIsWithin: Filter evaluates to true, if signal value is within a predefined(FilterMax, FilterMin) boundary. OneEveryN: Filter evaluates to true, once every N message occurrences.
[AUTOSAR] Description	The range of values is specified in the [17] specification, chapter 2.2.2, Reception Filtering.

1.6.1.3 Com/ComConfig/ComGwMapping/ComGwDestination/ComGwDestinationDescription/ComTransferProperty

[RTA-BSW] Description: Defines if a write access to this signal can trigger the transmission of the corresponding I-PDU. If the I-PDU is triggered, depends also on the transmission mode of the corresponding I-PDU.

PENDING: A write access to this signal never triggers the transmission of the corresponding I-PDU.

TRIGGERED: Depending on the transmission mode, a write access to this signal can trigger the transmission of the corresponding I-PDU.

TRIGGERED_ON_CHANGE: Depending on the transmission mode, a write access

to this signal can trigger the transmission of the corresponding I-PDU, but only in case the written value is different to the locally stored (last sent or initial value) in length or value.

TRIGGERED_ON_CHANGE_WITHOUT_REPETITION: Depending on the transmission mode, a write access to this signal can trigger the transmission of the corresponding I-PDU just once without a repetition, but only in case the written value is different to the locally stored (last sent or initial value) in length or value.

TRIGGERED_WITHOUT_REPETITION: Depending on the transmission mode, a write access to this signal can trigger the transmission of the corresponding I-PDU just once without a repetition.

[AUTOSAR] Description: Defines if a write access to this signal can trigger the transmission of the corresponding I-PDU. If the I-PDU is triggered, depends also on the transmission mode of the corresponding I-PDU.

1.6.1.4 Com/ComConfig/ComGwMapping/ComGwSource/ComGwSourceDescription/ComBitPosition

[RTA-BSW] Description	Starting position within the I-PDU. This parameter refers to the position in the I-PDU and not in the shadow buffer. If the endianness conversion is configured to Opaque the parameter ComBitPosition shall define the bit0 of the first byte like in little endian byte order. If signal endianness is BigEndian, then MSB(bit) shall be configured, otherwise LSB(bit) shall be configured.
[AUTOSAR] Description	Starting position within the I-PDU. This parameter refers to the position in the I-PDU and not in the shadow buffer. If the endianness conversion is configured to Opaque the parameter ComBitPosition shall define the bit0 of the first byte like in little endian byte order

1.6.1.5 Com/ComConfig/ComIPdu/ComFirstTimeout

[RTA-BSW] Parent Container	Com/ComConfig/ComIPdu/ComFirstTimeout
[AUTOSAR] Parent Container	Com/ComConfig/ComSignalGroup/ComFirstTimeout
[RTA-BSW] Description	Defines the length of the first deadline monitoring timeout period in seconds. This timeout is used immediately after start (or restart) of the deadline monitoring service. The timeout period of the successive periods is configured by ComTimeout.
[AUTOSAR] Description	Defines the length of the first deadline monitoring timeout period in seconds. This timeout is used immediately after start (or restart) of the deadline monitoring service. The timeout period of the successive periods is configured by ECUC_Com_00263.

1.6.1.6 Com/ComConfig/ComIPdu/ComIPduDirection

[RTA-BSW] Description	The direction defines if this I-PDU, and therefore the contributing signals and signal groups, shall be sent (ComIPduDirection: SEND) or received (-ComIPduDirection: RECEIVE).
[AUTOSAR] Description	The direction defines if this I-PDU, and therefore the contributing signals and signal groups, shall be sent or received.

1.6.1.7 Com/ComConfig/ComIPdu/ComIPduMainFunctionRef

[RTA-BSW] Description	Reference to the Com_MainFunctionRx/Com_MainFunctionTx this I-PDU belongs to. Tags: atp.Status=draft
[AUTOSAR] Description	Reference to the Com_MainFunctionRx/Com_MainFunctionTx this I-PDU belongs to.

1.6.1.8 Com/ComConfig/ComIPdu/ComIPduSignalProcessing

[RTA-BSW] Description	For the definition of the two modes Immediate and Deferred. DEFERRED: signal indication / confirmations are deferred for example to a cyclic task. IMMEDIATE: the signal indications / confirmations are performed in Com_RxIndication/Com_TxConfirmation.
[AUTOSAR] Description	For the definition of the two modes Immediate and Deferred.

1.6.1.9 Com/ComConfig/ComIPdu/ComIPduType

[RTA-BSW] Description	Defines if this I-PDU is a normal I-PDU that can be sent unfragmented (-ComIPduType: NORMAL) or if this is a large I-PDU that shall be sent via the Transport Protocol of the underlying bus. (ComIPduType: TP)
[AUTOSAR] Description	Defines if this I-PDU is a normal I-PDU that can be sent unfragmented or if this is a large I-PDU that shall be sent via the Transport Protocol of the underlying bus.

1.6.1.10 Com/ComConfig/ComIPdu/ComTimeout

[RTA-BSW] Parent Container	Com/ComConfig/ComIPdu/ComTimeout
[AUTOSAR] Parent Container	Com/ComConfig/ComSignalGroup/ComTimeout
[RTA-BSW] Description	Defines the length of the deadline monitoring timeout period in seconds. The period for the first timeout period can be configured separately by ComFirstTimeout.

[AUTOSAR] Description	Defines the length of the deadline monitoring timeout period in seconds. The period for the first timeout period can be configured separately by ECUC_Com_00183.
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1.6.1.11 Com/ComConfig/ComIPdu/ComTxIPdu/ComTxIPduClearUpdateBit

[RTA-BSW] Description	Defines when the update-bits of signals or signal groups, contained in this I-PDU, will be cleared. Confirmation: The update-bits are cleared when the transmission of the I-PDU was confirmed. Transmit: The update-bits are cleared directly after the invocation of PduR_ComTransmit. TriggerTransmit: The update-bits are cleared after the I-PDU was fetched via Com_TriggerTransmit.
[AUTOSAR] Description	Defines when the update-bits of signals or signal groups, contained in this I-PDU, will be cleared.

1.6.1.12 Com/ComConfig/ComIPdu/ComTxIPdu/ComTxModeFalse/ComTxMode/ComTxModeMode

[RTA-BSW] Enumeration Literals	DIRECT, MIXED, NONE, PERIODIC, MIXED_PHASE_SHIFT
[AUTOSAR] Enumeration Literals	DIRECT, MIXED, NONE, PERIODIC

1.6.1.13 Com/ComConfig/ComIPdu/ComTxIPdu/ComTxModeFalse/ComTxMode/ComTxModeNumberOfRepetitions

[RTA-BSW] Maximum Value	254
[AUTOSAR] Maximum Value	255

1.6.1.14 Com/ComConfig/ComIPdu/ComTxIPdu/ComTxModeTrue/ComTxMode/ComTxModeMode

[RTA-BSW] Enumeration Literals	DIRECT, MIXED, NONE, PERIODIC, MIXED_PHASE_SHIFT
[AUTOSAR] Enumeration Literals	DIRECT, MIXED, NONE, PERIODIC

1.6.1.15 Com/ComConfig/ComIPdu/ComTxIPdu/ComTxModeTrue/ComTxMode/ComTxModeNumberOfRepetitions

[RTA-BSW] Maximum Value	254
[AUTOSAR] Maximum Value	255

1.6.1.16 Com/ComConfig/ComIPduGroup/ComIPduGroupHandleId

[RTA-BSW] Introduction	The ComIPduGroupHandleId is required by the API calls to start and stop I-PDU Groups.
[AUTOSAR] Introduction	The ComIPduGroupHandleId is required by the API calls to start and stop I-PDU Groups. Range: 0 .. (ComSupportedIPduGroups-1)

1.6.1.17 Com/ComConfig/ComMainFunctionRx

[RTA-BSW] Description	Each element of this container defines one instance of Com_MainFunctionRx. Tags: atp.Status=draft
[AUTOSAR] Description	Each element of this container defines one instance of Com_MainFunctionRx.

1.6.1.18 Com/ComConfig/ComMainFunctionRx/ComMainRxPartitionRef

[RTA-BSW] Description	Reference to EcucPartition, where the according Com_MainFunction instance is assigned to. Tags: atp.Status=draft
[AUTOSAR] Description	Reference to EcucPartition, where the according Com_MainFunction instance is assigned to.

1.6.1.19 Com/ComConfig/ComMainFunctionRx/ComMainRxTimeBase

[RTA-BSW] Description	The period between successive calls to according instance of Com_MainFunctionRx in seconds. This parameter may be used by the COM generator to transform the values of the reception related timing configuration parameters of the COM module to internal implementation specific counter or tick values. The COM module's internal timing handling is implementation specific. Tags: atp.Status=draft
[AUTOSAR] Description	The period between successive calls to according instance of Com_MainFunctionRx in seconds. This parameter may be used by the COM generator to transform the values of the reception related timing configuration parameters of the COM module to internal implementation specific counter or tick values. The COM module's internal timing handling is implementation specific.

1.6.1.20 Com/ComConfig/ComMainFunctionTx

[RTA-BSW] Description	Each element of this container defines one instance of Com_MainFunctionTx. Tags: atp.Status=draft
[AUTOSAR] Description	Each element of this container defines one instance of Com_MainFunctionTx.

1.6.1.21 Com/ComConfig/ComMainFunctionTx/ComMainTxPartitionRef

[RTA-BSW] Description	Reference to EcucPartition, where the according Com_Main-Function instance is assigned to. Tags: atp.Status=draft
[AUTOSAR] Description	Reference to EcucPartition, where the according Com_Main-Function instance is assigned to.

1.6.1.22 Com/ComConfig/ComMainFunctionTx/ComMainTxTimeBase

[RTA-BSW] Description	The period between successive calls to according instance of Com_MainFunctionTx in seconds. This parameter may be used by the COM generator to transform the values of the reception related timing configuration parameters of the COM module to internal implementation specific counter or tick values. The COM module's internal timing handling is implementation specific. Tags: atp.Status=draft
[AUTOSAR] Description	The period between successive calls to according instance of Com_MainFunctionTx in seconds. This parameter may be used by the COM generator to transform the values of the reception related timing configuration parameters of the COM module to internal implementation specific counter or tick values. The COM module's internal timing handling is implementation specific.

1.6.1.23 Com/ComConfig/ComMainFunctionTx/ComPreparationNotification

[RTA-BSW] Description	This callback function indicates that the signals/signal groups to be sent via a dedicated Com_MainFunctionTx instance will now be prepared for transmission. Tags: atp.Status=draft
[AUTOSAR] Description	This callback function indicates that the signals/signal groups to be sent via a dedicated Com_MainFunctionTx instance will now be prepared for transmission.

1.6.1.24 Com/ComConfig/ComSignal/ComBitPosition

[RTA-BSW] Description	Starting position within the I-PDU. This parameter refers to the position in the I-PDU and not in the shadow buffer. If the endianness conversion is configured to Opaque the parameter ComBitPosition shall define the bit0 of the first byte like in little endian byte order. If signal endianness is BigEndian, then MSB(bit) shall be configured, otherwise LSB(bit) shall be configured.
[AUTOSAR] Description	Starting position within the I-PDU. This parameter refers to the position in the I-PDU and not in the shadow buffer. If the endianness conversion is configured to Opaque the parameter ComBitPosition shall define the bit0 of the first byte like in little endian byte order

1.6.1.25 Com/ComConfig/ComSignal/ComFilter/ComFilterAlgorithm

[RTA-BSW] Description	The range of values is specified below: Always: Filter evaluates to true. MaskedNewDiffersMaskedOld: Filter evaluates to true, if the masked signal value has changed. MaskedNewDiffersX: Filter evaluates to true, if masked signal value is not equal to a specific(FilterX) value. MaskedNewEqualsX: Filter evaluates to true, if masked signal value is equal to a specific(FilterX) value. Never: Filter evaluates to false. NewIsOutside: Filter evaluates to true, if signal value is outside a predefined(FilterMax, FilterMin) boundary. NewIsWithin: Filter evaluates to true, if signal value is within a predefined(FilterMax, FilterMin) boundary. OneEveryN: Filter evaluates to true, once every N message occurrences.
[AUTOSAR] Description	The range of values is specified in the [17] specification, chapter 2.2.2, Reception Filtering.

1.6.1.26 Com/ComConfig/ComSignal/ComRxDataTimeoutAction

[RTA-BSW] Description	This parameter defines the action performed upon expiration of the reception deadline monitoring timer. NONE: no replacement shall take place. REPLACE: signals shall be replaced by their ComSignalInitValue. SUBSTITUTE: signals shall be replaced by their ComTimeoutSubstitutionValue.
[AUTOSAR] Description	This parameter defines the action performed upon expiration of the reception deadline monitoring timer.

1.6.1.27 Com/ComConfig/ComSignal/ComTimeoutSubstitutionValue

[RTA-BSW] Introduction	In case the ComSignalType is UINT8, UINT16, UINT32, UINT64, SINT8, SINT16, SINT32, SINT64 the string shall be interpreted as defined in the chapter Integer Type in the AUTOSAR EcuC specification. In case the ComSignalType is FLOAT32, FLOAT64 the string shall be interpreted as defined in the chapter Float Type in the AUTOSAR EcuC specification. In case the ComSignalType is BOOLEAN the string shall be interpreted as defined in the chapter Boolean Type in the AUTOSAR EcuC specification. In case the ComSignal is a UINT8_N, UINT8_DYN the string shall be interpreted as a decimal representation of the characters separated by blanks, e.g. "97 98 100" means a string "abd", where the char "a" is in byte 0 (lowest address), "b" is in byte 1, and "d" is in byte 2 and (highest address). For the ComSignalType UINT8_DYN the dynamic length shall be set to the number of configured characters. An empty string "" shall be interpreted as 0-sized dynamic signal.
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[AUTOSAR] Introduction	In case of <code>UINT8_N</code> the default value is a string of length <code>ComSignalLength</code> with all bytes set to <code>0x00</code>. In case of <code>UINT8_DYN</code> the initial size shall be 0. In case the <code>ComSignalType</code> is <code>UINT8</code>, <code>UINT16</code>, <code>UINT32</code>, <code>UINT64</code>, <code>SINT8</code>, <code>SINT16</code>, <code>SINT32</code>, <code>SINT64</code> the string shall be interpreted as defined in the chapter Integer Type in the AUTOSAR EcuC specification. In case the <code>ComSignalType</code> is <code>FLOAT32</code>, <code>FLOAT64</code> the string shall be interpreted as defined in the chapter Float Type in the AUTOSAR EcuC specification. In case the <code>ComSignalType</code> is <code>BOOLEAN</code> the string shall be interpreted as defined in the chapter Boolean Type in the AUTOSAR EcuC specification. In case the <code>ComSignal</code> is a <code>UINT8_N</code>, <code>UINT8_DYN</code> the string shall be interpreted as a decimal representation of the characters separated by blanks, e.g. "97 98 100" means a string "abd", where the char "a" is in byte 0 (lowest address), "b" is in byte 1, and "d" is in byte 2 and (highest address). For the <code>ComSignalType</code> <code>UINT8_DYN</code> the dynamic length shall be set to the number of configured characters. An empty string "" shall be interpreted as 0-sized dynamic signal.
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1.6.1.28 Com/ComConfig/ComSignal/ComTransferProperty

[ADDED] Default Value	PENDING
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[RTA-BSW] Description: Defines if a write access to this signal can trigger the transmission of the corresponding I-PDU. If the I-PDU is triggered, depends also on the transmission mode of the corresponding I-PDU.

PENDING: A write access to this signal never triggers the transmission of the corresponding I-PDU.

TRIGGERED: Depending on the transmission mode, a write access to this signal can trigger the transmission of the corresponding I-PDU.

TRIGGERED_ON_CHANGE: Depending on the transmission mode, a write access to this signal can trigger the transmission of the corresponding I-PDU, but only in case the written value is different to the locally stored (last sent or initial value) in length or value.

TRIGGERED_ON_CHANGE_WITHOUT_REPETITION: Depending on the transmission mode, a write access to this signal can trigger the transmission of the corresponding I-PDU just once without a repetition, but only in case the written value is different to the locally stored (last sent or initial value) in length or value.

TRIGGERED_WITHOUT_REPETITION: Depending on the transmission mode, a

write access to this signal can trigger the transmission of the corresponding I-PDU just once without a repetition.

[AUTOSAR] Description: Defines if a write access to this signal can trigger the transmission of the corresponding I-PDU. If the I-PDU is triggered, depends also on the transmission mode of the corresponding I-PDU.

1.6.1.29 Com/ComConfig/ComSignalGroup/ComGroupSignal/ComBitPosition

[RTA-BSW] Description	Starting position within the I-PDU. This parameter refers to the position in the I-PDU and not in the shadow buffer. If the endianness conversion is configured to Opaque the parameter ComBitPosition shall define the bit0 of the first byte like in little endian byte order. If signal endianness is BigEndian, then MSB(bit) shall be configured, otherwise LSB(bit) shall be configured.
[AUTOSAR] Description	Starting position within the I-PDU. This parameter refers to the position in the I-PDU and not in the shadow buffer. If the endianness conversion is configured to Opaque the parameter ComBitPosition shall define the bit0 of the first byte like in little endian byte order

1.6.1.30 Com/ComConfig/ComSignalGroup/ComGroupSignal/ComFilter/ComFilterAlgorithm

[RTA-BSW] Description	The range of values is specified below: Always: Filter evaluates to true. MaskedNewDiffersMaskedOld: Filter evaluates to true, if the masked signal value has changed. MaskedNewDiffersX: Filter evaluates to true, if masked signal value is not equal to a specific(FilterX) value. MaskedNewEqualsX: Filter evaluates to true, if masked signal value is equal to a specific(FilterX) value. Never: Filter evaluates to false. NewsOutside: Filter evaluates to true, if signal value is outside a predefined(FilterMax, FilterMin) boundary. NewsWithin: Filter evaluates to true, if signal value is within a predefined(FilterMax, FilterMin) boundary. OneEveryN: Filter evaluates to true, once every N message occurrences.
[AUTOSAR] Description	The range of values is specified in the [17] specification, chapter 2.2.2, Reception Filtering.

1.6.1.31 Com/ComConfig/ComSignalGroup/ComGroupSignal/ComTimeout-SubstitutionValue

[RTA-BSW] Introduction	In case the ComSignalType is UINT8, UINT16, UINT32, UINT64, SINT8, SINT16, SINT32, SINT64 the string shall be interpreted as defined in the chapter Integer Type in the AUTOSAR EcuC specification. In case the ComSignalType is FLOAT32, FLOAT64 the string shall be interpreted as defined in the chapter Float Type in the AUTOSAR EcuC specification. In case the ComSignalType is BOOLEAN the string shall be interpreted as defined in the chapter Boolean Type in the AUTOSAR EcuC specification. In case the ComSignal is a UINT8_N, UINT8_DYN the string shall be interpreted as a decimal representation of the characters separated by blanks, e.g. "97 98 100" means a string "abd", where the char "a" is in byte 0 (lowest address), "b" is in byte 1, and "d" is in byte 2 and (highest address). For the ComSignalType UINT8_DYN the dynamic length shall be set to the number of configured characters. An empty string "" shall be interpreted as 0-sized dynamic signal.
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[AUTOSAR] Introduction	In case of <code>UINT8_N</code> the default value is a string of length <code>ComSignalLength</code> with all bytes set to <code>0x00</code>. In case of <code>UINT8_DYN</code> the initial size shall be 0. In case the <code>ComSignalType</code> is <code>UINT8</code>, <code>UINT16</code>, <code>UINT32</code>, <code>UINT64</code>, <code>SINT8</code>, <code>SINT16</code>, <code>SINT32</code>, <code>SINT64</code> the string shall be interpreted as defined in the chapter Integer Type in the AUTOSAR EcuC specification. In case the <code>ComSignalType</code> is <code>FLOAT32</code>, <code>FLOAT64</code> the string shall be interpreted as defined in the chapter Float Type in the AUTOSAR EcuC specification. In case the <code>ComSignalType</code> is <code>BOOLEAN</code> the string shall be interpreted as defined in the chapter Boolean Type in the AUTOSAR EcuC specification. In case the <code>ComSignal</code> is a <code>UINT8_N</code>, <code>UINT8_DYN</code> the string shall be interpreted as a decimal representation of the characters separated by blanks, e.g. "97 98 100" means a string "abd", where the char "a" is in byte 0 (lowest address), "b" is in byte 1, and "d" is in byte 2 and (highest address). For the <code>ComSignalType</code> <code>UINT8_DYN</code> the dynamic length shall be set to the number of configured characters. An empty string "" shall be interpreted as 0-sized dynamic signal.
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1.6.1.32 Com/ComConfig/ComSignalGroup/ComGroupSignal/ComTransferProperty

[RTA-BSW] Description	Optionally defines whether this group signal shall contribute to the <code>TRIGGERED_ON_CHANGE</code> transfer property of the signal group. If at least one group signal of a signal group has the "ComTransferProperty" configured all other group signals of that signal group shall have the attribute configured as well. PENDING: A change of the value of this group signal shall not be considered in the evaluation of the signal groups <code>ComTransferProperty</code>. TRIGGERED_ON_CHANGE: A change of the value of this group signal shall be considered in the evaluation of the signal groups <code>ComTransferProperty</code>.
[AUTOSAR] Description	Optionally defines whether this group signal shall contribute to the <code>TRIGGERED_ON_CHANGE</code> transfer property of the signal group. If at least one group signal of a signal group has the "ComTransferProperty" configured all other group signals of that signal group shall have the attribute configured as well.

1.6.1.33 Com/ComConfig/ComSignalGroup/ComRxDataTimeoutAction

[RTA-BSW] Description	This parameter defines the action performed upon expiration of the reception deadline monitoring timer. NONE: no replacement shall take place. REPLACE: signals shall be replaced by their ComSignalInitValue. SUBSTITUTE: signals shall be replaced by their ComTimeoutSubstitutionValue.
[AUTOSAR] Description	This parameter defines the action performed upon expiration of the reception deadline monitoring timer.

1.6.1.34 Com/ComConfig/ComSignalGroup/ComTransferProperty

[ADDED] Default Value	PENDING
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[RTA-BSW] Description: Defines if a write access to this signal can trigger the transmission of the corresponding I-PDU. If the I-PDU is triggered, depends also on the transmission mode of the corresponding I-PDU.

PENDING: A write access to this signal never triggers the transmission of the corresponding I-PDU.

TRIGGERED: Depending on the transmission mode, a write access to this signal can trigger the transmission of the corresponding I-PDU.

TRIGGERED_ON_CHANGE: Depending on the transmission mode, a write access to this signal can trigger the transmission of the corresponding I-PDU, but only in case the written value is different to the locally stored (last sent or initial value) in length or value.

TRIGGERED_ON_CHANGE_WITHOUT_REPETITION: Depending on the transmission mode, a write access to this signal can trigger the transmission of the corresponding I-PDU just once without a repetition, but only in case the written value is different to the locally stored (last sent or initial value) in length or value.

TRIGGERED_WITHOUT_REPETITION: Depending on the transmission mode, a write access to this signal can trigger the transmission of the corresponding I-PDU just once without a repetition.

[AUTOSAR] Description: Defines if a write access to this signal can trigger the transmission of the corresponding I-PDU. If the I-PDU is triggered, depends also on the transmission mode of the corresponding I-PDU.

1.6.1.35 Com/ComConfig/ComUserModule

[RTA-BSW] Description	Contains the configuration parameters of the Com user modules.
[AUTOSAR] Description	Contains the configuration parameters of the Com user modules. Tags: atp.Status=draft

1.6.1.36 Com/ComConfig/ComUserModule/ComUserModuleCnfRef

[RTA-BSW] Description	Reference to the Com user module configuration.
[AUTOSAR] Description	Reference to the Com user module configuration. Tags: atp.Status=draft

1.7 API Definition

List of supported API

Com_CopyRxData	Called to provide the received data of an IPDU segment (N-PDU) to the upper layer.
Com_CopyTxData	Called to acquire the transmit data of an IPDU segment (N-PDU).
Com_DeInit	Called by EcuM to de-Initializes the Com module.
Com_DisableReceptionDM	Disables reception Deadline Monitoring for the I-PDUs within the given I-PDU group.
Com_EnableReceptionDM	Enables reception Deadline Monitoring for the I-PDUs within the given I-PDU group.
Com_GetStatus	Returns the status of the AUTOSAR COM module.
Com_GetVersionInfo	Returns the version information of the called Com module.
Com_Init	Called by EcuM to initializes the Com module.
Com_InvalidateSignal	Invalidates the signal with the given SignalId by setting its value to its configured ComSignalDataInvalidValue.
Com_InvalidateSignalGroup	Invalidates all group signals of the signal group with the given SignalGroupId by setting their values to their configured ComSignalDataInvalidValues.
Com_IpduGroupStart	Starts a preconfigured I-PDU group.
Com_IpduGroupStop	Stops a preconfigured I-PDU group.
Com_MainFunctionRx_<shortName>	Performs the processing of the AUTOSAR COM module's receive processing that are not directly handled within the COM's functions invoked by the PDUR.
Com_MainFunctionTx_<shortName>	Performs the processing of the AUTOSAR COM module's transmission activities that are not directly handled within the COM's function invoked by the RTE.

Com_ReceiveDynSignal	Copies the data of the signal identified by SignalId to the location specified by SignalDataPtr and stores the length of the dynamical length signal at the position given by the Length parameter.
Com_ReceiveDynSignalWithMetaData	Copies the data of the signal identified by SignalId to the location specified by SignalDataPtr and stores the length of the dynamical length signal at the position given by the Length parameter and metadata is stored at MetaDataPtr.
Com_ReceiveSignal	Copies the data of the signal identified by SignalId to the location specified by SignalDataPtr.
Com_ReceiveSignalWithMetaData	Copies the data of the signal identified by SignalId to the location specified by SignalDataPtr and metadata is stored at MetaDataPtr.
Com_ReceiveSignalGroup	Copies the received signal group from the IPDU to the shadow buffer.
Com_ReceiveSignalGroupWithMetaData	Copies the signal group from the IPDU to the shadow buffer and metadata is stored at MetaDataPtr.
Com_ReceiveSignalGroupArray	Copies the received signal group array representation from the PDU to the location designated by SignalGroupArrayPtr.
Com_ReceiveSignalGroupArrayWithMetaData	Copies the received signal group array representation from PDU to the location designated by SignalGroupArrayPtr and metadata is stored at MetaDataPtr.
Com_RxIndication	Indication of a received IPDU from a lower layer communication interface module.
Com_SendDynSignal	Updates the signal object identified by SignalId with the signal referenced by the SignalDataPtr parameter.
Com_SendDynSignalWithMetaData	Updates the signal object identified by SignalId with the signal referenced by SignalDataPtr. The meta data is provided via the MetaDataPtr parameter.
Com_SendSignal	Updates the signal object identified by SignalId with the signal referenced by the SignalDataPtr parameter.
Com_SendSignalWithMetaData	Updates signal identified by SignalId with the signal referenced by SignalDataPtr and meta data provided via the MetadataPtr parameter.

Com_SendSignalGroup	Copies the content of the associated shadow buffer to the associated IPDU.
Com_SendSignalGroupWithMetaData	Copies the content of the associated shadow buffer to the associated I-PDU, meta data is provided via the MetaDataPtr parameter.
Com_SendSignalGroupArray	Copies the content of the provided Signal-GroupArrayPtr to the associated IPDU. The provided data shall correspond to the array representation of the signal group.
Com_SendSignalGroupArrayWithMetaData	Copies the content of the provided Signal-GroupArrayPtr to the associated I-PDU, meta data is provided via the MetaDataPtr parameter.
Com_StartOfReception	Called at the start of receiving an N-SDU. The N-SDU might be fragmented into multiple N-PDUs (FF with one or more following CFs) or might consist of a single N-PDU (SF).
Com_SwitchIpduTxMode	Sets the transmission mode of the IPDU referenced by PdulId to input Mode of this API.
Com_TpRxIndication	Called after an IPDU has been received via the TP API the result indicates whether the transmission was successful or not.
Com_TpTxConfirmation	Called after the IPDU has been transmitted on its network the result indicates whether the transmission was successful or not.
Com_TriggerIPDUSend	Triggers an IPDU for transmission.
Com_TriggerIPDUSendWithMetaData	Updates the internal metadata for the IPDU with the given ID by copying the metadata from the given position and with respect to the globally configured metadata length of the IPDU. Then the IPDU is triggered for transmission.
Com_TriggerTransmit	Checks whether the available data fits into the buffer size reported by PdulInfoPtr → SduLength. If it fits it shall copy its data into the buffer provided by PdulInfoPtr → SduDataPtr and update the length of the actual copied data in PdulInfoPtr → SduLength.
Com_TxConfirmation	The lower layer communication interface module confirms the transmission of an IPDU.

For Further information please refer to AUTOSAR_SWS_COM.pdf

1.8 Configuration Advice



NOTE

Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the "Configuration Generation" section of the RTA-CAR User Guide.

This section provides configuration advice for the Com module and it is divided into the following sub-sections:

- General advice
- APIs
- Callback Functions
- Configuration Parameters
- Callback Parameters
- Feature-specific Configuration advice
- Miscellaneous

1.8.1 General Advice

- The ShortNames configured across the ComIpduGroup, ComIpdu, ComSignal and ComSignalGroup containers must be unique, and these ShortNames cannot be used for the configuration of the notifications of signals or signal groups.
- The data invalid value configured for the sender side must be the same as configured for all receiver sides.
- Com_TriggerTransmit should not be called before the first call of PduR_-ComTransmit.
- The Initialize parameter of Com_IpduGroupStart should be TRUE for the first call after Com_Init is called.
- When Com_ReceiveSignalGroup is called, and if IpduGroup is stopped, then the last known value will be copied to the shadow buffer.
- When Init value is configured with the COM_TRIGGERED_ON_CHANGE transfer property for a signal, group signal or signal group, then the value to be sent by the application for any of these entities should be different from the configured init value. Failing to do this will result in the non-transmission of the message, even if triggered from the application.
- In the Pre-compile variant, ComSupportedIpduGroups should be equal to the number of ComIpduGroups configured.
- All COM timings (ComFirstTimeout, ComTimeout and ComTxModeTimePe-

riod) support microsecond resolution, so for example a sub-millisecond time of 3.125ms will not be rounded down to 3ms.

1.8.2 APIs

1.8.2.1 Com_ReceiveDynSignal

If the configured length (as in-parameter) is smaller than the received length of the dynamic length signal, Com_ReceiveDynSignal does not copy any data to the position referenced by SignalDataPtr.

1.8.2.2 Com_IsTxScheduled

Com_IsTxScheduled checks whether the Tx frame is scheduled to be sent in the coming cycle. The caller of this function queries the I-PDU transmission status by calling this function with the I-PDU handle Id for that particular I-PDU.

The caller of Com_IsTxScheduled is usually scheduled in the same task cycle as Com_MainfunctionTx. If Com_IsTxScheduled is called in a different task cycle as Com_MainfunctionTx, then note the following:

- ComTxScheduledInDiffTimebase should be set to true.
- The API signature of Com_IsTxScheduled is changed and an additional parameter is passed by the caller.

Tb: Time raster of Com_MainFunctionTx

Tc: Time raster of caller of Com_IsTxScheduled

The time raster passed by the caller is in ms, hence the ticks are converted into time in ms, Multiplying ticks by Tb will serve the purpose. After the conversion, the value obtained is the remaining time for I-PDU transmission, in ms. Here are three example cases:

Case 1 (Tb = Tc)

If Com_IsTxScheduled is called in the same raster as Com_MainFunctionTx, and if the remaining time is greater than caller time, the caller can wait for another raster to update I-PDU data, or the I-PDU data update can happen in the current raster.

Case 2 (Tb < Tc)

If Com_IsTxScheduled is called slower than Com_MainFunctionTx, and if the remaining time is greater than caller time, the caller can wait for another raster to update I-PDU data, otherwise the I-PDU data update can happen in the current raster.

Case 3 (Tb > Tc)

If Com_IsTxScheduled is called faster than Com_MainFunctionTx, the Com internal ticks would not have enough resolution to acknowledge the last raster before transmission, so when the Com ticks reach value 1, the caller can do the I-PDU data update in the current raster.

- Pre-requisites:

1. Ensure that the Com_IsTxScheduled service is used from a task cycle such that the caller's task cycle is in multiples of the mapped MainFunctionTx timeBase.
2. Ensure that the Com_IsTxScheduled service is invoked before the call to MainFunctionTx in the OS task.
3. Ensure that the Com_IsTxScheduled service is invoked only for TxIpdu configured with 'PERIODIC' transmission mode at least in one of the true/-false transmission mode configuration.

1.8.3 Callback Functions

1.8.3.1 Com_TriggerTransmit

This service is ideally invoked by time-triggered protocol layers like LIN and FlexRay. However, if the CAN protocol supports the use of TriggerTransmit, then the service for Metadata I-PDUs has to be supported.

1.8.3.2 Com_TpRxIndication

If Com_TpRxIndication is called with value E_NOT_OK for the parameter result, or when the I-PDU callout in the context of Com_TpRxIndication returns false, or when a configured IPDU counter mismatch occurs, the I-PDU is considered as being incorrectly received, and therefore all included signals are treated the same as invalidated signals (Com will execute all configured ComDataInvalidActions for all included signals and signal groups).

If Large data is discarded due to any of the above scenarios, and the Signal(-Group) is configured with UpdateBit and Timeout (non-zero), then its deadline monitoring timer will be reloaded if signal buffers are replaced with SignallnitValues.

The ComInvalidationForDiscardedTpRxIpdu parameter provides backward compatibility (the incorrectly received IPDU is discarded and InvalidActions are not invoked).

1.8.4 Configuration Parameters

1.8.4.1 ComIPduGroup

For flexibility in the usage of ComIPduGroups, unused/non-referenced ComIPduGroup configuration is now supported.

1.8.4.2 ComCommonIPduTxPeriodFeature

When a different Cycle time is required to be observed by the Com module, this feature can be configured to True and with the use of the functions a reduced busload can be achieved.

- API supported:

Com_ReduceBusload()

Com_RestoreBusload()

1.8.4.3 ComIPduControlViaCalibration

It has to be ensured that the Com_Cfg_SchM.h file includes the SchM_Com.h file. This is done by enabling the EcuCRbRteInUse EcuC parameter (under the EcuCRbSettings container of the EcuC module).

1.8.4.4 ComRxIpduTimeoutCalibration

If set to TRUE, the value of calibration variable for reception I-PDU is in terms of "Com ticks" (i.e. one tick for each time execution of Com_MainFunctionRx).

It has to be ensured that the Com_Cfg_SchM.h file includes the SchM_Com.h file. This is done by enabling the EcuCRbRteInUse EcuC parameter (under the EcuCRbSettings container of the EcuC module).

1.8.4.5 ComRxDualCycleSupport

This optional extension parameter enables/disables the dual cycle feature (update new IPDU based reception deadline value via the Com_SetRxIPduTimeoutTicks service).

If Com_SetRxIPduTimeoutTicks API is invoked with TimeoutTicks parameter equal to value zero(0), then Com uses configured ComTimeout or calibrated timeout value for deadline monitoring.

If Com_SetRxIPduTimeoutTicks API is invoked with TimeoutTicks parameter equal to COM_RXTIMER_TICKS_TO_STOP_IPDU_DM macro value, then its RX I-PDU deadline monitoring will be stopped, until new timeout ticks are provided.

Rx I-PDU deadline monitoring will not be restarted in the next Com_MainFunctionRx, unless TimeoutTicks parameter update differs from previously update ticks value.

1.8.4.6 ComEnableJitterFilter

If the scheduled operations would contain a jitter, in order to ensure that this jitter does not cause any unintended behaviour, this parameter can be set to True, which causes Com to add a value of 1 to avoid the jitter.

In the current implementation of the Com module, internal reception deadline monitoring timers are added with offset based on ComEnableJitterFilter switch, so the maximum supported range of TimeoutTicks is restricted to 65533.

1.8.4.7 ComFilterMin/ComFilterMax

ComFilterMin and ComFilterMax are used for Tx/Rx Filtering, and they must be configured for ComFilter algorithms like NEW_IS_OUTSIDE and NEW_IS_WITHIN.

1.8.4.8 ComTxIPduDeferredProcessing

The default value of this parameter has been set to TRUE in order to reduce bus

load due to too many transmission requests to the lower layer. If this feature is not required, then this parameter should be explicitly configured as FALSE.

1.8.4.9 ComMainFunction

The ComMainFunctionRx (container) is introduced to generate multiple Main-Functions for the reception side.

At least one Com_MainFunctionRx must be configured under the ComMainFunctionRx container for the processing of Rx-IPdus.

The ComMainFunctionTx (container) is introduced to generate multiple Main-Functions for the transmission side.

At least one Com_MainFunctionTx must be configured under the ComMainFunctionTx container for the processing of Tx-IPdus.

1.8.4.10 ComMainTxPartitionRef

This parameter refers to the EcucPartition where multiple Tx Com_MainFunction instances are assigned.

If this parameter is configured, then Com will fetch the coreId from EcucPartition, and core-specific memory sections will be generated, otherwise Com will generate default memory sections in Com_Memmap.h file.

1.8.4.11 ComMainRxPartitionRef

This parameter refers to the EcucPartition where multiple Rx Com_MainFunction instances are assigned.

If this parameter is configured, then Com will fetch the coreId from EcucPartition, and core-specific memory sections will be generated, otherwise Com will generate default memory sections in Com_Memmap.h file.

1.8.4.12 ComTxModeMode

The TxMode of a Transmit I-PDU has a separate mode called MIXED_PHASE_SHIFT, which is similar to the Autosar-defined MIXED mode, but the main difference is that the periodic timer would be reloaded when the event transmission of the I-PDU occurs.

1.8.4.13 ComRTESpecificSymbolicNames

This feature implemented as a compiler switch in c code is a temporary change until RTE supports AUTOSAR symbolic name generation for com-signal and com-signalgroup.

If enabled to TRUE, then RTE-specific symbolic names will be generated for Com Signals and Group Signals along with currently generating macros, otherwise only current macros will be generated.

1.8.4.14 ComRetryFailedTransmitRequests

If ComRetryFailedTransmitRequests is enabled, it may lead to multiple retries for the same failed transmission request, unless the Tx-ComIPdu is explicitly

stopped through the call to `Com_IpduGroupStop`.

If the `ComMinimumDelayTime` of an I-PDU is configured greater than 0, and if `ComRetryFailedTransmitRequests` is configured to `FALSE`, Com will load and start the minimum delay timer upon transmission of that I-PDU to the PDU Router via `PduR_ComTransmit`.

1.8.4.15 ComEnableMDTForCyclicTransmission

The default for `ComEnableMDTForCyclicTransmission` is `TRUE`, but if it is disabled:

- The minimum delay timer (MDT) will not be started or re-started for cyclic and repeated transmissions.
- Even though MDT is not monitored for PERIODIC transmissions, Com still validates the configured `ComMinimumDelayTime` for Tx ComIPdus with periodic transmission mode (as this is required in case of calls to `Com_TriggerIPduSend` or `Com_TriggerIPDUSendWithMetaData`).

1.8.4.16 ComEnableBufferForSignalProcessingImmediate

This parameter has been introduced to control the storing of the received frames into the local buffer even for immediate signal processing.

- If `ComEnableBufferForSignalProcessingImmediate` is set to `TRUE`, Com will reserve a local buffer to store the received frames even for immediate signal processing.
- If `ComEnableBufferForSignalProcessingImmediate` is set to `FALSE`, or if it is left unconfigured, Com will not reserve a local buffer to store the received frames for immediate signal processing.

Note: This feature is introduced for debugging purposes. The default is `FALSE`. If this feature is enabled, runtime will increase due to the update of received frames even for immediate signal processing.

1.8.4.17 ComTxIPduSortByCanId

Enables the sorting of `ComTxIPdu` based on cycle time (time period) and the configured `CanId` in the `CanIf` layer.

If the `CanIf` layer is a Pre-Compile configuration, then Com fetches the `CanId` and sorts based on cycle time (time period) and `CanId`. If the `CanIf` layer is a Post-Build selectable configuration, then Com displays a warning message and sorts only based on cycle time (time period). The Routing path is considered as `Com → PduR → CanIf`.

1.8.4.18 ComInitvalOptimization

If `ComInitvalOptimization` is enabled, Com supports the optimization of the init values. It does this by reducing the number of values by saving only unique init values in ROM, and by using index `uint8/uint16` instead of 32-bit values for each signal for fetching the correct value.

1.8.4.19 ComTimeout and ComUpdateBitPosition

When ComTimeout and ComUpdateBitPosition are configured, Com gives preference to [SWS_Com_00703] over [SWS_Com_00701] requirements.

1.8.4.20 ComIPduGroupControlCoreSpecificMemmap

If the switch is enabled, a core-specific Memmap will be generated for the IpduGroup-related RAM. (This assumes that the ComIPdus belonging to a specific ComIPduGroup have the same EcucPartition CoreID.)

1.8.5 Callback Parameters

1.8.5.1 ComNotification

ComNotification can be configured for all Tx and/or Rx signals and signal groups for which notification callback is required. The notification callback name, if generated in the Com_Cbk.h file, and a definition to it, has to be written in the application.

On the Tx side, once the I-PDU is transmitted, and a confirmation for the I-PDU is received, this ComNotification is called for all the signals/signal groups within the I-PDU which have configured it.

On the Rx side, after the I-PDU reception, all signals that are available for notification to the application are processed, the data is copied into the signal buffer, and a ComNotification is called.

1.8.5.2 ComErrorNotification (on the Tx side)

ComErrorNotification can be configured for all Tx I-PDUs for which Error Notification Callback is required. The error notification is called for a Tx I-PDU when a timeout occurs for that Tx I-PDU.

ComErrorNotification can be configured for all Tx Signals / Signal Groups for which Error Notification Callback is required. The error notification is called in the event of an error in transmission. In this context, an error is when the IpduGroup is stopped, the n-times transmission is still in progress, the timeout timer is started, and the confirmation of transmission for that I-PDU has not yet been received.

1.8.5.3 ComTimeoutNotification

Timeout notification can be configured for all Tx and Rx Signals and Signal Groups for which the Timeout Notification Callback is required.

On the Tx side, the timeout notification is called whenever a timeout occurs.

On the Rx side, on a signal with ComTimeoutNotification configured, timeout notification is called in the function Com_RxIndication as follows:

1. If the signal is configured with the update bit, the notification is called when a timeout occurs in the signal update of the signal/signal group
2. If the signal does not have the update bit configured, the notification is called when a timeout occurs in the I-PDU monitoring of the signal and the

timer times out for the I-PDU.

1.8.5.4 ComInvalidNotification (on the Rx side)

Invalid notification can be configured for all Rx Signals, Signal Groups and Group Signals. The notification is called individually when the signal or group signal is received with an invalid value. For signal groups, a separate notification is given when any group signal is received with an invalid value

1.8.5.5 ComIPduCallout

ComIPduCallout can be configured for all I-PDUs for which callouts are required to be called. A callout for a Tx I-PDU can be called either before the call to Pdur_-ComTransmit or in the context of Com_TriggerTransmit. A callout for an Rx I-PDU is called before Rx I-PDU processing is started. Further processing of the I-PDU is done only when the callout returns TRUE, otherwise the I-PDU is not processed further.

1.8.6 Feature-Specific Configuration Advice

This sub-section provides configuration advice for the following Com features -

- Signals
- Signal Groups
- Group Signals
- I-PDUs
- I-PDU Groups
- Filters
- Signal Gateway

1.8.6.1 Signals

- At least one Signal needs to be configured.
- The same ShortName cannot be used for two or more signals.
- Tx signals must be configured first, followed by the Rx signals.
- For the type UINT8_N:

Only the filter algorithms F_Always and F_Never can be configured.

Length would refer to the number of bytes in uint8_n.

Size must be equal to Length*8

Bitpos should be present at byte boundary (i.e. values must be 8, 16, multiples of 8 etc).

Endianness should be configured as COM_OPAQUE.

Length should not exceed IpduSize/8.

- The Size, BitPosition and UpdateBitPosition of the signal must not exceed

the size of the I-PDU.

- No two signals can be configured with the same UpdateBitPosition and the same IpduRef.
- The LSB should be configured as BitPosition for LittleEndian/Opaque Signals and MSB should be configured as BitPosition for BigEndian Signals.
- If InvalidAction is Notify, the Notification function should be configured for the ComInvalidNotification parameter. If InvalidAction is Replace, the Invalid Value and Init Value should be configured.
- If Rx TimeoutAction is configured as Replace, the ComTimeout or ComFirst-Timeout should not be configured as 0.
- Filters can be configured for signals with the TRIGGERED_ON_CHANGE transfer property.
- Float32 and Float64 signals are packed or unpacked in accordance with the IEEE-754 format. Filter configuration is not allowed for Float32 and Float64 signals.
- Note: If the upper layer reads a signal with an associated cleared update-bit, the last received value is re-turned.

Dynamic Signal Support

A dynamic length signal is a signal whose length can vary at run-time. The maximal length of a dynamic length signal needs to be specified at configuration time. A dynamic length signal has to be transmitted in a dynamic length I-PDU.

- Any signal, signal group or update bit belonging to the I-PDU must be configured before the dynamic signal is placed.
- Only one dynamic signal is allowed in an I-PDU.
- If the ComIPdu has dynamic signal configured, PduLength (in /ComIPdu/-ComPduIdRef) will be taken as the maximum length of the I-PDU.

Behaviour

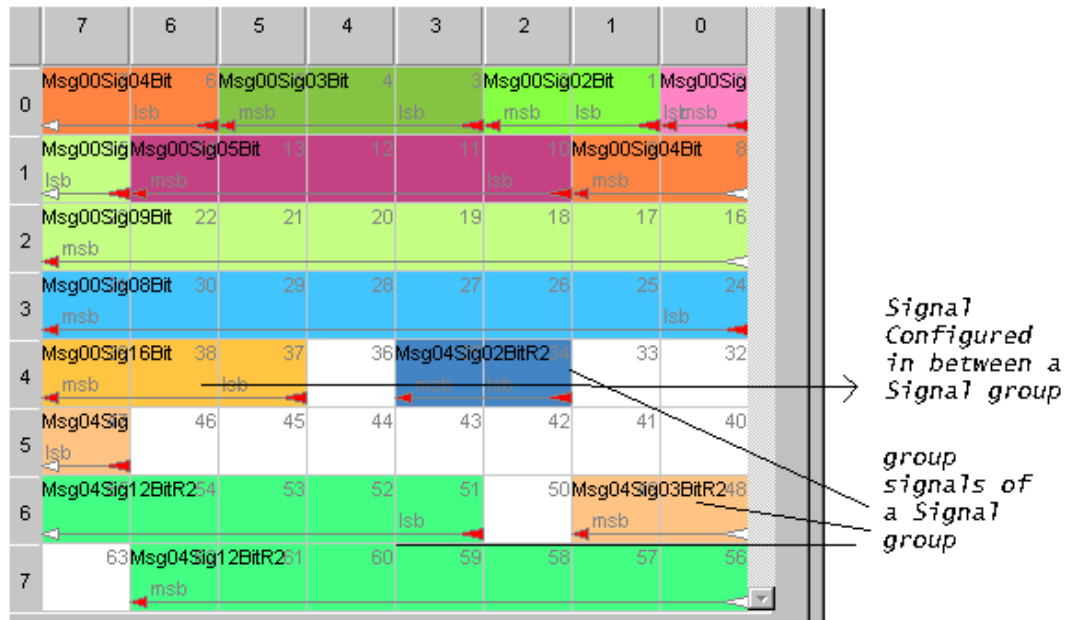
- Until the first request of Com_SendDynSignal, the length of the dynamic signal is taken as zero, and only the static part of the I-PDU is triggered from Com.
- Byte padding is not done for the dynamic part of the I-PDU. The byte padding fills the un-used areas, which do not occur in dynamic signal (as it is a uint8 array), and so the signal initialization should be enough.
- The previous dynamic signal will be retained for the current transmission - if the transmission of the I-PDU happens with size 9 (the dynamic part is size 5), then even if there wasn't an update from the application, the next I-PDU transmission will hold the same size 9.

1.8.6.2 Signal Groups

A signal group is just a logical entity. It is a collection of group signals which do not

necessarily need to be together - they can be scattered as a Com signal in the I-PDU.

The figure below illustrates how a signal group (Little Endian) with three group signals (Msg04Sig02BiR2, Msg04Sig03BiR2 and Msg04Sig12BiR2) is packed in the I-PDU. As you can see, any Com signal or an Update bit of the I-PDU may be configured in the gaps within the signal group.



Notes:

The behaviour of a Big Endian signal group is the same as the Little Endian Signal group.

The group signal's bit position and bit size are used in packing it into the I-PDU, and the signal group's attributes have no influence over the packing of a group signal.

General Configuration Advice for Signal Groups

- The same ShortName cannot be set for two or more signal groups.
- Group signals within the given signal group can be configured as ComSignal in an I-PDU without any continuation.
- The group signals can be scattered in the I-PDU.
- All the group signals within the given signal group should have the same Endianness.

Signal Group Parameter Dependencies

- If ComDataInvalidAction is configured as Notify, the ComInvalidNotification of Signal Group should be configured.
- If ComDataInvalidAction is configured as Replace, the ComSignalInitValue and ComSignalDataInvalidValue of the corresponding Group Signals should

be configured.

- If ComRxDataTimeOutAction is configured as Replace, the ComTimeout or ComFirstTimeout of the Signal Group should not be configured as 0.
- The same UpdateBitPosition and the same IpduRef cannot be configured for different Signal Groups.

1.8.6.3 Group Signals

- The same ShortName cannot be used for two or more Group Signals.
- The Number of Group Signals in a Signal Group can be more than 255.
- The size of the Group Signal must not exceed the size of the I-PDU and the size of the Signal Group.
- The BitPosition of a Group Signal must not exceed the size of the I-PDU.
- The initial value of Group Signal should not be equal to the invalid value (e.g. If the datatype of a signal is uint8 and the bit size is 6 bits, the possible range is 0...63).
- If type is uint8_n then:
 - Length would refer to the number of bytes.
 - Size must be equal to Length*8
 - Bitposition should be present at byte boundary (i.e values must be 8, 16, multiples of 8, etc.).
 - Endianness should be configured as COM_OPAQUE
 - Length should not exceed IpduSize.
 - Bitposition + Size should be less than the Size of I-PDU
- For Signals and Group Signals of the type uint8_n, only the filter algorithms ALWAYS and NEVER can be configured.
- For Signals and Group Signals of type Boolean only the following filter algorithms should be configured:
 - Always
 - Never
 - MaskedNewEqualsX
 - MaskedNewDiffersX
 - MaskedNewDiffersMaskedOld
 - OneEveryN
- Filters can be configured for Group Signals with the TRIGGERED_ON_CHANGE transfer property. The attribute ComTransferProperty can be configured with any of the available options.
- Filters cannot be configured to Group Signals with ComSignalType float32.
- Float 32 Group Signals are packed/unpacked in accordance with the IEEE-754 format.
- The endianness of the Group Signals inside a Signal Group can be different.

Parameter Dependencies

- If ComDataInvalidAction is configured as Notify, the ComInvalidNotification of the Group Signal should be configured.
- If ComDataInvalidAction is configured as Replace, the ComSignalDataInvalidValue of the Group Signal should be configured.
- ComInvalidNotification must be NULL_PTR for Tx group signals.

Configuration for Little Endian

The figure below illustrates the configuration of a Little Endian group signal. The LSB of the group signal should be configured as the bit position (e.g. 13), and the size of the group signal as the bit size (e.g. 12).

	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Byte 0	7	6	5	4	3	2	1	0
Byte 1	15 ← 2	14 ← 1	13 ← 0 LSB	12	11	10	9	8
Byte 2	23 ← 10	22 ← 9	21 ← 8	20 ← 7	19 ← 6	18 ← 5	17 ← 4	16 ← 3
Byte 3	31	30	29	28	27	26	25	24 MSB ← 11
Byte 4	39	38	37	36	35	34	33	32

Configuration for Big Endian

As illustrated below, always read the Big Endian group signal in the I-PDU buffer. The MSB of the group signal has to be configured as the bit position (e.g. 13), and the size of the Signal as the Bit size (e.g. 12).

	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Byte 0	7	6	5	4	3	2	1	0
Byte 1	15	14	13 ← 11 MSB	12 ← 10	11 ← 9	10 ← 8	9 ← 7	8 ← 6
Byte 2	23 ← 5	22 ← 4	21 ← 3	20 ← 2	19 ← 1	18 ← 0 LSB	17	16
Byte 3	31	30	29	28	27	26	25	24
Byte 4	39	38	37	36	35	34	33	32

1.8.6.4 I-PDUs

- At least one Tx I-PDU and one Rx I-PDU must be configured.
- The size of the I-PDU must be multiples of 8 bytes and no more than 255 bytes. The size cannot be 0.
- TxMode Repetition Period Factor should be configured to be greater than Minimum Delay Factor, and the Minimum Delay Factor must be less than TxModeTimePeriod (for both Transmission Mode True and Transmission Mode False).
- Com supports evaluation of Transmission Mode Selection for an I-PDU without ComIpdGroupRef configuration.
- If a direct I-PDU (in both True and False modes) is configured with a timeout factor, and if none of the entities (signals or signal groups) are configured with timeout factor, then Tx deadline monitoring is not done for the I-PDU.
- Update-bits must not be configured for signals that are transmitted within I-PDUs having the Direct transmission mode with $n > 1$, but they can be configured if the Direct transmission mode is $n = 0$.
- The attribute ComTxIpdClearUpdateBit is valid only for Tx I-PDUs, and it has to be set to the proper value depending on where the update bits need to be cleared.
- Setting the parameter ComTransferProperty of all signals inside a DIRECT I-PDU to Pending will result in the I-PDU not being sent.

I-PDU Parameter Dependencies

- If TxMode Mode is set to COM_PERIODIC or COM_MIXED, Tx Mode TimePeriod Factor is a mandatory field and must be configured.
- If TxMode Mode is set to COM_MIXED or COM_DIRECT, Repetition Period Factor and Tx Mode number of Repetitions are mandatory and must be configured.

1.8.6.5 I-PDU Groups

- I-PDUs must be configured under I-PDU Groups.
- An I-PDU can belong to any I-PDU Group, and it must belong to at least one I-PDU Group in order to be able to be started.
- An I-PDU is active (i.e. started) if at least one I-PDU Group to which it belongs is started, and an I-PDU is stopped if all the I-PDU Groups to which it belongs are stopped.
- While configuring I-PDU Group Group, the same I-PDU group name should not be configured under ComIpdGroupGroupRef.
- It is not possible to configure ComIpdGroupGroupRef for an I-PDU Group that does not have any I-PDUs referring to it. It has to be null.
- It is not possible for an I-PDU Group to include itself.

- The maximum number of I-PDU Groups is 65535 (pre-compile configurable in the parameter ComSupportedIPduGroups).
- If an I-PDU Group is referred by I-PDUs, and it is itself included in another I-PDU group, then the latter I-PDU group cannot be further included in another I-PDU Group (i.e. nesting level of 2). It can however include I-PDUs underneath it and then other I-PDU groups can refer it. E.g.

I-PDU Group "BUS1 RX" belongs to I-PDU group "BUS1"

I-PDU Group "BUS1" cannot itself be included in another I-PDU Group, but it can still have other I-PDU Groups included within it.

1.8.6.6 Filters

- The default Filter Algorithm is COM_ALWAYS.
- ComFilterMask has to be configured if the filter algorithm is configured as MASKED_NEW_DIFFERS_MASKED_OLD.
- ComFilterMin and ComFilterMax have to be configured if the NEW_IS_WITHIN or NEW_IS_OUTSIDE filter algorithms are configured.
- ComFilterOffset and ComFilterPeriod have to be configured if the filter algorithm ONE EVERY_N is configured.
- ComFilterMask and ComFilterX have to be configured if the MASKED_NEW_DIFFERS_X or MASKED_NW_EQUALS_X filter algorithm is configured.
- Negative values are not allowed for ComFilterMin, ComFilterMax and ComFilterX.

1.8.6.7 Signal Gateway

To function correctly, a Gateway I-PDU should have similar signal groups configured on the Rx side and be routed to the Tx side. In this context, "similar signal groups" means the following:

- The number of group signals in both the signal groups should be the same.
- The bit size of the group signals on both sides should be the same.
- The signal type of the group signals should also be the same.

1.8.7 Miscellaneous

1.8.7.1 Dynamic signal support over Non-Tp interface

If the ComIPdu has dynamic signal configured, the PduLength (in ComIPdu/ComPduIdRef) will be taken as the maximum length of the I-PDU.

Com sends dynamic length I-PDUs via the generic PduR APIs, and these I-PDUs cannot be sent over a Tp interface. For sending a dynamic length signal, the RTE needs to specify the actual length at the send call. The corresponding API is Com_SendDynSignal.

1.8.7.2 Multiple MainFunctions

Instead of processing all Rx/Tx IpDus in one Com_MainFunction, you can split the processing for some ComIPdus, segregating them into multiple Com_MainFunctionRx/Tx.

ComIPduMainFunctionRef (in ComConfig/ComIPdu) can be configured for each of the ComIPdus that need to be processed in a separate Com_MainFunction.

- If ComIPduMainFunctionRef is configured for Rx-IPdu, then Rx-IPdu (time-out/deferred processing) shall be processed in the newly generated Com_MainFunctionRx_<shortName> function.
- If ComIPduMainFunctionRef is configured for Tx-IPdu, then Tx-IPdu (time-out/deferred processing) shall be processed in the newly generated Com_MainFunctionTx_<shortName> function.

1.8.7.3 Post-Build Support

The Com module supports Post-Build Selectable. ComSignal or ComSignalGroup can either be part of a Tx ComIPdu or Rx ComIPdu in any of the configured pre-defined variants.

Post-Build Loadable is not currently supported, so the Com module has to be rebuilt if any of the parameter values is changed or if a new container is added.

1.9 Integration Advice

The Com module supports generation of Com_Cfg_BSWMD.arxml file. This file can be fed to ISOLAR-AB tool to generate the RTE header files.

1.9.1 Init Functions

Com is initialized by Com_Init, which is called by EcuM during the system initialization. Com_Init must not pre-empt any other Com module functions. During execution of Com_Init, Com will do the following:

- Initialise each I-PDU. It does this by setting not-used areas of I-PDU with the ComTxIPduUnusedAreasDefault value (if any), setting each signal with its respective configured ComSignalInitValue, and clearing all update-bits.
- Initialise the shadow buffer of a signal group on the sender side.
- Initialise, with the configured init values, all Rx-SignalGroup Array and Secondary GroupSignal Buffers on the receiver side.

1.9.2 Delnit Functions

The Com module is shut-down using Com_Delnit, which de-initializes all the RAM areas of the previously activated configset. After deinitialization, no communication via Com is possible, and all previously started I-PDU groups are stopped.

If Com_Init is called more than once, Com_Deinit must first be called before each subsequent call to Com_Init.

Com_Init and Com_Deinit cannot interrupt each other, or any other Com functions (API, Scheduler, Interrupt).

The following points should also be observed to ensure the proper functioning of the Com module in an interrupt environment.

- Com_InvalidateShadowSignal and Com_UpdateShadowSignal cannot interrupt each other for the same Id.
- On single core and multi-core systems, callbacks are high priority processes and hence cannot be interrupted (interrupt protection is not yet fully complete for Multi-core systems).

1.9.3 Scheduled Functions

Com supports the following functions, all of which are called cyclically.

- Com_MainFunctionRx_<shortName>. For receive processing.
- Com_MainFunctionTx_<shortName>. For transmit processing.

A call to any of these functions will return with a call to Com_Init if Com was not previously initialized.

The Com_MainFunctionRx_<shortName> and Com_MainFunctionTx_<shortName> functions are scheduled according to the ComMainRxTimeBase and ComMainTxTimeBase configuration parameters respectively.

1.9.4 Com in an Interrupt Environment

To ensure the proper functioning of the Com module in an interrupt environment, Com_RecvSignalGroup and Com_ReceiveShadowSignal must not interrupt each other.

1.9.4.1 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the ENTER and EXIT resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The following is a list of all exclusive areas that this module might use regardless of the user configuration. It is the job of the integrator to ensure that the implementation of these exclusive areas is correctly defined. This implementation can be done using the RTE exclusive area configuration wizard or by manual implementation of the corresponding BSW API calls to ensure exclusivity.

The Com module uses the following OS resources:

- Com_RxIpduProtArea
- Com_TxIpduProtArea
- Com_RxSigBuff
- Com_TxIpduProtArea_INVALIDATESIGNAL

- Com_TxIpduProtArea_MAINFUNCTIONTX
- Com_TxIpduProtArea_SENDDYNSIGNAL
- Com_TxIpduProtArea_SEND_DYNGROUPSIGNAL
- Com_TxIpduProtArea_SENDIPDU_DATA
- Com_TxIpduProtArea_SENDIPDU_TXFLAGS
- Com_TxIpduProtArea_SENDSIGNAL
- Com_TxIpduProtArea_SENDSIGNALGRP
- Com_TxIpduProtArea_TRIGGERTRANSMIT
- Com_TxIpduProtArea_TRIGGERIPDUSEND
- Com_TxIpduProtArea_TRIGGERIPDUSENDWITHMETADATA
- Com_TxIpduProtArea_SIGTXCHANGEMODE
- Com_TxIpduProtArea_SIGGRPTXCHANGEMODE
- Com_TxIpduProtArea_TXCONFIRMATION
- Com_TxIpduProtArea_IPDUGROUPSTART
- Com_TxIpduProtArea_SWITCHTXIPDU
- Com_RxIpduProtArea_RECEIVE_METADATA
- Com_TxIpduProtArea_TRANSMIT_METADATA
- Com_RxIPdu_RECEIVE_METADATA
- Com_TxIPdu_TRANSMIT_METADATA
- Com_TxGrpSigBuffer
- Com_RxPduBuffer
- Com_RxSigToTicks
- Com_RxSigGrpToTicks
- Com_RxSigBuff_MAINFUNCTIONRX
- Com_RxSigBuff_RXINDICATION
- Com_RxSigBuff_RECEIVESIGNAL
- Com_RxSigDynBuff
- Com_RxGrpSigDynBuff
- Com_RxGrpSigSecBuff
- Com_RxSigGrpArrayBuff
- Com_RxGrpSigBuff
- Com_RxIPduProcess
- Com_RxIndication
- Com_TxIPdu

If Rx or Tx ComMainFunctions are configured to generate multiple MainFunctions, they shall be scheduled in different tasks to parallel-process ComIPdus. In this

type of scheduled environment, the Com service APIs will be called in a sequential order, which means that they do not interrupt each other and resource locks are not required.

1.10 Hardware Requirements

Table 1.101.Com Hardware Requirements

Support	Usage
FLASH	28082 bytes
RAM	25373 bytes

Usage calculated based on a sample configuration.

2 Efficient COM for Large Data

Module name	LdCom
Package	RTA-COM
Module ID	49
AUTOSAR Version	22-11

2.1 Introduction

The COM module is designed to work with the signal base concept, in which specific signals are “concreted” into Ipdu, cyclically updated, and exchanged between ECUs. The advantage of this method is that ASWs work directly with target signals.

The signal base utilities used by RTE and COM, which include composing/decomposing signals, conversion, signal update management and timeout monitoring, are useful and necessary for systems working directly with signals, and where the volume of signals fits within the constraints of the MCU memory resource.

However, in newer vehicle architectures, large amounts of data is now being exchanged (e.g. streamed video) that does not require signal base functionality. For this reason, the Large Data Com module (LdCom) has been introduced as an efficient Com for Large Data. It focuses on spontaneous, non-cyclic communication without serializing, filtering and conversion, resulting in a highly-efficient implementation.

The diagram below shows how the LdCom module is placed between RTE and the PDU Router within the AUTOSAR Layered Architecture.

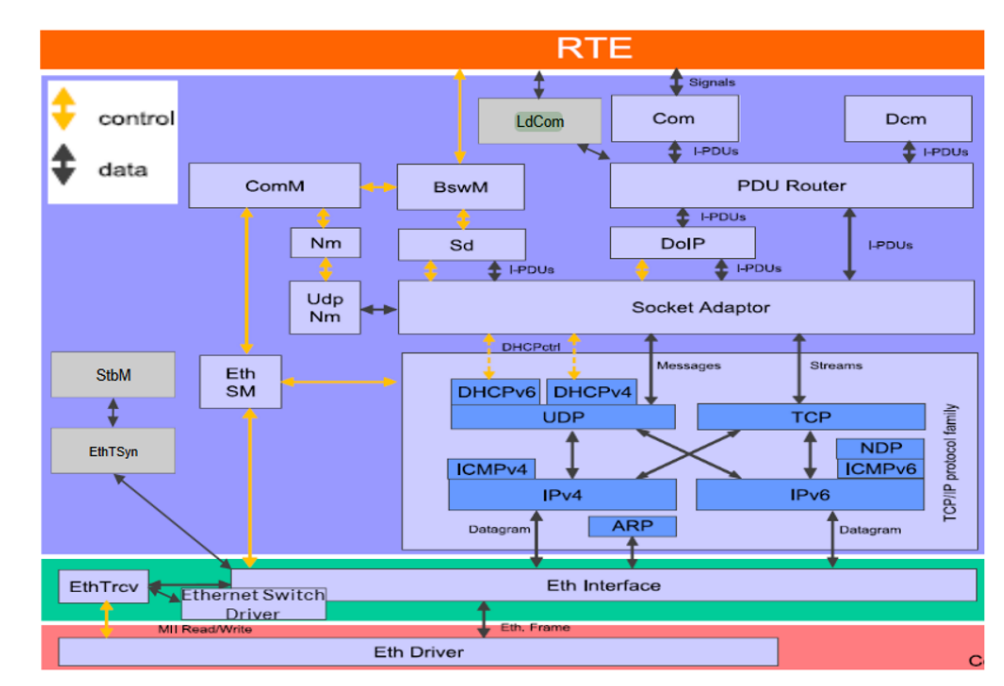


Figure 2.1.Autosar Architecture with LdCom

Module Configuration

This module is not currently supported by the RTA-BSW Configuration Generator (ConfGen) and must be manually configured.

2.2 Supported Features

- Provision of signal oriented data interface for the RTE
- Provision of received signals to RTE
- Support of large and dynamic length data types
- Support of IF and TP based communication
- Provision of PDU oriented data interface towards PduR

LdCom is independent of the upper layers and is considered an optional feature of the Com module.

2.2.1 Post Build Loadable Support

Configuration Container LdComlPdu is a Post Build Loadable parameter. The new Container can be added at Post Build Time, but removal of the Container is not currently supported.

2.3 Extensions

2.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

2.3.1.1 LdCom/LdComConfig/LdComIPdu/LdComRxCopyRxData

Short Name	LdComRxCopyRxData
Description	Only on receiver side: Name of Rte_LdComCbkJCopyRxData callback function to be called.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.2 LdCom/LdComConfig/LdComIPdu/LdComRxIndication

Short Name	LdComRxIndication
Description	Only on receiver side: Name of Rte_LdComCbkJRxIndication callback function to be called.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.3 LdCom/LdComConfig/LdComIPdu/LdComRxStartOfReception

Short Name	LdComRxStartOfReception
Description	Only on receiver side: Name of Rte_LdComCbkJStartOfReception callback function to be called.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.4 LdCom/LdComConfig/LdComIPdu/LdComTpRxIndication

Short Name	LdComTpRxIndication
Description	Only on receiver side: Name of Rte_LdComCbkJTpRxIndication callback function to be called.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.5 LdCom/LdComConfig/LdComIPdu/LdComTpTxConfirmation

Short Name	LdComTpTxConfirmation
Description	Only on sender side: Name of Rte_LdComCbKTpTxConfirmation callback function to be called.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.6 LdCom/LdComConfig/LdComIPdu/LdComTxConfirmation

Short Name	LdComTxConfirmation
Description	Only on sender side: Name of Rte_LdComCbKTxConfirmation callback function to be called.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.7 LdCom/LdComConfig/LdComIPdu/LdComTxCopyTxData

Short Name	LdComTxCopyTxData
Description	Only on sender side: Name of Rte_LdComCbKCopyTxData callback function to be called.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.8 LdCom/LdComConfig/LdComIPdu/LdComTxTriggerTransmit

Short Name	LdComTxTriggerTransmit
Description	Only on sender side: Name of Rte_LdComCbKTriggerTransmit callback function to be called. If defined TriggerTransmit has to be supported for this signal.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

2.3.1.9 LdCom/LdComGeneral/LdComUserCbKHeaderFile

Short Name	LdComUserCbKHeaderFile
------------	------------------------

Description	Defines the header files for callback functions which shall be included by the LDCOM module.
Multiplicity	0..*
String Length Range	1..32
Default Value	-
Type	ECUC-STRING-PARAM-DEF

2.3.2 Additional Services

The following API functions are available in version v12.7.0 in addition to those defined in AR v22-11

- There are no additional API defined in this version.

2.4 Limitations

The following features are not supported for version v12.7.0

- The *LdComUserCbkHeaderFile* is a non-AUTOSAR configuration parameter that provides the header file with corresponding declarations. To do so, add the header file to *LdCom.LdComGeneral.LdComUserCbkHeaderFile* and use this parameter when specifying a non-LdComUser use case.

2.5 Exclusions

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- There are no known exclusions in this version.

2.6 Deviations

2.6.1 Configuration Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

2.6.1.1 LdCom/LdComConfig/LdComIPdu

[RTA-BSW] Description	Contains the configuration parameters of the LdComs signal (IPdu) inside LdCom.
[AUTOSAR] Description	Contains the configuration parameters of the LdCom's signal (IPdu) inside LdCom.

2.6.1.2 LdCom/LdComConfig/LdComUserModule

[RTA-BSW] Description	Contains the configuration parameters of the LdCom user modules.
-----------------------	--

[AUTOSAR] Description	Contains the configuration parameters of the LdCom user modules. Tags: atp.Status=draft
-----------------------	--

2.6.1.3 LdCom/LdComConfig/LdComUserModule/LdComUserModuleCnfRef

[RTA-BSW] Description	Reference to the LdCom user module configuration.
[AUTOSAR] Description	Reference to the LdCom user module configuration. Tags: atp.Status=draft

2.7 API Definition

Table 2.14.LdCom Supported API functions

LdCom_Init	Initializes internal and external interfaces and variables of the LdCom module.
LdCom_DeInit	Reset the LdCom module to the uninitialized state.
LdCom_GetVersionInfo	Returns the version information of the LdCom module.
LdCom_Transmit	Initiate a transmission of a signal using either IF or TP.
LdCom_CopyTxData	Called to acquire the transmit data of an I-PDU segment (N-PDU).
LdCom_CopyRxData	Called to provide the received data of an I-PDU segment (N-PDU) to the upper layer.
LdCom_RxIndication	Indication of a received I-PDU from a lower layer communication interface module.
LdCom_TpRxIndication	Called after an I-PDU has been received via the TP API and indicates whether transmission was successful or not.
LdCom_TxConfirmation	The lower layer communication interface module confirms the transmission of an IPDU.
LdCom_TpTxConfirmation	Called after the I-PDU has been transmitted on its network and indicates whether transmission was successful or not.
LdCom_StartOfReception	Called at the start of receiving an N-SDU.
LdCom_TriggerTransmit	Checks whether the available data fits into the buffer size reported by PduInfoPtr → SduLength.

For further information on these functions, please refer to AUTOSAR_SWS_LargeDataCOM.pdf

2.8 Configuration Advice



NOTE

Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the "Configuration Generation" section of the RTA-CAR User Guide.

2.8.1 General Hints

- The LdCom module supports both Pre-compile and Post-build loadable variants. The implemented configuration variant should be configured accordingly.
- At least one LdComIPdu should be configured. LdCom supports both If and Tp PDUs.
- If LdComUser is configured, then the LdComIPdus should be linked to a single LdComUserIpdu as a minimum.

2.8.2 TX LdComIPdu Configuration

Shown below is a sample configuration for TX Pdu which is managed by LdCom.

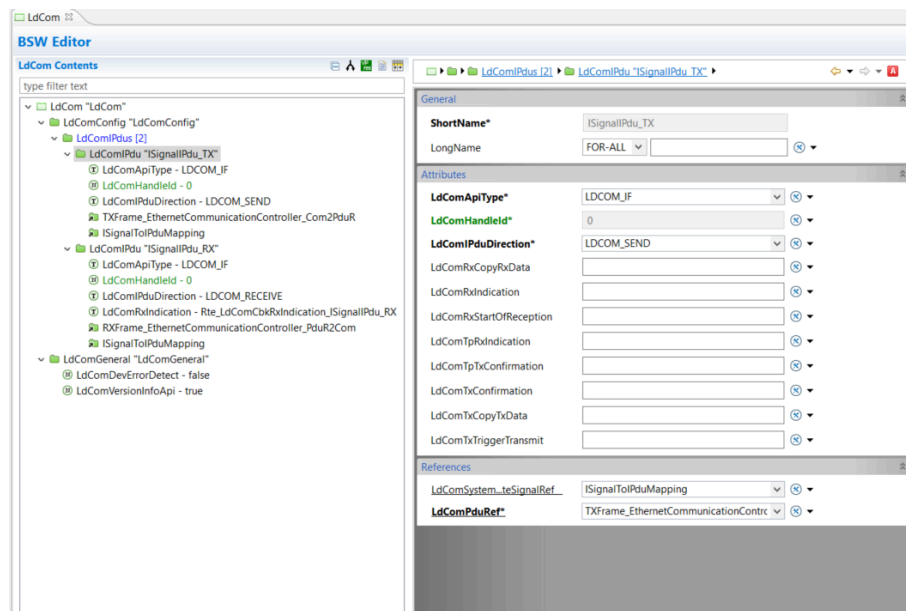


Figure 2.2. Sample Configuration for TX LdComIPdu

Note that LdCom is connected with PduR through the PDU defined by EcuC (LdComPduRef).

2.8.3 RX LdComIPdu Configuration

Shown below is a sample configuration for RX Pdu which is managed by LdCom.

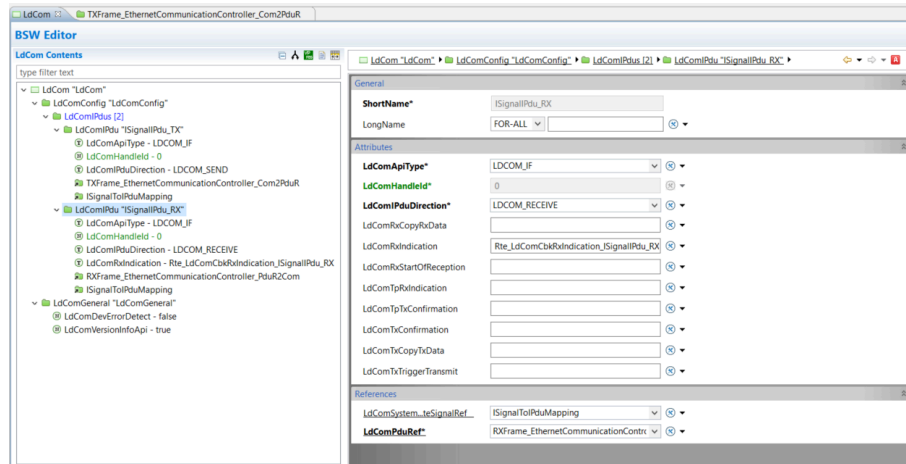


Figure 2.3. Sample Configuration for RX LdComIPdu

LdComRxIndication is required for each IPdu, and mapping with available callback function, which is implemented by the upper layer (in this example RTE is being used).

2.8.4 LdComRxIndication Configuration in algo.properties

The default value for the configuration of the LdComRxIndication parameter can be set in the algo.properties file as follows:

```
manprop_IsLdComNotificationGen=true    # enable this feature
manprop_IsLdComNotificationGen=false    # disable this feature
```

2.9 Integration Advice

2.9.1 Init Functions

LdCom is initialized via LdCom_Init.

2.9.2 Delnit Functions

LdCom_Delnit de-initializes the LdCom module.

2.9.3 Scheduled Functions

LdCom does not have a scheduled function.

2.9.4 LdCom Integration

AUTOSAR states that LdCom “supports communication of linear opaque byte wise data”, and this is also mentioned in the RTA-RTE release notes regarding transmission of transformed signals with LdCom:

- When a transforming ISignal is mapped via an ISignalToPduMapping to an LdComIPdu, the generated Rte_Write API calls LdCom_Transmit to transmit the transformed byte array.
- RTA-RTE now raises error 3031 when an ISignal that does not reference a DataTransformation is mapped via an ISignalToPduMapping to an LdComIPdu.

This requirement is also applicable on the RX side, transforming ISignals are required.

This means that, if these interfaces are supported by RTE, RTE must be configured to generate the relevant interfaces to LdCom.

Based on this understanding, LdCom integration should follow the general advice below, based on specific use cases.

2.9.4.1 General advice

- LdCom must be integrated with the RTA-Com package (Communication Stack). It requires at least the PDU Router (PduR) to be integrated along with it.
- Corresponding routing paths should be configured across the Communication stack.
- Components that are not part of the RTA-Com Package, but which have dependencies with LdCom, must also consider LdCom related configuration:
 - PDUs must be configured in EcuC.
 - Interfaces to LdCom should be configured in RTE if RTE supported LdCom APIs.
- Rte callback declaration generation in *LdCom_Cfg.h* has been removed. Hence, to provide the header file with corresponding declarations, add it to the *LdCom.LdComGeneral.LdComUserCbkHeaderFile* configuration parameter and use the latter when specifying a non-LdComUser use case.

2.9.5 LdCom with Transformer

If LdCom is integrated with the transformer, the RTA-BSW solution for working with a transformer should be followed as below:

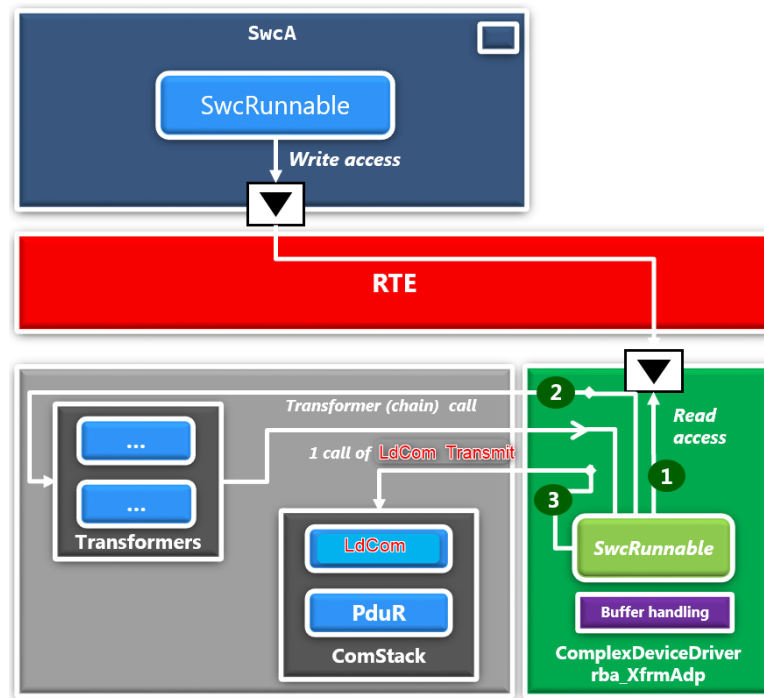


Figure 2.4. Transformer Architecture with LdCom through transforming adapter

Only the SomelpXf transformer can work with both Com and LdCom, so please refer to the detailed integration workflow for the SomelpXf Adapter to integrate transforming signals. Then the two points below should be followed for the communication between the transformer adapter and LdCom:

- All relevant ISignals of the transformer need to be configured to work with LdCom via the `rba_SysElem_UseLdCom` parameter in the configuration of the `rba_SysElem` module (in `rba_SysElem_IPdu/rba_SysElem_ISignalIPdu`).
- All callback functions from LdCom, which are declared in `LdCom_Cfg.h`, need to be routed to the relevant upper layer, which in this case would be the transformer adapter. For this reason, the relevant APIs need to be implemented in `integration/LdCom_Appl.c`

2.9.6 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the ENTER and EXIT resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

LdCom does not use OS Resources.

2.10 Hardware Requirements

Table 2.15.LdCom Hardware Requirements

Support	Usage
FLASH	1990 bytes
RAM	9 bytes

Usage calculated based on a sample configuration.

3 Communication Manager

Module name	ComM
Package	RTA-COM
AUTOSAR Module ID	012
AUTOSAR Version	22-11

3.1 Introduction

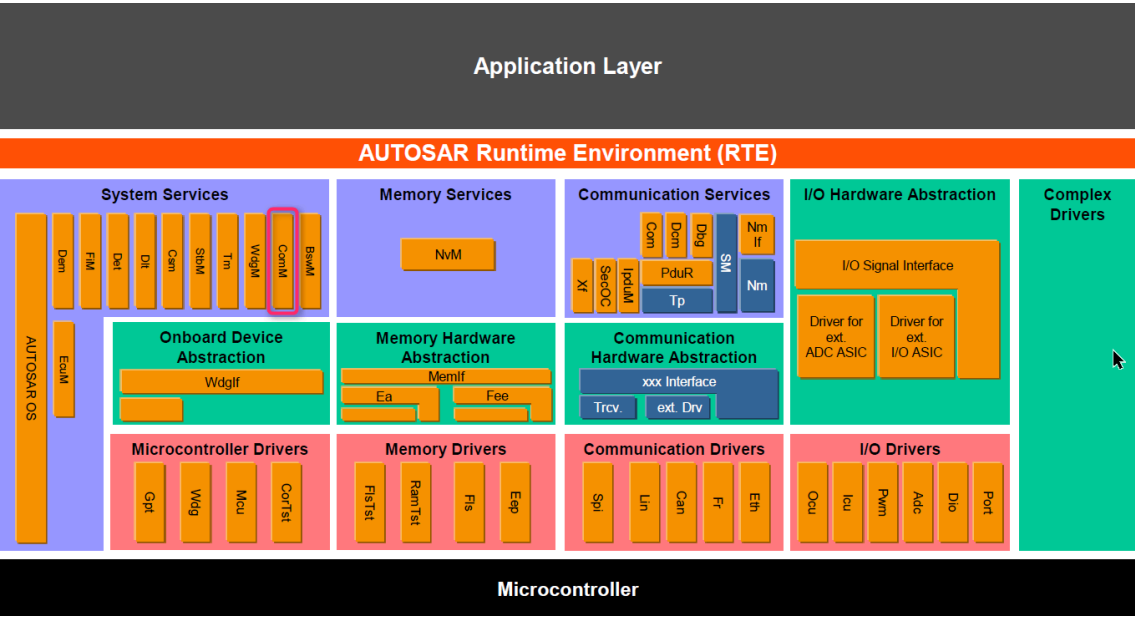


Figure 3.1.Communication Manager

The Communication Manager (ComM) is a resource manager which encapsulates the control of the underlying communication services. ComM collects and coordinates bus communication access requests.

Module Configuration

This module can be generated by the RTA-BSW Configuration Generator. For any unsupported automatic configuration, the module can be manually configured by the user.

3.2 Supported Features

3.2.1 Partial Network Cluster (PNC) Management

ComM supports the Partial Networking Cluster (PNC) Management feature, which involves waking up an ECU only when it is required. When an ECU is not required to do anything, it can be disconnected from the network in order to reduce power consumption. ComM implements a state machine for each PNC to represent that PNC's

communication mode, and each PNC has its own state. The following main states and sub-states are supported.

PNC main states	Sub-States
PNC_FULL_COMMUNICATION	- PNC_REQUESTED - PNC_READY_SLEEP - PNC_PREPARE_SLEEP
PNC_NO_COMMUNICATION	None

3.2.2 ComM Channel State Machine

In order to control more than one bus channel, ComM implements one state machine for every communication channel. To execute the state machine, each channel has its own ComM_MainFunction_<Channel_ShortName> which should be scheduled periodically by the SchM or RTE.

This table shows the 3 main states (and sub-states) in the ComM state machine.

Communication Mode	Message Transmission	Message Reception	Sub-State
COMM_FULL_COMMUNICATION	ON	ON	- COMM_FULL_COM_NETWORK_REQUESTED - COMM_FULL_COM_READY_SLEEP
COMM_SILENT_COMMUNICATION	OFF	ON	None
COMM_NO_COMMUNICATION	OFF	OFF	- COMM_NO_COM_NO_PENDING_REQUEST - COMM_NO_COM_REQUEST_PENDING

3.2.3 Bus Communication Control

ComM's primary function is to simplify the control of the underlying communication service (Can, Lin, FlexRay, Ethernet) so that users do not require knowledge of the hardware in order to control the communication availability. In this context, users can be the BswM module, a runnable entity, an SWC or a group of SWCs.

A user will make a request to ComM for the required Communication Mode, and when that Communication Mode is entered as a result of a successful request, the communication capability in the corresponding bus can be switched ON or OFF.

A user can only request `COMM_FULL_COMMUNICATION` or `COMM_NO_COMMUNICATION` communication modes (`COMM_SILENT_COMMUNICATION` and sub-modes/sub-states are only used for network synchronization) and the request is performed via a call to `ComM_RequestComMode`. The request is configurable in a `BswMAction` (in the `BswM` module if the user is `Bswm`), or in a `Client-Server Port` if the user is an `SWC`.

`ComM` will coordinate multiple independent user requests according to the "highest wins" strategy, so a request for `COMM_FULL_COMMUNICATION` will overrule `COMM_NO_COMMUNICATION`.

The actual bus state is controlled by the `Bus State Manager` modules (e.g. `CanSM`, `LinSM` etc), and `ComM` is responsible for forwarding the `Communication Mode` request to relevant `Bus State Manager` module.

3.2.4 Network Management Dependencies

`ComM` uses the `Network Management (Nm)` module to synchronize the control of communication capabilities in the network. See the `Integration Advice` for more details.

3.2.5 Communication Inhibition

`ComM` supports the `Communication Inhibition (limitation)` feature. See the `Configuration Advice` section for more details on enabling this feature.

Irrespective of inhibition status, user requests will always be stored, and the counter (for channel and PNC) will also be incremented. Inhibition is achieved by ignoring the user request counters in the channel and PNC state machines.

Inhibition requested at channel level is translated to PNC and user. The Inhibition counter is maintained at both PNC and user level, which is incremented upon request of inhibition and decremented when inhibition is released.

Ignoring of user requests will be done in the respective main functions (channel and PNC) by not considering the user request counter.

All the direct users of a channel are inhibited by ignoring the user request counter for the channel depending on the state of the channel and the inhibition type (prevent wake up or limit to no com) in the channel state machine.

Indirect user requests (user → PNC → channel) are inhibited at the PNC level by ignoring the user count for the PNC depending on the PNC state and inhibition type (prevent wake up or limit to no com) in PNC state machine. Thus this request is not propagated to all linked channels.

All the other requests (external request in ERA and EIRA, DCM request, Wake up from `EcuM`, and `Nm`) are not affected because of inhibition.

`ECU group classification` acts as a mask only. It decides if the inhibition request needs to be processed or not but it doesn't affect the requested inhibition status.

3.2.6 ComMManageReference

The `ComMManageReference` extension parameter has been introduced to

support Managed Channels and Managing Channels.

- Managed Channel (aka Slave Channel): A ComM channel that references another ComM channel by the ECUC parameter ComMManageReference (No Interaction with Nm). The ComMNmVariant should always be "LIGHT". The ComMPncGatewayType of a Managed Channel is neither set to COMM_GATEWAY_TYPE_ACTIVE nor COMM_GATEWAY_TYPE_PASSIVE.
- Managing channel (aka Master Channel): A ComM channel that is referenced from at least one other channel by the ECUC parameter ComMManageReference (Interacts with Nm). The ComMNmVariant should always be "FULL". A Managing Channel will always be referenced by a Managed Channel. User requests and Passive requests will be propagated from the Managed Channel to the Managing Channel.

Communication Inhibition is handled independently for Managing and Managed channels. Limitation is not propagated from Managed Channel to Managing Channel or vice-versa, but the propagated user request from the Managed Channel to the Managing Channel will not be cancelled.

When there is a Managed Channel request propagation to the Managing Channel, and Managing Channel is in inhibition, the global inhibition counter will not be incremented, as this propagation is treated as an indirect user request. The propagated user request will be processed only after the inhibition is disabled on the Managing Channel.

If the Managing Channel is in inhibition, it will be moved to "no communication". If there is a user request on the Managed Channel, the request will be propagated to its Managing Channel. The Managed Channel will be moved to the "network requested" state and continues to be in this state until the user requests for "no communication" and ComMTMinFullComModeDuration expires. Once inhibition is removed on the Managing Channel, it will be moved to "full communication".

Managed and Managing Channels process ComM_EcuM_WakeUpIndication requests only if their current state is "no communication".

3.2.7 Post-Build Selectable

The Post-Build Selectable feature is supported for the ComMPncEnabled parameter, which sets the PNC (Partial Network Cluster) feature to either Enabled/Disabled. In addition, the ComMChannelPerPncs and ComMUserPerPncs containers respectively refer to the Channel and User required by the PNC.

3.3 Extensions

3.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

3.3.1.1 ComM/ComMConfigSet/ComMPnc/ComMPncComSignal

Short Name	ComMPncComSignal
------------	------------------

Description	Represents the PncComSignals which are used to communicate the EIRA and ERA status of this PNC.
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

3.3.1.2 ComM/ComMConfigSet/ComMPnc/ComMPncComSignal/ComMPncComSignalChannelRef

Short Name	ComMPncComSignalChannelRef
Description	Reference to the ComMChannel which is used to determine whether this PncComSignal shall participate in the active or passive role (via the parameter ComMPncGatewayType of the ComMChannel).
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

3.3.1.3 ComM/ComMConfigSet/ComMPnc/ComMPncComSignal/ComMPncComSignalDirection

Short Name	ComMPncComSignalDirection
Description	Indicates the communication direction of this PncComSignal.
Multiplicity	1..1
Default Value	-
Enumeration Literals	RX, TX
Type	ECUC-ENUMERATION-PARAM-DEF

3.3.1.4 ComM/ComMConfigSet/ComMPnc/ComMPncComSignal/ComMPncComSignalKind

Short Name	ComMPncComSignalKind
Description	Indicates whether this PncComSignal represents EIRA or ERA PNC information.
Introduction	This parameter ComMPncComSignalKind is optional and shall be ignored when ComMPncComSignalDirection equals TX.
Multiplicity	0..1
Default Value	-
Enumeration Literals	EIRA, ERA
Type	ECUC-ENUMERATION-PARAM-DEF

3.3.1.5 ComM/ComMConfigSet/ComMPnc/ComMPncComSignal/ComMPncComSignalRef

Short Name	ComMPncComSignalRef
Description	Reference to the ComSignal which is used to transport the partial network channel request information.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

3.3.1.6 ComM/ComMConfigSet/ComMPnc/ComMPncMainFunctionPeriod

Short Name	ComMPncMainFunctionPeriod
Description	Specifies the period in seconds that the Partial network MainFunction has to be triggered with.
Multiplicity	1..1
Value Range	0..INF
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

3.3.1.7 ComM/ComMConfigSet/ComMUser/ComMUserDisableRtePort

Short Name	ComMUserDisableRtePort
Description	This parameter disables the generation of p-ports to the application. True : ComM disables the generation of p-ports.False : ComM enables the generation of p-ports.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.8 ComM/ComMGeneral/ComMRteEnabled

Short Name	ComMRteEnabled
Description	Enables or disables support of RTE.False: RTE is disabled.True: RTE is enabled.
Multiplicity	1..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.2 Additional Services

The following API are available in version v12.7.0 in addition to those defined in AR

v22-11

3.3.2.1 ComM_Nm_TransmissionFailure

Syntax	ComM_Nm_TransmissionFailure(NetworkHandleType Channel)
Parameters	Channel Handle to identify a communication channel.
Description	Indicates that NM message transmission has failed.

3.3.2.2 ComM_Nm_NetworkTimeoutException

Syntax	ComM_Nm_NetworkTimeoutException(NetworkHandleType Channel)
Parameters	Channel Handle to identify a communication channel.
Description	Indicates that NM-Timeout time has expired when Network is still not released.

3.3.2.3 ComM_GetCurrentPNCComMode

Syntax	Std_ReturnType ComM_GetCurrentPNCComMode(ComM_UserHandleType User, ComM_ModeType * ComMode)
Parameters	User Handle of the user who requests a mode ComMode ComM_ModeType
Return Values	E_OK: Successfully returned Communication Mode from Bus State Manager E_NOT_OK: Return of Communication Mode from Bus State Manager failed COMM_E_MULTIPLE_PNC_ASSIGNED: As multiple PNCs are assigned to the affected user, the current mode of the PNC could not be provided COMM_E_NO_PNC_ASSIGNED: As no PNC is assigned to the affected user, the current mode of the PNC could not be provided COMM_E_UNINIT: ComM is not initialized
Description	Returns the current Communication Mode of the corresponding PNC to which the affected user is assigned.

3.3.2.4 ComM_Nm_ForwardSynchronizedPncShutdown

Syntax	void ComM_Nm_ForwardSynchronizedPncShutdown(NetworkHandleType Channel_u8)
Parameter	Channel_u8 Channel Id
Description	Notification that the Synchronization Shutdown can be started.

3.4 Limitations

- Storing of the ComMFullCommRequestNotificationEnable and ComMPncEnable parameters in NvM is not supported.
- The current implementation of PNC supports only the use case "CBV + Pnc Info" but not the use case "CBV + NodeId + Pnc Info". See the Configuration

Advice for a workaround.

- The default value `COMM_GATEWAY_TYPE_ACTIVE` of the `ComMPncGatewayType` parameter has been removed (if the parameter is not configured, it is now regarded as not set). This is to differentiate between the Gateway coordinated and non-coordinated channels.

The following AUTOSAR-defined features are not currently supported:

- Post Build support
- Memory Mapping
- RTE support

3.5 Exclusions

The following features from AR v22-11 are not supported in version v12.7.0.

API functions not currently implemented:

- `ComM_GetPncToChannelMapping`
- `ComM_UpdatePncToChannelMapping`
- `ComM_ResetPncToChannelMapping`
- `ComM_PnLearningRequest`
- `ComM_UpdatePncMembership`

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- `ComM/ComMConfigSet/ComMChannel/ComMChannelPartitionRef`
- `ComM/ComMConfigSet/ComMChannel/ComMDynamicPncToChannelMappingEnabled`
- `ComM/ComMConfigSet/ComMPnc/ComMChannelPerPnc`
- `ComM/ComMConfigSet/ComMPnc/ComMChannelPerTxOnlyPnc`
- `ComM/ComMConfigSet/ComMPnc/ComMPncEthIfSwitchPortGroupRef`
- `ComM/ComMConfigSet/ComMPnc/ComMUserPerPnc`
- `ComM/ComMConfigSet/ComMUser/ComMUserEcucPartitionRef`
- `ComM/ComMGeneral/ComMDynamicPncToChannelMappingSupport`

3.6 Deviations

3.6.1 Configuration Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

3.6.1.1 ComM/ComMConfigSet/ComMChannel/ComMManageReference

[RTA-BSW] Description	Representst the reference between a ComMChannel with role managing channel and a ComMChannel with role managed channel.
[AUTOSAR] Description	Represents the reference between a ComMChannel with role managing channel and a ComMChannel with role managed channel.
[RTA-BSW] Upper Multiplicity	1
[AUTOSAR] Upper Multiplicity	*

3.6.1.2 ComM/ComMConfigSet/ComMChannel/ComMNetworkManagement/ComMNmVariant

[RTA-BSW] Introduction	Shall be harmonized with NM configuration. Dependency: ComMNmVariant shall be NONE if ComMBusType = COMM_BUS_TYPE_INTERNAL. ComMNmVariant shall be LIGHT for managed channels. ComMNmVariant shall be FULL for managing channels.
[AUTOSAR] Introduction	Shall be harmonized with NM configuration.
[RTA-BSW] Enumeration Literals	FULL, LIGHT, SLAVE_ACTIVE, SLAVE_PASSIVE, NONE, PASSIVE
[AUTOSAR] Enumeration Literals	FULL, LIGHT, NONE, PASSIVE, SLAVE_ACTIVE, SLAVE_PASSIVE

3.6.1.3 ComM/ComMConfigSet/ComMChannel/ComMNetworkManagement/ComMPncNmRequest

[RTA-BSW] Description	If this parameter equals true then every time a FULL Communication is requested due to a change in the PNC state machine to COMM_PNC_REQUESTED Nm shall be called using the API Nm_NetworkRequest.
[AUTOSAR] Description	If this parameter equals true, then Nm shall be requested again by calling Nm_NetworkRequest under either the following conditions:
[REMOVED] Introduction	- every time a FULL Communication is requested due to a change in the PNC state machine to COMM_PNC_REQUESTED - if a shut-down for a PNC coincides with a PNC request of the same PNC

3.6.1.4 ComM/ComMConfigSet/ComMChannel/ComMPncGatewayType

[REMOVED] Default Value	COMM_GATEWAY_TYPE_ACTIVE
-------------------------	--------------------------

3.6.1.5 ComM/ComMConfigSet/ComMChannel/ComMTMinFullComModeDuration

[RTA-BSW] Parent Container	ComM/ComMConfigSet/ComMChannel/ComMTMinFullComModeDuration
[AUTOSAR] Parent Container	ComM/ComMGeneral/ComMTMinFullComModeDuration
[RTA-BSW] Description	Minimum time duration in seconds, spent in the COMM_FULL_COMMUNICATION sub-state COMM_FULL_COM_NETWORK_REQUESTED. This parameter per channel context is optional. If configured, this parameter value overrules the ComMTMinFullComModeDuration parameter in general ComM config for the corresponding ComMChannel. If not configured, the ComMTMinFullComModeDuration value per channel is considered from the parameter ComMTMinFullComModeDuration in general ComM config.
[AUTOSAR] Description	Minimum time duration in seconds, spent in the COMM_FULL_COMMUNICATION sub-state COMM_FULL_COM_NETWORK_REQUESTED.
[RTA-BSW] Lower Multi- plicity	0
[AUTOSAR] Lower Multi- plicity	1
[REMOVED] Default Value	5

3.6.1.6 ComM/ComMConfigSet/ComMChannel/ComMWakeupSleepRequestEnabled

[ADDED] Introduc- tion	Dependency: Only applicable if ComMBusType equals COMM_BUS_TYPE_ETH and the used Ethernet hardware (e.g. PHY, Ethernet switch) is compatible with the OA TC10 specification.
---------------------------	--

3.6.1.7 ComM/ComMConfigSet/ComMPnc

[RTA-BSW] Upper Multiplicity	56
[AUTOSAR] Upper Multiplicity	504

3.6.1.8 ComM/ComMConfigSet/ComMPnc/ComMPncId

[RTA-BSW] Maximum Value	63
[AUTOSAR] Maximum Value	511

3.6.1.9 ComM/ComMConfigSet/ComMUser

[RTA-BSW] Upper Multiplicity	255
[AUTOSAR] Upper Multiplicity	65635

3.6.1.10 ComM/ComMGeneral/ComMSynchronizedPncShutdownEnabled

[RTA-BSW] Introduction	False: synchronized PNC shutdown is disabled True: synchronized PNC shutdown is enabled NOTE: This is only possible for ECU that has the role of an top-level PNC coordinator or intermediate PNC within the PNC network
[AUTOSAR] Introduction	FALSE: synchronized PNC shutdown is disabled TRUE: synchronized PNC shutdown is enabled NOTE: This is only possible for ECU that has the role of an top-level PNC coordinator or intermediate PNC within the PNC network

3.6.1.11 ComM/ComMGeneral/ComMVersionInfoApi

[RTA-BSW] Description	Switches the possibility to read the published information with the service ComM_GetPublishedInformation().
[AUTOSAR] Description	Switches the possibility to read the version information with the service ComM_GetVersionInfo().

3.7 API Definition

List of supported AUTOSAR API

Table 3.27.ComM API Definition

ComM_Init	Initializes the AUTOSAR Communication Manager and restarts the internal state machines.
ComM_DeInit	This API de-initializes the AUTOSAR Communication Manager.
ComM_GetStatus	Return the status as COMM_INIT if it is initialized; else returns COMM_UNINIT.
ComM_RequestComMode	Requesting of a Communication Mode by a user.
ComM_GetMaxComMode	Function to query the maximum allowed Communication Mode of the corresponding user.
ComM_GetRequestedComMode	Function to query the currently requested Communication Mode of the corresponding user.
ComM_GetCurrentComMode	Function to query the current communication mode of the user.
ComM_GetVersionInfo	Get the version information (Module Id; Vendor Id; Vendor specific version numbers) of the module.

ComM_LimitChannelToNoComMode	Changes the inhibition status for the channel for changing from COMM_NO_COMMUNICATION to a higher Communication Mode.
ComM_LimitECUToNoComMode	Changes the inhibition status for the ECU (=all channels) for changing from COMM_NO_COMMUNICATION to a higher Communication Mode.
ComM_PreventWakeUp	Changes the inhibition status COMM_NO_WAKEUP for the corresponding channel.
ComM_SetECUGroupClassification	Changes the ECU Group Classification status.
ComM_ReadInhibitCounter	This function returns the amount of rejected COMM_FULL_COMMUNICATION user requests.
ComM_ResetInhibitCounter	This function resets the Inhibited COMM_FULL_COMMUNICATION request Counter.
ComM_GetInhibitionStatus	Return the inhibition (wakeup inhibition and limit to no communication status) status of the ComM channel.
ComM_Nm_NetworkStartIndication	Indicate that NM-message has been received in the Bus Sleep Mode.
ComM_Nm_NetworkMode	Notify that the network management has entered.
ComM_Nm_PrepareBusSleepMode	Notify that the network management has entered Prepare Bus-Sleep Mode.
ComM_Nm_BusSleepMode	Notify that the network management has entered Bus-Sleep Mode
ComM_Nm_RestartIndication	Is called when at least on one of the coordinated busses NM is restarted; while NmIf has started to shut down the coordinated busses.
ComM_DCM_ActiveDiagnostic	Give indication of active diagnostic by the DCM.
ComM_DCM_InactiveDiagnostic	Give indication of inactive diagnostic by the DCM.
ComM_EcuM_WakeUpIndication	Notify a wake up on the corresponding channel.
ComM_EcuM_PNCWakeUpIndication	Notify a wake up on the corresponding Partial Network Cluster.
ComM_CommunicationAllowed	Used by EcuM or BswM to indicate to ComM when communication is allowed.
ComM_BusSM_ModeIndication	Give an indication of the actual bus mode by the corresponding Bus State Manager.
ComM_BusSM_BusSleepMode	Notify corresponding Bus State Manager of Bus-Sleep Mode.
ComM_MainFunction	Performs the processing of ComM activities that are not directly initiated by the calls (e.g. from the RTE).

For Further information please refer to AUTOSAR_SWS_COMManager.pdf

3.8 Configuration Advice



NOTE

Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the "Configuration Generation" section of the RTA-CAR User Guide.

3.8.1 General

- ComMMainFunctionPeriod should be configured as a small enough value to act as the counting block for other time dependent configuration parameters.
- The supported minimum ComMMainFunctionPeriod (and ComMPncMainFunctionPeriod) configuration value is 1 micro second i.e. 0.000001s.
- All ComM timings for ComMMainFunctionPeriod (and ComMPncMainFunctionPeriod) support microsecond resolution, so for example a sub-millisecond time of 3.125ms will not be rounded down to 3ms.
- ComM expects at least one ComMUser to be configured. If no user is expected to be configured in the project, you should configure a dummy ComMUser and an internal channel, and then assign the user to the channel. This will not affect functionality as the channel is never requested for a mode transition.
- ComMUserIdentifier must be configured for all ComMUsers. A partial configuration of ComMUserIdentifier is not permitted.
- ComMResetAfterForcingNoComm can only be enabled if BswM supports the BswM_ComM_InitiateReset API.
- The Communication Inhibition status is dependent on the configuration of the ComMEcuGroupClassification parameter. If ComMEcuGroupClassification is not set, LimitToNoCom, PreventWakeup and LimitEcuToNoCom requests are not processed and return E_NOT_OK.
- All Tx signals belonging to a channel should be mapped to the same ComSignal (across PNs), and Tx signals should be unique for each PN channel.
- The current implementation of PNC supports only the use case of "CBV + Pnc Info" but not the use case "CBV + NodeId + Pnc Info". However, if you want to configure NodeId, then you need to apply the offset manually. This can be done as follows, using a calculation connected to SWS_ComM_00825:

```
Configured ComMPncId = Wanted ComMPncId - 8 * (<PNC Vector Offset> - 1)
```

In order to better illustrate how this should work, here is a mapping for a use case where <PNC Vector Offset> is equal to 2.

NM PDU format		AR 4.1.1+ PNC		Example usage	RTA-BSW	
Byte No.	Bit No.	ComMPncld No.	Req		Wanted ComMPncld	Configured ComMPncld (Wanted ComMPncld - 8)
Byte 0	Bit 7	Not used	ECUC_CanNm_00060	CBV	Not used	Not used
	Bit 6					
	Bit 5					
	Bit 4					
	Bit 3					
	Bit 2					
	Bit 1					
	Bit 0					
Byte 1	Bit 7	ComMPncld 15	SWS_ComM_00825 ECUC_ComM_00874	NID	Not used	Not used
	Bit 6	ComMPncld 14				
	Bit 5	ComMPncld 13				
	Bit 4	ComMPncld 12				
	Bit 3	ComMPncld 11				
	Bit 2	ComMPncld 10				
	Bit 1	ComMPncld 9				
	Bit 0	ComMPncld 8				
Byte 2	Bit 7	ComMPncld 23		PNI 1	23	15
	Bit 6	ComMPncld 22			22	14
	Bit 5	ComMPncld 21			21	13
	Bit 4	ComMPncld 20			20	12
	Bit 3	ComMPncld 19			19	11
	Bit 2	ComMPncld 18			18	10
	Bit 1	ComMPncld 17			17	9
	Bit 0	ComMPncld 16			16	8
Byte 3	Bit 7	ComMPncld 31		PNI 2	31	23
	Bit 6	ComMPncld 30			30	22
	Bit 5	ComMPncld 29			29	21
	Bit 4	ComMPncld 28			28	20
	Bit 3	ComMPncld 27			27	19
	Bit 2	ComMPncld 26			26	18
	Bit 1	ComMPncld 25			25	17
	Bit 0	ComMPncld 24			24	16
Byte 4	Bit 7	ComMPncld 39		PNI 3	39	31
	Bit 6	ComMPncld 38			38	30
	Bit 5	ComMPncld 37			37	29
	Bit 4	ComMPncld 36			36	28
	Bit 3	ComMPncld 35			35	27
	Bit 2	ComMPncld 34			34	26
	Bit 1	ComMPncld 33			33	25
	Bit 0	ComMPncld 32			32	24
Byte 5	Bit 7	ComMPncld 47		PNI 4	47	39
	Bit 6	ComMPncld 46			46	38
	Bit 5	ComMPncld 45			45	37
	Bit 4	ComMPncld 44			44	36
	Bit 3	ComMPncld 43			43	35
	Bit 2	ComMPncld 42			42	34
	Bit 1	ComMPncld 41			41	33
	Bit 0	ComMPncld 40			40	32
Byte 6	Bit 7	ComMPncld 55		PNI 5	55	47
	Bit 6	ComMPncld 54			54	46
	Bit 5	ComMPncld 53			53	45
	Bit 4	ComMPncld 52			52	44
	Bit 3	ComMPncld 51			51	43
	Bit 2	ComMPncld 50			50	42
	Bit 1	ComMPncld 49			49	41
	Bit 0	ComMPncld 48			48	40
Byte 7	Bit 7	ComMPncld 63		PNI 6	63	55
	Bit 6	ComMPncld 62			62	54
	Bit 5	ComMPncld 61			61	53
	Bit 4	ComMPncld 60			60	52
	Bit 3	ComMPncld 59			59	51
	Bit 2	ComMPncld 58			58	50
	Bit 1	ComMPncld 57			57	49
	Bit 0	ComMPncld 56			56	48

Figure 3.2.Sample Mapping

- If the LIN module is configured as SLAVE, then the corresponding Lin Channels should have ComMNmVariant set as SLAVE_ACTIVE.

3.8.2 Partial Network Cluster (PNC) Management

ComMPncSupport (in ComMGeneral) needs to be TRUE to enable PNC support for ComM.

PNCs should be either fully Gateway coordinated (all assigned channels have a gateway type) or not Gateway coordinated at all (assigned to multiple channels, but all channels have no gateway type). A mix of coordinated and not coordinated ComMChannels is not allowed in a PNC. A non-Gateway coordinated PNC should not have ERA Rx signals configured to it.

ComMPncIds should be configured manually, with a minimum value of 8 and a maximum value of 63. For the PNC feature to work, an appropriate mask value has to be configured in the respective BusNm(s) based on the ComMPncId.

ComMPncIds should be configured in the range between PNC vector offset up to PNC vector length values configured across all BusNms.

Configure ComMPncs as each PNC map to one or more ComMChannel as well as one or more ComMUser.

ComM expects that the PNC bit vector (ComMPncComSignalRef) is configured as an array of type uint8_n (ComSignalType).

PnInfo should be configured with same PNC vector length and PNC vector offset values across all BusNms

Referred ComSignals in ComMPncComSignal should be of PNC vector length configured across all BusNms and should start with the defined start position (ComBitPosition = 8 * PNC vector offset).

ComM uses two types of bit vector, named EIRA and ERA, to exchange PNC status information. The sections below provide specific configuration advice for EIRA and ERA bit vectors.

If the Synchronized PNC Shutdown feature is required, you need to do the following:

- Set NmSynchronizedPncShutdownEnabled to TRUE (in the Nm configuration) as well as the ComMSynchronizedPncShutdownEnabled parameter.
- Set FrNmSynchronizedPncShutdownEnabled (for FlexRay), CanNmSynchronizedPncShutdownEnabled (for Can), UdpNmSynchronizedPncShutdownEnabled (for UdpNm) to TRUE for the required channels.
- Before code generation, verify that the configuration value of ComMMainFunctionPeriod and ComMPncMainFunctionPeriod are not set to zero.

3.8.3 EIRA Configuration advice

The aggregated state of external/internal requested PNs is called External and Internal Requests Array (EIRA) and is used by every ECU that has to switch I-PDU Groups because of the activity of Partial Networks (PN). This information will be

sent from CanNm to Com via PduR.

- All EIRA signals belonging to a bustype (across PNs) should refer to same ComSignal.
- Each ComMPnc can have a number of EIRA Tx ComMPncComSignals, or no EIRA Tx ComMPncComSignals at all, in which case the application has to take care of sending the ComSignal.
- Each ComMPnc can have a number of EIRA Rx ComMPncComSignals, a number of unique BusType channels that are mapped to the ComMPnc, which means that there should be 1 ComMPncComSignal for each bus, or no EIRA Rx ComMPncComSignal at all, in which case the application has to take care of the transition of the ComMPnc state machine (PNC_NO_COMMUNICATION to PNC_READY_SLEEP, PNC_READY_SLEEP to PNC_PREPARE_SLEEP, and PNC_PREPARE_SLEEP to PNC_READY_SLEEP).
- EIRA Rx ComMPncComSignals should be configured for different bustype channels which are mapped to the ComMPnc. For example, if three channels are mapped to the ComMPnc as CANA, CANB (bustype CAN) and FR (bustype Flexray) then two EIRA Rx ComMPncComSignals can be configured for the ComMPnc.
- ComMPncComSignalChannelRef should be configured for each ComMPncComSignal with ComMPncComSignalKind set to EIRA and ComMPncComSignalDirection set to Tx. If ComMPncComSignalKind is set to EIRA, and ComMPncComSignalDirection is set to Rx, then there is no need to configure ComMPncComSignalChannelRef.
- Configure Tx and Rx ComSignals in the Com module, and map these signals to ComMPncSignals for each PNC. These ComSignals need to be configured with ComSignalType as boolean, ComBitSize as 1, and ComBitposition set according to the ComMPncId.
- Set the ComSignalEndiannes for both Tx and Rx ComSignals as LITTLE ENDIAN.
- Configure ComFilter for each ComMPncSignals in Com as below:
 - ComFilterAlgorithm: MASKED_NEW_DIFFERS_MASKED_OLD
 - ComFilterMask: 1
 - ComFilterMax: -1
 - ComFilterMin: -1
 - ComFilterOffset: -1
 - ComFilterPeriod: -1
 - ComFilterX: -1
- Map Tx ComSignals to appropriate Tx Ipdu, and map Rx ComSignals to appropriate Rx Ipdu, via ComIpduSignalRef (in ComIpdu).
- The example below illustrates both users being mapped on both of the PNCs.

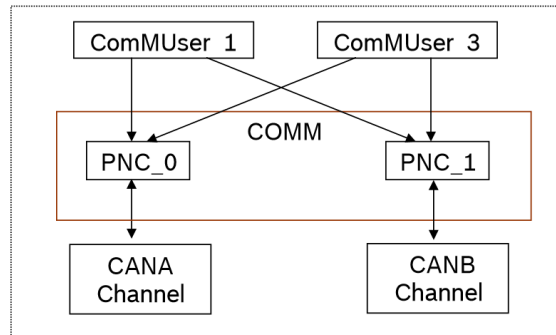


Figure 3.3.Example EIRA Scenario

3.8.4 ERA Configuration advice

The aggregated state of external requested PNs is called External Request Array (ERA) and is used by the Gateways to collect only the external PN requests. This information will be sent from CanNm to Com via PduR.

- All ERA signals belonging to a channel (across PNs) should refer to same ComSignal, and ERA signals should be unique for each PN channel.
- ComMPncGatewayEnabled must be set to TRUE to enable PNC Gateway support for ComM.
- If ComMPncGatewayEnabled is enabled, a number of ComMChannels mapped to the ComMPnc (via ComMPncComSignal) can be configured with ComMPncComSignalKind of ERA and ComMPncComSignalDirection of Rx. A ComMPncComSignal with ComMPncComSignalKind of ERA should not be configured with ComMPncComSignalDirection as Tx.
- Configure Rx ComSignals in the Com module, and map these signals to the ComMPncSignals for each PNC. These ComSignals needs to be configured with ComSignalType as boolean, ComBitSize as 1, and ComBitposition set according to the ComMPncId.
- Set the ComSignalEndiannes for the Rx ComSignals as LITTLE ENDIAN.
- Configure ComFilter for each ComMPncSignals in Com as below:
 - ComFilterAlgorithm: MASKED_NEW_DIFFERS_MASKED_OLD
 - ComFilterMask: 1
 - ComFilterMax: -1
 - ComFilterMin: -1
 - ComFilterOffset: -1
 - ComFilterPeriod: -1
 - ComFilterX: -1
- Map Rx ComSignals to appropriate Rx Ipdu via ComIPduSignalRef (in ComIpdu).
- The example below illustrates both users being mapped on both of the PNCs.

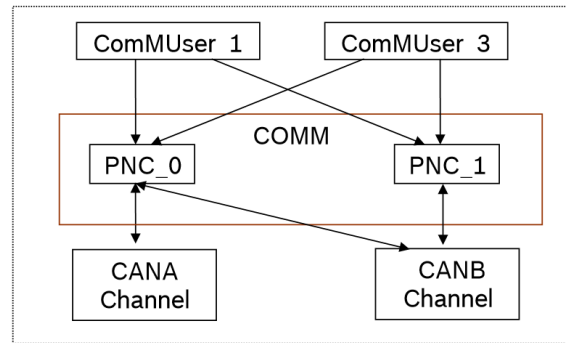


Figure 3.4. Example ERA scenario

3.9 Integration Advice

3.9.1 Init Functions

ComM is initialised by a call to ComM_Init by EcuM or BswM.

If it is the first ECU initialization, ComMNoWakeUp (for all channels) and ComMEcuGroupClassification (global to ECU) will be initialized with default/configured values, otherwise these values will be fetched from NvM. ComMNoCom (for all channels) will be initialized with the default/configured value.

The Inhibition counter needs to be initialized to '0' if it is the first ECU initialization, otherwise it will be fetched from NvM.

3.9.2 Delnit Functions

ComM_Delnit is called by EcuM or BswM at the shutdown phase. ComM_Delnit should be only called if all channels controlled by ComM are in the NO_COMMUNICATION state, and before ECU shutdown. The call to ComM_Delnit will be ignored if a channel controlled by ComM is not in NO_COMMUNICATION state.

3.9.3 Scheduled Functions

Channel and PNC Main functions are generated based on Short names.

- A function named ComM_MainFunction_<Channel_ShortName>() is generated in ComM_Main.c and ComM_Main.h for each ComMChannel.
- A function named ComM_MainFunc_PNC_<PNC_ShortName>() for each PNC should be scheduled before the ComM_MainFunction_<Channel_ShortName>() functions, and can be scheduled based on the <bus.type> (e.g. a ComMPnc on Can can be scheduled in a 10ms proc).

These functions should be called from a task in SchM or RTE with periods equal to ComMMainFunctionPeriod or ComMPncMainFunctionPeriod.

ComM_BSWMD.arxml is used by RTEGen to specify the requirements for scheduling the required functions with OS tasks.

3.9.4 NVRAM Manager (NvM) dependency

The NVRAM Manager (NvM) has to be initialized in order to have "Direct Access" to ComM's stored parameters.

An NvM block has to be configured in NvM and it should be referenced by ComM-GlobalNvMBlockDescriptor (in the ComMGeneral container) to use the NvM feature.

3.9.5 Network management (Nm) dependency

ComM uses the Network Management (Nm) module to synchronize the control of communication capabilities in the network. The Network Management variant can be configured via the ComMNmVariant parameter (in ComMChannel/ComMNetworkManagement), and this configuration plays an important role in deciding what ComM does in each state. Here are some examples:

- If ComMNmVariant=FULL and the user has requested COMM_FULL_-COMMUNICATION, ComM will request the network by a call to Nm_NetworkRequest.
- If ComMNmVariant=FULL and the ComM state machine is in COMM_FULL_COM_READY_SLEEP, ComM will request to release the network by a call to Nm_NetworkRelease.
- If ComMNmVariant=LIGHT or ComMNmVariant=NONE, ComM will omit calls of NM services.
- If ComMNmVariant=LIGHT, in sub-state COMM_FULL_-COM_READY_SLEEP, the timer for ComMNmLightTimeout (in ComMChannel/ComMNetworkManagement) will be started. After this timer has expired, the ComM channel state machine will switch to COMM_NO_COMMUNICATION.

Any combinations of ComMNmVariant (Full/Passive/Light/None) are allowed for an ECU. No users, Tx signals or Rx ERA signals can be configured for a passive channel.

The ComMNmPassiveModeEnable parameter is obsolete and maintained only for backward compatibility. As the Nm module has a dependency on ComMNmPassiveModeEnable, you should not configure it as TRUE if any of the ComMChannels have their ComMNmVariant configured as FULL.

3.9.6 BswM declaration

The following APIs are accessible by BswM. If the declaration is not provided by ComM the build would fail.

- ComM_GetCurrentComMode
- ComM_GetCurrentPNCComMode
- ComM_GetInhibitionStatus
- ComM_GetMaxComMode

- ComM_GetRequestedComMode
- ComM_LimitChannelToNoComMode
- ComM_LimitECUToNoComMode
- ComM_PreventWakeUp
- ComM_ReadInhibitCounter
- ComM_RequestComMode
- ComM_ResetInhibitCounter
- ComM_SetECUGroupClassification

3.9.7 ComM SWC Description

ComM_SWCD.arxml is used by RTEGen to specify ComM services requested by SWCs.

ComM_SWCD.arxml provides:

- ComM Service-SWC which should be mapped to ECU Instance in "SWC to ECU mapping".
- Definition of Client-Server Interfaces for ComM services that can be used by SWCs.
- Provided-Ports for each ComMUser, so that integrators can use to call ComM services from SWC (Ex: RequestComMode: request of a Communication Mode by a user).

3.9.8 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the ENTER and EXIT resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The following is a list of all exclusive areas that this module might use regardless of the user configuration. It is the job of the integrator to ensure that the implementation of these exclusive areas is correctly defined. This implementation can be done using the RTE exclusive area configuration wizard or by manual implementation of the corresponding BSW API calls to ensure exclusivity.

The ComM module uses the following OS resources:

- ComM_PNCTxSigNoNest
- ComM_PNC
- ComM_EIRANoNest
- ComM_ERANoNest
- ComM_PNCWakeUpNoNest
- ComM_User

- ComM_ChannelNoNest
- ComM_Limitation
- ComM_Channel
- ComM_FullComRequestors

3.10 Hardware Requirements

Table 3.28.ComM Hardware Requirements

Support	Usage
FLASH	6123 bytes
RAM	956 bytes

Usage calculated based on a sample configuration.

4 I-Pdu Multiplexer

Module name	IpduM
Package	RTA-COM
AUTOSAR Module ID	52
AUTOSAR Version	22-11

4.1 Introduction

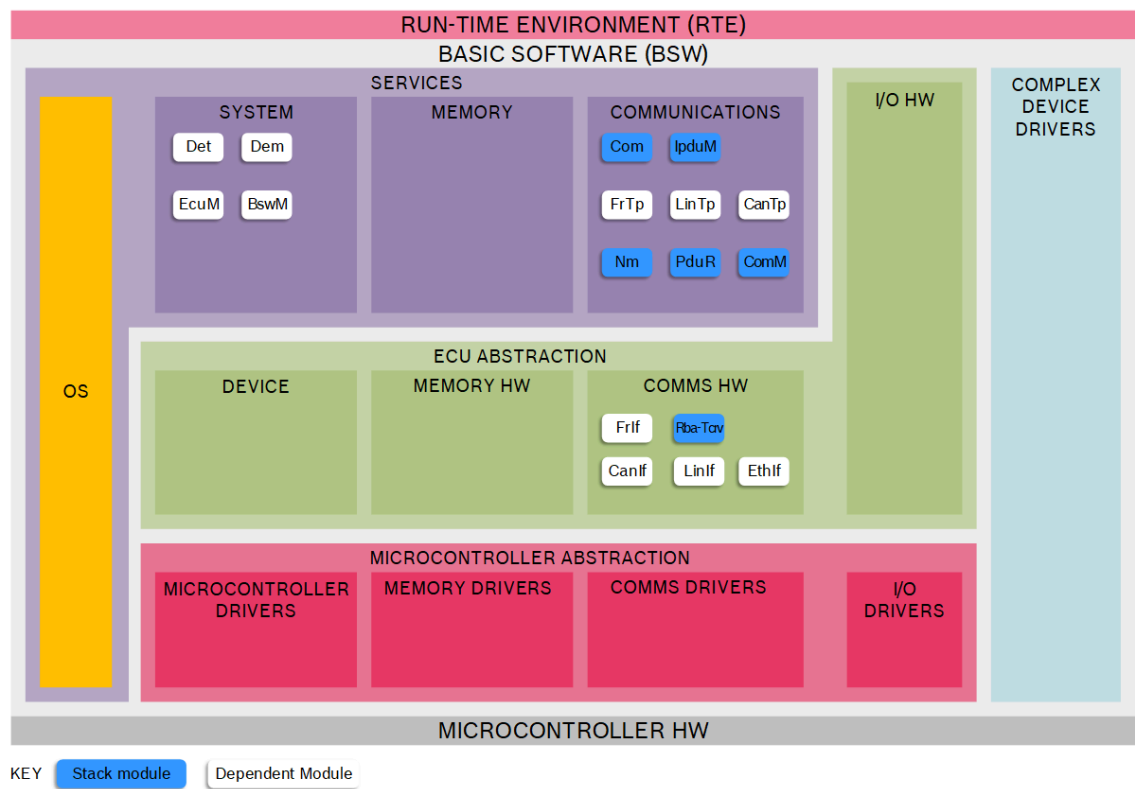


Figure 4.1.Communication Architecture

The I-PDU Multiplexer module (IpduM) is located in the upper communication service layer. The diagram above shows the position of IpduM in the AUTOSAR layered architecture.

The IpduM module provides the following functionality:

- Multiplexing and De-multiplexing of I-PDUs: IpduM supports multiplexing (combining) of I-PDUs for transmission, and de-multiplexing (separating) of I-PDUs for reception operations. A multiplexed I-PDU consists of two separate COM I-PDUs, one with a static part and one with a dynamic part.
- Container Mapping: IpduM supports the combining of multiple similar I-PDUs into a single 'Container PDU' for transmission, and separation of a Container PDU into individual I-PDUs for reception operations.

- Transmit/Receive operations: These operations may be explicitly requested by the calling module (PduR), or triggered by changes in the PDU contents.

Module Configuration

This module can be generated by the RTA-BSW Configuration Generator.

For any unsupported automatic configuration, the module can be manually configured by the user.

4.2 Supported Features

4.2.1 General

The IpduM module supports the following approaches of multiplexing several I-PDUs into one, as defined by the AUTOSAR specification.

- I-PDU Multiplexing
- Multiple-PDU-to-Container Mapping

See the sections below for more details on these two approaches.

4.2.1.1 I-PDU Multiplexing Features

- Transmission Request
- Transmission Trigger
- Just-In-Time update of Parts
- Transmission Confirmation
- Reception
- IpduMRxDirectComInvocation

4.2.1.2 Multiple-PDU-to-Container Mapping Features

- Transmission Request
- Transmission Trigger
- Last-is-Best semantics
- Queued Transmission
- Transmission Confirmation
- Reception
- Dynamic and Static Container Layouts

4.2.2 I-PDU Multiplexing

The IpduM module supports the combining (multiplexing) of two Com I-PDUs into a single IpduM PDU. A multiplexed I-PDU consists of a static part and a dynamic part, each of which corresponds to a single Com I-PDU. The static part consists of zero or more signals or signal groups (S0, S1, S2 etc), and the dynamic part

consists of the selector field and one or more signals or signal groups (D0, D1, D2 etc). Each of these parts may be divided into segments. The dynamic part may correspond to one or more layouts, each of which is specified by the value of the selector field, whose position and length can be configured as required.

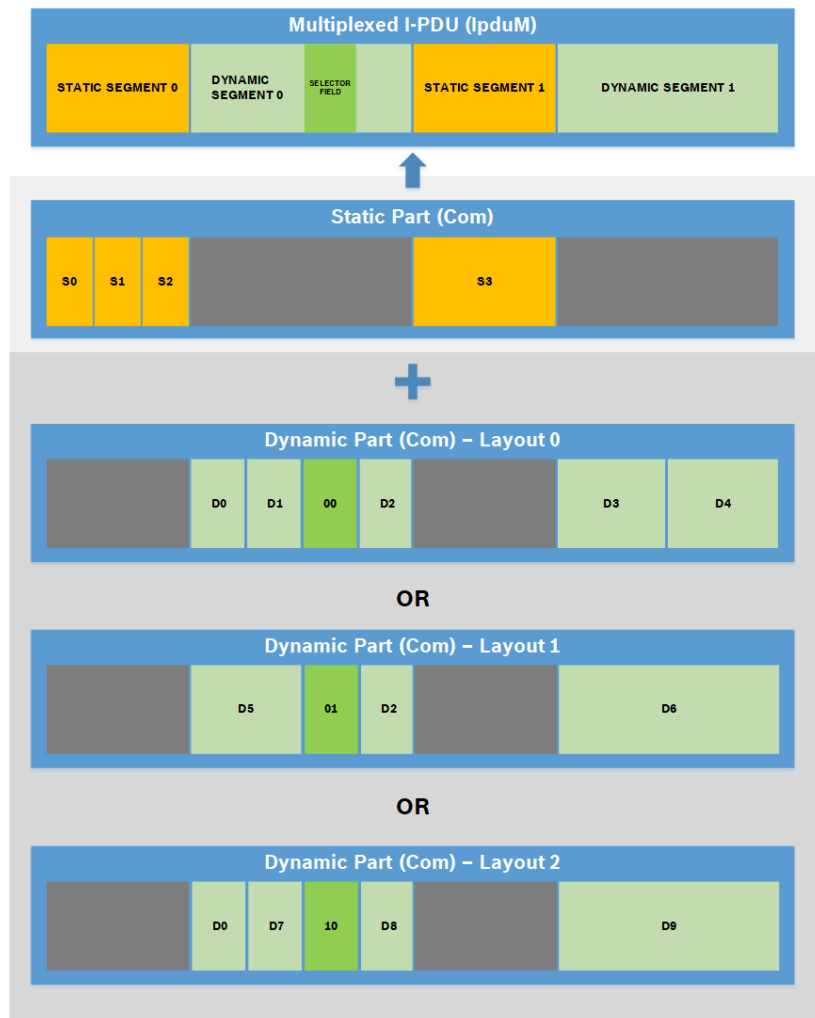


Figure 4.2. Multiplexed I-PDU Example

The diagram above shows an example of a multiplexed I-PDU formed from two Com module I-PDUs, one of which contains a static part and one which contains a dynamic part in one of three possible layouts (0,1,2). Note that both of the source PDUs must have the same segment position and segment length, regardless of layout. The selector field must be included in one dynamic segment only (dynamic segment 0 in this example).

Multiplexed transmission requests may be initiated by one of two mechanisms:

- From the PduR module, by calling IpduM_Transmit.
- From the IpduM module, by calling the API function on detection of a change in the dynamic or static part of an I-PDU (referred to as a 'triggered' request'). Triggered requests may be enabled or disabled through the

IpduMTxTriggerMode configuration parameter.

IpduM also supports disassembly of multiplexed I-PDUs received from the PduR module. It disassembles the static and dynamic parts of the multiplexed I-PDU received from the PduR into separate I-PDUs. PduR is then responsible for routing these separate I-PDUs to their destinations.

4.2.3 Multiple-PDU-to-Container Handling

The IpduM module supports container mapping operations. For transmission, it can collect multiple small I-PDUs into one Container I-PDU, which is then transferred via PduR as one large I-PDU. For reception, a Container PDU may be received from PduR and unpacked into several smaller I-PDUs.

The Container layout can be either dynamically defined (as per AUTOSAR) using headers in front of the contained I-PDUs, or statically (see below) without headers but with defined static positions for contained I-PDUs.

IpduM supports “Triggered”, “Last-Is-Best” and “Queued” transmission modes for Container PDUs.

4.2.3.1 Static Container Layout

To enable the static container layout, the IpduMContainerHeaderSize of the Container PDU has to be configured to IPDUM_HEADERTYPE_NONE, and then IpduM will statically place the contained I-PDUs within the Container PDU according to their configured IpduMContainedTxPduOffset.

On transmission of Container PDUs with static container layout and IpduMContainerTxTriggerMode set as IPDUM_DIRECT, the Contained I-PDU is packed as per the IpduMContainedPduOffset configured for it inside the Container. The Container is triggered when the Trigger Condition of the Static Container is satisfied.

On reception of a static container, the Contained I-PDU is extracted based on the configured ContainedTxPdu Offset. When the Contained I-PDU is not updated, the Unused Area Pattern configured in place of the Contained I-PDU is given to the upper layer in IpduM_RxIndication.

4.3 Extensions

4.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

4.3.1.1 IpduM/IpduMConfig/IpduMContainerTxPdu/IpduMContainerTxDisableLastIsBestProcessing

Short Name	IpduMContainerTxDisableLastIsBestProcessing
Description	Switch to Enable or Disable the LAST_IS_BEST Processing of all Contained Pdu's inside Container Tx PDU.

Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

4.3.2 Additional Services

The following API functions are available in version v12.7.0 in addition to those defined in AR v22-11

- There are no additional APIs defined in this version.

4.4 Limitations

4.4.1 General

- The IpduMPduUpdateBitPosition configuration parameter is not supported for a Container PDU with IpduMContainerHeaderSize set to IPDUM_HEAD-ERTYPE_NONE.
- The IpduMRxDirectComInvocation configuration parameter is supported only for I-PDU Multiplexing and not for Multiple-PDU-to-Container Mapping Multiplexing Type.
- Changing the Lower Multiplicity of the IpduMRxDynamicPart and IpduMRxDynamicSegment parameters from 1 to 0 is not supported. These are mandatory parameters.
- Changing the Lower Multiplicity of the IpduMTxConfirmationPduld from 1 to 0 is also not currently supported.

4.5 Exclusions

The following features from AR v22-11 are not supported in version v12.7.0.

API functions not currently implemented:

- IpduM_MainFunctionRx

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- IpduM/IpduMConfig/IpduMMaxTxBufferSize
- IpduM/IpduMConfig/IpduMMaxTxPathwayCnt
- IpduM/IpduMConfig/IpduMTxPathway/IpduMTxRequest/IpduMTxDynamicPart/IpduMTxDynamicConfirmation
- IpduM/IpduMConfig/IpduMTxPathway/IpduMTxRequest/IpduMTxStaticPart/IpduMTxStaticConfirmation
- IpduM/IpduMGeneral/IpduMStaticPartExists

4.6 Deviations

4.6.1 Configuration Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

4.6.1.1 IpduM/IpduMConfig/IpduMContainedRxPdu/IpduMContainedPduOffset

[RTA-BSW] Description	Static offset (in bytes) of the ContainedPdu. Tags: atp.Status=draft
[AUTOSAR] Description	Static offset (in bytes) of the ContainedPdu. Tags: atp.Status=draft

4.6.1.2 IpduM/IpduMConfig/IpduMContainedRxPdu/IpduMPduUpdateBit-Position

[RTA-BSW] Description	This value specifies where the PDU's Update-Bit is stored in the Container PDU (bit location of PDU's Update-Bit in the Container PDU). Tags: atp.Status=draft
[AUTOSAR] Description	This value specifies where the PDU's Update-Bit is stored in the Container PDU (bit location of PDU's Update-Bit in the Container PDU). Tags: atp.Status=draft

4.6.1.3 IpduM/IpduMConfig/IpduMContainedTxPdu/IpduMContainedPduOffset

[RTA-BSW] Description	Static offset (in bytes) of the ContainedPdu. Tags: atp.Status=draft
[AUTOSAR] Description	Static offset (in bytes) of the ContainedPdu. Tags: atp.Status=draft

4.6.1.4 IpduM/IpduMConfig/IpduMContainedTxPdu/IpduMPduUpdateBit-Position

[RTA-BSW] Description	This value specifies where the PDU's Update-Bit is stored in the Container PDU (bit location of PDU's Update-Bit in the Container PDU). Tags: atp.Status=draft
[AUTOSAR] Description	This value specifies where the PDU's Update-Bit is stored in the Container PDU (bit location of PDU's Update-Bit in the Container PDU). Tags: atp.Status=draft

4.6.1.5 IpduM/IpduMConfig/IpduMContainerRxPdu/IpduMContainerQueueSize

[RTA-BSW] Description	Defines a local queue for handling of each ContainerPdu. Defined in number of instances of this ContainerPdu. Ex: If this parameter is unconfigured or configured to 1 implies there will be one instance of this container Pdu If this parameter configured to 2 implies there will be one instance of this container pdu plus one queued instance. If this paramter configured to 3 implies there will be one instance of this container pdu plus two queued instance and so on.
[AUTOSAR] Description	Defines a local queue for handling of each ContainerPdu.

4.6.1.6 IpduM/IpduMConfig/IpduMContainerRxPdu/IpduMMainFunctionRxRef

[RTA-BSW] Description	Reference to the IpduM_MainFunctionRx instance this container PDU belongs to. Tags: atp.Status=draft
[AUTOSAR] Description	Reference to the IpduM_MainFunctionRx instance this container PDU belongs to.

4.6.1.7 IpduM/IpduMConfig/IpduMContainerTxPdu/IpduMMainFunctionTxRef

[RTA-BSW] Description	Reference to the IpduM_MainFunctionTx instance this container PDU belongs to. Tags: atp.Status=draft
[AUTOSAR] Description	Reference to the IpduM_MainFunctionTx instance this container PDU belongs to.

4.6.1.8 IpduM/IpduMConfig/IpduMContainerTxPdu/IpduMUnusedAreas-Default

[RTA-BSW] Description	IpduM fills not updated areas of the Container PDU with this byte-pattern. Tags: atp.Status=draft
[AUTOSAR] Description	IpduM fills not updated areas of the Container PDU with this byte-pattern. Tags: atp.Status=draft

4.6.1.9 IpduM/IpduMConfig/IpduMMainFunctionRx

[RTA-BSW] Description	Each element of this container defines one instance IpduM_MainFunctionRx, in case multi-core distribution feature is active. Tags: atp.Status=draft
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[AUTOSAR] Description	Each element of this container defines one instance IpduM_MainFunctionRx, in case multi-core distribution feature is active.
-----------------------	--

4.6.1.10 IpduM/IpduMConfig/IpduMMainFunctionRx/IpduMMainRxPartition-Ref

[RTA-BSW] Description	Reference to EcucPartition, where the according IpduM_MainFunction instance is assigned to. Tags: atp.Status=draft
[AUTOSAR] Description	Reference to EcucPartition, where the according IpduM_MainFunction instance is assigned to.

4.6.1.11 IpduM/IpduMConfig/IpduMMainFunctionRx/IpduMMainRxTime-Base

[RTA-BSW] Description	The period between successive calls to according instance of IpduM_MainFunctionRx in seconds. This parameter may be used by the IpduM generator to transform the values of the reception related timing configuration parameters of the IpduM module to internal implementation specific counter or tick values. The IpduM module's internal timing handling is implementation specific. Tags: atp.Status=draft
[AUTOSAR] Description	The period between successive calls to according instance of IpduM_MainFunctionRx in seconds. This parameter may be used by the IpduM generator to transform the values of the reception related timing configuration parameters of the IpduM module to internal implementation specific counter or tick values. The IpduM module's internal timing handling is implementation specific.

4.6.1.12 IpduM/IpduMConfig/IpduMMainFunctionTx

[RTA-BSW] Description	Each element of this container defines one instance IpduM_MainFunctionTx, in case multi-core distribution feature is active (mutual exclusive to ComTimeBase). Tags: atp.Status=draft
[AUTOSAR] Description	Each element of this container defines one instance IpduM_MainFunctionTx, in case multi-core distribution feature is active (mutual exclusive to ComTimeBase).

4.6.1.13 IpduM/IpduMConfig/IpduMMainFunctionTx/IpduMMainTxPartition-Ref

[RTA-BSW] Description	Reference to EcucPartition, where the according IpduM_MainFunction instance is assigned to. Tags: atp.Status=draft
[AUTOSAR] Description	Reference to EcucPartition, where the according IpduM_MainFunction instance is assigned to.

4.6.1.14 IpduM/IpduMConfig/IpduMMainFunctionTx/IpduMMainTxTime-

BaseIpduMMainTxTimeBase

[RTA-BSW] Description	The period between successive calls to IpduM_MainFunctionTx in seconds. This parameter may be used by the IpduM generator to transform the values of the reception related timing configuration parameters of the IpduM module to internal implementation specific counter or tick values. The IpduM module's internal timing handling is implementation specific. Tags: atp.Status=draft
[AUTOSAR] Description	The period between successive calls to IpduM_MainFunctionTx in seconds. This parameter may be used by the IpduM generator to transform the values of the reception related timing configuration parameters of the IpduM module to internal implementation specific counter or tick values. The IpduM module's internal timing handling is implementation specific.
[RTA-BSW] Maximum Value	65.535
[AUTOSAR] Maximum Value	INF

4.6.1.15 IpduM/IpduMConfig/IpduMRxPathway/IpduMRxIndication/IpduMRxDynamicPart

[RTA-BSW] Lower Multiplicity	1
[AUTOSAR] Lower Multiplicity	0

4.6.1.16 IpduM/IpduMConfig/IpduMRxPathway/IpduMRxIndication/IpduMRxDynamicSegment

[RTA-BSW] Lower Multiplicity	1
[AUTOSAR] Lower Multiplicity	0

4.6.1.17 IpduM/IpduMConfig/IpduMTxPathway/IpduMTxRequest/IpduMTxConfirmationPduld

[RTA-BSW] Description	Handle Id used by the PduR for confirmation (IpduM_TxConfirmation) and for TriggerTransmit (IpduM_TriggerTransmit).
[AUTOSAR] Description	Handle Id used by the PduR for confirmation (IpduM_TxConfirmation) and for TriggerTransmit (IpduM_TriggerTransmit).
[RTA-BSW] Lower Multiplicity	1
[AUTOSAR] Lower Multiplicity	0

4.7 API Definition

List of supported API functions.

Table 4.20. IpduM API Functions

IpduM_Init	Initializes the module
IpduM_GetVersionInfo	Returns the version information of the module
IpduM_Transmit	Called by the PduR module to request a transmission of an I-PDU
IpduM_RxIndication	Called by the PduR module to identify the I-PDU ID for the destination of the static and dynamic parts
IpduM_TxConfirmation	Called by the PduR module to translate transmission confirmations from multiplexed or container I-PDUs to their original source I-PDUs
IpduM_TriggerTransmit	Called by the PduR module to trigger the transmission of an I-PDU
IpduM_MainFunctionTx	Called by the PduR module to perform the processing of the reception activities that are not directly handled within the calls from PduR

For further information please refer to the document AUTOSAR_SWS_IPDUMultiplexer.pdf

4.8 Configuration Advice



NOTE

Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the "Configuration Generation" section of the RTA-CAR User Guide.

4.8.1 Timeout Operations

The cycle time, which defines the frequency of invocation of the IpduM main functions by the BSW Scheduler, is defined by setting the IpduMRxTimeBase and IpduMTxTimeBase configuration parameters. Note that the latter value is also used as the time base for transmit confirmation operations (see also the Integration Advice) when you are configuring the following parameters:

- IpduMTxConfirmationTimeout
- IpduMContainerTxConfirmationTimeout
- IpduMContainerTxSendTimeout
- IpduMContainedTxPduSendTimeout

4.8.2 Processing I-PDUs with Dynamic Parts

When transmitting multiplexed I-PDUs with dynamic parts, which may contain multiple dynamic segments, you should configure one instance of `IPduMTxDynamicPart` (indexed by the value of `IPduMTxDynamicHandleId`) for each possible layout within a dynamic segment. This will allow the selector field value to correctly specify the layout for each of the incoming I-PDU's dynamic segments. The value of `IPduMTxDynamicPduRef` should match the reference of the corresponding destination routing path in the `PduR` module (i.e. the value of `PduR-DestPduRef`).

When receiving multiplexed I-PDUs with dynamic parts, you should configure one instance of `IPduMRxDynamicPart` for each possible value of the selector bit field (indexed by the value of `IPduMRxSelectorValue`). The value of `IPduMOutgoingDynamicPduRef` should match the reference of the source routing path in the `PduR` module (i.e. the value of `PduRSrcPduRef`).

4.8.3 Processing I-PDUs with Static Parts

If you wish to transmit or receive multiplexed I-PDUs containing static parts, set `IPduMStaticPartExists` to `TRUE`. If you do not require this feature, you should set `IPduMStaticPartExists` to `FALSE` in order to optimise the performance of the resulting code after compilation.

When transmitting multiplexed I-PDUs with static parts, you should configure one instance of `IPduMTxStaticPart` (indexed by `IPduMTxStaticHandleId`) for each static segment within the static part of the I-PDU. The value of `IPduMTxStaticPduRef` should match the reference of the corresponding destination routing path in the `PduR` module (i.e. the value of `PduRDestPduRef`).

When receiving multiplexed I-PDUs with static parts, you only need to configure one instance of `IPduMRxStaticPart`. The value of `IPduMOutgoingStaticPduRef` should match the reference of the corresponding source routing path in the `PduR` module (i.e. the value of `PduRSrcPduRef`).

4.8.4 Last-Is-Best Processing

When `IpduMContainedTxPduPriorityHandling` is set to `TRUE` for Contained I-Pdus (Priority Handling is enabled), the Contained I-Pdus are packed based on the configured Priority Value. When the Trigger Condition is satisfied, the container is packed based on the Priority Value of the Contained I-Pdus.

`IpduMContainedTxPduPriority` does not need to be configured for Contained I-PDUs belonging to `IPDUM_DIRECT` containers (Priority Handling of Contained I-PDUs is not supported for `IPDUM_DIRECT` containers). `IpduMContainerTxDisableLastIsBestProcessing` also does not need to be configured for `IPDUM_DIRECT` containers.

The Contained I-PDU data inside the container is replaced with the latest I-PDU Data received from Upper layer for Last is Best Semantic Pdus belonging to `IPDUM_DIRECT` containers, whereas for `IPDUM_TRIGGERTRANSMIT` containers, all Contained I-PDUs is replaced with the latest data from Upper Layer.

When `IpduMContainerTxDisableLastIsBestProcessing` set as `TRUE` for `IPDUM_TRIGGERTRANSMIT` containers, the Priority Handling for Contained I-Pdus mapped to the container is disabled (the Contained I-Pdus are packed based on the order of the received Contained Pdus).

4.8.5 IpduMContainerTXPdu

When using `IpduMContainerTXPdu`, the value of `IpduMContainerHeaderSize` should be taken into consideration when configuring `IpduMContainedPduHeaderId`:

- `IPDUM_HEADERTYPE_SHORT`: The value of `IpduMContainedPduHeaderId` should be in the range of 24 bits.
- `IPDUM_HEADERTYPE_LONG`: The value of `IpduMContainedPduHeaderId` should be in the range of 32 bits.

(Note: `IpduMMainFunctionTxRef` is mandatory if `IpduMContainerTxPdu` is present. The Time Base for Timeout Trigger Condition is taken from the `IpduMMainTxTimeBase` configuration parameter.)

4.8.6 Contained Pdu

The Contained Pdu of Highest Offset can be of a variable length for a static container. This is supported only for `IPDUM_DIRECT` containers.

4.8.7 Post-Build Selectable

`IpduM` supports Post-Build Selectable.

- When Post-Build Selectable is configured, unique short names must be given for `IpduMTxDynamicPart`, `IpduMTxStaticPart`, `IpduMTxRequest` and `IpduMRxIndication` across variants as Symbolic name is generated, which should be unique.
- When Post-Build Selectable is configured, unique short names should also be provided for `IpduMContainedTxPdu` and `IpduMContainerTxPdu`.

4.9 Integration Advice

4.9.1 Init Functions

`IpduM` is initialized by `IpduM_Init`, which initializes all module-related global variables and the buffers of the `IpduM` I-PDUs. Both the `Com` and `PduR` modules must be initialized before `IpduM_Init` is called.

4.9.2 Delnit Functions

No `Delnit` is required.

4.9.3 Scheduled Functions

The `IpduM` main function has been split into two parts, `IpduM_MainFunctionTx` and `IpduM_MainFunctionRx`. These functions perform the processing of the

IpduM activities that are not directly initiated by the calls from PduR.

IpduM_MainFunctionTx and IpduM_MainFunctionRx are both called at a cyclic interval, defined by the configuration parameters IpduMTxTimeBase and IpduM-RxTimeBase respectively. These two parameters replace the AUTOSAR-specified configuration parameter IpduMConfigurationTimebase.

4.9.4 Com Module Interactions

The IpduM module relies on the configuration of the Com module.

- For each dynamic layout, an I-PDU needs to be configured in Com, and these I-PDUs must contain the correct selector field value, which can be initialized by configuring them as signals that are initialized with an init value but are never written after initialization. The IpduM module itself does not set the selector bit field.
- The Rx- and Tx-Confirmation path can be optimized from the IpduM module directly to the Com module, i.e. to call the Com module API functions directly from the IpduM module without routing through the PduR module. This can be achieved by setting IpduMRxDirectComInvocation to TRUE.
- In cases where a multiplexed I-PDU is triggered for sending by updating only one part (either the dynamic or static part), the non-triggering part might be overwritten if updated multiple times between two transmissions, even if IpduMTxConfirmationTimeout is set. This may happen because the timeout timer is only started if the triggering part is updated. It may therefore be useful to configure the IpduM transmission confirmation timeout separately from the transmission deadline monitoring timeouts for the single Com I-PDUs in the Com module configuration.

4.9.5 Transmit Timeout

Setting a value of 0 in any of the following configuration parameters will cause IpduM to omit any transmission confirmation operations for multiplexed or container I-PDUs.

- IpduMTxConfirmationTimeout
- IpduMContainerTxConfirmationTimeout
- IpduMContainerTxSendTimeout
- IpduMContainedTxPduSendTimeout

Under normal circumstances, new transmission requests are blocked until confirmation of the current request has been received, but setting any of the above parameters to 0 will disable such blocking.

4.9.6 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the ENTER and EXIT resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in

the OS with the correct task/ISR allocation.

The following is a list of all exclusive areas that this module might use regardless of the user configuration. It is the job of the integrator to ensure that the implementation of these exclusive areas is correctly defined. This implementation can be done using the RTE exclusive area configuration wizard or by manual implementation of the corresponding BSW API calls to ensure exclusivity.

The IpduM module uses the following OS resources:

- IpduM_MulticoreLockTxMpdu
- IpduM_LockTxDynContainerBufferProperties
- IpduM_LockTxStaticContainerBufferProperties
- IpduM_LockRxDynContainerBufferProperties
- IpduM_LockRxStaticContainerBufferProperties

4.10 Hardware Requirements

Table 4.21. IpduM Hardware Requirements

Support	Usage
FLASH	10960 bytes
RAM	5304 bytes

Usage calculated based on a sample configuration.

5 Mirror

Module name	Mirror
Package	RTA-COM
Module ID	N/A
AUTOSAR Version	22-11

5.1 Introduction

The Bus Mirroring module implements an AUTOSAR standard specification that replicates the traffic and state of internal (source) buses to an external (destination) bus. Therefore, a Tester connected to the external bus can monitor internal buses (which are not directly accessible) for debugging purposes.

5.1.1 Overview

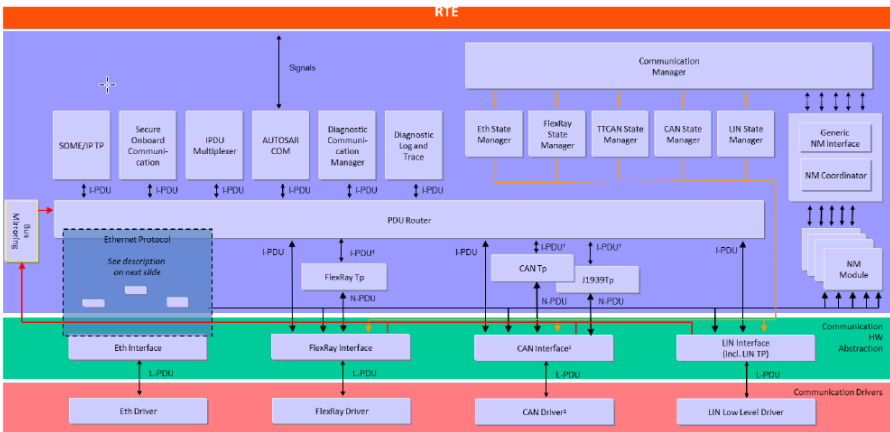


Figure 7.1: AUTOSAR BSW architecture showing the Bus Mirroring module

Figure 5.1.Mirror overview

5.1.2 Interaction

This implementation offers the Bus Mirroring scenario between the Can and Ethernet (IP) bus interfaces (i.e. not with FrIf and/or LinIf). It also interacts with the PduR, Dcm, StbM and Det modules.

Module Configuration

This module can be generated by the RTA-BSW Configuration Generator. For any unsupported automatic configuration, the module can be manually configured by the user.

5.2 Supported Features

- A filtering mechanism is used to control the frames being transmitted from the source bus to the destination bus and prevent the latter from overload-

ing.overloading.



NOTE

Please see Section 5.8.1 for more details on filtering.

- A Tester connected via the service port shall receive unmodified frames from the source bus.
- The Bus Mirroring module defines a protocol for a Header (including a timestamp) and Data (including a timestamp, network Id/type/state plus payload) when packing smaller frames into one larger frame of the destination bus.



NOTE

Please refer either to Section 5.8.3 or section 7.4.2 of AUTOSAR_SWS_Bus-Mirroring.pdf for more details on the Bus Mirroring protocol.

- The Bus Mirroring module does not support a source bus that has a frame size larger than that of the destination bus.
- The Bus Mirroring module cannot influence the traffic on the source bus; it only copies the (transmitted and received) frames that the source bus provides.

5.3 Extensions

5.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

5.3.1.1 Mirror/MirrorGeneral/MirrorRbFetchDataItemTimeStamp

Short Name	MirrorRbFetchDataItemTimeStamp
Description	Source for fetching TimeStamp Field of DataItem.
Multiplicity	0..1
Default Value	MIRROR_STBM_TIMESTAMP
Enumeration Literals	MIRROR_STBM_TIMESTAMP, MIRROR_MCU_TICKS, MIRROR_CANIF_HARDWARE_TIMESTAMP
Type	ECUC-ENUMERATION-PARAM-DEF

5.4 Limitations

Version v12.7.0 has following limitations:

- MirrorMainPartitionRef and MirrorEcucPartitionRef are not currently supported.

- GetStaticFilterState is not currently supported (signature mismatch between the RTE and Mirror API declaration).
- Only one instance of Mirror_MainFunction is currently supported.
- Only MirrorSourceNetworkCan is currently supported as MirrorSourceNetwork.
- Only MirrorDestNetworkIp is currently supported as MirrorDestNetwork.
- Only one CanSMController is supported i.e only the first CanSMController-CanIfCtrlCfg is enabled by the Bus Mirroring module to report the CanFrame. In other words, there is a 1:1 mapping between the MirrorSourceNetwork and CanIfCtrlCfg (rather than a 1:n mapping).

5.5 Exclusions

Mirror_AddLinRangeFilter Mirror_AddLinMaskFilter Mirror_AddFlexRayFilter
 Mirror_GetNetworkHandle Mirror_ReportLinFrame Mirror_ReportFlexRayFrame
 Mirror_ReportFlexRayChannelStatus

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- Mirror/MirrorConfigSet/MirrorDestNetwork/MirrorDestNetworkCan
- Mirror/MirrorConfigSet/MirrorDestNetwork/MirrorDestNetworkCan/
MirrorStatusCanId
- Mirror/MirrorConfigSet/MirrorDestNetwork/MirrorDestNetworkCanXL
- Mirror/MirrorConfigSet/MirrorDestNetwork/MirrorDestNetworkCanXL/Mirror-
ComMNetworkHandleRef
- Mirror/MirrorConfigSet/MirrorDestNetwork/MirrorDestNetworkCanXL/
MirrorDestPdu
- Mirror/MirrorConfigSet/MirrorDestNetwork/MirrorDestNetworkCanXL/
MirrorDestPdu/MirrorDestPduld
- Mirror/MirrorConfigSet/MirrorDestNetwork/MirrorDestNetworkCanXL/
MirrorDestPdu/MirrorDestPduRef
- Mirror/MirrorConfigSet/MirrorDestNetwork/MirrorDestNetworkCanXL/
MirrorDestPdu/MirrorDestPduUsesTriggerTransmit
- Mirror/MirrorConfigSet/MirrorDestNetwork/MirrorDestNetworkCanXL/
MirrorDestQueueSize
- Mirror/MirrorConfigSet/MirrorDestNetwork/MirrorDestNetworkCanXL/
MirrorDestTransmissionDeadline
- Mirror/MirrorConfigSet/MirrorDestNetwork/MirrorDestNetworkCanXL/Mirror-
NetworkId
- Mirror/MirrorConfigSet/MirrorDestNetwork/MirrorDestNetworkCdd
- Mirror/MirrorConfigSet/MirrorDestNetwork/MirrorDestNetworkFlexRay

- Mirror/MirrorConfigSet/MirrorDestNetwork/MirrorDestNetworkFlexRay/
MirrorDestPduFlexRay
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkCan/
MirrorSourceCanMaskBasedIdMapping
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkCan/
MirrorSourceCanMaskBasedIdMapping/MirrorSourceCanMaskBasedIdMap-
pingDestBaseId
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkCan/
MirrorSourceCanMaskBasedIdMapping/MirrorSourceCanMaskBasedIdMap-
pingSourceCanIdCode
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkCan/
MirrorSourceCanMaskBasedIdMapping/MirrorSourceCanMaskBasedIdMap-
pingSourceCanIdMask
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkCan/
MirrorSourceCanSingleIdMapping
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkCan/
MirrorSourceCanSingleIdMapping/MirrorSourceCanSingleIdMappingDest-
CanId
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkCan/
MirrorSourceCanSingleIdMapping/MirrorSourceCanSingleIdMappingSource-
CanId
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkFlexRay
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkFlexRay/
MirrorSourceFlexRayFilter
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkFlexRay/
MirrorSourceFlexRayFilter/MirrorSourceFlexRayFilterChannelAssignment
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkFlexRay/
MirrorSourceFlexRayFilter/MirrorSourceFlexRayFilterCycleRepetition
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkFlexRay/
MirrorSourceFlexRayFilter/MirrorSourceFlexRayFilterId
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkFlexRay/
MirrorSourceFlexRayFilter/MirrorSourceFlexRayFilterLowerBaseCycle
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkFlexRay/
MirrorSourceFlexRayFilter/MirrorSourceFlexRayFilterLowerSlot
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkFlexRay/
MirrorSourceFlexRayFilter/MirrorSourceFlexRayFilterUpperBaseCycle
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkFlexRay/
MirrorSourceFlexRayFilter/MirrorSourceFlexRayFilterUpperSlot
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkLin
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkLin/
MirrorSourceLinFilter
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkLin/

MirrorSourceLinFilter/MirrorSourceLinFilterMask

- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkLin/
MirrorSourceLinFilter/MirrorSourceLinFilterMask/MirrorSourceLinFilterId
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkLin/
MirrorSourceLinFilter/MirrorSourceLinFilterMask/MirrorSourceLinFilterLinId-
Code
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkLin/
MirrorSourceLinFilter/MirrorSourceLinFilterMask/MirrorSourceLinFilterLinId-
Mask
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkLin/
MirrorSourceLinFilter/MirrorSourceLinFilterRange
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkLin/
MirrorSourceLinFilter/MirrorSourceLinFilterRange/MirrorSourceLinFilterId
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkLin/
MirrorSourceLinFilter/MirrorSourceLinFilterRange/MirrorSourceLinFilter-
Lower
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkLin/
MirrorSourceLinFilter/MirrorSourceLinFilterRange/MirrorSourceLinFil-
terUpper
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkLin/
MirrorSourceLinToCanBaselId
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkLin/
MirrorSourceLinToCanIdMapping
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkLin/
MirrorSourceLinToCanIdMapping/MirrorSourceLinToCanIdMappingCanId
- Mirror/MirrorConfigSet/MirrorSourceNetwork/MirrorSourceNetworkLin/
MirrorSourceLinToCanIdMapping/MirrorSourceLinToCanIdMappingLinId
- Mirror/MirrorGeneral/MirrorEcucPartitionRef
- Mirror/MirrorGeneral/MirrorMainFunction
- Mirror/MirrorGeneral/MirrorMainFunction/MirrorMainPartitionRef

5.6 Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

5.6.1 Mirror/MirrorGeneral/MirrorMainFunction/MirrorMainFunctionPeriod

[RTA-BSW] Description	Execution cycle of the respective Mirror_MainFunction instance in seconds. Tags: atp.Status=draft
[AUTOSAR] Description	Execution cycle of the respective Mirror_MainFunction instance in seconds.

5.7 API Definition

List of supported API functions

Table 5.4. Mirror Supported API functions

Mirror_Init. "Initializes the Bus Mirroring module."	
Mirror_DeInit	Resets the Bus Mirroring module to the uninitialized state.
Mirror_GetVersionInfo	Returns the version information.
Mirror_GetStaticFilterState	Returns the state of a pre-configured filter.
Mirror_SetStaticFilterState	Sets the state of a pre-configured filter.
Mirror_AddCanRangeFilter	Creates a CAN ID range filter.
Mirror_AddCanMaskFilter	Creates a CAN ID mask filter.
Mirror_RemoveFilter	Removes a CAN filter that was added at runtime.
Mirror_IsMirrorActive	Returns the global mirroring state.
Mirror_Offline	Disables mirroring i.e. source buses reset to disabled; queued messages purged; destination bus reset to default destination; pre-configured filters disabled; filters added at runtime removed.
Mirror_GetDestNetwork	Returns the currently selected destination bus.
Mirror_SwitchDestNetwork	Changes destination bus to the given ComM channel. Previously active destination bus and all source buses are disabled.
Mirror_IsSourceNetworkStarted	Returns the state of a source bus.
Mirror_StartSourceNetwork	Activates a source bus.
Mirror_StopSourceNetwork	De-activates a source bus.
Mirror_GetNetworkType	Returns the network type of the given network.
Mirror_GetNetworkId	Returns the network ID of the given network.
Mirror_ReportCanFrame	Reports a received or transmitted CAN frame.
Mirror_TxConfirmation	The lower layer communication interface module confirms the transmission of a PDU, or the failure to transmit a PDU.
Mirror_TriggerTransmit	The upper layer module checks if available data fits into buffer size reported by PduInfoPtr->SduLength. If so, data is copied into buffer provided by PduInfoPtr->SduDataPtr and length of actual copied data is updated in PduInfoPtr->SduLength. If not, it returns <i>E_NOT_OK</i> without changing PduInfoPtr.
Mirror_MainFunction	Main function of Bus Mirroring module. Used for scheduling and timeout supervision.

For further information, please refer to AUTOSAR_SWS_BusMirroring.pdf

5.8 Configuration Advice

5.8.1 Filtering

The Bus Mirroring module implements a filtering mechanism to prevent overloading the destination bus. The filter can be configured statically (via `MirrorSourceCanFilter`) or added dynamically at run-time for the source bus.

Upon initialization, all statically configured filters are disabled, and no dynamic filters are available.

Statically configured filters can be explicitly activated and deactivated using `Mirror_SetStaticFilterState`. Dynamic filters can be added at run-time, using `Mirror_AddCanMaskFilter` and removed again by calling `Mirror_RemoveFilter`, with the *filter ID* returned by `Mirror_AddCanMaskFilter`.

Filters are also deactivated/removed when mirroring is stopped or when the destination network is changed.

While a filter is active (i.e. statically configured and activated by `Mirror_SetStaticFilterState`, or dynamically added using `Mirror_AddCanMaskFilter`) all frames from the source bus that match the filter shall be mirrored.



NOTE

This means that no frames from a source bus are mirrored when no filters are active.

When a statically configured filter is deactivated by `Mirror_SetStaticFilterState`, or a dynamically added filter is removed by `Mirror_RemoveFilter`, frames that have been accepted before the deactivation/removal shall still be mirrored to the destination bus.

5.8.1.1 CAN Frame Filters

A CAN mask filter statically configured as `MirrorSourceCanFilterMask` matches the reported *canId*, if this *canId* masked by the `MirrorSourceCanFilterCanIdMask` equals the `MirrorSourceCanFilterCanIdCode`.

A CAN mask filter dynamically added by a call to `Mirror_AddCanMaskFilter` matches the reported *canId*, if this *canId* masked by the *mask* equals the *id*.

A CAN range filter statically configured as `MirrorSourceCanFilterRange` matches the reported *canId*, if the value of this *canId* is $\geq$ the `MirrorSourceCanFilterLower` and $\leq$ the `MirrorSourceCanFilterUpper`.

A CAN range filter dynamically added by a call to `Mirror_AddCanRangeFilter` matches the reported *canId*, if the value of this *canId* is $\geq$ the *lowerId* and $\leq$ the *upperId*.

5.8.2 Status Acquisition

The Bus Mirroring module shall poll the status of each active CAN source bus by

cyclically calling `CanIf_GetControllerMode` and `CanIf_GetTrcvMode` from the `Mirror_MainFunction`. If the returned `ControllerModePtr` is `CAN_CS_STARTED` and the returned `TransceiverModePtr` is `CANTRCV_TRCVMODE_NORMAL`, the reported CAN source bus state shall be set to online, otherwise to offline.

If the bus is online, the Bus Mirroring module shall call `CanIf_GetControllerErrorState`, and if the returned `ErrorStatePtr` is `CAN_ERRORSTATE_PASSIVE` or `CAN_ERRORSTATE_BUSOFF`, the reported CAN source bus state shall be set to error passive or bus-off, respectively.

Additionally, if the bus is online, the Bus Mirroring module shall also call `CanIf_GetControllerTxErrorCounter` and add the returned `TxErrorCounterPtr` to the reported CAN source bus state.

**NOTE**

The APIs `CanIf_GetControllerMode` and `CanIf_GetControllerErrorState` expect a `ControllerId`, and `CanIf_GetTrcvMode` expects a `TransceiverId`, but a *network ID* is required to report the status to the destination bus. The translation of the first two to the latter can be determined at generation time by following the references from the `CanIfCtrlId` and `CanTrcvChannelId`, respectively, to the `MirrorNetworkId` through the ECU configuration via `MirrorComMNetworkHandleRef`.

5.8.3 Mirroring Protocol

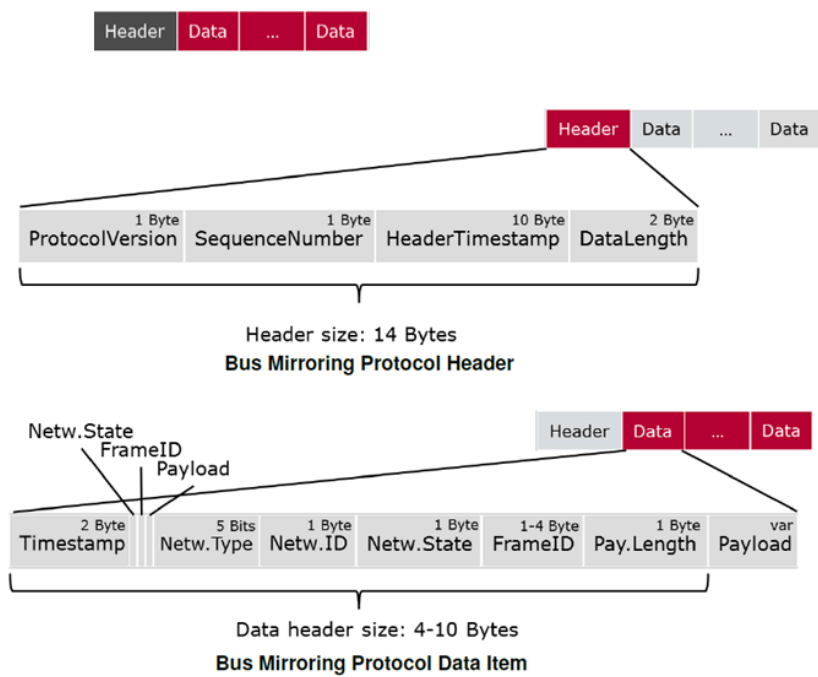


Figure 5.2.Mirroring protocol

5.8.3.1 Creation

When the Bus Mirroring module is initialized, or when `Mirror_SwitchDestNetwork` is called to activate the Ethernet/IP (`MirrorDestNetworkIp`) destination bus, it shall activate a new destination frame buffer and reset the `SequenceNumber` to 0.

When the first data item is added to an empty destination frame buffer, the Bus Mirroring module shall first write the header to the buffer. The `ProtocolVersion` field shall be set to 1, the `SequenceNumber` to the incremented `SequenceNumber` of the last destination frame, the `HeaderTimestamp` filled with the information returned by `StbM_GetCurrentTime`, and the `DataLength` field set to 0.

If the optional configuration parameter `MirrorDestTransmissionDeadline` is set, the Bus Mirroring module shall start the transmission timeout timer.

When a source frame has been received, the Bus Mirroring module shall create a new data item and place it at the end of the currently active destination frame buffer, and add the size of the new data item to the header field `DataLength`.

The `Timestamp` field of the new data item shall be set to the difference between the timestamp in the header and the current time acquired either using `StbM_GetCurrentTime`, `CanIf_GetCurrentTime` or `Mcu_Rb_GetSysTicks()`. The

FrameIDAvailable and PayloadAvailable bits shall be set to 1, and the fields NetworkType, NetworkID, FrameID, PayloadLength, and Payload shall be set according to the received source frame.

If the source bus state has changed since the last transmission of a source frame, the NetworkStateAvailable bit shall be set to 1 and the NetworkState field to the source bus state. Otherwise, the NetworkStateAvailable bit shall be set to 0 and the NetworkState field omitted.

When the source bus state has changed, and if no source frame is received from the same source bus within one *Main Function* cycle, the Bus Mirroring module shall create a new data item and place it at the end of the currently active destination frame buffer, and add the size of the new data item to the header field DataLength.

The Timestamp field of the new data item shall be set to the difference between the timestamp in the header and the current time acquired either using StbM_GetCurrentTime, CanIf_GetCurrentTime or Mcu_Rb_GetSysTicks(). The NetworkStateAvailable bit shall be set to 1, the fields NetworkType and NetworkID set according to the source bus (a CAN bus has a NetworkType of 1), and the NetworkState field set to the source bus state.

Depending on the source bus state, the FrameIDAvailable shall be set to 1 or 0. In the first case, the FrameID shall be set according to the source bus, and in the latter case the FrameID is omitted.

The PayloadAvailable bit shall be set to 0, and the fields PayloadLength and Payload are omitted.

5.8.3.2 Queueing

When a data item does not fit in the remaining space of the currently active destination frame buffer, the Bus Mirroring module shall place this buffer in the queue and activate a new destination frame buffer. The data item shall then be placed in the new buffer.

When the relative timestamp of a data item exceeds 655:35 ms, the Bus Mirroring module shall place the currently active destination frame buffer in the queue and activate a new destination frame buffer. The data item shall then be placed in the new buffer.

If the optional configuration parameter MirrorDestTransmissionDeadline is configured and the transmission timeout expires, the Bus Mirroring module shall place the currently active destination frame buffer in the queue and activate a new destination frame buffer.

The size of the queue for the serialized destination frames is determined by the configuration parameter MirrorDestQueueSize; the size of the queue elements is determined by the PduLength of the Pdu referenced by MirrorDestPduRef.

If a destination frame cannot be placed in the queue because the queue is already full, the Bus Mirroring module shall drop that destination frame, report the runtime error *MIRROR_E_QUEUE_OVERRUN*, and set (to 1) the *Frames Lost* bit of the NetworkState of the next data item created in the currently active destination frame buffer.

5.8.3.3 Transmission

To initiate the transmission of a queued serialized destination frame, the Bus Mirroring module shall call `PduR_MirrorTransmit` with `PduInfoPtr->MetaDataPtr` set to the `NULL_PTR` and `PduInfoPtr->SduLength` set to the written part of the destination frame. If `MirrorDestPduUsesTriggerTransmit` is enabled, `PduInfoPtr->SduDataPtr` shall be set to the `NULL_PTR`, otherwise to the used part of the queued destination frame.

A `NULL_PTR` for `PduInfoPtr->SduDataPtr` ensures that the destination bus interface module (SoAd) fetches the destination frame using `Mirror_TriggerTransmit`.

If the `PduR_MirrorTransmit` returns `E_NOT_OK`, the Bus Mirroring module shall immediately remove the destination frame from the queue, report the runtime error `MIRROR_E_TRANSMIT_FAILED`, and set (to 1) the *Frames Lost* bit of the `NetworkState` of the next data item created in the currently active destination frame buffer.

The Bus Mirroring module shall initiate the transmission of queued serialized destination frames from the `Mirror_MainFunction` and from the `Mirror_TxConfirmation` callback. This ensures that queued destination frames are transmitted as fast as possible.

The Bus Mirroring module shall not transmit the next serialized destination frame before the previous destination frame has been confirmed by a call to `Mirror_TxConfirmation`.

When `Mirror_TriggerTransmit` is called for a serialized destination frame, the Bus Mirroring module shall copy the used part of the queued destination frame to `PduInfoPtr->SduDataPtr` and update `PduInfoPtr->SduLength` accordingly.

If the `PduInfoPtr->SduLength` provided by `Mirror_TriggerTransmit` is too small for the currently transmitted serialized destination frame, the Bus Mirroring module shall remove the destination frame from the queue, report the runtime error `MIRROR_E_TRANSMIT_FAILED`, set (to 1) the *Frames Lost* bit of the `NetworkState` of the next data item created in the currently active serialized destination frame buffer, and return `E_NOT_OK` to stop this transmission.

When `Mirror_TxConfirmation` is called to report the successful or failed transmission of a serialized destination frame, the Bus Mirroring module shall remove the destination frame from the queue.

If the `Mirror_TxConfirmation` reports the failed transmission of a serialized destination frame (result is `E_NOT_OK`), the Bus Mirroring module shall report the runtime error `MIRROR_E_TRANSMIT_FAILED`, and set (to 1) the *Frames Lost* bit of the `NetworkState` of the next data item created in the currently active destination frame buffer.

Please see the RTA-Can Stack Reference Guide for more details.

5.8.4 Dcm For Bus Mirroring

The Bus Mirroring module is controlled by a Diagnostic Control Application

through the dedicated (service) port API via the Dcm configuration.



NOTE

Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the "Configuration Generation" section of the RTA-CAR User Guide.

5.9 Integration Advice

5.9.1 Init Functions

The C initialization code, also known as start-up code, initializes global and static variables. It must be executed before any service call.

The Bus Mirroring module initializes each source frame during execution of the API `Mirror_Init`, plus internal data not yet initialized by the start-up code e.g. C structs.

`Mirror_Init` shall not enable inter-ECU communication.

Regarding the initialization of unused areas of an I-PDU, the module shall fill/set the values of filters (either static or dynamic).

Upon initialization, the Bus Mirroring module remains inactive until the source bus is enabled (via `Mirror_StartSourceNetwork`).

5.9.2 Delnit Functions

The Bus Mirroring module provides the API `Mirror_DeInit` for de-initialization of the Mirror layer. After de-initialization, no communication via this module is possible.

5.9.3 Scheduled Functions

Only one instance of `Mirror_MainFunction` is currently supported.

5.9.4 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the ENTER and EXIT resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The following is a list of all exclusive areas that this module might use regardless of the user configuration. It is the job of the integrator to ensure that the implementation of these exclusive areas is correctly defined. This implementation can be done using the RTE exclusive area configuration wizard or by manual implementation of the corresponding BSW API calls to ensure exclusivity.

The Mirror module uses the following OS resources:

- Mirror_DestNetwork
- Mirror_StaticFilter
- Mirror_DynamicFilter

5.10 Hardware Requirements

Information relating to the hardware resources required by this module is not currently available.

6 Network Management Interface

Module name	Nm
Package	RTA-COM
AUTOSAR Module ID	29
AUTOSAR Version	22-11

6.1 Introduction

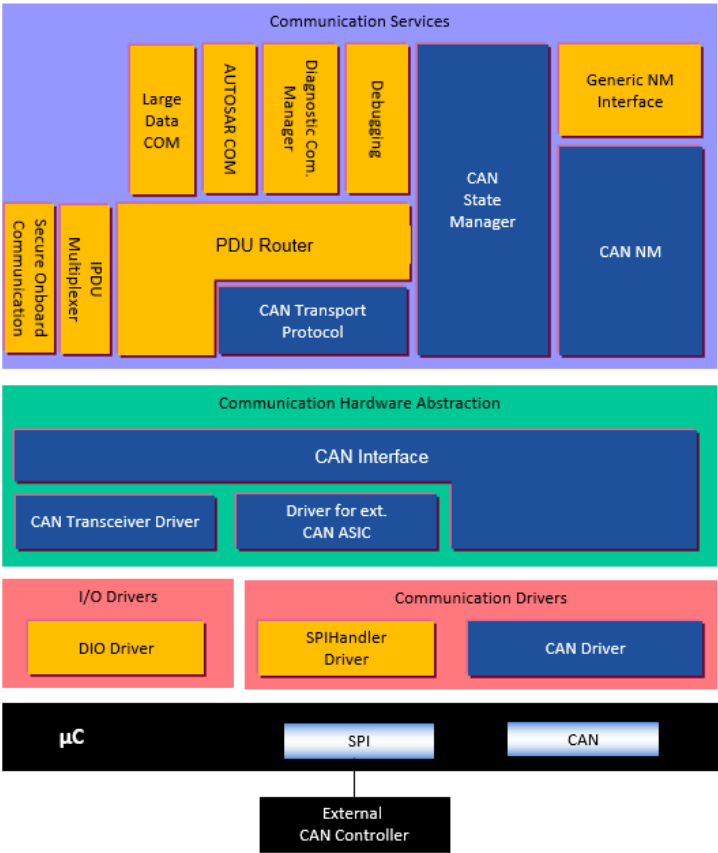


Figure 6.1.Communication stack

The Network Management Interface module (Nm) is a bus-independent adaptation layer between the Communication Manager (ComM) and the bus-specific Network Management modules (e.g. CanNm, FrNm, LinNm).

The primary functions of the Nm module are:

- Converting ComM's generic function calls to bus-specific functions of the bus-specific NM layer.
- Converting callback functions of the bus-specific NM layer to generic call-backs to ComM.

Module Configuration

This module can be generated by the RTA-BSW Configuration Generator.

For any unsupported automatic configuration, the module can be manually configured by the user.

6.2 Supported Features

6.2.1 AUTOSAR Features

The Nm module supports all of the following features described in the AUTOSAR specification for the Network Management Interface:

- Base Functionality
- Standard Services
- Communication Control Services
- Extra Services
- Standard Call-back notifications
- Extra Call-back notifications

6.2.2 Bus-Specific Nm Layers

In addition to the “official” AUTOSAR NM modules (CanNm, FrNm, LinNm), the Nm module also supports generic, bus-specific NM layers. Any component which implements the required provided interfaces and uses the provided callback functions of Nm can be used as a bus-specific NM<BusNm>. The AUTOSAR bus-NMs are then explicitly supported since they implement the interfaces of <BusNm>.

6.2.3 Synchronized Pnc Shutdown

A Synchronized Pnc Shutdown feature is provided via the Nm_RequestSynchronizedPncShutdown and Nm_ForwardSynchronizedPncShutdown extension APIs. See the Configuration Advice section for more information.

6.3 Extensions

6.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

6.3.1.1 Nm/NmChannelConfig/NmShutdownDelayTimer

Short Name	NmShutdownDelayTimer
Description	This parameter defines the time in seconds which the NM Coordination algorithm shall delay the release of this channel with.
Multiplicity	1..1

Value Range	0..Inf
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

6.3.1.2 Nm/NmGlobalConfig/NmGlobalConfigLinNm

Short Name	NmGlobalConfigLinNm
Description	This is a sub container of "NmGlobalConfig" which contains all global LinNm configuration parameters of the Nm Interface
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

6.3.1.3 Nm/NmGlobalConfig/NmGlobalConfigLinNm/LinNmCoordinatorSyncSupport

Short Name	LinNmCoordinatorSyncSupport
Description	Enables/disables the coordinator synchronization support
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

6.3.1.4 Nm/NmGlobalConfig/NmGlobalConfigLinNm/LinNmMainFunctionPeriod

Short Name	LinNmMainFunctionPeriod
Description	Call cycle in seconds of LinNm_MainFunction
Multiplicity	0..1
Value Range	0..255
Default Value	false
Type	ECUC-FLOAT-PARAM-DEF

6.3.1.5 Nm/NmGlobalConfig/NmGlobalConfigLinNm/LinNmPassiveModeEnabled

Short Name	LinNmPassiveModeEnabled
Description	Pre-processor switch for enabling support of the Passive Mode of the LinN
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

6.3.1.6 Nm/NmGlobalConfig/NmGlobalConfigLinNm/LinNmState-ChangeIndEnabled

Short Name	LinNmStateChangeIndEnabled
Description	Pre-processor switch for enabling the Network Management state change notification
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

6.3.1.7 Nm/NmGlobalConfig/NmGlobalConfigLinNm/LinNmTimeoutTime

Short Name	LinNmTimeoutTime
Description	Network Timeout after passive start-up. It denotes the time in seconds how long the NM shall stay in Network Mode in case of passive start-up before transition into Bus-Sleep Mode is initiated
Multiplicity	0..1
Value Range	0..65.535
Default Value	false
Type	ECUC-FLOAT-PARAM-DEF

6.3.1.8 Nm/NmGlobalConfig/NmGlobalConstants/NmNumberOfClusters

Short Name	NmNumberOfClusters
Description	Number of NM clusters allowed within one ECU
Multiplicity	0..1
Value Range	0..255
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

6.3.1.9 Nm/NmGlobalConfig/NmGlobalConstants/NmRemoteSleep-CancelCallback

Short Name	NmRemoteSleepCancelCallback
Description	Upper layer callback that is called to indicate cancellation of Remote Sleep situation after BusNm receives NM messages/Votes
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

6.3.1.10 Nm/NmGlobalConfig/NmGlobalConstants/NmRemoteSleepIndCallback

Short Name	NmRemoteSleepIndCallback
Description	Upper layer callback that is called to indicate Remote Sleep situation detected by BusNm
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

6.3.1.11 Nm/NmGlobalConfig/NmGlobalConstants/NmRxCallbackHeader

Short Name	NmRxCallbackHeader
Description	This is the header file that declares the prototype of NmRxIndicationCallback
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

6.3.1.12 Nm/NmGlobalConfig/NmGlobalConstants/NmRxIndicationCallback

Short Name	NmRxIndicationCallback
Description	Application callback that is called when Nm is indicated about a reception by BusNm. This should be of type "void Callback_Name(NetworkHandleType)". If no callback is configured, NULL_PTR will be assigned
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

6.3.1.13 Nm/NmGlobalConfig/NmGlobalConstants/NmStateChangeIndCallback

Short Name	NmStateChangeIndCallback
Description	Upper layer callback that is called to indicate state change notifications
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

6.3.2 Additional Services

The following API are available in version v12.7.0 in addition to those defined in AR v22-11

6.3.2.1 Nm_RequestSynchronizedPncShutdown

Syntax	Std_ReturnType Nm_RequestSynchronizedPncShutdown(NetworkHandleType NetworkHandle, PNCHandleType PncId)
Parameters In	NetworkHandle Identification of the NM-channel. PncId Identification of the Pnc which is requested for a synchronized shutdown across the PNC network topology.
Description	Forwards request for a synchronized PNC shutdown of a particular PNC given by PncId to the affected <Bus>Nm by calling <Bus>Nm_RequestSynchronizedPncShutdown. Only valid if NmStandardBusType is not set to NM_BUSNM_LOCALNM (e.g. CanNm_RequestSynchronizedPncShutdown function is called for NM_BUSNM_CANNM).

6.3.2.2 Nm_NetworkTimeoutException

Syntax	void Nm_NetworkTimeoutException(const NetworkHandleType NetworkHandle)
Parameters In	NetworkHandle Identification of the NM-channel.
Description	NM callback interface notifying expiration of NM-Timeout time.

6.4 Limitations

The following features are not supported for version v12.7.0:

- The NM Coordinator functionality is not currently supported.
- Nm currently supports the Passive Mode Feature only in the CAN Stack (CanNm) since other BusNms do not yet support Passive Mode.
- The parameter NmGenericBusNmShutdownTime is not supported yet.

6.5 Exclusions

The following features from AR v22-11 are not supported in version v12.7.0.

API functions not currently implemented:

```
Nm_PncBitVectorRxIndication          Nm_PncBitVectorTxIndication
Nm_PncBitVectorTxConfirmation
```

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- Nm/NmChannelConfig/NmBusType/NmGenericBusNmConfig/NmGenericBusNmShutdownTime
- Nm/NmChannelConfig/NmDynamicPncToChannelMappingEnabled

- Nm/NmChannelConfig/NmPnEnabled
- Nm/NmChannelConfig/NmPnEraCalcEnabled
- Nm/NmChannelConfig/NmPnFilterMaskByte
- Nm/NmChannelConfig/NmPnFilterMaskByte/NmPnFilterMaskByteIndex
- Nm/NmChannelConfig/NmPnFilterMaskByte/NmPnFilterMaskByteValue
- Nm/NmChannelConfig/NmPncBitVectorLength
- Nm/NmGlobalConfig/NmEcucPartitionRef
- Nm/NmGlobalConfig/NmGlobalFeatures/NmDynamicPncToChannelMappingSupport
- Nm/NmGlobalConfig/NmGlobalFeatures/NmGlobalCoordinatorTime
- Nm/NmGlobalConfig/NmGlobalFeatures/NmPartialNetworkSupportEnabled
- Nm/NmGlobalConfig/NmGlobalProperties/NmPnEraCalcEnabled
- Nm/NmGlobalConfig/NmGlobalProperties/NmPnResetTime
- Nm/NmGlobalConfig/NmGlobalProperties/NmPnShutdownMessageRetransmissionDuration
- Nm/NmGlobalConfig/NmGlobalProperties/NmPncBitVectorOffset

6.6 Deviations

6.6.1 Configuration Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

6.6.1.1 Nm/NmChannelConfig/NmBusType

[ADDED] Description	NmBusType
[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

6.6.1.2 Nm/NmChannelConfig/NmBusType/NmGenericBusNmConfig

[ADDED] Description	NmGenericBusNmConfig
---------------------	----------------------

6.6.1.3 Nm/NmChannelConfig/NmBusType/NmGenericBusNmConfig/NmGenericBusNmPrefix

[RTA-BSW] Description	The prefix which identifies the generic <BusNm>. This will be used to determine the API name to be called by Nm for the provided interfaces of the <BusNm>. This string will be used for the module prefix before the “_” character in the API call name.
-----------------------	---

[AUTOSAR] Description	The prefix which identifies the generic <Bus>Nm. This will be used to determine the API name to be called by Nm for the provided interfaces of the <Bus>Nm. This string will be used for the module prefix before the “_” character in the API call name.
[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

6.6.1.4 Nm/NmChannelConfig/NmBusType/NmStandardBusNmConfig

[ADDED] Description	NmStandardBusNmConfig
---------------------	-----------------------

6.6.1.5 Nm/NmChannelConfig/NmBusType/NmStandardBusNmConfig/NmStandardBusType

[RTA-BSW] Description	Identifies the bus type of the channel for standard AUTOSAR <BusNm>s and is used to determine which set of API calls to be called by Nm for the <BusNm>s.
[AUTOSAR] Description	Identifies the bus type of the channel for standard AUTOSAR <Bus>Nms and is used to determine which set of API calls to be called by Nm for the <Bus>Nms.
[RTA-BSW] Enumeration Literals	NM_BUSNM_CANNM, NM_BUSNM_FRNM, NM_BUSNM_J1939NM, NM_BUSNM_LINNM, NM_BUSNM_UDPNM
[AUTOSAR] Enumeration Literals	NM_BUSNM_CANNM, NM_BUSNM_FRNM, NM_BUSNM_J1939NM, NM_BUSNM_LOCALNM, NM_BUSNM_UDPNM

6.6.1.6 Nm/NmChannelConfig/NmChannelSleepMaster

[RTA-BSW] Introduction	If this parameter is set to TRUE, the Nm shall assume that the channel is always ready to go to sleep and that no calls to Nm_RemoteSleepIndication or Nm_RemoteSleepCancellation will be made from the <BusNm> representing this channel. If this parameter is set to FALSE, the Nm shall not assume that the network is ready to sleep until a call has been made to Nm_RemoteSleepCancellation.
[AUTOSAR] Introduction	If this parameter is set to TRUE, the Nm shall assume that the channel is always ready to go to sleep and that no calls to Nm_RemoteSleepIndication or Nm_RemoteSleepCancellation will be made from the <Bus>Nm representing this channel. If this parameter is set to FALSE, the Nm shall not assume that the network is ready to sleep until a call has been made to Nm_RemoteSleepCancellation.

6.6.1.7 Nm/NmChannelConfig/NmComMChannelRef

[RTA-BSW] Lower Multiplicity	0
------------------------------	---

[AUTOSAR] Lower Multiplicity	1
------------------------------	---

6.6.1.8 Nm/NmChannelConfig/NmStateReportEnabled

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

6.6.1.9 Nm/NmChannelConfig/NmSynchronizingNetwork

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

6.6.1.10 Nm/NmGlobalConfig

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

6.6.1.11 Nm/NmGlobalConfig/NmGlobalConstants

[ADDED] Description	NmGlobalConstants
[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

6.6.1.12 Nm/NmGlobalConfig/NmGlobalConstants/NmNumberOfChannels

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

6.6.1.13 Nm/NmGlobalConfig/NmGlobalFeatures

[ADDED] Description	NmGlobalFeatures
[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

6.6.1.14 Nm/NmGlobalConfig/NmGlobalFeatures/NmBusSynchronizationEnabled

[RTA-BSW] Description	Pre-processor switch for enabling bus synchronization support of the <BusNm>s. This feature is required for NM Coordinator nodes only.
[AUTOSAR] Description	Pre-processor switch for enabling bus synchronization support of the <Bus>Nms. This feature is required for NM Coordinator nodes only.

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

6.6.1.15 Nm/NmGlobalConfig/NmGlobalFeatures/NmCarWakeUpRxEnabled

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

6.6.1.16 Nm/NmGlobalConfig/NmGlobalFeatures/NmComControlEnabled

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

6.6.1.17 Nm/NmGlobalConfig/NmGlobalFeatures/NmCoordinatorSupport-Enabled

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

6.6.1.18 Nm/NmGlobalConfig/NmGlobalFeatures/NmCoordinatorSyncSupport

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

6.6.1.19 Nm/NmGlobalConfig/NmGlobalFeatures/NmPduRxIndicationEnabled

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

6.6.1.20 Nm/NmGlobalConfig/NmGlobalFeatures/NmRemoteSleepIndEnabled

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

6.6.1.21 Nm/NmGlobalConfig/NmGlobalFeatures/NmStateChangeIndEnabled

[RTA-BSW] Lower Multiplicity	0
------------------------------	---

[AUTOSAR] Lower Multiplicity	1
------------------------------	---

6.6.1.22 Nm/NmGlobalConfig/NmGlobalFeatures/NmSynchronizedPnc-ShutdownEnabled

[REMOVED] Introduction	FALSE: synchronized PNC shutdown is disabled TRUE: synchronized PNC shutdown is enabled
------------------------	---

6.6.1.23 Nm/NmGlobalConfig/NmGlobalFeatures/NmUserDataEnabled

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

6.6.1.24 Nm/NmGlobalConfig/NmGlobalProperties

[ADDED] Description	NmGlobalProperties
[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

6.6.1.25 Nm/NmGlobalConfig/NmGlobalProperties/NmDevErrorDetect

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

6.6.1.26 Nm/NmGlobalConfig/NmGlobalProperties/NmVersionInfoApi

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1
[REMOVED] Default Value	false

6.7 API Definition

List of supported API

Nm_Init	Initializes the Nm module.
Nm_MainFunction	Implements the processes of the Nm module that need fixed cyclic scheduling.
Nm_BusSleepMode	Notification that the network management has entered Bus-Sleep Mode.
Nm_CheckRemoteSleepIndication	Checks whether or not remote sleep indication takes place.
Nm_CarWakeUpIndication	Indicates reception of a Car Wake Up request.

Nm_CoordReadyToSleepCancellation	Cancels an indication when the NM Coordinator Sleep Ready bit in the Control Bit Vector is reset to 0.
Nm_CoordReadyToSleepIndication	Sets an indication when the NM Coordinator Sleep Ready bit in the Control Bit Vector is set.
Nm_DisableCommunication	Disables the NM PDU transmission ability.
Nm_EnableCommunication	Enables the NM PDU transmission ability.
Nm_GetLocalNodeIdentifier	Gets the node identifier configured for the local node.
Nm_GetNodeIdentifier	Gets the node identifier out of the last successfully received NM message.
Nm_GetPduData	Gets the complete PDU data out of the most recently received NM message.
Nm_GetState	Returns the state of the Network Management.
Nm_GetUserData	Gets user data out of the last successfully received NM message.
Nm_GetVersionInfo	Returns the version information of Nm module.
Nm_NetworkMode	Notification that the network management has entered Network Mode.
Nm_NetworkRequest	Calls the <BusNm>_NetworkRequest bus specific function.
Nm_NetworkRelease	Calls the <BusNm>_NetworkRelease bus specific function.
Nm_NetworkStartIndication	Notification that a NM message has been received in the Bus-Sleep Mode.
Nm_PassiveStartUp	Calls the <BusNm>_PassiveStartUp function.
Nm_PduRxIndication	Notification that a NM message has been received.
Nm_PrepareBusSleepMode	Notification that the network management has entered Prepare Bus-Sleep Mode.
Nm_RemoteSleepCancellation	Notification that the network management has detected that not all other nodes on the network are ready to enter Bus-Sleep Mode.
Nm_RemoteSleepIndication	Notification that the network management has detected that all other nodes on the network are ready to enter Bus-Sleep Mode.
Nm_RepeatMessageIndication	Indicates that an NM message with the Repeat Message Request Bit set has been received.
Nm_RepeatMessageRequest	Sets the Repeat Message Request Bit for NM messages transmitted next on the bus.
Nm_SetUserData	Sets user data for NM messages transmitted next on the bus.
Nm_StateChangeNotification	Notification that the state of the lower layer <BusNm> has changed.

Nm_SynchronizationPoint	Notification to the NM Coordinator functionality that this is a suitable point in time to initiate the coordinated shutdown.
Nm_SynchronizeMode	Notification that the NM has entered Synchronize Mode.
Nm_TxTimeoutException	Indicates that an attempt to send an NM message failed.
Nm_ForwardSynchronizedPncShutdown	Notification that the network management has received a PN shutdown message on a particular NM-channel.

For Further information please refer to AUTOSAR_SWS_NetworkManagementInterface.pdf

6.8 Configuration Advice



NOTE

Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the "Configuration Generation" section of the RTA-CAR User Guide.

6.8.1 Nm Coordinator functionality

The Coordinator functionality is not supported. The NmCoordinatorSupportEnabled parameter is not used and NM_COORDINATOR_SUPPORT_ENABLED is always generated as STD_OFF.

6.8.2 Configuration for Lin Channels

- NmGlobalConfigLinNm is an optional container which needs to be configured if you need to configure LinNm channels.
- LinNmPassiveModeEnabled should always be configured as FALSE because Nm does not support the Passive Mode feature.
- NmChannelSleepMaster should be configured as TRUE for LIN channels.

6.8.3 Nm_CarWakeUpIndication

This function is called by a <Bus>Nm to indicate reception of a Car Wake Up (CWU) request.

- If the <bus>Nm calls Nm_CarWakeUpIndication and NmCarWakeUpCallout is defined (in /NmGlobalConfig/NmGlobalFeatures/), Nm calls the callout function defined by NmCarWakeUpCallout with nmNetworkHandle as the parameter.

- If the <bus>Nm calls Nm_CarWakeUpIndicationIf and NmCarWakeUpCallout is not defined, Nm calls BswM_CarWakeUpIndication with nmNet-workHandle as the parameter.

6.8.4 Nm_StateChangeNotification

This function notifies that the state of the lower layer <BusNm> has changed.

When NmStateReportEnabled is set to TRUE, Nm_StateChangeNotification calls Com_SendSignal passing NmStateReportSignalRef as the Com_SignalIdType. NmStateReportSignalRef points to a 6 bit signal called Network Management State (NMS), and this NMS needs to be configured in Com.

6.8.5 Sample Nm Configuration

There is not enough information for Nm in the DBC file, so the Nm channel needs to be configured manually in the System Template (for input into Configuration generation). This example shows where the manual changes are made to the System Template.

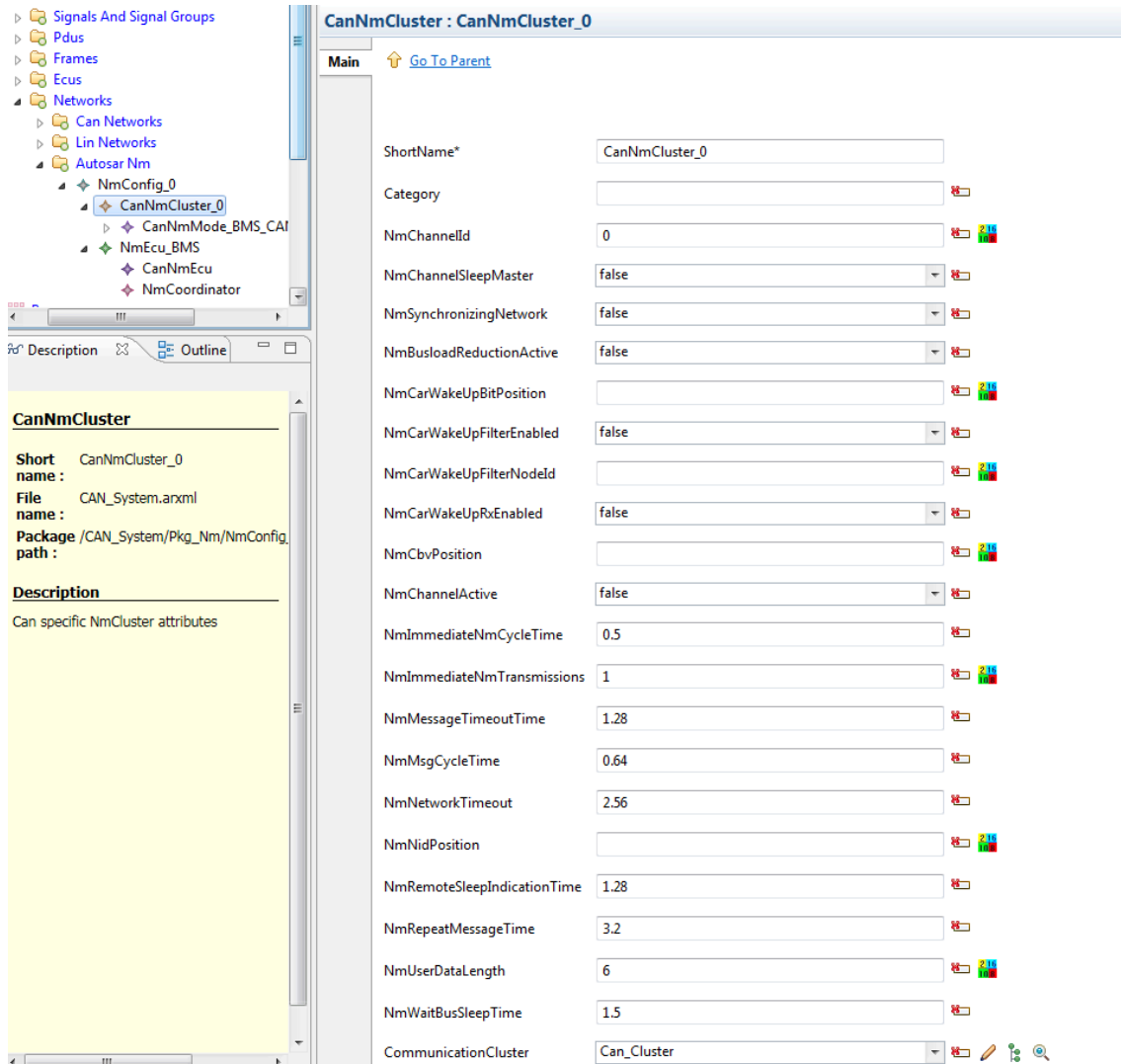


Figure 6.2. Sample Configuration

6.8.6 Synchronized Pnc Shutdown

The `nmSynchronizedPncShutdownEnabled` parameter (in `nmGlobalFeatures`) should only be set to TRUE if the corresponding parameters in `ComM` and `BusNm` (`CanNm`, `FrNm` and `UdpNm`) have also been set to TRUE. A combination of values (some TRUE, some FALSE) is not allowed.

6.9 Integration Advice

6.9.1 Init Functions

As the Coordinator functionality is not yet supported, `Nm_Init` is implemented only as a stub function. Calling `Nm_Init` will have no effect.

6.9.2 Delnit Functions

No specific handling necessary.

6.9.3 Scheduled Functions

As the Coordinator functionality is not yet supported, Nm_MainFunction is implemented only as a stub function. Calling Nm_MainFunction will have no effect.

6.9.4 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the ENTER and EXIT resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The following is a list of all exclusive areas that this module might use regardless of the user configuration. It is the job of the integrator to ensure that the implementation of these exclusive areas is correctly defined. This implementation can be done using the RTE exclusive area configuration wizard or by manual implementation of the corresponding BSW API calls to ensure exclusivity.

The Nm module uses the following OS resources:

- Nm_CallbacksNoNest
- Nm_MainFunctionNoNest
- Nm_ChangeStateNoNest
- Nm_GetStateNoNest

6.10 Hardware Requirements

Table 6.44.Nm Hardware Requirements

Support	Usage
FLASH	1090 bytes
RAM	0 bytes

Usage calculated based on a sample configuration.

7 PDU Router

Module name	PduR
Package	RTA-COM
AUTOSAR Module ID	51
AUTOSAR Version	22-11

7.1 Introduction

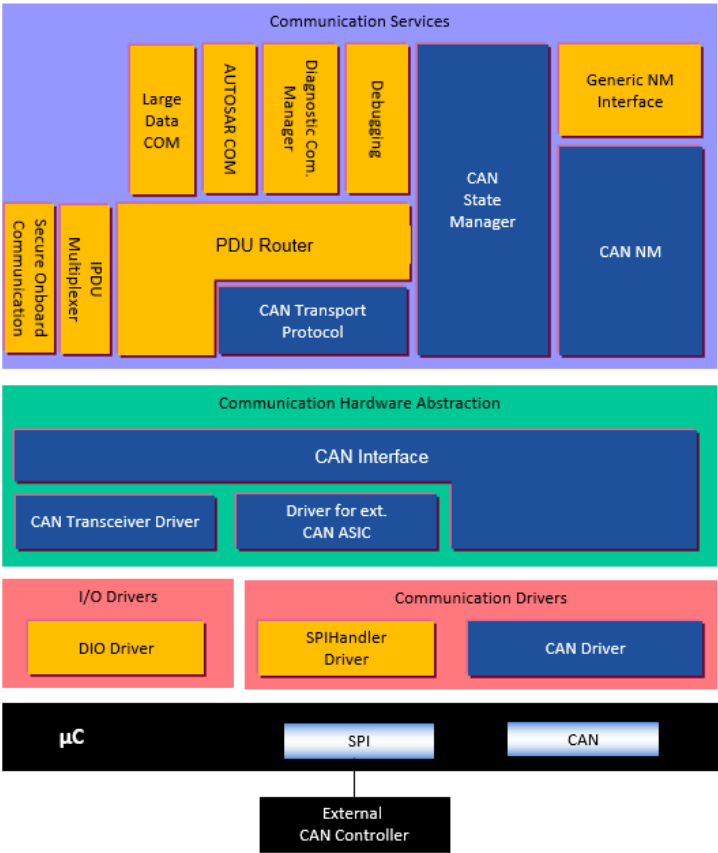


Figure 7.1.AUTOSAR Communication stack

The PDU Router (PduR) belongs to the Communication Stack. It is located above the lower layer Interface (IF) and Transport Protocol (TP) modules, and below upper layer modules, such as Com and Dcm.

The PduR provides services for I-PDU routing between the following types of module:

- Communication Interface modules (e.g. Com, LdCom, LinIf, CanIf, FrIf).
- Transport Protocol modules (e.g. LinTp, CanTp, FrTp, J1939Tp).
- Complex driver modules (Cdd).

- Network Management Modules (CanNm, FrNm, J1939Nm, UdpNm).

From the ECU point of view, the PduR performs three different classes of operation relating to the routing on an I-PDU:

- Pdu Reception. The PduR receives I-PDUs from lower layer modules and forwards them to upper layer modules.
- Pdu Transmission. The PduR transmits I-PDUs to lower layer modules at the request of upper layer modules.
- Pdu Gateway. The PduR supports the "gatewaying" of I-PDUs from one source bus to one or more destination buses, through the lower layer modules.

The PduR does not modify any contents of the I-PDUs that it receives or transmits, or that it handles via the Gateway. It simply routes those I-PDUs using I-PDU identifier-based static routing tables.

Module Configuration

This module can be generated by the RTA-BSW Configuration Generator.

For any unsupported automatic configuration, the module can be manually configured by the user.

7.2 Supported Features

7.2.1 Reception

When an I-PDU is received from a lower layer module, PduR will propagate it according to its routing path.

The ability to request reception cancellations for I-PDUs can be optionally enabled.

PduR allows Singlecast mode (1 to 1) and Multicast mode (1 to n) for the I-PDU delivery from Communication Interface modules and/or Transport Protocol modules to local modules.

7.2.2 Transmission

Any I-PDU transmission operation performed by PduR onto a lower layer module is asynchronous, and it is always triggered by a Pdu transmit request from an upper layer source. The transmission is considered finalized when a TxConfirmation is received on the upper layer.

The ability to request transmission cancellations for I-PDUs can be optionally enabled.

PduR allows the transmission in Singlecast mode (1 to 1) and Multicast mode (1 to n) from local modules to Communication Interface modules and/or Transport Protocol modules.

7.2.3 I-PDU Gateway

PduR supports the gatewaying of I-PDUs from one source bus to one or more

destination buses. In this context, PduR is both a receiver and a transmitter at the same time, and in some cases also provides buffering for the I-PDU.

Supported features of the I-PDU Gateway include:

Communication Interface modules

- I-PDUs may be gatewayed from a source Communication Interface module to one or more destination Communication Interface modules. A FIFO can be configured for each destination Communication Interface module (not the local module). PduR supports Immediate FIFO, where a transmission request is made directly when it is received, and Deferred FIFO, where the I-PDU is transmitted to the destination when the transmission of the last I-PDU is confirmed. A last-is-best buffer/no buffer can also be used.
- I-PDUs may also be gatewayed (n to 1) i.e. from multiple Communication Interface modules to one Communication Interface module using a last-is-best buffer/FIFO buffer/no buffer. Only one source will be enabled.
- An I-PDU may be received by one or more upper modules at the same time as it is gatewayed to one or more Communication Interface modules using a last-is-best/FIFO/no buffer.

Transport Protocol modules

- I-PDUs may be gatewayed from a source Transport Protocol to one or more destination Transport Protocol modules. PduR supports Gatewaying on-the-fly, where it starts transmission to the destination Transport Protocol module when a specific threshold is reached. PduR also supports the Buffer Pooling mechanism for Transport Protocol Gatewaying. See the Configuration Advice for more details.
- I-PDUs may also be gatewayed (n to 1) i.e. from multiple Transport Protocol modules to one Transport Protocol module using a buffer. Only one source will be enabled.
- An I-PDU may be received by one or more upper modules at the same time as it is gatewayed to one or more Transport Protocol modules using a last-is-best/FIFO buffer.
- DoIP V4 is supported for Transport Protocol Gateway routing paths for CanTp and DoIP modules only.
 - In case of N:1 Transport Protocol gateway routing paths involving N number of DoIP sources and 1 CanTp destination, then the DoIP_Tp_StatusIndication function will be called N number of times. Note: The order of calls is not assured to be same as order of DoIP request calls.
 - When the status type is RX_ST_TYPE_E, then the PduR buffer address is provided in DoIP_Tp_StatusIndication function only in case the PduR_CanTpStartOfReception call is accepted by the PduR.

7.2.4 Zero Cost Operation

PduR supports Zero Cost Operation, in which the communication modules directly above or below it can directly call each other without using PduR functions (i.e. an ECU has only a single upper layer and a single lower layer, so no I-PDU routing is needed).

Zero Cost Operation is supported as long as the following conditions are all true:

1. There is only one interface module
2. No PDU gateway functionality is needed
3. No multicast PDU is configured
4. There is a maximum of one upper layer module that communicates via the interface (IF) module
5. No IPDUM is used
6. There is a maximum of one upper layer module that communicates via a TP module.

7.2.5 Routing path groups

PduR supports Routing Path Groups, which is a group of I-PDUs that can be disabled and enabled during runtime. As the group contains the destination I-PDUs and not the routing path itself, this makes it possible to enable or disable I-PDUs for a specific bus. The enabling and disabling of routing path groups is typically done by the BswM module.

7.2.6 Mode Dependent Routing Control

Whenever there is message transmit request from the upper layer, the message is transmitted to one (or more, in the case of multicast) lower layers as per the statically configured routing paths in the respective routing table for the Pdu.

Similarly, during reception of a message from any lower layer, the same message is forwarded to respective upper layers as well as any other lower layer (in case of multicast gateway) as per the statically configured routing path(s) in the respective routing table for the Pdu.

Mode Dependent Routing Control allows the enabling/disabling of this routing to the destination layer based on the mode of the ECU, which is controlled by BswM.

During certain modes of ECU monitored in BswM, it is possible to control message transmissions on a specific bus (destination), and to enable/disable routing to a certain number of destination layers out of the total number of destination layers that have been configured for a routing path, in the case of 1:n routing.

See the Configuration Advice for more information.

7.2.7 Complex Driver interaction

Complex Drivers (Cdd) can be used as upper or lower layer modules for the PduR, since the PduR configuration allows the Cdd module to be configured as any

other Bsw module from the PduRs point of view.

For further information please refer to AUTOSAR_SWS_PDURouter.pdf

7.2.8 Post-Build Selectable

PduR currently supports Post-Build Selectable, but only Handle ID generation is supported. In a future implementation, all containers and parameters will be supported as PBS.

7.2.9 Bus Mirroring

PduR facilitates the use of Can to Ethernet Bus Mirroring, which replicates the traffic and state of internal (source) buses to an external (destination) bus. Therefore, a Tester connected to the external bus can monitor internal buses (which are not directly accessible) for debugging purposes.

7.3 Extensions

7.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

7.3.1.1 PduR/PduRBswModules/PduRRbInvalidHandlers

Short Name	PduRRbInvalidHandlers
Description	Container for each PduRBswModule to indicate whether Invalid Handlers should be supported or not for any feature. This is relevant only for a lower layer module. This container need not be configured
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

7.3.1.2 PduR/PduRBswModules/PduRRbInvalidHandlers/PduRRbIHCancel-Receive

Short Name	PduRRbIHCancelReceive
Description	Specifies for a PduRBswModule, if call to Invalid Handler needs to be generated instead of actual CancelReceive API. Value TRUE: Call to Invalid Handler is generated. Value FALSE: Call to Actual Module-Name_CancelReceive API is generated.
Multiplicity	1..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

7.3.1.3 PduR/PduRBSwModules/PduRRbInvalidHandlers/PduRRbIHCancelTransmit

Short Name	PduRRbIHCancelTransmit
Description	Specifies for a PduRBSwModule, if call to Invalid Handler needs to be generated instead of actual CancelTransmit API. Value TRUE: Call to Invalid Handler is generated. Value FALSE: Call to Actual ModuleName_CancelTransmit API is generated
Multiplicity	1..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

7.3.1.4 PduR/PduRBSwModules/PduRReleaseRxBuffer

Short Name	PduRReleaseRxBuffer
Description	Specifies if BSW module supports the ReleaseRxBuffer API or not. Value true the API is supported. Tags: atp.Status=draft
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

7.3.1.5 PduR/PduRBSwModules/PduRUseTag

Short Name	PduRUseTag
Description	This parameter, if set to true, enables the usage of the tag (<up>) in the following API calls:
Introduction	* PduR_<Up>CancelReceive * PduR_<Up>CancelTransmit Example: If used by COM and the parameter is enabled the PduR_ComCancelTransmit is used. The background is that upper layer modules differ in usage of this tag (e.g. COM is using the tag, DCM is not).
Multiplicity	0..0
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

7.3.1.6 PduR/PduRGeneral/PduRRbDemSupport

Short Name	PduRRbDemSupport
Description	This container contains switches for Dem Monitoring Control.

Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

7.3.1.7 PduR/PduRGeneral/PduRRbDemSupport/PduRRbInitFailed

Short Name	PduRRbInitFailed
Description	Monitoring Control Switch for error PDUR_E_INIT_FAILED. This parameter shall be seen as an input requirement to the configuration generator.
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

7.3.1.8 PduR/PduRGeneral/PduRRbDemSupport/PduRRbPduInstanceLost

Short Name	PduRRbPduInstanceLost
Description	Monitoring Control Switch for error PDUR_E_PDU_INSTANCE_LOST. This parameter shall be seen as an input requirement to the configuration generator.
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

7.3.1.9 PduR/PduRGeneral/PduRRbDisablePostBuildSelectableVariants

Short Name	PduRRbDisablePostBuildSelectableVariants
Description	This parameter enables and disables a PostBuild Selectable support in PostBuild variant. This parameter will not affect the precompile variant. False : PduR will be PBS in PBS environment. True : PduR will be Pre-Compile in PBS environment
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

7.3.1.10 PduR/PduRGeneral/PduRRbDsmReInitEnabled

Short Name	PduRRbDsmReInitEnabled
Long Name	PduRRbDSMReInitEnabled
Description	Enables or disables the reporting of DEM events to DSM after the Re-initialisation of DSM has occurred.

Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

7.3.1.11 PduR/PduRGeneral/PduRRbDsmReInitMainFunctionPeriod

Short Name	PduRRbDsmReInitMainFunctionPeriod
Description	Specifies the period in seconds that the MainFunction has to be triggered with. Comment: this main function is used for DsmReinit feature.
Multiplicity	0..1
Value Range	0.001..1
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

7.3.1.12 PduR/PduRGeneral/PduRRbGwTpWithLimitedBuffer

Short Name	PduRRbGwTpWithLimitedBuffer
Description	Configuration parameter to enable or disable PDU Router support for gateway Tp using limited buffer resources; Note : When this parameter is set true need to configure PduRTpThreshold.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

7.3.1.13 PduR/PduRGeneral/PduRRbIFGatewayOperation

Short Name	PduRRbIFGatewayOperation
Description	Configuration parameter to enable or disable PDU Router IF Gateway from an interface module to lower layer interface modules; if PDUR_ZERO_COST_OPERATION is enabled, this parameter has to be disabled. This parameter shall be seen as an input requirement to the configuration generator.
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

7.3.1.14 PduR/PduRGeneral/PduRRbMulticastFromTpSupport

Short Name	PduRRbMulticastFromTpSupport
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Description	Configuration parameter to enable or disable PDU Router support for multicasts from a TP module to upper layer modules and lower layer TP modules; if PDUR_ZERO_COST_OPERATION is enabled, this parameter has to be disabled. This parameter shall be seen as an input requirement to the configuration generator. For Example : LoTp (Src) → P-duR → (Dest1)LoTp, (Dest2)UpTp
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

7.3.1.15 PduR/PduRGeneral/PduRRbMulticastToIfSupport

Short Name	PduRRbMulticastToIfSupport
Description	Configuration parameter to enable or disable PDU Router support for multicasts from an upper layer module to interface modules; if PDUR_ZERO_COST_OPERATION is enabled, this parameter has to be disabled. This parameter shall be seen as an input requirement to the configuration generator.
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

7.3.1.16 PduR/PduRGeneral/PduRRbMulticastToTpSupport

Short Name	PduRRbMulticastToTpSupport
Description	Configuration parameter to enable or disable PDU Router support for multicasts from an upper layer module to TP modules; if PDUR_ZERO_COST_OPERATION is enabled, this parameter has to be disabled. This parameter shall be seen as an input requirement to the configuration generator.
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

7.3.1.17 PduR/PduRGeneral/PduRRbQueueingSupport

Short Name	PduRRbQueueingSupport
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Description	Configuration parameter to enable or disable PDU Router Queueing Feature. If PDUR_ZERO_COST_OPERATION is enabled, this parameter has to be disabled.
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

7.3.1.18 PduR/PduRGeneral/PduRRbTPGatewayOperation

Short Name	PduRRbTPGatewayOperation
Description	Configuration parameter to enable or disable PDU Router TP gateway from a TP module to lower layer TP modules; if PDUR_ZERO_COST_OPERATION is enabled, this parameter has to be disabled. This parameter shall be seen as an input requirement to the configuration generator.
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

7.3.1.19 PduR/PduRRoutingPaths/PduRRbDemEventParameterRefs

Short Name	PduRRbDemEventParameterRefs
Description	Container for the references to DemEventParameter elements which shall be invoked using the API Dem_ReportErrorStatus API in case the corresponding error occurs. The EventId is taken from the referenced DemEventParameter's DemEventId value. The standardized errors are provided in the container and can be extended by vendor specific error references.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

7.3.1.20 PduR/PduRRoutingPaths/PduRRbDemEventParameterRefs/ PDUR_E_INIT_FAILED

Short Name	PDUR_E_INIT_FAILED
Description	Reference to configured DEM event to report PDU Router initialization failed (PDU Router changed to PDUR_REDUCED state).
Multiplicity	0..1
Default Value	-

Type	ECUC-SYMBOLIC-NAME-REFERENCE-DEF
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7.3.1.21 PduR/PduRRoutingPaths/PduRRbDemEventParameterRefs/ PDUR_E_PDU_INSTANCE_LOST

Short Name	PDUR_E_PDU_INSTANCE_LOST
Description	Reference to configured DEM event to report Loss of a PDU instance (FIFO flushed because of an overrun).
Multiplicity	0..1
Default Value	-
Type	ECUC-SYMBOLIC-NAME-REFERENCE-DEF

7.3.1.22 PduR/PduRRoutingPaths/PduRRbDemEventParameterRefs/PduR- RbDynCreate_E_INIT_FAILED

Short Name	PduRRbDynCreate_E_INIT_FAILED
Long Name	Dynamically create PDUR_E_INIT_FAILED DemEvent
Description	Selects the way how the PDUR_E_INIT_FAILED DemEvent is created. If not set or set to Disabled, no such DemEvent is automatically created. If the dynamic creation of this DemEvent is enabled, the reference to this DemEvent (in parameter PDUR_E_INIT_FAILED) is also automatically created and shall therefore not be set manually.
Multiplicity	0..1
Default Value	Disabled
Enumeration Literals	Disabled, Standard
Type	ECUC-ENUMERATION-PARAM-DEF

7.3.1.23 PduR/PduRRoutingPaths/PduRRbDemEventParameterRefs/PduR- RbDynCreate_E_PDU_INSTANCE_LOST

Short Name	PduRRbDynCreate_E_PDU_INSTANCE_LOST
Long Name	Dynamically create PDUR_E_PDU_INSTANCE_LOST DemEvent
Description	Selects the way how the PDUR_E_PDU_INSTANCE_LOST DemEvent is created. If not set or set to Disabled, no such DemEvent is automatically created. If the dynamic creation of this DemEvent is enabled, the reference to this DemEvent (in parameter PDUR_E_PDU_INSTANCE_LOST) is also automatically created and shall therefore not be set manually.
Multiplicity	0..1

Default Value	Disabled
Enumeration Literals	Disabled, Standard
Type	ECUC-ENUMERATION-PARAM-DEF

7.3.1.24 PduR/PduRRoutingPaths/PduRRbRoutingPathSeqGroup

Short Name	PduRRbRoutingPathSeqGroup
Description	This container groups routing paths which are to be part of particular sequence routing .
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

7.3.1.25 PduR/PduRRoutingPaths/PduRRbTpBufferPool

Short Name	PduRRbTpBufferPool
Description	Specifies a pool of buffers used for gateway through TP.
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

7.3.1.26 PduR/PduRRoutingPaths/PduRRbTpBufferPool/PduRRbTpPool-Buffer

Short Name	PduRRbTpPoolBuffer
Description	Specifies a buffer used for gateway through TP.
Multiplicity	1..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

7.3.1.27 PduR/PduRRoutingPaths/PduRRbTpBufferPool/PduRRbTpPool-Buffer/PduRRbLengthOfTpBufferInPool

Short Name	PduRRbLengthOfTpBufferInPool
Description	Length of the TP buffer in number of bytes
Multiplicity	1..1
Value Range	1..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

7.3.1.28 PduR/PduRRoutingPaths/PduRRbTpBufferPool/PduRRbTpPool-

Buffer/PduRRbTpPoolBufferCopies

Short Name	PduRRbTpPoolBufferCopies
Description	The number of copies required for this buffer
Multiplicity	1..1
Value Range	1..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

7.3.1.29 PduR/PduRRoutingPaths/PduRRbTpBufferPool/PduRRbTpPool-HandleId

Short Name	PduRRbTpPoolHandleId
Description	Unique Handle Id for accessing Buffer Pools
Multiplicity	1..1
Value Range	0..65534
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

7.3.1.30 PduR/PduRRoutingPaths/PduRRoutingPath/PduRDestTxBufferRef

Short Name	PduRDestTxBufferRef
Description	Reference to a buffer in the PduR. This buffer is required for communication interface gatewaying, and for transport protocol gatewaying.
Multiplicity	0..*
Default Value	-
Type	ECUC-CHOICE-REFERENCE-DEF

7.3.1.31 PduR/PduRRoutingPaths/PduRRoutingPath/PduRRbDestTpBuffer-Ref

Short Name	PduRRbDestTpBufferRef
Description	Reference to a buffer that is allocated in the PduRRbTpBufferPool. Having a global (for PduR) list of buffers allows reuse and hence less memory consumption.
Multiplicity	0..1
Default Value	-
Type	ECUC-CHOICE-REFERENCE-DEF

7.3.1.32 PduR/PduRRoutingPaths/PduRRoutingPath/PduRRbFirstFrameBufLength

Short Name	PduRRbFirstFrameBufLength
Long Name	PduRFirstFrameBufferLength
Description	Length of the buffer in bytes required to store the First frame while gatewaying CanTP Frames. Set this value specific to the type of Can frame: Default value is set as per Standard Can message which has FF length 6 bytes. For CanFd frames, FF length is typically 58, hence this value is to be set to 58 in those cases. Size is limited to a maximum of 60 bytes because it is only meant to store FirstFrame; not the consecutive frames.
Multiplicity	0..1
Value Range	6..60
Default Value	6
Type	ECUC-INTEGER-PARAM-DEF

7.3.1.33 PduR/PduRRoutingPaths/PduRRoutingPath/PduRRbRoutingPathSeqGroupRef

Short Name	PduRRbRoutingPathSeqGroupRef
Description	Reference to routing path Sequence Configuration parameter to maintain transmit sequence across the routing paths in PDU Router TP gateway operation. In this case, Pdu Tx of one RP is controlled if another RP (where same reference is configured) pdu is already in transmission. Tx of this RP pdu shall be done in TxConf API..
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

7.3.1.34 PduR/PduRRoutingPaths/PduRRoutingPath/PduRRbTpGatewayQueueDepth

Short Name	PduRRbTpGatewayQueueDepth
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Description	This parameter specifies whether queue is required in Tp Gatewaying or not. If configured as 2 then, queuing is enabled. Max queue depth can be 2. By default the parameter value is 1. NOTE: Configure parameter TpGatewayQueueDepth if queueing feature is required. Configure PduRQueueDepth if FIFO TP feature is required.
Multiplicity	0..1
Minimum Value	1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

7.3.1.35 PduR/PduRRoutingPaths/PduRRoutingPath/PduRRbTpHandle-FirstFrame

Short Name	PduRRbTpHandleFirstFrame
Description	Configuration parameter to make sure first Frame is always accepted in PDU Router TP gateway operaton. In this case, after Rx of first frame, FC wait shall be sent by Tp module before sending consecutive frame. Gateway-on-the-fly shall be disabled if this is enabled.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

7.3.1.36 PduR/PduRRoutingPaths/PduRTxBuffer

Short Name	PduRTxBuffer
Description	Specifies a buffer used for gatewaying via communication interfaces or transport protocols.
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

7.3.2 Additional Services

The following API are available in version v12.7.0 in addition to those defined in AR v22-11

- There are no additional API defined in this version.

7.4 Limitations

7.4.1 General

- Mode Dependent Routing Control only supports paths involving interface (IF) modules, transport protocol (TP) modules are not supported for this feature.
- Postbuild Loadable Configuration Variant is not supported. Both the Pre-Compile and Post-Build Selectable variants are supported in the RTA-BSW implementation of the PduR module.
- The Change Transport Protocol Parameter API (PduR_<Up>ChangeParameter) is not supported.
- The Change Parameter API function (PduRChangeParameterApi) is not supported.
- Disabling of Pdu (using RPG) for Cancel Reception, TriggerTransmit and CopyTxData API is not supported.
- When TriggerTransmit is called, the latest data present in the buffer will always be provided, even if the default values are copied to it by the PduR_DisbaleRouting API.
- PduR supports the SomelpTp component if PduRZeroCostOperation is set. The Multicast Tx, If Gateway and Tp Gateway features are not supported for the SomelpTp component.
- Communication interface is supported for the DoIP module in PduR, but the Rx path and trigger transmission is not supported for DoIPIf.
- A Datatype mismatch currently exists in the Det_ReportError API between instancelid (uint8) and module-Id (uint16) where the ID of the calling module is sent as the Instance ID while calling Det_ReportError. As the highest module ID does not currently exceed 255, module IDs are sent as Instance IDs without any typecasting.
- PduR_Init can be called by both BswM or EcuM, so the Instance ID of 0u is passed while calling the Det_ReportError function. 0u is also passed while calling Det_ReportError in the PduR_GetVersionInfo and PduR_GetConfigurationId APIs.
- Since PduR doesn't support a one-fits-all approach, the signature of the PDUR_REPORT_ERROR API is kept the same as per the AR45 implementation in the Txconfirmation API, which is under the AR42 compiler switch.
- The IpduM and SomelpTp modules do not support the PduR_<Up>CancelTransmit API. PduR does not support PduR_<Up>CancelReceive for the SecOC module.
- Note the following when PduRCommunicationInterface and PduRTransportProtocol are both configured as TRUE in PduRBswModule for a Cdd module:
 - Because PduR cannot differentiate PduR_d< <CddShortName> >-Transmit API for IF and TP, a PDUR_E_TP_TX_REQ_REJECTED error is

reported to DET when PduR_d< < CddShortName > > Transmit returns E_NOT_OK when the Cdd can act as both IF and TP or only TP.

- RxIndication APIs are differentiated as "PduR_< < ModuleName > > RxIndication" for IF and "PduR_< < ModuleName > > TpRxIndication" for TP, and TxConfirmation APIs are differentiated as "PduR_< < ModuleName > > TxConfirmation" for IF and "PduR_< < ModuleName > > TpTxConfirmation" for TP.
- As the SD Module expects Transmit() and Reception api, TxConfirmation() api and TriggerTransmit are not supported because SdInstanceTxPdu doesn't have TxPduId parameter. With this Id 0 and PduR_Sd_Null_TxConfirmation() will always be generated.
- For the rba_EbDataClient component, the Rx, CancelReceive and CancelTransmit APIs are not supported.
- In the Multicast Multiframe TP gateway Routing Path, TpThreshold should not be configured. When the Multicast Multiframe TP gateway Routing Path is configured, PduRRbGwTpWithLimitedBuffer should be disabled.
- For the IdsM component, the Rx, CancelReceive and CancelTransmit APIs are not supported.
- For the CanTp component, PduR_CanTpCancelTransmit is not supported.
- If a Routing Path is disabled and DefaultValue is also configured, then the PduR_Lolf_TriggerTransmit API will be processed without checking if DisableRouting is called with TRUE or FALSE.
- If PduRDisableRouting is called for the Gateway Routing Path and the default values are not configured, then E_NOT_OK will be returned for the PduR_Lolf_TriggerTransmit function.
- For n:1 Routing Paths, if PduRRoutingPathGroupRef is configured for only some of the Routing Paths and the specific Routing Path is disabled, then the Pdur_Lolf_TriggerTransmit API is called and PduR will disable all the 'n' routing paths. The same applies for the PduR_LoTpCopyTxData API.

7.4.2 Multicast

- No update-bits can be used.
- No sequence counters can be used.
- PduR doesn't guarantee the confirmation of all matching requests. For example, if PduR receives 2 multicast transmit requests from the upper layer, and if a confirmation is later received from the lower layer, PduR will be unable to determine whether the confirmation relates to the first or second transmission.
- Multicast Tp Transmission (i.e. source UpTp to Multiple destination LoTps) is

not supported.

7.4.3 Gateway

- Multicast transport protocol gateway involving any upper layer is not supported.
- Gateway for LdCom Module is not supported
- Routing of I-Pdus between Communication Interface modules with different rate conversion is not supported.



NOTE

This can be done via the COM module, using signal gateways. In this case the I-PDU has to be routed to the COM module.

- Cancel transmission can not be used for I-Pdus that are gatewayed from one bus to another.
- No routing between communication interface and transport protocol layers is allowed
- If a PDU is transmitted to multiple lower transport protocol layers, the PDU must be a single frame
- Gateway-on-the-fly shall be disabled if the HandleFirstFrame feature is enabled.
- Gateway-on-the-fly is applicable only when the buffer pool (PduRRbDestTpBufferRef) is configured for the routing path.
- PduRFirstFrameBufferLength shall be configured depending on the size of the first frame expected to be received.
- For n:m Tp Gateway routing paths, PduRRbDestTpBufferRef cannot be configured. Use PduRTxBuffer (the dedicated buffer) instead.

7.4.4 PduRDefaultValueElementBytePosition

- PduRDefaultValueElementBytePosition is only supported with uint8 values.

7.4.5 Bus Mirroring

- Mirror Multicast Tx cannot be tested (in Continuous Testing) because the Mirror module does not support the configuration of multicast channels. However, this feature is supported from PduR.

7.5 Exclusions

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- PduR/PduRBswModules/PduRBswModuleRef
- PduR/PduRBswModules/PduRCopyRxData
- PduR/PduRBswModules/PduRCopyTxData
- PduR/PduRBswModules/PduRRetransmission
- PduR/PduRBswModules/PduRRxIndication
- PduR/PduRBswModules/PduRStartOfReception
- PduR/PduRBswModules/PduRTpRxIndication
- PduR/PduRBswModules/PduRTpTransmit
- PduR/PduRBswModules/PduRTpTxConfirmation
- PduR/PduRBswModules/PduRTransmit
- PduR/PduRDemEventParameterRefs
- PduR/PduRDemEventParameterRefs/PDUR_ROUTING_PATH_BUFFER-
_OVERFLOW
- PduR/PduRGeneral/PduRMetaDataSupport
- PduR/PduRRoutingPaths/PduRBuffer
- PduR/PduRRoutingPaths/PduRDestPdu/PduRTransmissionConfirmation
- PduR/PduRRoutingPaths/PduRMaxRoutingPathCnt
- PduR/PduRRoutingPaths/PduRMaxRoutingPathGroupCnt
- PduR/PduRRoutingPaths/PduRRoutingPath/PduRDestBufferRef
- PduR/PduRRoutingPaths/PduRSrcPdu/PduRSourcePduBlockSize
- PduR/PduRRoutingPaths/PduRSrcPdu/PduRSrcPduUpTxConf

7.6 Deviations

7.6.1 Configuration Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

7.6.1.1 PduR/PduRRoutingPaths/PduRRoutingPath/PduRDestPduRRef

[ADDED] Description	PduRDestPduRRef
---------------------	-----------------

7.6.1.2 PduR/PduRRoutingPaths/PduRRoutingPath/PduRQueueDepth

[ADDED] Default Value	1
-----------------------	---

7.6.1.3 PduR/PduRRoutingPaths/PduRRoutingPath/PduRRoutingPath-GroupRef

[RTA-BSW] Description	Reference to routing path destinations.
[AUTOSAR] Description	Reference to routing paths.
[RTA-BSW] Upper Multiplicity	1
[AUTOSAR] Upper Multiplicity	*

7.6.1.4 PduR/PduRRoutingPaths/PduRRoutingPath/PduRSrcPduRRef

[ADDED] Description	PduRSrcPduRRef
---------------------	----------------

7.6.1.5 PduR/PduRRoutingPaths/PduRRoutingPathGroup

[RTA-BSW] Description	This container groups routing path destinations. Destinations are used instead of routing paths since a routing path can be 1:n. It is desirable to be able to enable/disable a specific bus (i.e. a destination) rather than a routing path. Of course it is possible to create groups that covers specific routing paths as well.
[AUTOSAR] Description	This container groups routing paths. By this grouping, it is possible to switch all routings related to one network, or to one kind of PDUs. PduR-RoutingPaths link one source with one destination. Enabling and disabling of routing path groups is done using the PduR API.
[ADDED] Introduction	Enabling and disabling of routing path groups are made using the PduR API

7.6.1.6 PduR/PduRRoutingPaths/PduRTxBuffer/PduRPduMaxLength

[RTA-BSW] Parent Container	PduR/PduRRoutingPaths/PduRTxBuffer/PduRPduMaxLength
[AUTOSAR] Parent Container	PduR/PduRRoutingPaths/PduRBuffer/PduRPduMaxLength
[RTA-BSW] Description	Length of the Tx buffer in bytes. This parameter limits the size of buffered routed PDUs.
[AUTOSAR] Description	Length of the PDU buffer in bytes. This parameter limits the size of buffered routed PDUs.

7.7 API Definition

List of supported API:

API function	Comment
PduR_Init	Initializes the PDU Router

PduR_GetVersionInfo	Returns the version information of this module.
PduR_GetConfigurationId	Returns the unique identifier of the post-build time configuration of the PDU Router
PduR_EnableRouting	Enables the routing path table specified on the parameter.
PduR_DisableRouting	Disables the routing path table specified at the parameter list. The single buffers of that routing path can be reset or hold their values.

Configurable interface definitions for upper layer modules:

API function	Comment
PduR_<Up>Transmit	Requests transmission of an I-PDU.
PduR_<Up>CancelTransmit	Request for cancellation of an ongoing transmission of a communication interface module I-PDU.
PduR_<Up>CancelReceive	Requests for cancellation of an ongoing reception of an I-Pdu in transport protocol



NOTE

<Up> can be any of Com, Dcm, IpduM, J1939Rm, J1939Dcm or Cdd

Configurable interface definitions for lower layer modules:

API function	Comment
PduR_<Lo>RxIndication	Indication of a received I-PDU
PduR_<Lo>TxConfirmation	Confirmation of the transmission of the I-PDU by the lower layer module
PduR_<Lo>TriggerTransmit	Lower layer module requests a buffer of the I-PDU for transmission.



NOTE

<Lo> can be any of CanIf, CanNm, FrIf, FrNm, LinIf, IpduM or CddIf

Configurable interface definitions for transport protocol modules:

API function	Comment
PduR_<LoTp>CopyRxData	The transport protocol module calls this function every time a new N-PDU has to be sent to the upper layer.
PduR_<LoTp>RxIndication	Called by the transport protocol module after an I-PDU has been received successfully or when an error occurred. It is also used to confirm cancellation of an I-PDU.

PduR_<LoTp>StartOfReception	This function will be called by the transport protocol module at the start of receiving an I-PDU. The I-PDU might be fragmented into multiple N-PDUs (FF with one or more following CFs) or might consist of a single N-PDU (SF).
PduR_<LoTp>CopyTxData	This function is called by the transport protocol module to query the transmit data of an I-PDU segment. Each call to this function copies the next part of the transmit data until TpDataState indicates TP_DATARETRY. In this case the API restarts to copy the data beginning at the location indicated by TpTxDataCnt.
PduR_<LoTp>TxConfirmation	This function is called by a transport protocol module after the I-PDU has been transmitted on its network.

**NOTE**

<LoTp> can be any of CanTp, FrTp, LinTp, J1939Tp or CddTp

For further information please refer to AUTOSAR_SWS_PDURouter.pdf

7.8 Configuration Advice

**NOTE**

Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the "Configuration Generation" section of the RTA-CAR User Guide.

7.8.1 General

- CanIfTxPduUserTxConfirmationUL (in CanIf/CanIfInitCfg/CanIfTxPduCfg) must be configured to PduR for all TxPdus that are referred by PduR, otherwise the validation checks will fail.
- If only PduRComIfUpperLayerContribution is configured in Cdd then its Pdu reference and Cdd under PduRBSwModules needn't be configured in PduR.
- The PduRDefaultValue feature should be configured only for gateway IF Routing Paths.

7.8.2 Reference BswModules Table

The following list is a non-exhaustive but recommended set of parameters to be set depending on the Bsw module to be configured for Pdu Routing:

PduR Bsw Module Short Name ¹	PduRCommunica- tionInterface	PduRTrans- portProtocol	PduR Upper Module	PduR Lower Module	PduRTrigger Transmit
Com	True	False	True	False	True
Dcm	False	True	True	False	True
IpduM ²	True	False	True	True	True
CanIf	True	False	False	True	False
CanNm	True	False	False	True	True
CanTp	False	True	False	True	False
CddUpIf	True	False	True	False	False
CddUpTp	False	True	True	False	False
CddLoIf	True	False	False	True	False
CddLoTp	False	True	False	True	False
FrIf	True	False	False	True	True
FrNm	True	False	False	True	True
FrTp	False	True	False	True	True
LinIf	True	False	False	True	True
LinTp	False	True	False	True	True
UdpNm	True	False	False	True	True
SoAdIf ³	True	False	False	True	True/False
SoAdTp ³	False	True	False	True	False
SecOc	True	True	True	True	True/False

7.8.3 Routing Paths

- In a RoutingPath, if PduRQueueDepth > 1 (i.e. FIFO TP is enabled) then PDUR_STORE_AND_FORWARD and PDUR_QUEUEING_SUPPORT should not be enabled, and PduRRbDestTpBufferRef should not refer to PduRRbTp-BufferPool but instead refer to the PduRTpBuffer.
- In a Gateway (n:1) with FIFO TP support, PduRQueueDepth must be the same for all the Routing Path sources that are referred to the same destination.
- The PduLength of the Destination Pdu should be greater than or equal to the length of a Source Pdu under a RoutingPath.
- If PduRMulticastToTpSupport (in PduRGeneral) is set to TRUE, then at least one Transport Protocol multicast routing path must be configured. The same applies to PduRRbMulticastToIfSupport and the configuration of Communication Interface multicast.

¹ The code generation for PduR assumes that AUTOSAR naming conventions for the BswModules short names is followed.

² For this configuration, the IpduM module is considered both upper and lower layer module.

³ Unlike other modules (i.e. where the short name matches the module name), the short name for the SoAd module must be SoAdIf or SoAdTp.

- In n:m IF routingPath, for the same referred destination, the configured buffer should be same.
- In n:m TP RoutingPath, there should be one common destination among all the Routing Paths, and the configured buffer should be common across all the destinations.
 - Supported Configuration example (Dest1 is common)
 - RP1: Src1, Dest1
 - RP2: Src2, Dest1, Dest2
 - RP3: Dest1, Dest3
 - Un-supported Configuration example (Dest1 and Dest2 are common, but not in all routing paths):
 - RP1: Src1, Dest1
 - RP2: Src2, Dest1, Dest2
 - RP3: Dest2

7.8.4 Mode Dependent Routing Control

Mode Dependent Routing Control is optionally activated in the PDUR_MODE_DEPENDENT_ROUTING macro, and it is configured according to the following rules:

- For each Routing Path Group (RPG), PduRIsEnableAtInit can be configured as TRUE or FALSE to enable or disable Routing Control.
- A Routing Path Group can have 1 or more Routing Path Destinations from 1 or more Routing Paths grouped together, but a single Routing Path Destination can be referred by only 1 Routing Path Group (i.e. there must be a unique RoutingPathGroupId for each RoutingPathGroup configured).
- Each Routing Path Group can be configured as a 1:n RoutingPathDestination, which contains a reference to an existing Routing Path from the routing table and the respective Routing Path's pduDestPdu.
- No more than one Routing Path Group can have the same pduDestPduRef.

7.8.5 Buffer Pooling (Transport Protocol Gateway)

PduR supports the Buffer Pooling mechanism for Transport Protocol Gatewaying.

- PduRRbTpPoolBuffer (in PduRRoutingPaths/PduRRbTpBufferPool) configures the multiple buffers, whose length are configured via PduRRbLengthOfTpBufferInPool.
- PduRRbTpBufferPool needs to be referenced via PduRRbDestTpBufferRef (in PduRRoutingPaths/PduRRoutingTable/PduRRoutingPath/PduRDestPdu)
- If PduRRbTpBufferPool is not configured, then PduRRbQueueingSupport (in PduGeneral) needs to be set to False.
- PduRRbTpPoolBufferCopies (in PduRRoutingPaths/PduRRbTpBufferPool/PduRRbTpPoolBuffer) can be used to state the number of copies of the same buffer.
- In case of n:1 and n:m the same buffer should be configured in PduRRoutingPath. It is also not recommend to configure PduRRbTpBufferPool in n:m

RoutingPaths, as in some cases this may not work. For example [RP1:src1 ->dest1RP2:src1:->dest3RP3:src2:->dest1RP4:src3:dest1 RP5:src2:-dest2]

7.8.6 PduRRblInvalidHandlers

The PduRRblInvalidHandlers container (in PduRBswModules) allows you to specify for which module, and for which API, invalid handlers are to be generated.

For example, if PduRCancelTransmit is set to TRUE, you can select PduRRblHCancelTransmit to TRUE, which will cause PduR scripts to generate a call to PduR_IH_CancelTransmit_Func instead of [ModuleName]_CancelTransmit API.

The PduRRblHCancelReceive and PduRRblHCancelTransmit parameters are useful in generating invalid handlers to avoid compilation errors when the lower layer does not support the corresponding feature.

7.8.7 SecOC in Gateway ECU

A more efficient way of configuring the Gateway ECU is to omit SecOC from the transmission process, so that PDU data packets are re-routed directly through the PduR layer to CanIf. Please see section 9.8 *Configuration Advice/SecOC in Gateway ECU* in the *RTA-Sec_Stack_Reference_Guide_EN.pdf* for more details.

7.9 Integration Advice

7.9.1 Init Functions

PduR is initialized by PduR_Init, which should be called before the interfaced modules.

Please notice that buffers for Transport Protocol gateways are not initialized at generation time, but will be initialized during module initialization.

7.9.2 Delnit Functions

No Delnit function is defined for this module.

7.9.3 Scheduled Functions

As any PDU Router operation is triggered by another communication module, the PDU Router does not require scheduled functions.

7.9.4 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the ENTER and EXIT resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The following is a list of all exclusive areas that this module might use regardless of the user configuration. It is the job of the integrator to ensure that the implementation of these exclusive areas is correctly defined. This implementation can

be done using the RTE exclusive area configuration wizard or by manual implementation of the corresponding BSW API calls to ensure exclusivity.

The PduR module uses the following OS resources:

- PduR_LockIFGatewayTx
- PduR_LockIFGatewayTxConf
- PduR_LockIFGatewayTriggerTxConf
- PduR_LockEnableRoutingNoNest
- PduR_LockIFGatewayTriggerTxNoNest
- PduR_LockIFGatewayTriggerTx
- PduR_LockDisableRoutingNoNest
- PduR_LockMCTxConfNoNest
- PduR_LockMcTpTxCopyTxData
- PduR_LockMcTpTxConf
- PduR_LockTpGwSessionPtr

7.10 Hardware Requirements

Table 7.49.PduR Hardware Requirements

Support	Usage
FLASH	11912 bytes
RAM	8272 bytes

Usage calculated based on a sample configuration.

8 ComBased Transformer

Module name	ComXf
Package	RTA-COM
AUTOSAR Module ID	175
AUTOSAR Version	22-11

8.1 Introduction

Prior to AUTOSAR 4.2, the only way to send large complex data over the network was to individually map each element of the data to a system signal. This required a great deal of effort, especially when mapping many complex data to their corresponding system signals. There were also runtime issues due to the fact that the RTE, for a single data transmission, had to perform a `Com_SendSignal` call for each mapped system signal.

These issues have been resolved in AUTOSAR 4.2 with the introduction of a new communication mechanism called “data transformation”. This mechanism enables the transmission (through Transformers) of large complex data as byte arrays via Sender/Receiver and Client/Server communication.

Transformers employ a data transformation mechanism to linearize and transform data, and they are executed by the RTE for any inter-ECU communication that has been configured to be transformed.

Multiple Transformers can be concatenated into transformer chains. The first transformer in the chain receives the data from the RTE, and the RTE allocates the buffer and calculates the required buffer size using the attribute “BufferComputation”.

Each subsequent transformer in the chain uses as input the output of the preceding transformer.

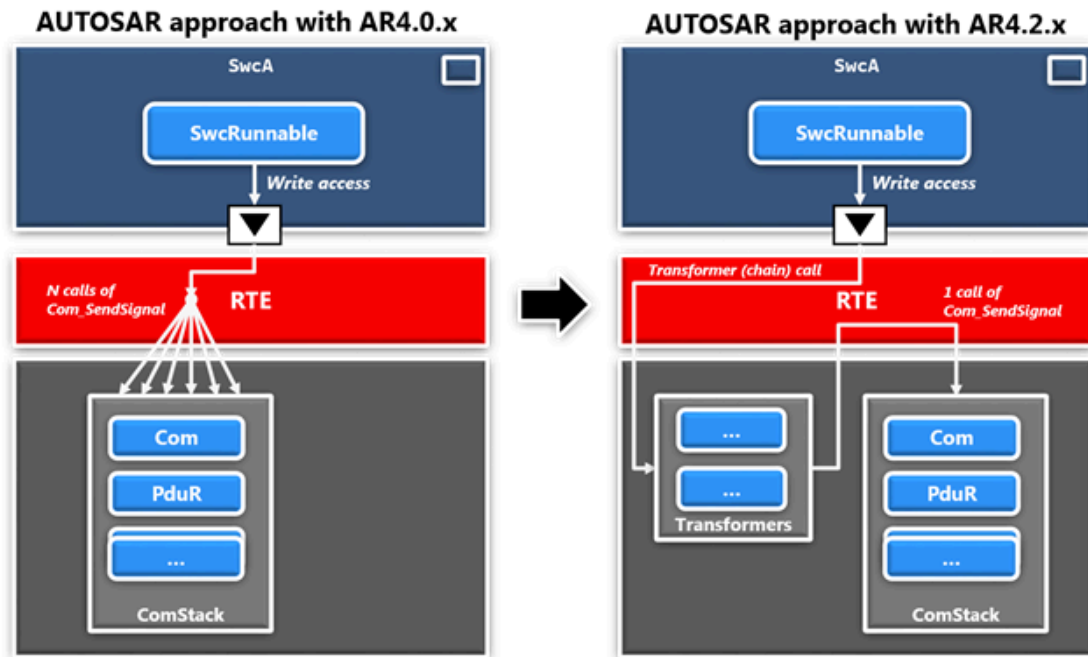


Figure 8.1. Data Transmission with and without Transformers

The COM Based Transformer (ComXf) performs the same actions as the COM module with regard to signal packing, including endianness conversion and sign extension. Other COM features, such as data filtering or deadline monitoring, are not in the scope of the COM Based Transformer.

Module Configuration

This module's configuration parameters are included in the configuration of another module. See the Configuration Advice section for more details.

8.2 Supported Features

8.2.1 Functionality Outline

A SWC initiates an inter-ECU communication, which is configured to be transformed, and then passes the data through the RTE to ComXf.

On the **sender** side (known as "forward transformation"), ComXf will -

1. Serialize the data of the SWC and bring them into a uint8-array representation based on the ComSignalGroup description inside the COM module configuration.
2. Forward the uint8-array representation to the COM module, via the Com_SendSignalGroupArray API, to be placed inside the respective IPdu. The COM module may analyze the provided uint8-array (depending on the configuration of the Transmission Mode Selection) and trigger the respective transmission mode. The IPdu is sent, via the Com stack, over the bus to the receiver(s).

On the **receiver** side (known as “inverse transformation”) -

1. The RTE on the receiver's side executes the transformer chain in the reverse order.
2. The ComXf of the receiver deserializes the linear data from the IPdu back into the original data structure. These data are handed over to the receiving SWC.

In both the sender and receiver scenarios, from the SWC's point-of-view, it is totally transparent as to whether the data have been transformed or not.

Note that in a Transformer Chain, ComXf is always the first transformer on the transmission side, and the last one on reception side.

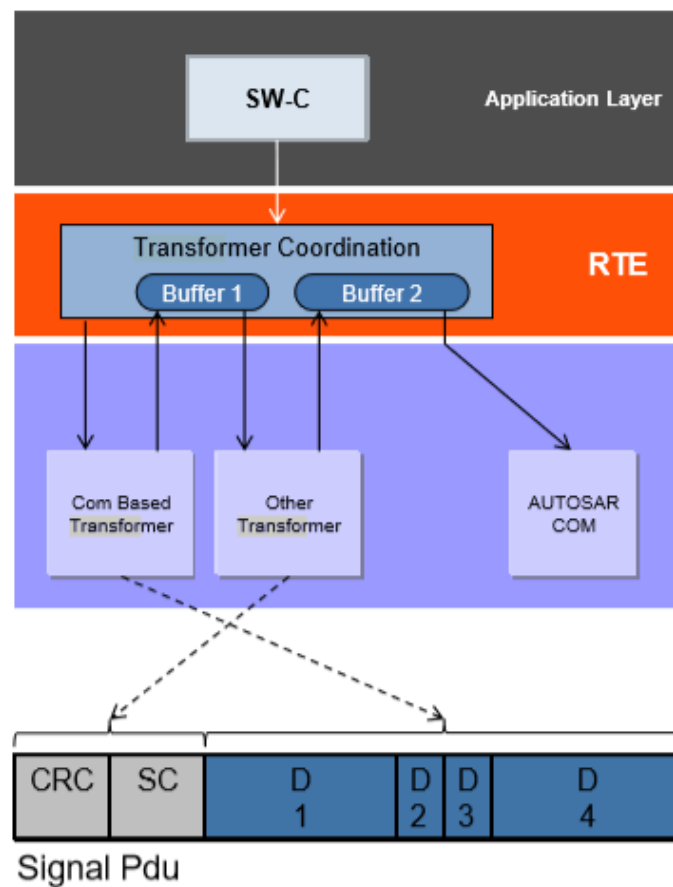


Figure 8.2.COM Based Transformation

8.3 Extensions

8.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

- No known extension parameters are defined in this version.

8.3.2 Additional Services

The following API are available in version v12.7.0 in addition to those defined in AR v22-11

- There are no additional API defined in this version.

8.4 Limitations

Version v12.7.0 has following limitations.

- Currently, RTA-BSW does not support automatic detection and selection of **ComXf** in the CodeGen Launch Configuration window, so you need to make sure this module is selected whenever the rba_SysElem module has been configured.
- The ComXf_<transformerId> and ComXf_Inv_<transformerId> APIs are implemented according to AR4.2.2 Specification (AR v22-11 API signature is not supported).
- The buffer (the out parameter) given to the serializer should be initialized with all zeros to avoid unexpected results due to bitwise operations.
- The serialized data should not exceed the value 7FFFFFFF.
- A *Bufferlength* of type *uint32* is supported and testing is done up to 2^{18} value due to tool constraints.
- Signed signals of 16 bits (SINT8 and SINT16) are supported/tested. If the signals configured are of type SINT32 or SINT64, the deserialized output might be wrong/malformed due to the integer limit in Java.

8.5 Exclusions

List of excluded API functions

Table 8.2.ComXf Excluded API functions

ComXf_Init	Initializes the Transformer.
ComXf_DeInit	Deinitializes the transformer.

For further information on these functions, please refer to AUTOSAR_SWS_COMBasedTransformer.pdf

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- There are no known exclusions in this version.

8.6 Deviations

- There are no known deviations in this version.

8.7 API Definition

List of supported API functions

Table 8.3.ComXf Supported API functions

ComXf_<transformerId>	Transforms a S/R communication using the serialization of COM Based Transformer.
ComXf_Inv_<transformerId>	Deserializes a S/R communication using the deserialization of COM Based Transformer.
ComXf_GetVersionInfo	Returns the version information of the called transformer module.

For further information on these functions, please refer to AUTOSAR_SWS_COMBasedTransformer.pdf

8.8 Configuration Advice



NOTE

Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the “Configuration Generation” section of the RTA-CAR User Guide.

8.8.1 Dependencies on other Modules

ComXf has the following dependencies on other modules.

COM

ComXf reads the COM module configuration, which contains the detailed description of the communication matrix. Information such as the start position, endianness, signal length and signal type that is configured inside the COM module, is also required by ComXf.

rba_SysElem

rba_SysElem imports configuration parameters from the System Extract and stores them in Ecuc Value File.

rba_Xfrm

rba_Xfrm contains the header TransformerTypes.h (which includes StdTypes.h), Xfrm_EcucParamDef.xml for general Transformer Values (i.e. which Signal/SignalGroup is used with which Transformer Mechanism), and

also a memcpy api called `rba_Xfrm_MemCopy`.

8.9 Integration Advice

8.9.1 Init Functions

There are no specific Initialization actions required.

8.9.2 Delnit Functions

There are no specific shutdown actions required.

8.9.3 Scheduled Functions

ComXf does not have a scheduled function.

8.9.4 ComXf and SomelpXf

If your project includes both ComXf and SomelpXf, they should both be integrated at the same release level.

8.9.5 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the ENTER and EXIT resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The ComXf module does not use any exclusive OS resources.

8.10 Hardware Requirements

Information relating to the hardware resources required by this module is not currently available.

9 rba_SysElem

Module name	Rba_SysElem
Package	RTA-COM
Module ID	N/A
AUTOSAR Version	22-11

9.1 Introduction

The Rba_SysElem module holds all relevant information from the AUTOSAR system level regarding DataTransformations (SystemSignals, ISignals, DataMappings, DataTypes,...) in the form of ECU configuration values for the later BSW code-generation of modules ComXf, SomelpXf and Rba_Xfrm.

In RTA-BSW, Rba_SysElem provides a EcucParamdef to get Transformer relevant configuration parameters from the SystemExtract. There is no C-Code in this Component.

RTA-BSW ConfGen supports automatic generation of Rba_SysElem, along with Rba_Xfrm module Ecuc values from an ECU Extract or System Extract.

Module Configuration

This module can be generated by the RTA-BSW Configuration Generator.

For any unsupported automatic configuration, the module can be manually configured by the user.

9.2 Supported Features

- Imports configuration parameters from System Extract and stores them in Ecuc Value File.
- Advanced Validation Feature gives hints on values that are incorrectly configured or missing.

9.3 Extensions

9.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

9.3.1.1 rba_SysElem

Short Name	rba_SysElem
Long Name	System Elements
Description	System Elements
Multiplicity	0..1
Default Value	-

Type	ECUC-MODULE-DEF
------	-----------------

9.3.1.2 rba_SysElem/rba_SysElem_SystemModel

Short Name	rba_SysElem_SystemModel
Long Name	System Model
Description	This container is added to have an additional hierarchical level. This is needed due to a bug that no ChoiceContainer is support as top level Container directly under the module Container.
Multiplicity	1..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.3 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_-DataMappings

Short Name	rba_SysElem_DataMappings
Long Name	DataMappings
Description	Container for DataMappings
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.4 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_-DataMappings/rba_SysElem_DataMapping

Short Name	rba_SysElem_DataMapping
Long Name	DataMapping
Description	Mapping of port elements (data elements and parameters) to frames and signals.
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.5 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_-DataMappings/rba_SysElem_DataMapping/rba_SysElem_-ClientServerToSignalMapping

Short Name	rba_SysElem_ClientServerToSignalMapping
Long Name	ClientServerToSignalMapping

Description	This element maps the ClientServerOperation to call- and return-SystemSignals. The serialization is defined by the referenced SerializationTechnology
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.6 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_-DataMappings/rba_SysElem_DataMapping/rba_SysElem_-ClientServerToSignalMapping/rba_SysElem_CallSignalRef

Short Name	rba_SysElem_CallSignalRef
Long Name	CallSignal Reference
Description	Reference to the callSignal to which the IN and INOUT ArgumentDataPrototypes are mapped.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.7 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_-DataMappings/rba_SysElem_DataMapping/rba_SysElem_-ClientServerToSignalMapping/rba_SysElem_ClientServerOperationRef

Short Name	rba_SysElem_ClientServerOperationRef
Long Name	ClientServerOperation Reference
Description	Reference to a ClientServerOperation, which is mapped to a call System-Signal and a return SystemSignal
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.8 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_-DataMappings/rba_SysElem_DataMapping/rba_SysElem_-ClientServerToSignalMapping/rba_SysElem_ReturnSignalRef

Short Name	rba_SysElem_ReturnSignalRef
Long Name	ReturnSignal Reference
Description	Reference to the returnSignal to which the OUT and INOUT ArgumentDataPrototypes are mapped
Multiplicity	0..1
Default Value	-

Type	ECUC-REFERENCE-DEF
------	--------------------

9.3.1.9 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_-DataMappings/rba_SysElem_DataMapping/rba_SysElem_CommunicationDirection

Short Name	rba_SysElem_CommunicationDirection
Long Name	CommunicationDirection
Description	Describes the communication direction.
Multiplicity	0..1
Default Value	-
Enumeration Literals	in, out
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.10 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_-DataMappings/rba_SysElem_DataMapping/rba_SysElem_SenderReceiverToSignalGroupMapping

Short Name	rba_SysElem_SenderReceiverToSignalGroupMapping
Long Name	SenderReceiverToSignalGroupMapping
Description	Mapping of a sender receiver communication data element with a composite datatype to a signal group.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.11 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_-DataMappings/rba_SysElem_DataMapping/rba_SysElem_SenderReceiverToSignalGroupMapping/rba_SysElem_SystemSignalGroupRef

Short Name	rba_SysElem_SystemSignalGroupRef
Long Name	SignalGroup Reference
Description	Reference to the signal group, which contain all primitive datatypes of the composite type.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.12 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_-DataMappings/rba_SysElem_DataMapping/rba_SysElem_SenderReceiverToSignalGroupMapping/rba_SysElem_TopLevelTypeMapping

Short Name	rba_SysElem_TopLevelTypeMapping
Long Name	Top Level Type Mapping Reference
Description	References the Type Mapping used by this SignalGroupMapping
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.13 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_-DataMappings/rba_SysElem_DataMapping/rba_SysElem_SenderReceiverToSignalGroupMapping/rba_SysElem_TypeMapping

Short Name	rba_SysElem_TypeMapping
Long Name	TypeMapping
Description	The CompositeTypeMapping maps the the ApplicationArrayElements and ApplicationRecordElements to Signals of the SignalGroup.
Multiplicity	1..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.14 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_-DataMappings/rba_SysElem_DataMapping/rba_SysElem_SenderReceiverToSignalGroupMapping/rba_SysElem_TypeMapping/rba_SysElem_SenderRecArrayTypeMapping

Short Name	rba_SysElem_SenderRecArrayTypeMapping
Long Name	SenderRecArrayTypeMapping
Description	If the ApplicationCompositeDataType is an Array, the "ArrayTypeMapping" will be used.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.15 `rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_-DataMappings/rba_SysElem_DataMapping/rba_SysElem_SenderReceiverToSignalGroupMapping/rba_SysElem_-TypeMapping/rba_SysElem_SenderRecArrayTypeMapping/rba_SysElem_ArrayElementMapping`

Short Name	<code>rba_SysElem_ArrayElementMapping</code>
Long Name	<code>ArrayElementMapping</code>
Description	Each <code>ApplicationArrayElement</code> must be mapped on a <code>SystemSignal</code> .
Multiplicity	1..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.16 `rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_-DataMappings/rba_SysElem_DataMapping/rba_SysElem_SenderReceiverToSignalGroupMapping/rba_SysElem_-TypeMapping/rba_SysElem_SenderRecArrayTypeMapping/rba_SysElem_ArrayElementMapping/rba_SysElem_Index`

Short Name	<code>rba_SysElem_Index</code>
Long Name	<code>Index</code>
Description	This index can be used to ensure the correct order of the <code>ArrayElementMapping</code> . This is more a fallback in case a file split and merge would swap the order
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.17 `rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_-DataMappings/rba_SysElem_DataMapping/rba_SysElem_SenderReceiverToSignalGroupMapping/rba_SysElem_-TypeMapping/rba_SysElem_SenderRecArrayTypeMapping/rba_SysElem_ArrayElementMapping/rba_SysElem_SystemSignalRef`

Short Name	<code>rba_SysElem_SystemSignalRef</code>
Long Name	<code>System Signal Reference</code>
Description	Reference to the system signal used to carry the primitive <code>ApplicationArrayElement</code> .
Multiplicity	0..1
Default Value	-

Type	ECUC-REFERENCE-DEF
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9.3.1.18 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_-DataMappings/rba_SysElem_DataMapping/rba_SysElem_SenderReceiverToSignalGroupMapping/rba_SysElem_TypeMapping/rba_SysElem_SenderRecArrayTypeMapping/rba_SysElem_ArrayElementMapping/rba_SysElem_TypeMappingRef

Short Name	rba_SysElem_TypeMappingRef
Long Name	Type Mapping Reference
Description	References to Type Mapping that is a child of this ArrayElementMapping
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.19 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_-DataMappings/rba_SysElem_DataMapping/rba_SysElem_SenderReceiverToSignalGroupMapping/rba_SysElem_TypeMapping/rba_SysElem_SenderRecRecordTypeMapping

Short Name	rba_SysElem_SenderRecRecordTypeMapping
Long Name	SenderRecRecordTypeMapping
Description	If the ApplicationCompositeDataType is a Record, the "RecordTypeMapping" will be used.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.20 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_-DataMappings/rba_SysElem_DataMapping/rba_SysElem_SenderReceiverToSignalGroupMapping/rba_SysElem_TypeMapping/rba_SysElem_SenderRecRecordTypeMapping/rba_SysElem_RecordElementMapping

Short Name	rba_SysElem_RecordElementMapping
Long Name	RecordElementMapping
Description	Each ApplicationRecordElement must be mapped on a SystemSignal.
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.21 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_-DataMappings/rba_SysElem_DataMapping/rba_SysElem_SenderReceiverToSignalGroupMapping/rba_SysElem_TypeMapping/rba_SysElem_SenderRecRecordTypeMapping/rba_SysElem_RecordElementMapping

lem_SenderReceiverToSignalGroupMapping/rba_SysElem_-
TypeMapping/rba_SysElem_SenderRecRecordTypeMapping/
rba_SysElem_RecordElementMapping/rba_SysElem_Implemen-
tationRecordElementRef

Short Name	rba_SysElem_ImplementationRecordElementRef
Long Name	Implementation Record Element Reference
Description	Reference to the SubElement the Record
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.22 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_-
DataMappings/rba_SysElem_DataMapping/rba_SysE-
lem_SenderReceiverToSignalGroupMapping/rba_SysElem_-
TypeMapping/rba_SysElem_SenderRecRecordTypeMapping/
rba_SysElem_RecordElementMapping/rba_SysElem_SystemSig-
nalRef

Short Name	rba_SysElem_SystemSignalRef
Long Name	System Signal Reference
Description	Reference to the system signal used to carry the primitive Application RecordElement.
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.23 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_-
DataMappings/rba_SysElem_DataMapping/rba_SysE-
lem_SenderReceiverToSignalGroupMapping/rba_SysElem_-
TypeMapping/rba_SysElem_SenderRecRecordTypeMapping/
rba_SysElem_RecordElementMapping/rba_SysElem_TypeMap-
pingRef

Short Name	rba_SysElem_TypeMappingRef
Long Name	Type Mapping Reference
Description	References to Type Mapping that is a child of this RecordElementMapping
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.24 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_-DataMappings/rba_SysElem_DataMapping/rba_SysElem_SenderReceiverToSignalGroupMapping/rba_SysElem_VariableDataPrototypeRef

Short Name	rba_SysElem_VariableDataPrototypeRef
Long Name	VariableDataPrototype Reference
Description	Reference to a data element with a composite datatype which is mapped to a signal group.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.25 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_-DataMappings/rba_SysElem_DataMapping/rba_SysElem_SenderReceiverToSignalMapping

Short Name	rba_SysElem_SenderReceiverToSignalMapping
Long Name	SenderReceiverToSignalMapping
Description	Mapping of a sender receiver communication data element with a primitive datatype to a signal.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.26 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_-DataMappings/rba_SysElem_DataMapping/rba_SysElem_SenderReceiverToSignalMapping/rba_SysElem_SystemSignalRef

Short Name	rba_SysElem_SystemSignalRef
Long Name	SystemSignal Reference
Description	Reference to the system signal used to carry the data element.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.27 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_-DataMappings/rba_SysElem_DataMapping/rba_SysElem_SenderReceiverToSignalMapping/rba_SysElem_VariableDataPrototypeRef

Short Name	rba_SysElem_VariableDataPrototypeRef
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Long Name	VariableDataPrototype Reference
Description	Reference to the data element, which ought to be sent over the Communication bus.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.28 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_DataTransformations

Short Name	rba_SysElem_DataTransformations
Long Name	DataTransformations
Description	Container for DataTransformations
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.29 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_DataTransformations/rba_SysElem_DataTransformation

Short Name	rba_SysElem_DataTransformation
Long Name	DataTransformation
Description	A DataTransformation represents a transformer chain. It is an ordered list of transformers.
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.30 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_DataTransformations/rba_SysElem_DataTransformation/rba_SysElem_ExecuteDespiteDataUnavailability

Short Name	rba_SysElem_ExecuteDespiteDataUnavailability
Long Name	ExecuteDespiteDataUnavailability
Description	Specifies whether the transformer is executed even if no input data are available.
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

9.3.1.31 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_DataTransformations/rba_SysElem_DataTransformation/rba_SysElem_ExecuteDespiteDataUnavailability

lem_TransformationTechnologyrba_SysElem_Transformation-Technology

Short Name	rba_SysElem_TransformationTechnology
Long Name	TransformationTechnology
Description	A TransformationTechnology is a transformer inside a transformer chain.
Multiplicity	1..255
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.32 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_Data-Transformations/rba_SysElem_DataTransformation/rba_SysElem_TransformationTechnology/rba_SysElem_Index

Short Name	rba_SysElem_Index
Long Name	Index
Description	This index is used to ensure the correct order of the TransformationTechnologies inside a DataTransformation independent of the used tools in the workflow.
Multiplicity	1..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.33 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_Data-Transformations/rba_SysElem_DataTransformation/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationTechnologyRef

Short Name	rba_SysElem_TransformationTechnologyRef
Long Name	TransformationTechnology Reference
Description	Reference to the TransformationTechnology Configuration Container.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.34 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_E2EProfileCompatibilityProps

Short Name	rba_SysElem_E2EProfileCompatibilityProps
Long Name	E2EProfileCompatibilityProps

Description	E2EProfileCompatibilityProps collects settings for configuration of the E2E state machine
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.35 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_E2EProfileCompatibilityProps/rba_SysElem_TransitToInvalidExtended

Short Name	rba_SysElem_TransitToInvalidExtended
Long Name	TransitToInvalidExtended
Description	E2E State machine behavior concerning transition from NODATA/INIT to INVALID. value=0 (false): no direct transition from NODATA to INVALID, no transition from INIT to INVALID due to counter-related faults (Autosar R19-11 or former behavior) value=1 (true): direct transition from NODATA to INVALID covered, transition from INIT to INVALID due to counter-related faults covered (state machine extended)
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

9.3.1.36 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_IPdus

Short Name	rba_SysElem_IPdus
Long Name	IPdus
Description	Container for IPdus
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.37 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_IPdus/rba_SysElem_IPdu

Short Name	rba_SysElem_IPdu
Long Name	IPdu
Description	The IPdu (Interaction Layer Protocol Data Unit) element is used to sum up all Pdus that are routed by the PduR.
Multiplicity	0..*
Default Value	-
Type	ECUC-CHOICE-CONTAINER-DEF

9.3.1.38 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_IPdu/rba_SysElem_IPdu/rba_SysElem_ISignalIPdu

Short Name	rba_SysElem_ISignalIPdu
Long Name	ISignalIPdu
Description	Represents the IPdus handled by Com. The ISignalIPdu assembled and disassembled in AUTOSAR COM consists of one or more signals. In case no multiplexing is performed this IPdu is routed to/from the Interface Layer. A maximum of one dynamic length signal per IPdu is allowed.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.39 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_IPdu/rba_SysElem_IPdu/rba_SysElem_ISignalIPdu/rba_SysElem_ISignalToPduMapping

Short Name	rba_SysElem_ISignalToPduMapping
Long Name	ISignalToPduMapping
Description	Definition of SignalToIPduMappings included in the SignalIPdu.
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.40 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_IPdu/rba_SysElem_IPdu/rba_SysElem_ISignalIPdu/rba_SysElem_ISignalToPduMapping/rba_SysElem_ISignalGroupRef

Short Name	rba_SysElem_ISignalGroupRef
Long Name	ISignalGroup Reference
Description	Reference to an ISignalGroup that is mapped into the ISignalIPdu. If an ISignalToIPduMapping for an ISignalGroup is defined, only the UpdateIndication BitPosition and the transferProperty is relevant. The startPosition and the packingByteOrder shall be ignored. Each ISignal contained in the ISignalGroup shall be mapped into an IPdu by an own ISignalToIPduMapping. The references to the ISignal and to the ISignalGroup in an ISignalToIPduMapping are mutually exclusive.
Multiplicity	0..1

Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.41 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_IPdus/ rba_SysElem_IPdu/rba_SysElem_ISignalIPdu/rba_SysE- lem_ISignalToPduMapping/rba_SysElem_ISignalRef

Short Name	rba_SysElem_ISignalRef
Long Name	ISignal Reference
Description	Reference to a ISignal that is mapped into the ISignalIPdu. Each ISignal contained in the ISignalGroup shall be mapped into an IPdu by an own ISignal-ToIPduMapping. The references to the ISignal and to the ISignalGroup in an ISignalToIPduMapping are mutually exclusive.
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.42 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_IPdus/ rba_SysElem_IPdu/rba_SysElem_ISignalIPdu/rba_SysE- lem_ISignalToPduMapping/rba_SysElem_ISignalToPduMapping- ForeignRef

Short Name	rba_SysElem_ISignalToPduMappingForeignRef
Long Name	ISignalToPduMapping Foreign Reference
Description	Reference to the ISignalToIPduMapping in the SystemExtract.
Multiplicity	1..1
Default Value	-
Type	ECUC-FOREIGN-REFERENCE-DEF

9.3.1.43 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_IPdus/ rba_SysElem_IPdu/rba_SysElem_ISignalIPdu/rba_SysE- lem_ISignalToPduMapping/rba_SysElem_PackingByteOrder

Short Name	rba_SysElem_PackingByteOrder
Long Name	PackingByteOrder

Description	This parameter defines the order of the bytes of the signal and the packing into the SignalIPdu. The byte ordering "Little Endian" (MostSignificantByteLast), "Big Endian" (MostSignificantByteFirst) and "Opaque" can be selected. For opaque data endianness conversion shall be configured to Opaque. The value of this attribute impacts the absolute position of the signal into the SignalIPdu (see the startPosition attribute description). For an ISignalGroup the packingByteOrder is irrelevant and shall be ignored.
Multiplicity	0..1
Default Value	-
Enumeration Literals	mostSignificantByteFirst, mostSignificantByteLast, opaque
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.44 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_IPdu/rba_SysElem_IPdu/rba_SysElem_ISignalIPdu/rba_SysElem_ISignalToPduMapping/rba_SysElem_StartPosition

Short Name	rba_SysElem_StartPosition
Long Name	StartPosition
Description	This parameter is necessary to describe the bitposition of a signal within an SignalIPdu. It denotes the least significant bit for "Little Endian" and the most significant bit for "Big Endian" packed signals within the IPdu (see the description of the packingByteOrder attribute). In AUTOSAR the bit counting is always set to "sawtooth" and the bit order is set to "Decreasing". The bit counting in byte 0 starts with bit 0 (least significant bit). The most significant bit in byte 0 is bit 7. Please note that the way the bytes will be actually sent on the bus does not impact this representation: they will always be seen by the software as a byte array. If a mapping for the ISignalGroup is defined, this attribute is irrelevant and shall be ignored.
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.45 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_IPdu/rba_SysElem_IPdu/rba_SysElem_ISignalIPdu/rba_SysElem_Length

Short Name	rba_SysElem_Length
Long Name	Length
Description	Pdu length in bytes. In case of dynamic length IPdus (containing a dynamical length signal), this value indicates the maximum data length. It should be noted that in former AUTOSAR releases (Rel 2.1, Rel 3.0, Rel 3.1, Rel 4.0 Rev. 1) this parameter was defined in bits. The Pdu length of zero bytes is allowed.
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.46 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_IPdu/rba_SysElem_IPdu/rba_SysElem_ISignalIPdu/rba_SysElem_MetaDataLength

Short Name	rba_SysElem_MetaDataLength
Long Name	MetaDataLength
Description	Number of additional bytes of MetaData in the PDU data field. The Meta-Data contains auxiliary information for the PDU, e.g. the CAN ID.
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.47 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_IPdu/rba_SysElem_IPdu/rba_SysElem_ISignalIPdu/rba_SysElem_UnusedBitPattern

Short Name	rba_SysElem_UnusedBitPattern
Long Name	UnusedBitPattern
Description	AUTOSAR COM and AUTOSAR IPDUM are filling not used areas of an IPDU with this bit-pattern. This attribute is mandatory to avoid undefined behavior. This byte-pattern will be repeated throughout the IPdu.
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.48 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_IPdus/rba_SysElem_IPdu/rba_SysElem_ISignalIPdu/rba_SysElem_UseLdCom

Short Name	rba_SysElem_UseLdCom
Long Name	UseLdCom
Description	This parameter specifies, if the constraints are fulfilled to transfer this ISignal with LdCom. Following constraints shall be checked: TPS_SYST_02015 to TPS_SYST_02028, TPS_SYST_03001 If LdCom is integrated in the ECU shall not be checked.
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

9.3.1.49 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignalGroups

Short Name	rba_SysElem_ISignalGroups
Long Name	ISignalGroups
Description	Container for ISignalGroups
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.50 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignalGroups/rba_SysElem_ISignalGroup

Short Name	rba_SysElem_ISignalGroup
Long Name	ISignalGroup
Description	SignalGroup of the Interaction Layer. The RTE supports a "signal fan-out" where the same System Signal Group is sent in different SignalIPdus to multiple receivers. An ISignalGroup refers to a set of ISignals that shall always be kept together. A ISignalGroup represents a COM Signal Group.
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.51 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignalGroups/rba_SysElem_ISignalGroup/rba_SysElem_ComBasedSig-

nalGroupTransformationRefrba_SysElem_ComBasedSignal- GroupTransformationRef

Short Name	rba_SysElem_ComBasedSignalGroupTransformationRef
Long Name	ComBasedSignalGroupTransformation Reference
Description	Optional reference to a DataTransformation which represents the transformer chain that is used to transform the data that shall be placed inside this ISignalGroup based on the COMBasedTransformer approach.
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.52 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignal- Groups/rba_SysElem_ISignalGroup/rba_SysElem_ISignalGroup- ForeignRef

Short Name	rba_SysElem_ISignalGroupForeignRef
Long Name	ISignalGroup Foreign Reference
Description	Reference to this ISignalGroup in the SystemExtract.
Multiplicity	1..1
Default Value	-
Type	ECUC-FOREIGN-REFERENCE-DEF

9.3.1.53 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignal- Groups/rba_SysElem_ISignalGroup/rba_SysElem_ISignalRef

Short Name	rba_SysElem_ISignalRef
Long Name	ISignal Reference
Description	Reference to a set of ISignals that shall always be kept together.
Multiplicity	0..*
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.54 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignal- Groups/rba_SysElem_ISignalGroup/rba_SysElem_SystemSignal- GroupRef

Short Name	rba_SysElem_SystemSignalGroupRef
Long Name	System Signal Group Reference
Description	Reference to the SystemSignalGroup that is defined on VFB level and that is supposed to be transmitted in the ISignalGroup.

Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.55 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignal-Groups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationComSpecProps

Short Name	rba_SysElem_TransformationComSpecProps
Long Name	TransformationComSpecProps
Description	Specifies port specific configuration properties for transformer attributes
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.56 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignal-Groups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationComSpecProps/rba_SysElem_EndToEndTransformationComSpecProps

Short Name	rba_SysElem_EndToEndTransformationComSpecProps
Long Name	EndToEndTransformationComSpecProps
Description	Specifies port specific configuration properties for EndToEnd transformer attributes
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.57 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignal-Groups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationComSpecProps/rba_SysElem_EndToEndTransformationComSpecProps/rba_SysElem_ClearToInvalid

Short Name	rba_SysElem_ClearToInvalid
Long Name	ClearToInvalid
Description	Clear monitoring window on transition from state Valid to state Invalid
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

9.3.1.58 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignal-Groups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationComSpecProps/rba_SysElem_EndToEndTransformationCom-

SpecProps/rba_SysElem_DisableEndToEndCheck

Short Name	rba_SysElem_DisableEndToEndCheck
Long Name	DisableEndToEndCheck
Description	Disables/Enables the E2E check. The E2Eheader is removed from the payload independent from the setting of this attribute
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

9.3.1.59 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignalGroups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationComSpecProps/rba_SysElem_EndToEndTransformationComSpecProps/rba_SysElem_MaxDeltaCounter

Short Name	rba_SysElem_MaxDeltaCounter
Long Name	MaxDeltaCounter
Description	Maximum allowed gap between two counter values of two consecutive checks
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.60 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignalGroups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationComSpecProps/rba_SysElem_EndToEndTransformationComSpecProps/rba_SysElem_MaxErrorStateInit

Short Name	rba_SysElem_MaxErrorStateInit
Long Name	MaxErrorStateInit
Description	Maximal number of checks in which ProfileStatus equal to E2E_P_ERROR was determined, within the last WindowSize checks, for the state E2E_SM_INIT. The minimum value is 0
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.61 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignalGroups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationComSpecProps/rba_SysElem_EndToEndTransformationComSpecProps/rba_SysElem_MaxErrorStateInit

SpecProps/rba_SysElem_MaxErrorStateInvalid

Short Name	rba_SysElem_MaxErrorStateInvalid
Long Name	MaxErrorStateInvalid
Description	Maximal number of checks in which ProfileStatus equal to E2E_P_ERROR was determined, within the last WindowSize checks, for the state E2E_SM_INVALID. The minimum value is 0
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.62 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignalGroups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationComSpecProps/rba_SysElem_EndToEndTransformationComSpecProps/rba_SysElem_MaxErrorStateValid

Short Name	rba_SysElem_MaxErrorStateValid
Long Name	MaxErrorStateValid
Description	Maximal number of checks in which ProfileStatus equal to E2E_P_ERROR was determined, within the last WindowSize checks, for the state E2E_SM_VALID. The minimum value is 0
Multiplicity	1..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.63 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignalGroups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationComSpecProps/rba_SysElem_EndToEndTransformationComSpecProps/rba_SysElem_MaxNoNewOrRepeatedData

Short Name	rba_SysElem_MaxNoNewOrRepeatedData
Long Name	MaxNoNewOrRepeatedData
Description	The maximum amount of missing or repeated Data which the receiver does not expect to exceed under normal communication conditions.
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.64 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignalGroups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationComSpecProps/rba_SysElem_EndToEndTransformationComSpecProps/rba_SysElem_MinOkStateInit

Short Name	rba_SysElem_MinOkStateInit
Long Name	MinOkStateInit
Description	Minimal number of checks in which ProfileStatus equal to E2E_P_OK was determined, within the last WindowSize checks, for the state E2E_SM_INIT. The minimum value is 1
Multiplicity	0..1
Minimum Value	1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.65 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignalGroups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationComSpecProps/rba_SysElem_EndToEndTransformationComSpecProps/rba_SysElem_MinOkStateInvalid

Short Name	rba_SysElem_MinOkStateInvalid
Long Name	MinOkStateInvalid
Description	Minimal number of checks in which ProfileStatus equal to E2E_P_OK was determined, within the last WindowSize checks, for the state E2E_SM_INVALID. The minimum value is 1
Multiplicity	0..1
Minimum Value	1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.66 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignalGroups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationComSpecProps/rba_SysElem_EndToEndTransformationComSpecProps/rba_SysElem_MinOkStateValid

Short Name	rba_SysElem_MinOkStateValid
Long Name	MinOkStateValid
Description	Minimal number of checks in which ProfileStatus equal to E2E_P_OK was determined, within the last WindowSize checks, for the state E2E_SM_VALID. The minimum value is 1
Multiplicity	0..1

Minimum Value	1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.67 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignalGroups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationComSpecProps/rba_SysElem_EndToEndTransformationComSpecProps/rba_SysElem_SyncCounterInit

Short Name	rba_SysElem_SyncCounterInit
Long Name	SyncCounterInit
Description	Number of Data required for validating the consistency of the counter that must be received with a valid counter (i.e. counter within the allowed lock-in range) after the detection of an unexpected behavior of a received counter.
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.68 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignalGroups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationComSpecProps/rba_SysElem_EndToEndTransformationComSpecProps/rba_SysElem_WindowSize

Short Name	rba_SysElem_WindowSize
Long Name	WindowSize
Description	Size of the monitoring window for the E2E state machine
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.69 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignalGroups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationComSpecProps/rba_SysElem_EndToEndTransformationComSpecProps/rba_SysElem_WindowSizeInit

Short Name	rba_SysElem_WindowSizeInit
Long Name	WindowSizeInit
Description	Size of the monitoring window of state Init for the E2E state machine

Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.70 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignalGroups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationComSpecProps/rba_SysElem_EndToEndTransformationComSpecProps/rba_SysElem_WindowSizeInvalid

Short Name	rba_SysElem_WindowSizeInvalid
Long Name	WindowSizeInvalid
Description	Size of the monitoring window of state Invalid for the E2E state machine
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.71 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignalGroups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationComSpecProps/rba_SysElem_EndToEndTransformationComSpecProps/rba_SysElem_WindowSizeValid

Short Name	rba_SysElem_WindowSizeValid
Long Name	WindowSizeValid
Description	Size of the monitoring window of state Valid for the E2E state machine
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.72 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignalGroups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationComSpecProps/rba_SysElem_EndToEndTransformationComSpecProps/rba_SysElem_e2eProfileCompatibilityPropsRef

Short Name	rba_SysElem_e2eProfileCompatibilityPropsRef
Long Name	e2eProfileCompatibilityPropsRef
Description	Reference to additional settings for the E2E state machine
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.73 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignal-Groups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationISignalProps

Short Name	rba_SysElem_TransformationISignalProps
Long Name	TransformationISignalProps
Description	TransformationISignalProps holds all the attributes for the different TransformationTechnologies that are ISignal specific.
Multiplicity	0..*
Default Value	-
Type	ECUC-CHOICE-CONTAINER-DEF

9.3.1.74 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignal-Groups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationISignalProps/rba_SysElem_EndToEndTransformationISignal-Props

Short Name	rba_SysElem_EndToEndTransformationISignalProps
Long Name	EndToEndTransformationISignalProps
Description	The container EndToEndTransformationISignalProps specifies ISignalGroup specific configuration properties for E2E transformer attributes.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.75 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignal-Groups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationISignalProps/rba_SysElem_EndToEndTransformationISignal-Props/rba_SysElem_CSErrorReaction

Short Name	rba_SysElem_CSErrorReaction
Long Name	CsErrorReaction
Description	Defines whether the transformer chain of client/server communication coordinates an autonomous error reaction together with the RTE or whether any error reaction is the responsibility of the application.
Multiplicity	0..1
Default Value	-
Enumeration Literals	applicationOnly, autonomous
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.76 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignal-

Groups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationISignalProps/rba_SysElem_EndToEndTransformationISignalProps/rba_SysElem_DataID

Short Name	rba_SysElem_DataID
Long Name	DataID
Description	This represents a unique numerical identifier. Note: ID is used for protection against masquerading. The details concerning the maximum number of values (this information is specific for each E2E profile) applicable for this attribute are controlled by a semantic constraint that depends on the category of the EndToEndProtection.
Multiplicity	0..*
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.77 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignalGroups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationISignalProps/rba_SysElem_EndToEndTransformationISignalProps/rba_SysElem_DataLength

Short Name	rba_SysElem_DataLength
Long Name	DataLength
Description	Length of Data in bits.
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.78 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignalGroups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationISignalProps/rba_SysElem_EndToEndTransformationISignalProps/rba_SysElem_MaxDataLength

Short Name	rba_SysElem_MaxDataLength
Long Name	MaxDataLength
Description	Maximum length of Data in bits.
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.79 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignalGroups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationISignalProps/rba_SysElem_EndToEndTransformationISignalProps/rba_SysElem_MinDataLength

Short Name	rba_SysElem_MinDataLength
Long Name	MinDataLength
Description	Minimum length of Data in bits.
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.80 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignalGroups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationISignalProps/rba_SysElem_EndToEndTransformationISignalProps/rba_SysElem_SourceID

Short Name	rba_SysElem_SourceID
Long Name	SourceID
Description	This attribute represents a unique numerical identifier identifying the source of a certain transmission. In case of C/S communication, this ID uniquely identifies the client
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.81 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignalGroups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationISignalProps/rba_SysElem_EndToEndTransformationISignalProps/rba_SysElem_TransformationTechnologyRef

Short Name	rba_SysElem_TransformationTechnologyRef
Long Name	TransformationTechnology Reference
Description	Reference to the TransformationTechnology description that contains transformer specific and ISignal independent configuration properties.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.82 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignalGroups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationISignalProps/rba_SysElem_UserDefinedTransformationISignalProps

Short Name	rba_SysElem_UserDefinedTransformationISignalProps
Long Name	UserDefinedTransformationISignalProps
Description	The UserDefinedTransformationISignalProps is used to specify ISignalGroup specific configuration properties for custom transformers.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.83 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignalGroups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationISignalProps/rba_SysElem_UserDefinedTransformationISignalProps/rba_SysElem_CSErrorReaction

Short Name	rba_SysElem_CSErrorReaction
Long Name	CsErrorReaction
Description	Defines whether the transformer chain of client/server communication coordinates an autonomous error reaction together with the RTE or whether any error reaction is the responsibility of the application.
Multiplicity	0..1
Default Value	-
Enumeration Literals	applicationOnly, autonomous
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.84 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignalGroups/rba_SysElem_ISignalGroup/rba_SysElem_TransformationISignalProps/rba_SysElem_UserDefinedTransformationISignalProps/rba_SysElem_TransformationTechnologyRef

Short Name	rba_SysElem_TransformationTechnologyRef
Long Name	TransformationTechnology Reference
Description	Reference to the TransformationTechnology description that contains transformer specific and ISignal independent configuration properties.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.85 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals

Short Name	rba_SysElem_ISignals
Long Name	ISignals
Description	Container for ISignals
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.86 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/rba_SysElem_ISignal

Short Name	rba_SysElem_ISignal
Long Name	ISignal
Description	Signal of the Interaction Layer. The RTE supports a "signal fan-out" where the same System Signal is sent in different SignalPdu to multiple receivers. To support the RTE "signal fan-out" each SignalPdu contains ISignals. If the same System Signal is to be mapped into several SignalPdu there is one ISignal needed for each ISignal-ToIPduMapping. ISignals describe the Interface between the Precompile configured RTE and the potentially Post-build configured Com Stack (see ECUC Parameter Mapping). In case of the SystemSignalGroup an ISignal must be created for each SystemSignal contained in the SystemSignalGroup.
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.87 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/rba_SysElem_ISignal/rba_SysElem_DataTransformationRef

Short Name	rba_SysElem_DataTransformationRef
Long Name	DataTransformation Reference
Description	Optional reference to a DataTransformation which represents the transformer chain that is used to transform the data that shall be placed inside this ISignal.
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.88 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/ rba_SysElem_ISignal/rba_SysElem_DataTypePolicy

Short Name	rba_SysElem_DataTypePolicy
Long Name	DataTypePolicy
Description	With the aggregation of SwDataDefProps an ISignal specifies how it is represented on the network. This representation follows a particular policy. Note that this causes some redundancy which is intended and can be used to support flexible development methodology as well as subsequent integrity checks. If the policy "networkRepresentationFromComSpec" is chosen the network representation from the ComSpec that is aggregated by the PortPrototype shall be used. If the "override" policy is chosen the requirements specified in the PortInterface and in the ComSpec are not fulfilled by the networkRepresentationProps. In case the System Description doesn't use a complete Software Component Description (VFB View) the "legacy" policy can be chosen.
Multiplicity	1..1
Default Value	-
Enumeration Literals	legacy, networkRepresentationFromComSpec, override, transformingISignal
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.89 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/ rba_SysElem_ISignal/rba_SysElem_ISignalForeignRef

Short Name	rba_SysElem_ISignalForeignRef
Long Name	ISignal Foreign Reference
Description	Reference to this ISignal in the SystemExtract.
Multiplicity	1..1
Default Value	-
Type	ECUC-FOREIGN-REFERENCE-DEF

9.3.1.90 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/ rba_SysElem_ISignal/rba_SysElem_InitValue

Short Name	rba_SysElem_InitValue
Long Name	InitValue

Description	Optional definition of a ISignal's initValue in case the System Description doesn't use a complete Software Component Description (VFB View). This supports the inclusion of legacy system signals.
Multiplicity	0..1
Default Value	-
Type	ECUC-CHOICE-CONTAINER-DEF

9.3.1.91 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/rba_SysElem_ISignal/rba_SysElem_InitValue/rba_SysElem_ArrayValueSpecification

Short Name	rba_SysElem_ArrayValueSpecification
Long Name	ArrayValueSpecification
Description	Specifies the values for an array
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.92 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/rba_SysElem_ISignal/rba_SysElem_InitValue/rba_SysElem_ArrayValueSpecification/rba_SysElem_NumericalValueSpecification

Short Name	rba_SysElem_NumericalValueSpecification
Long Name	NumericalValueSpecification
Description	A numerical ValueSpecification which is intended to be assigned to a Primitive data element. Note that the numerical value is a variant, it can be computed by a formula.
Multiplicity	1..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.93 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/rba_SysElem_ISignal/rba_SysElem_InitValue/rba_SysElem_ArrayValueSpecification/rba_SysElem_NumericalValueSpecification/rba_SysElem_NumericalValue

Short Name	rba_SysElem_NumericalValue
Long Name	Value
Description	This is the value itself.
Multiplicity	1..1
Default Value	-

Type	ECUC-FLOAT-PARAM-DEF
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9.3.1.94 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/rba_SysElem_ISignal/rba_SysElem_InitValue/rba_SysElem_NumericalValueSpecification

Short Name	rba_SysElem_NumericalValueSpecification
Long Name	NumericalValueSpecification
Description	A numerical ValueSpecification which is intended to be assigned to a Primitive data element. Note that the numerical value is a variant, it can be computed by a formula.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.95 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/rba_SysElem_ISignal/rba_SysElem_InitValue/rba_SysElem_NumericalValueSpecification/rba_SysElem_NumericalValue

Short Name	rba_SysElem_NumericalValue
Long Name	Value
Description	This is the value itself.
Multiplicity	1..1
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

9.3.1.96 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/rba_SysElem_ISignal/rba_SysElem_InitValue/rba_SysElem_TextValueSpecification

Short Name	rba_SysElem_TextValueSpecification
Long Name	TextValueSpecification
Description	The purpose of TextValueSpecification is to define the labels that correspond to enumeration values.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.97 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/ rba_SysElem_ISignal/rba_SysElem_InitValue/rba_SysElem_- TextValueSpecification/rba_SysElem_TextValue

Short Name	rba_SysElem_TextValue
Long Name	Value
Description	This is the value itself. Note that vt uses the operator to separate the values for the different bitfield masks in case that the semantics of the related DataPrototype is described by means of a BITFIELD_TEXTTABLE in the associated CompuMethod.
Multiplicity	1..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

9.3.1.98 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/ rba_SysElem_ISignal/rba_SysElem_Length

Short Name	rba_SysElem_Length
Long Name	Length
Description	Size of the signal in bits. The size needs to be derived from the mapped VariableDataPrototype according to the mapping of primitive DataTypes to BaseTypes as used in the RTE. Indicates maximum size for dynamic length signals. The ISignal length of zero bits is allowed.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.99 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/ rba_SysElem_ISignal/rba_SysElem_SafetySignalRole

Short Name	rba_SysElem_SafetySignalRole
Long Name	SafetySignalRole
Description	Safety signal role has to be added to the sgd (special group data) in GM specific E2E Profiles.
Multiplicity	0..1
Default Value	-
Enumeration Literals	ARC, CRC, CS, PV
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.100 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/ rba_SysElem_ISignal/rba_SysElem_SystemSignalRef

Short Name	rba_SysElem_SystemSignalRef
Long Name	SystemSignal Reference
Description	Reference to the System Signal that is supposed to be transmitted in the ISignal.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.101 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/ rba_SysElem_ISignal/rba_SysElem_TransformationCom- SpecProps

Short Name	rba_SysElem_TransformationComSpecProps
Long Name	TransformationComSpecProps
Description	Specifies port specific configuration properties for transformer attributes
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.102 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/ rba_SysElem_ISignal/rba_SysElem_TransformationCom- SpecProps/rba_SysElem_EndToEndTransformationCom- SpecProps

Short Name	rba_SysElem_EndToEndTransformationComSpecProps
Long Name	EndToEndTransformationComSpecProps
Description	Specifies port specific configuration properties for EndToEnd transformer attributes
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.103 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/ rba_SysElem_ISignal/rba_SysElem_TransformationCom- SpecProps/rba_SysElem_EndToEndTransformationCom- SpecProps/rba_SysElem_ClearToInvalid

Short Name	rba_SysElem_ClearToInvalid
Long Name	ClearToInvalid

Description	Clear monitoring window on transition from state Valid to state Invalid
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

9.3.1.104 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/ rba_SysElem_ISignal/rba_SysElem_TransformationCom- SpecProps/rba_SysElem_EndToEndTransformationCom- SpecProps/rba_SysElem_DisableEndToEndCheck

Short Name	rba_SysElem_DisableEndToEndCheck
Long Name	DisableEndToEndCheck
Description	Disables/Enables the E2E check. The E2Eheader is removed from the payload independent from the setting of this attribute
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

9.3.1.105 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/ rba_SysElem_ISignal/rba_SysElem_TransformationCom- SpecProps/rba_SysElem_EndToEndTransformationCom- SpecProps/rba_SysElem_MaxDeltaCounter

Short Name	rba_SysElem_MaxDeltaCounter
Long Name	MaxDeltaCounter
Description	Maximum allowed gap between two counter values of two consecutive checks
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.106 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/ rba_SysElem_ISignal/rba_SysElem_TransformationCom- SpecProps/rba_SysElem_EndToEndTransformationCom- SpecProps/rba_SysElem_MaxErrorStateInit

Short Name	rba_SysElem_MaxErrorStateInit
Long Name	MaxErrorStateInit
Description	Maximal number of checks in which ProfileStatus equal to E2E_P_ERROR was determined, within the last WindowSize checks, for the state E2E_SM_INIT. The minimum value is 0
Multiplicity	0..1

Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.107 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/rba_SysElem_ISignal/rba_SysElem_TransformationCom-SpecProps/rba_SysElem_EndToEndTransformationCom-SpecProps/rba_SysElem_MaxErrorStateInvalid

Short Name	rba_SysElem_MaxErrorStateInvalid
Long Name	MaxErrorStateInvalid
Description	Maximal number of checks in which ProfileStatus equal to E2E_P_ERROR was determined, within the last WindowSize checks, for the state E2E_SM_INVALID. The minimum value is 0
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.108 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/rba_SysElem_ISignal/rba_SysElem_TransformationCom-SpecProps/rba_SysElem_EndToEndTransformationCom-SpecProps/rba_SysElem_MaxErrorStateValid

Short Name	rba_SysElem_MaxErrorStateValid
Long Name	MaxErrorStateValid
Description	Maximal number of checks in which ProfileStatus equal to E2E_P_ERROR was determined, within the last WindowSize checks, for the state E2E_SM_VALID. The minimum value is 0
Multiplicity	1..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.109 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/rba_SysElem_ISignal/rba_SysElem_TransformationCom-SpecProps/rba_SysElem_EndToEndTransformationCom-SpecProps/rba_SysElem_MaxNoNewOrRepeatedData

Short Name	rba_SysElem_MaxNoNewOrRepeatedData
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Long Name	MaxNoNewOrRepeatedData
Description	The maximum amount of missing or repeated Data which the receiver does not expect to exceed under normal communication conditions.
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.110 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/rba_SysElem_ISignal/rba_SysElem_TransformationComSpecProps/rba_SysElem_EndToEndTransformationComSpecProps/rba_SysElem_MinOkStateInit

Short Name	rba_SysElem_MinOkStateInit
Long Name	MinOkStateInit
Description	Minimal number of checks in which ProfileStatus equal to E2E_P_OK was determined, within the last WindowSize checks, for the state E2E_SM_INIT. The minimum value is 1
Multiplicity	0..1
Minimum Value	1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.111 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/rba_SysElem_ISignal/rba_SysElem_TransformationComSpecProps/rba_SysElem_EndToEndTransformationComSpecProps/rba_SysElem_MinOkStateInvalid

Short Name	rba_SysElem_MinOkStateInvalid
Long Name	MinOkStateInvalid
Description	Minimal number of checks in which ProfileStatus equal to E2E_P_OK was determined, within the last WindowSize checks, for the state E2E_SM_INVALID. The minimum value is 1
Multiplicity	0..1
Minimum Value	1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.112 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/rba_SysElem_ISignal/rba_SysElem_TransformationComSpecProps/rba_SysElem_EndToEndTransformationCom-

SpecProps/rba_SysElem_MinOkStateValid

Short Name	rba_SysElem_MinOkStateValid
Long Name	MinOkStateValid
Description	Minimal number of checks in which ProfileStatus equal to E2E_P_OK was determined, within the last WindowSize checks, for the state E2E_SM_VALID. The minimum value is 1
Multiplicity	0..1
Minimum Value	1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.113 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/
rba_SysElem_ISignal/rba_SysElem_TransformationCom-
SpecProps/rba_SysElem_EndToEndTransformationCom-
SpecProps/rba_SysElem_SyncCounterInit

Short Name	rba_SysElem_SyncCounterInit
Long Name	SyncCounterInit
Description	Number of Data required for validating the consistency of the counter that must be received with a valid counter (i.e. counter within the allowed lock-in range) after the detection of an unexpected behavior of a received counter.
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.114 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/
rba_SysElem_ISignal/rba_SysElem_TransformationCom-
SpecProps/rba_SysElem_EndToEndTransformationCom-
SpecProps/rba_SysElem_WindowSize

Short Name	rba_SysElem_WindowSize
Long Name	WindowSize
Description	Size of the monitoring window for the E2E state machine
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.115 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/
rba_SysElem_ISignal/rba_SysElem_TransformationCom-
SpecProps/rba_SysElem_EndToEndTransformationCom-
SpecProps/rba_SysElem_WindowSizeInit

Short Name	rba_SysElem_WindowSizeInit
Long Name	WindowSizeInit
Description	Size of the monitoring window of state Init for the E2E state machine
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.116 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/
rba_SysElem_ISignal/rba_SysElem_TransformationCom-
SpecProps/rba_SysElem_EndToEndTransformationCom-
SpecProps/rba_SysElem_WindowSizeInvalid

Short Name	rba_SysElem_WindowSizeInvalid
Long Name	WindowSizeInvalid
Description	Size of the monitoring window of state Invalid for the E2E state machine
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.117 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/
rba_SysElem_ISignal/rba_SysElem_TransformationCom-
SpecProps/rba_SysElem_EndToEndTransformationCom-
SpecProps/rba_SysElem_WindowSizeValid

Short Name	rba_SysElem_WindowSizeValid
Long Name	WindowSizeValid
Description	Size of the monitoring window of state Valid for the E2E state machine
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.118 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/
rba_SysElem_ISignal/rba_SysElem_TransformationCom-
SpecProps/rba_SysElem_EndToEndTransformationCom-

SpecProps/rba_SysElem_e2eProfileCompatibilityPropsRef

Short Name	rba_SysElem_e2eProfileCompatibilityPropsRef
Long Name	e2eProfileCompatibilityPropsRef
Description	Reference to additional settings for the E2E state machine
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.119 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/
rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps

Short Name	rba_SysElem_TransformationISignalProps
Long Name	TransformationISignalProps
Description	TransformationISignalProps holds all the attributes for the different TransformationTechnologies that are ISignal specific.
Multiplicity	0..*
Default Value	-
Type	ECUC-CHOICE-CONTAINER-DEF

9.3.1.120 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/
rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/
rba_SysElem_EndToEndTransformationISignalProps

Short Name	rba_SysElem_EndToEndTransformationISignalProps
Long Name	EndToEndTransformationISignalProps
Description	The container EndToEndTransformationISignalProps specifies ISignal specific configuration properties for E2E transformer attributes.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.121 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/
rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/
rba_SysElem_EndToEndTransformationISignalProps/rba_SysElem_CSErrorReaction

Short Name	rba_SysElem_CSErrorReaction
Long Name	CsErrorReaction
Description	Defines whether the transformer chain of client/server communication coordinates an autonomous error reaction together with the RTE or whether any error reaction is the responsibility of the application.

Multiplicity	0..1
Default Value	-
Enumeration Literals	applicationOnly, autonomous
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.122 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/rba_SysElem_EndToEndTransformationISignalProps/rba_SysElem_DataID

Short Name	rba_SysElem_DataID
Long Name	DataID
Description	This represents a unique numerical identifier. Note: ID is used for protection against masquerading. The details concerning the maximum number of values (this information is specific for each E2E profile) applicable for this attribute are controlled by a semantic constraint that depends on the category of the EndToEndProtection.
Multiplicity	0..*
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.123 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/rba_SysElem_EndToEndTransformationISignalProps/rba_SysElem_DataLength

Short Name	rba_SysElem_DataLength
Long Name	DataLength
Description	Length of Data in bits.
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.124 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/
rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/
rba_SysElem_EndToEndTransformationISignalProps/rba_SysE-
lem_MaxDataLength

Short Name	rba_SysElem_MaxDataLength
Long Name	MaxDataLength
Description	Maximum length of Data in bits.
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.125 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/
rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/
rba_SysElem_EndToEndTransformationISignalProps/rba_SysE-
lem_MinDataLength

Short Name	rba_SysElem_MinDataLength
Long Name	MinDataLength
Description	Minimum length of Data in bits.
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.126 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/
rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/
rba_SysElem_EndToEndTransformationISignalProps/rba_SysE-
lem_SourceID

Short Name	rba_SysElem_SourceID
Long Name	SourceID
Description	This attribute represents a unique numerical identifier identifying the source of a certain transmission. In case of C/S communication, this ID uniquely identifies the client
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.127 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/
rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/
rba_SysElem_EndToEndTransformationISignalProps/rba_SysE-
lem_TransformationTechnologyRef

Short Name	rba_SysElem_TransformationTechnologyRef
Long Name	TransformationTechnology Reference
Description	Reference to the TransformationTechnology description that contains transformer specific and ISignal independent configuration properties.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.128 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/
rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/
rba_SysElem_SOMEIPTransformationISignalProps

Short Name	rba_SysElem_SOMEIPTransformationISignalProps
Long Name	SOMEIPTransformationISignalProps
Description	The container SOMEIPTransformationISignalProps specifies ISignal specific configuration properties for SOME/IP transformer attributes.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.129 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/
rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/
rba_SysElem_SOMEIPTransformationISignalProps/rba_SysE-
lem_CSErrorReaction

Short Name	rba_SysElem_CSErrorReaction
Long Name	CsErrorReaction
Description	Defines whether the transformer chain of client/server communication coordinates an autonomous error reaction together with the RTE or whether any error reaction is the responsibility of the application.
Multiplicity	0..1
Default Value	-
Enumeration Literals	applicationOnly, autonomous
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.130 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/

rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/
rba_SysElem_SOMEIPTransformationISignalProps/rba_SysElem_ImplementsLegacyStringSerialization

Short Name	rba_SysElem_ImplementsLegacyStringSerialization
Long Name	ImplementsLegacyStringSerialization
Description	This attribute indicates that Strings in the SOME/IP message shall NOT be serialized according to the SOME/IP specification for Strings. If this attribute is set to true, BOM and null-termination shall NOT be added in the serialization for Strings in the payload. If this attribute is set to false (or not set) BOM and null-termination shall be added in the serialization for Strings in the payload according to the SOME/IP specification for Strings. This Parameter is considered, if the Project is based on AR45/R19-11 AUTOSAR Version or newer.
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

9.3.1.131 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/
rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/
rba_SysElem_SOMEIPTransformationISignalProps/rba_SysElem_ImplementsSOMEIPStringHandling

Short Name	rba_SysElem_ImplementsSOMEIPStringHandling
Long Name	ImplementsSOMEIPStringHandling
Description	This attribute indicates whether Strings in the SOME/IP message shall be processed according to the SOME/IP specification for Strings. This attribute has been introduced due to compatibility reasons for AUTOSAR before R4.3. If this attribute is set to true Strings in the payload shall be handled according to the SOME/IP specification on Strings. If this attribute is set to false (or not set) no special handling for Strings in the payload shall be performed. This Parameter is considered, if the Project is based on AR4.2, AR4.3 or AR4.4 AUTOSAR Versions.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

9.3.1.132 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/
rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/
rba_SysElem_SOMEIPTransformationISignalProps/rba_SysElem_ImplementsSOMEIPStringHandling

lem_InterfaceVersionrba_SysElem_InterfaceVersion

Short Name	rba_SysElem_InterfaceVersion
Long Name	InterfaceVersion
Description	The interface version the SOME/IP transformer shall use.
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.133 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/rba_SysElem_SOMEIPTransformationISignalProps/rba_SysElem_IsDynamicLengthFieldSize

Short Name	rba_SysElem_IsDynamicLengthFieldSize
Long Name	IsDynamicLengthFieldSize
Description	This attribute shall be used to determine the wire type in the context of using the TLV encoding
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

9.3.1.134 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/rba_SysElem_SOMEIPTransformationISignalProps/rba_SysElem_MessageType

Short Name	rba_SysElem_MessageType
Long Name	MessageType
Description	The Message Type which shall be placed into the SOME/IP header.
Multiplicity	0..1
Default Value	-
Enumeration Literals	request, requestNoReturn, notification, response, error
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.135 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/rba_SysElem_SOMEIPTransformationISignalProps/rba_SysElem_ProtobufFileNames

Short Name	rba_SysElem_ProtobufFileNames
Long Name	ProtobufFileNames

Description	Specifies the names of the .proto files used. These .proto files are expected to contain all the messages included in the ProtobufMessageName of ArgumentDataPrototype and VariableDataPrototype.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

9.3.1.136 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/rba_SysElem_SOMEIPTransformationISignalProps/rba_SysElem_SessionHandlingSR

Short Name	rba_SysElem_SessionHandlingSR
Long Name	SessionHandlingSR
Description	Defines whether the SOME/IP transformer shall use session handling for Sender/Receiver communication.
Multiplicity	0..1
Default Value	-
Enumeration Literals	sessionHandlingActive, sessionHandlingInactive
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.137 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/rba_SysElem_SOMEIPTransformationISignalProps/rba_SysElem_SizeOfArrayLengthFields

Short Name	rba_SysElem_SizeOfArrayLengthFields
Long Name	SizeOfArrayLengthFields
Description	The size of all length fields (in Bytes) of fixed-size arrays in the SOME/IP message. The data type of the length field for an array is determined by the value of this parameter: - uint8 if the value equals 1 - uint16 if the value equals 2 - uint32 if the value equals 4 - no length field if the parameter is not configured
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.138 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/rba_SysElem_SOMEIPTransformationISignalProps/rba_SysElem_

SizeOfStructLengthFieldsrba_SysElem_SizeOfStructLengthFields

Short Name	rba_SysElem_SizeOfStructLengthFields
Long Name	SizeOfStructLengthFields
Description	The size of all length fields (in Bytes) of structs in the SOME/IP message. The data type of the length field for a struct is determined by the value of this parameter: - uint8 if the value equals 1 - uint16 if the value equals 2 - uint32 if the value equals 4 - no length field if the parameter is not configured
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.139 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/rba_SysElem_SOMEIPTransformationISignalProps/rba_SysElem_SizeOfUnionLengthFields

Short Name	rba_SysElem_SizeOfUnionLengthFields
Long Name	SizeOfUnionLengthFields
Description	The size of all length fields (in Bytes) of unions in the SOME/IP message. The data type of the length field for an union is determined by the value of this parameter: - uint8 if the value equals 1 - uint16 if the value equals 2 - uint32 if the value equals 4 - no length field if the parameter is not configured
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.140 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/rba_SysElem_SOMEIPTransformationISignalProps/rba_SysElem_TlvIdDefinition

Short Name	rba_SysElem_TlvIdDefinition
Long Name	TlvIdDefinition
Description	The container represents the ability to define the tlvDataId
Multiplicity	0..*
Default Value	-

Type	ECUC-PARAM-CONF-CONTAINER-DEF
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9.3.1.141 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/
rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/
rba_SysElem_SOMEIPTransformationISignalProps/rba_SysE-
lem_TlvIdDefinition/rba_SysElem_TLVDataId

Short Name	rba_SysElem_TLVDataId
Long Name	TLV DataId
Description	This attribute represents the definition of the value of the TlvDataId
Multiplicity	1..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.142 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/
rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/
rba_SysElem_SOMEIPTransformationISignalProps/rba_SysE-
lem_TlvIdDefinition/rba_SysElem_TlvArgumentRef

Short Name	rba_SysElem_TlvArgumentRef
Long Name	TlvArgument Reference
Description	This reference assigns a tlvDataId to a given argument of a ClientServerOperation. ClientServerOperationArgumentDataPrototype.
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.143 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/
rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/
rba_SysElem_SOMEIPTransformationISignalProps/rba_SysE-
lem_TlvIdDefinition/rba_SysElem_TlvImplementationDataType-
ElementRef

Short Name	rba_SysElem_TlvImplementationDataTypeElementRef
Long Name	TlvImplementationDataTypeElement Reference
Description	This reference associates the definition of a TLV data id with a given ImplementationDataTypeElement. ImplementationDataTypeElement.
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.144 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/
rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/
rba_SysElem_SOMEIPTransformationISignalProps/rba_SysElem_
TransformationTechnologyRef

Short Name	rba_SysElem_TransformationTechnologyRef
Long Name	TransformationTechnology Reference
Description	Reference to the TransformationTechnology description that contains transformer specific and ISignal independent configuration properties.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.145 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/
rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/
rba_SysElem_UserDefinedTransformationISignalProps

Short Name	rba_SysElem_UserDefinedTransformationISignalProps
Long Name	UserDefinedTransformationISignalProps
Description	The UserDefinedTransformationISignalProps is used to specify ISignal specific configuration properties for custom transformers.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.146 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/
rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/
rba_SysElem_UserDefinedTransformationISignalProps/rba_SysE-
lem_CSErrorReaction

Short Name	rba_SysElem_CSErrorReaction
Long Name	CsErrorReaction
Description	Defines whether the transformer chain of client/server communication coordinates an autonomous error reaction together with the RTE or whether any error reaction is the responsibility of the application.
Multiplicity	0..1
Default Value	-
Enumeration Literals	applicationOnly, autonomous
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.147 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ISignals/
rba_SysElem_ISignal/rba_SysElem_TransformationISignalProps/

rba_SysElem_UserDefinedTransformationISignalProps/rba_SysElem_TransformationTechnologyRef

Short Name	rba_SysElem_TransformationTechnologyRef
Long Name	TransformationTechnology Reference
Description	Reference to the TransformationTechnology description that contains transformer specific and ISignal independent configuration properties.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.148 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes

Short Name	rba_SysElem_ImplementationDataTypes
Long Name	Implementation Data Types
Description	Container for Implementation Data Types
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.149 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType

Short Name	rba_SysElem_ImplementationDataType
Long Name	Implementation Data Type
Description	Describes a reusable data type on the implementation level. This will typically correspond to a typedef in C-code.
Multiplicity	0..*
Default Value	-
Type	ECUC-CHOICE-CONTAINER-DEF

9.3.1.150 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_ArrayImplDataType

Short Name	rba_SysElem_ArrayImplDataType
Long Name	Array Implementation Data Type
Description	Implementation Data Type of Category: Array
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.151 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_ArrayImplDataType/rba_SysElem_Category

Short Name	rba_SysElem_Category
Long Name	Category
Description	Categories of the Implementation Data Types.
Multiplicity	0..1
Default Value	-
Enumeration Literals	VALUE, TYPE_REFERENCE, DATA_REFERENCE, FUNCTION_REFERENCE, STRUCTURE, UNION, ARRAY
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.152 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_ArrayImplDataType/rba_SysElem_DynArrayLength

Short Name	rba_SysElem_DynArrayLength
Long Name	Dynamic array length
Description	Specifies if the Array is dynamical or not.
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

9.3.1.153 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_ArrayImplDataType/rba_SysElem_DynArray-SizeProfile

Short Name	rba_SysElem_DynArraySizeProfile
Long Name	Dynamic Array Size Profile
Description	Specifies the profile which the array will follow in case this data type is a variable size array.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

9.3.1.154 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_ArrayImplDataType/rba_SysElem_ImplDataTypeRef

Short Name	rba_SysElem_ImplDataTypeRef
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Long Name	Implementation Data Type Reference
Description	Reference to the Implementation Data Type
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.155 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_ArrayImplDataType/rba_SysElem_ImplementationDataTypeForeignRef

Short Name	rba_SysElem_ImplementationDataTypeForeignRef
Long Name	Implementation Data Type Foreign Ref
Description	References the Original IMPLEMENTATION-DATA-TYPE from System Extract from Software-AR-Packages
Multiplicity	0..1
Default Value	-
Type	ECUC-FOREIGN-REFERENCE-DEF

9.3.1.156 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_ArrayImplDataType/rba_SysElem_NumArrayElem

Short Name	rba_SysElem_NumArrayElem
Long Name	Number of array elements
Description	Number of Elements in the Array.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.157 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_ArrayImplDataType/rba_SysElem_arraySizeSemantics

Short Name	rba_SysElem_arraySizeSemantics
Long Name	Array Size Semantics
Description	Specifies the semantics which the array will follow in case this data type is a variable size array.
Multiplicity	0..1
Default Value	-

Enumeration Literals	FIXED_SIZE, VARIABLE_SIZE
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.158 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_DynamicArrayImplDataType

Short Name	rba_SysElem_DynamicArrayImplDataType
Long Name	Dynamic Array Implementation Data Type
Description	Implementation Data Type of the Category: STRUCTURE. Handling of Variable-Size Array Datatypes
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.159 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_DynamicArrayImplDataType/rba_SysElem_DynArray_Payload

Short Name	rba_SysElem_DynArray_Payload
Long Name	DynArray_Payload
Description	PayLoad of a Variable-Size Array Data Type to indicate the maximum number of elements of the array
Multiplicity	1..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.160 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_DynamicArrayImplDataType/rba_SysElem_DynArray_Payload/rba_SysElem_ImplDataTypeRef

Short Name	rba_SysElem_ImplDataTypeRef
Long Name	Implementation Data Type Reference
Description	Reference to the Implementation Data Type
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.161 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/

rba_SysElem_DynamicArrayImplDataType/rba_SysElem_DynArray_Size

Short Name	rba_SysElem_DynArray_Size
Long Name	DynArray_Size
Description	Size Indicator of a Variable-Size Array Data Type holds the number of valid elements of the array
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.162 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_DynamicArrayImplDataType/rba_SysElem_DynArray_Size/rba_SysElem_ImplDataTypeRef

Short Name	rba_SysElem_ImplDataTypeRef
Long Name	Implementation Data Type Reference
Description	Reference to the Implementation Data Type
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.163 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_DynamicArrayImplDataType/rba_SysElem_ImplementationDataTypeForeignRef

Short Name	rba_SysElem_ImplementationDataTypeForeignRef
Long Name	Implementation Data Type Foreign Ref
Description	References the Original IMPLEMENTATION-DATA-TYPE from System Extract from Software-AR-Packages
Multiplicity	0..1
Default Value	-
Type	ECUC-FOREIGN-REFERENCE-DEF

9.3.1.164 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_DynamicArrayImplDataType/rba_SysElem_arraySizeHandling

Short Name	rba_SysElem_arraySizeHandling
Long Name	Array Size Handling

Description	Specifies the array size handling for Dynamic Arrays.
Multiplicity	0..1
Default Value	-
Enumeration Literals	ALL_INDICES_DIFFERENT_ARRAY_SIZE, ALL_INDICES_SAME_ARRAY_SIZE, INHERITED_FROM_ARRAY_ELEMENT_TYPE_SIZE
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.165 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_DynamicArrayImplDataType/rba_SysElem_dynamicArraySizeProfile

Short Name	rba_SysElem_dynamicArraySizeProfile
Long Name	Dynamic Array Size Profile
Description	Specifies the profile for Dynamic Arrays.
Multiplicity	0..1
Default Value	-
Enumeration Literals	VSA_LINEAR, VSA_SQUARE, VSA_RECTANGULAR, VSA_FULLY_FLEXIBLE
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.166 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_DynamicArrayImplDataType/rba_SysElem_maxNumberOfElements

Short Name	rba_SysElem_maxNumberOfElements
Long Name	Maximum Number of array elements
Description	Maximum Number of Elements in the Array.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.167 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_PrimitiveImplDataType

Short Name	rba_SysElem_PrimitiveImplDataType
Long Name	Primitive Implementation Data Type
Description	Implementation Data Type of Category: Primitive
Multiplicity	0..1
Default Value	-

Type	ECUC-PARAM-CONF-CONTAINER-DEF
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9.3.1.168 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_PrimitiveImplDataType/rba_SysElem_BaseDataType

Short Name	rba_SysElem_BaseDataType
Long Name	Base data type
Description	Is this Implementation Data Type a Base Data Type?
Multiplicity	1..1
Default Value	-
Enumeration Literals	boolean, uint8, uint16, uint32, uint64, sint8, sint16, sint32, sint64, float32, float64
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.169 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_PrimitiveImplDataType/rba_SysElem_Category

Short Name	rba_SysElem_Category
Long Name	Category
Description	The Category of this Implementation Data Type: - TYPE-REFERENCE - VALUE - ARRAY
Multiplicity	1..1
Default Value	-
Enumeration Literals	VALUE, TYPE_REFERENCE, DATA_REFERENCE, FUNCTION_REFERENCE, STRUCTURE, UNION, ARRAY
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.170 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_PrimitiveImplDataType/rba_SysElem_ImplementationDataTypeForeignRef

Short Name	rba_SysElem_ImplementationDataTypeForeignRef
Long Name	Implementation Data Type Foreign Ref
Description	References the Original IMPLEMENTATION-DATA-TYPE from System Extract from Software-AR-Packages
Multiplicity	0..1
Default Value	-
Type	ECUC-FOREIGN-REFERENCE-DEF

9.3.1.171 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_RecordImplDataType

Short Name	rba_SysElem_RecordImplDataType
Long Name	Record Implementation Data Type
Description	Implementation Data Type of the Category: RECORD
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.172 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_RecordImplDataType/rba_SysElem_Category

Short Name	rba_SysElem_Category
Long Name	Category
Description	Category of the RECORD
Multiplicity	0..1
Default Value	-
Enumeration Literals	VALUE, TYPE_REFERENCE, DATA_REFERENCE, FUNCTION_REFERENCE, STRUCTURE, UNION, ARRAY
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.173 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_RecordImplDataType/rba_SysElem_ImplementationDataTypeForeignRef

Short Name	rba_SysElem_ImplementationDataTypeForeignRef
Long Name	Implementation Data Type Foreign Ref
Description	References the Original IMPLEMENTATION-DATA-TYPE from System Extract from Software-AR-Packages
Multiplicity	0..1
Default Value	-
Type	ECUC-FOREIGN-REFERENCE-DEF

9.3.1.174 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_RecordImplDataType/rba_SysElem_SubElement

Short Name	rba_SysElem_SubElement
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Long Name	SubElement
Description	SubElement inside a Record Datatype
Multiplicity	1..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.175 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_RecordImplDataType/rba_SysElem_SubElement/rba_SysElem_BaseDataType

Short Name	rba_SysElem_BaseDataType
Long Name	Base data type
Description	Is this Implementation Data Type a Base Data Type?
Multiplicity	0..1
Default Value	-
Enumeration Literals	boolean, uint8, uint16, uint32, uint64, sint8, sint16, sint32, sint64, float32, float64
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.176 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_RecordImplDataType/rba_SysElem_SubElement/rba_SysElem_ImplDataTypeRef

Short Name	rba_SysElem_ImplDataTypeRef
Long Name	Implementation Data Type Reference
Description	Reference to the Implementation Data Type
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.177 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_RecordImplDataType/rba_SysElem_SubElement/rba_SysElem_Index

Short Name	rba_SysElem_Index
Long Name	Index
Description	This index is used to ensure the correct order of the SubElements inside a Record Datatype independent of the used tools in the workflow.
Multiplicity	1..1

Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.178 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_RecordImplDataType/rba_SysElem_SubElement/rba_SysElem_isOptional

Short Name	rba_SysElem_isOptional
Long Name	Optional Element
Description	Specifies if the Element is Optional in the Structure.
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

9.3.1.179 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_RecordImplDataType/rba_SysElem_isStructWithOptionalElement

Short Name	rba_SysElem_isStructWithOptionalElement
Long Name	Optional Element Structure
Description	Specifies if the Structure is Optional Element Structure.
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

9.3.1.180 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_StringImplDataType

Short Name	rba_SysElem_StringImplDataType
Long Name	String Implementation Data Type
Description	Implementation Data Type of Category: String
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.181 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_StringImplDataType/rba_SysElem_ImplDataTypeRef

Short Name	rba_SysElem_ImplDataTypeRef
Long Name	Implementation Data Type Reference
Description	Reference to the Implementation Data Type
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.182 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_StringImplDataType/rba_SysElem_ImplementationDataTypeForeignRef

Short Name	rba_SysElem_ImplementationDataTypeForeignRef
Long Name	Implementation Data Type Foreign Ref
Description	References the Original IMPLEMENTATION-DATA-TYPE from System Extract from Software-AR-Packages
Multiplicity	0..1
Default Value	-
Type	ECUC-FOREIGN-REFERENCE-DEF

9.3.1.183 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_StringImplDataType/rba_SysElem_StringEncoding

Short Name	rba_SysElem_StringEncoding
Long Name	String Encoding
Description	String Encoding
Multiplicity	1..1
Default Value	-
Enumeration Literals	UTF_8, UTF_16
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.184 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_StringImplDataType/rba_SysElem_swFillCharacter

Short Name	rba_SysElem_swFillCharacter
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Long Name	swFillCharacter
Description	Filler character for text parameter to pad up to the maximum length swMax- TextSize.
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.185 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/ rba_SysElem_StringImplDataType/rba_SysElem_swMaxTextSize

Short Name	rba_SysElem_swMaxTextSize
Long Name	swMaxTextSize
Description	Specifies the maximum text size in characters
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.186 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/ rba_SysElem_UnionImplDataType

Short Name	rba_SysElem_UnionImplDataType
Long Name	Union Implementation Data Type
Description	Implementation Data Type of the Category: UNION
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.187 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/ rba_SysElem_UnionImplDataType/rba_SysElem_Category

Short Name	rba_SysElem_Category
Long Name	Category
Description	Category of the Union Implementation Data Type
Multiplicity	0..1
Default Value	-
Enumeration Literals	VALUE, TYPE_REFERENCE, DATA_REFERENCE, FUNCTION_REFERENCE, STRUCTURE, UNION, ARRAY
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.188 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_UnionImplDataType/rba_SysElem_ImplementationDataTypeForeignRef

Short Name	rba_SysElem_ImplementationDataTypeForeignRef
Long Name	Implementation Data Type Foreign Ref
Description	References the Original IMPLEMENTATION-DATA-TYPE from System Extract from Software-AR-Packages
Multiplicity	0..1
Default Value	-
Type	ECUC-FOREIGN-REFERENCE-DEF

9.3.1.189 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_UnionImplDataType/rba_SysElem_SubElement

Short Name	rba_SysElem_SubElement
Long Name	SubElement
Description	SubElement of the Union datatype
Multiplicity	1..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.190 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_UnionImplDataType/rba_SysElem_SubElement/rba_SysElem_BaseDataType

Short Name	rba_SysElem_BaseDataType
Long Name	Base data type
Description	Is this Implementation Data Type a Base Data Type?
Multiplicity	0..1
Default Value	-
Enumeration Literals	boolean, uint8, uint16, uint32, uint64, sint8, sint16, sint32, sint64, float32, float64
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.191 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_UnionImplDataType/rba_SysElem_SubElement/rba_SysElem_ImplDataTypeRef

Short Name	rba_SysElem_ImplDataTypeRef
Long Name	Implementation Data Type Reference
Description	Reference to the Implementation Data Type
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.192 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_ImplementationDataTypes/rba_SysElem_ImplementationDataType/rba_SysElem_UnionImplDataType/rba_SysElem_SubElement/rba_SysElem_Index

Short Name	rba_SysElem_Index
Long Name	Index
Description	This index is used to ensure the correct order of the SubElements inside a Record Datatype independent of the used tools in the workflow.
Multiplicity	1..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.193 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_PortInterfaces

Short Name	rba_SysElem_PortInterfaces
Long Name	PortInterfaces
Description	Container for PortInterfaces
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.194 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_PortInterfaces/rba_SysElem_PortInterface

Short Name	rba_SysElem_PortInterface
Long Name	PortInterface
Description	Required or Provided Interface of the Port

Multiplicity	0..*
Default Value	-
Type	ECUC-CHOICE-CONTAINER-DEF

9.3.1.195 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_PortInterfaces/rba_SysElem_PortInterface/rba_SysElem_ClientServerInterface

Short Name	rba_SysElem_ClientServerInterface
Long Name	ClientServerInterface
Description	A client/server interface declares a number of operations that can be invoked on a server by a client.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.196 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_PortInterfaces/rba_SysElem_PortInterface/rba_SysElem_ClientServerInterface/rba_SysElem_ApplicationError

Short Name	rba_SysElem_ApplicationError
Long Name	ApplicationError
Description	Application errors that are defined as part of this interface.
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.197 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_PortInterfaces/rba_SysElem_PortInterface/rba_SysElem_ClientServerInterface/rba_SysElem_ApplicationError/rba_SysElem_ApplicationErrorForeignRef

Short Name	rba_SysElem_ApplicationErrorForeignRef
Long Name	ApplicationError Foreign Reference
Description	Reference to the ApplicationError
Multiplicity	1..1
Default Value	-
Type	ECUC-FOREIGN-REFERENCE-DEF

9.3.1.198 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_PortInterfaces/rba_SysElem_PortInterface/rba_SysElem_ClientServerInterface/rba_SysElem_ApplicationError/rba_SysElem_ErrorCode

Short Name	rba_SysElem_ErrorCode
Long Name	ErrorCode
Description	Possible errors that may be raised by the referring operation.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.199 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_PortInterfaces/rba_SysElem_PortInterface/rba_SysElem_ClientServerInterface/rba_SysElem_ClientServerInterfaceForeignRef

Short Name	rba_SysElem_ClientServerInterfaceForeignRef
Long Name	ClientServerInterface Foreign Reference
Description	Reference to the Client-Server Interface.
Multiplicity	1..1
Default Value	-
Type	ECUC-FOREIGN-REFERENCE-DEF

9.3.1.200 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_PortInterfaces/rba_SysElem_PortInterface/rba_SysElem_ClientServerInterface/rba_SysElem_ClientServerOperation

Short Name	rba_SysElem_ClientServerOperation
Long Name	ClientServerOperation
Description	ClientServerOperation(s) of this ClientServerInterface.
Multiplicity	1..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.201 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_PortInterfaces/rba_SysElem_PortInterface/rba_SysElem_ClientServerInterface/rba_SysElem_ClientServerOperation/rba_SysElem_ArgumentDataPrototype

Short Name	rba_SysElem_ArgumentDataPrototype
Long Name	ArgumentDataPrototype
Description	An argument of this ClientServerOperation
Multiplicity	0..*

Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.202 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_PortInterfaces/rba_SysElem_PortInterface/rba_SysElem_ClientServerInterface/rba_SysElem_ClientServerOperation/rba_SysElem_ArgumentDataPrototype/rba_SysElem_DatatypeRef

Short Name	rba_SysElem_DatatypeRef
Long Name	Datatype Reference
Description	References to the implementation Datatype
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.203 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_PortInterfaces/rba_SysElem_PortInterface/rba_SysElem_ClientServerInterface/rba_SysElem_ClientServerOperation/rba_SysElem_ArgumentDataPrototype/rba_SysElem_Direction

Short Name	rba_SysElem_Direction
Long Name	Direction
Description	This attribute specifies the direction of the argument prototype. - in: The argument value is passed to the callee. - out: The argument value is passed from the callee to the caller. - inout: The argument value is passed to the callee but also passed back from the callee to the caller.
Multiplicity	0..1
Default Value	-
Enumeration Literals	in, out, inout
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.204 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_PortInterfaces/rba_SysElem_PortInterface/rba_SysElem_ClientServerInterface/rba_SysElem_ClientServerOperation/rba_SysElem_ArgumentDataPrototype/rba_SysElem_Index

Short Name	rba_SysElem_Index
Long Name	Index

Description	The 'argument's of the ClientServerOperation is a ordered list. The index can be used to ensure the correct order. Background: Because the sequence of the 'arguments' get created in correct order by using the rba_SysElem-importer script, so the index is more a fallback in case a file split and merge would swap or mix the order.
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.205 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_PortInterfaces/rba_SysElem_PortInterface/rba_SysElem_ClientServerInterface/rba_SysElem_ClientServerOperation/rba_SysElem_ArgumentDataPrototype/rba_SysElem_ProtobufMessageName

Short Name	rba_SysElem_ProtobufMessageName
Long Name	ProtobufMessageName
Description	Specifies the name of the protobuf message that is used for the Argument-DataPrototype.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

9.3.1.206 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_PortInterfaces/rba_SysElem_PortInterface/rba_SysElem_ClientServerInterface/rba_SysElem_ClientServerOperation/rba_SysElem_ArgumentDataPrototype/rba_SysElem_ServerArgumentImplPolicy

Short Name	rba_SysElem_ServerArgumentImplPolicy
Long Name	ServerArgumentImplPolicy

Description	This defines how the argument type of the servers RunnableEntity is implemented. If the attribute is not defined this has the same semantics as if the attribute is set to the value useArgumentType for primitive arguments and structures and to the value useArrayBaseType for arrays. - useArgumentType: The argument type of the RunnableEntity is derived from the AutosarDataType of the ArgumentPrototype. - useArrayBaseType: The argument type of the RunnableEntity is derived from the AutosarDataType of the elements of the array that corresponds to the ArgumentPrototype. This represents the base type of the array in C. - useVoid: The argument type of the RunnableEntity is void.
Multiplicity	0..1
Default Value	-
Enumeration Literals	useArgumentType, useArrayBaseType, useVoid
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.207 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_PortInterfaces/rba_SysElem_PortInterface/rba_SysElem_ClientServerInterface/rba_SysElem_ClientServerOperation/rba_SysElem_ClientServerOperationForeignRef

Short Name	rba_SysElem_ClientServerOperationForeignRef
Long Name	ClientServerOperation Foreign Reference
Description	Reference to this ClientServerOperation in the SystemExtract.
Multiplicity	1..1
Default Value	-
Type	ECUC-FOREIGN-REFERENCE-DEF

9.3.1.208 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_PortInterfaces/rba_SysElem_PortInterface/rba_SysElem_ClientServerInterface/rba_SysElem_ClientServerOperation/rba_SysElem_PossibleErrors

Short Name	rba_SysElem_PossibleErrors
Long Name	PossibleErrors
Description	tbd
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.209 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_PortIn-

terfaces/rba_SysElem_PortInterface/rba_SysElem_ClientServer-
Interface/rba_SysElem_ClientServerOperation/rba_SysElem_-
PossibleErrors/rba_SysElem_ApplicationErrorRef

Short Name	rba_SysElem_ApplicationErrorRef
Long Name	ApplicationError Reference
Description	tbd
Multiplicity	1..*
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.210 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_PortIn-
terfaces/rba_SysElem_PortInterface/rba_SysElem_ClientServer-
Interface/rba_SysElem_IsService

Short Name	rba_SysElem_IsService
Long Name	IsService
Description	This flag is set if the Port Interface is used for communication with a ServiceSwComponentType
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

9.3.1.211 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_PortIn-
terfaces/rba_SysElem_PortInterface/rba_SysElem_Sender-
ReceiverInterface

Short Name	rba_SysElem_SenderReceiverInterface
Long Name	SenderReceiverInterface
Description	A sender/receiver interface declares a number of data elements to be sent and received.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.212 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_PortIn-
terfaces/rba_SysElem_PortInterface/rba_SysElem_Sender-
ReceiverInterface/rba_SysElem_SenderReceiverInterfaceFor-
eignRef

Short Name	rba_SysElem_SenderReceiverInterfaceForeignRef
Long Name	SenderReceiverInterface Foreign Reference
Description	Reference to this SenderReceiverInterface in the SystemExtract.

Multiplicity	1..1
Default Value	-
Type	ECUC-FOREIGN-REFERENCE-DEF

9.3.1.213 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_PortInterfaces/rba_SysElem_PortInterface/rba_SysElem_Sender-ReceiverInterface/rba_SysElem_VariableDataPrototype

Short Name	rba_SysElem_VariableDataPrototype
Long Name	VariableDataPrototype
Description	A VariableDataPrototype is used to contain values in an ECU application. This means that most likely a VariableDataPrototype allocates "static" memory on the ECU. In some cases optimization strategies might lead to a situation where the memory allocation can be avoided.
Multiplicity	1..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.214 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_PortInterfaces/rba_SysElem_PortInterface/rba_SysElem_Sender-ReceiverInterface/rba_SysElem_VariableDataPrototype/rba_SysElem_DatatypeRef

Short Name	rba_SysElem_DatatypeRef
Long Name	Datatype Reference
Description	Reference to the Datatype.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.215 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_PortInterfaces/rba_SysElem_PortInterface/rba_SysElem_Sender-ReceiverInterface/rba_SysElem_VariableDataPrototype/rba_SysElem_ProtobufMessageName

Short Name	rba_SysElem_ProtobufMessageName
Long Name	ProtobufMessageName
Description	Specifies the name of the protobuf message that is used for the VariableDataPrototype.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

9.3.1.216 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_PortInterfaces/rba_SysElem_PortInterface/rba_SysElem_SenderReceiverInterface/rba_SysElem_VariableDataPrototype/rba_SysElem_VariableDataPrototypeForeignRef

Short Name	rba_SysElem_VariableDataPrototypeForeignRef
Long Name	VariableDataPrototype Foreign Reference
Description	Reference to this VariableDataPrototype in the SystemExtract.
Multiplicity	1..1
Default Value	-
Type	ECUC-FOREIGN-REFERENCE-DEF

9.3.1.217 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_SystemSignalGroups

Short Name	rba_SysElem_SystemSignalGroups
Long Name	System Signal Groups
Description	Container for System Signal Groups
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.218 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_SystemSignalGroups/rba_SysElem_SystemSignalGroup

Short Name	rba_SysElem_SystemSignalGroup
Long Name	System Signal Group
Description	A signal group refers to a set of signals that must always be kept together. A signal group is used to guarantee the atomic transfer of AUTOSAR composite data types. The SystemSignalGroup defines a signal grouping on VFB level. On cluster level the Signal grouping is described by the ISignalGroup element.
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.219 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_SystemSignalGroups/rba_SysElem_SystemSignalGroup/rba_SysElem_SystemSignal

Short Name	rba_SysElem_SystemSignal
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Long Name	SystemSignal
Description	Reference to a set of SystemSignals that must always be kept together.
Multiplicity	0..*
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.220 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_SystemSignalGroups/rba_SysElem_SystemSignalGroup/rba_SysElem_SystemSignalGroupForeignRef

Short Name	rba_SysElem_SystemSignalGroupForeignRef
Long Name	System Signal Group Foreign Reference
Description	Reference to this SystemSignalGroup in the SystemExtract.
Multiplicity	1..1
Default Value	-
Type	ECUC-FOREIGN-REFERENCE-DEF

9.3.1.221 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_SystemSignalGroups/rba_SysElem_SystemSignalGroup/rba_SysElem_TransformingSystemSignalRef

Short Name	rba_SysElem_TransformingSystemSignalRef
Long Name	Transforming SystemSignal Reference
Description	Optional reference to the SystemSignal which shall contain the transformed (linear) data.
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.222 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_SystemSignals

Short Name	rba_SysElem_SystemSignals
Long Name	System Signals
Description	Container for SystemSignals
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.223 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_System-Signals/rba_SysElem_SystemSignal

Short Name	rba_SysElem_SystemSignal
Long Name	System Signal
Description	The system signal represents the communication system's view of data exchanged between SW components which reside on different ECUs. The system signals allow to represent this communication in a flattened structure, with exactly one system signal defined for each data element prototype sent and received by connected SW component instances.
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.224 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_System-Signals/rba_SysElem_SystemSignal/rba_SysElem_DynamicLength

Short Name	rba_SysElem_DynamicLength
Long Name	DynamicLength
Description	The length of dynamic length signals is variable in run-time. Only a maximum length of such a signal is specified in the configuration (attribute length in ISignal element).
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

9.3.1.225 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_System-Signals/rba_SysElem_SystemSignal/rba_SysElem_SystemSignal-ForeignRef

Short Name	rba_SysElem_SystemSignalForeignRef
Long Name	System Signal Foreign Reference
Description	Reference to this SystemSignal in the SystemExtract.
Multiplicity	1..1
Default Value	-
Type	ECUC-FOREIGN-REFERENCE-DEF

9.3.1.226 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies

Short Name	rba_SysElem_TransformationTechnologies
Long Name	Transformation Technologies
Description	Container for TransformationTechnology
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.227 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology

Short Name	rba_SysElem_TransformationTechnology
Long Name	TransformationTechnology
Description	A TransformationTechnology is a transformer inside a transformer chain.
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.228 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_BufferProperties

Short Name	rba_SysElem_BufferProperties
Long Name	BufferProperties
Description	Configuration of the buffer properties the transformer needs to work.
Multiplicity	1..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.229 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_BufferProperties/rba_SysElem_CompuScale

Short Name	rba_SysElem_CompuScale
Long Name	CompuScale
Description	If the transformer changes the size of the data, the CompuScale can be used to specify a rule to derive the size of the output data based on the size of the input data.
Multiplicity	0..1

Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.230 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_BufferProperties/rba_SysElem_CompuScale/rba_SysElem_Denominator

Short Name	rba_SysElem_Denominator
Long Name	Denominator
Description	Part of the Buffer Computation
Multiplicity	1..1
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

9.3.1.231 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_BufferProperties/rba_SysElem_CompuScale/rba_SysElem_Numerator1

Short Name	rba_SysElem_Numerator1
Long Name	Numerator1
Description	Part of the Buffer Computation
Multiplicity	1..1
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

9.3.1.232 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_BufferProperties/rba_SysElem_CompuScale/rba_SysElem_Numerator2

Short Name	rba_SysElem_Numerator2
Long Name	Numerator2
Description	Part of the Buffer Computation
Multiplicity	1..1
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

9.3.1.233 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_BufferProperties/rba_SysElem_HeaderLength

Short Name	rba_SysElem_HeaderLength
Long Name	HeaderLength
Description	Defines the length of the header (in bits) this transformer will add in front of the data.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.234 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_BufferProperties/rba_SysElem_InPlace

Short Name	rba_SysElem_InPlace
Long Name	InPlace
Description	If set, the transformer uses the input buffer as output buffer.
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

9.3.1.235 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_NeedsOriginalData

Short Name	rba_SysElem_NeedsOriginalData
Long Name	NeedsOriginalData
Description	Specifies whether this transformer gets access to the SWC's original data.
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

9.3.1.236 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_Protocol

Short Name	rba_SysElem_Protocol
Long Name	Protocol
Description	Specifies the protocol that is implemented by this transformer.
Multiplicity	1..1

Default Value	-
Type	ECUC-STRING-PARAM-DEF

9.3.1.237 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription

Short Name	rba_SysElem_TransformationDescription
Long Name	TransformationDescription
Description	A transformer can be configured with transformer specific parameters which are represented by the TransformerDescription.
Multiplicity	0..1
Default Value	-
Type	ECUC-CHOICE-CONTAINER-DEF

9.3.1.238 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription/rba_SysElem_EndToEndTransformationDescription

Short Name	rba_SysElem_EndToEndTransformationDescription
Long Name	EndToEndTransformationDescription
Description	EndToEndTransformationDescription holds these attributes which are profile specific and have the same value for all E2E transformers.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.239 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription/rba_SysElem_EndToEndTransformationDescription/rba_SysElem_ClearToInvalid

Short Name	rba_SysElem_ClearToInvalid
Long Name	ClearToInvalid
Description	Clear monitoring window on transition from state Valid to state Invalid
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

9.3.1.240 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription/rba_SysElem_EndToEndTransformationDescription/rba_SysElem_ClearToInvalid

gy/rba_SysElem_TransformationDescription/rba_SysElem_End-
ToEndTransformationDescription/rba_SysElem_CounterOffset

Short Name	rba_SysElem_CounterOffset
Long Name	Counter Offset
Description	Offset of the counter in the Data[] array in bits.
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.241 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_Trans-
formationTechnologies/rba_SysElem_TransformationTechnolo-
gy/rba_SysElem_TransformationDescription/rba_SysElem_End-
ToEndTransformationDescription/rba_SysElem_CrcOffset

Short Name	rba_SysElem_CrcOffset
Long Name	CRC Offset
Description	Offset of the CRC in the Data[] array in bits.
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.242 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_Trans-
formationTechnologies/rba_SysElem_TransformationTechnolo-
gy/rba_SysElem_TransformationDescription/rba_SysElem_End-
ToEndTransformationDescription/rba_SysElem_DataIdMode

Short Name	rba_SysElem_DataIdMode
Long Name	Data ID Mode
Description	This attribute describes the inclusion mode that is used to include the implicit two-byte Data ID in the one-byte CRC.
Multiplicity	0..1
Default Value	-
Enumeration Literals	all16Bit, alternating8Bit, lower8Bit, lower12Bit
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.243 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_Trans-
formationTechnologies/rba_SysElem_TransformationTechnolo-
gy/rba_SysElem_TransformationDescription/rba_SysElem_End-
ToEndTransformationDescription/rba_SysElem_DataIdNibbleOff-

setrba_SysElem_DataIdNibbleOffset

Short Name	rba_SysElem_DataIdNibbleOffset
Long Name	Data ID Nibble Offset
Description	Offset of the Data ID nibble in the Data[] array in bits.
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.244 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription/rba_SysElem_EndToEndTransformationDescription/rba_SysElem_MaxDeltaCounter

Short Name	rba_SysElem_MaxDeltaCounter
Long Name	Maximum Delta Counter
Description	Maximum allowed gap between two counter values of two consecutive checks.
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.245 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription/rba_SysElem_EndToEndTransformationDescription/rba_SysElem_MaxErrorStateInit

Short Name	rba_SysElem_MaxErrorStateInit
Long Name	Maximum Error State Init
Description	Maximal number of checks in which ProfileStatus equal to E2E_P_ERROR was determined, within the last WindowSize checks, for the state E2E_SM_INIT.
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.246 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription/rba_SysElem_End-

ToEndTransformationDescription/rba_SysElem_MaxErrorStateInvalid

Short Name	rba_SysElem_MaxErrorStateInvalid
Long Name	Maximum Error State Invalid
Description	Maximal number of checks in which ProfileStatus equal to E2E_P_ERROR was determined, within the last WindowSize checks, for the state E2E_SM_INVALID.
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.247 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription/rba_SysElem_EndToTransformationDescription/rba_SysElem_MaxErrorStateValid

Short Name	rba_SysElem_MaxErrorStateValid
Long Name	Maximum Error State Valid
Description	Maximal number of checks in which ProfileStatus equal to E2E_P_ERROR was determined, within the last WindowSize checks, for the state E2E_SM_VALID.
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.248 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription/rba_SysElem_EndToTransformationDescription/rba_SysElem_MaxNoNewOrRepeatedData

Short Name	rba_SysElem_MaxNoNewOrRepeatedData
Long Name	Maximum Number Of New Or Repeated Data
Description	The maximum allowed amount of consecutive failed counter checks.
Multiplicity	0..1
Minimum Value	0
Default Value	-

Type	ECUC-INTEGER-PARAM-DEF
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9.3.1.249 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription/rba_SysElem_EndToEndTransformationDescription/rba_SysElem_MinOkStateInit

Short Name	rba_SysElem_MinOkStateInit
Long Name	Minimum OK State Init
Description	Minimal number of checks in which ProfileStatus equal to E2E_P_OK was determined, within the last WindowSize checks, for the state E2E_SM_INIT.
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.250 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription/rba_SysElem_EndToEndTransformationDescription/rba_SysElem_MinOkStateInvalid

Short Name	rba_SysElem_MinOkStateInvalid
Long Name	Minimum OK State Invalid
Description	Minimal number of checks in which ProfileStatus equal to E2E_P_OK was determined, within the last WindowSize checks, for the state E2E_SM_INVALID.
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.251 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription/rba_SysElem_EndToEndTransformationDescription/rba_SysElem_MinOkStateValid

Short Name	rba_SysElem_MinOkStateValid
Long Name	Minimum OK State Valid
Description	Minimal number of checks in which ProfileStatus equal to E2E_P_OK was determined, within the last WindowSize checks, for the state E2E_SM_VALID.

Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.252 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription/rba_SysElem_EndToEndTransformationDescription/rba_SysElem_Offset

Short Name	rba_SysElem_Offset
Long Name	Offset
Description	Offset of the E2E header in the Data[] array in bits.
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.253 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription/rba_SysElem_EndToEndTransformationDescription/rba_SysElem_ProfileBehavior

Short Name	rba_SysElem_ProfileBehavior
Long Name	Profile Behaviour
Description	Behavior of the check functionality
Multiplicity	0..1
Default Value	-
Enumeration Literals	PRE_R4_2, R4_2
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.254 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription/rba_SysElem_EndToEndTransformationDescription/rba_SysElem_ProfileName

Short Name	rba_SysElem_ProfileName
Long Name	Profile Name
Description	Definition of the E2E profile.
Multiplicity	1..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

9.3.1.255 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription/rba_SysElem_EndToEndTransformationDescription/rba_SysElem_SyncCounterInit

Short Name	rba_SysElem_SyncCounterInit
Long Name	Sync Counter Init
Description	Number of checks required for validating the consistency of the counter that must be received with a valid counter (i.e. counter within the allowed lock-in range) after the detection of an unexpected behavior of a received counter.
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.256 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription/rba_SysElem_EndToEndTransformationDescription/rba_SysElem_UpperHeaderBitsToShift

Short Name	rba_SysElem_UpperHeaderBitsToShift
Long Name	Upper Header Bits To Shift
Description	This attribute describes the number of upper-header bits to be shifted. value = 0 or not present: shift of upper header is NOT performed. value > 0: the E2E Transformer on the protect-side, takes the first upperHeaderBitsToShift bits from the upper buffer (e.g. SOME/IP header part generated by SOME/IP transformer) and shifts them towards the lower bytes and bits within the Data[] for the length of the E2E header (e.g. 12 bytes in case of E2E Profile 4). This means the shift distance is fixed - it depends on the E2E header size - what is configured here is the number of bits that are to be shifted. This option is defined because the Some/IP header generated by SOME/IP transformer must be, due to compatibility between non-protected and E2E-protected communication, at the same position, which is before E2E header.
Multiplicity	1..1
Minimum Value	0

Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.257 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription/rba_SysElem_EndToEndTransformationDescription/rba_SysElem_WindowSize

Short Name	rba_SysElem_WindowSize
Long Name	Window Size
Description	Size of the monitoring window for the E2E state machine
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.258 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription/rba_SysElem_EndToEndTransformationDescription/rba_SysElem_WindowSizeInit

Short Name	rba_SysElem_WindowSizeInit
Long Name	WindowSizeInit
Description	Size of the monitoring window of state Init for the E2E state machine
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.259 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription/rba_SysElem_EndToEndTransformationDescription/rba_SysElem_WindowSizeInvalid

Short Name	rba_SysElem_WindowSizeInvalid
Long Name	WindowSizeInvalid
Description	Size of the monitoring window of state Invalid for the E2E state machine
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.260 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription/rba_SysElem_EndToEndTransformationDescription/rba_SysElem_WindowSizeValid

Short Name	rba_SysElem_WindowSizeValid
Long Name	WindowSizeValid
Description	Size of the monitoring window of state Valid for the E2E state machine
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.261 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription/rba_SysElem_EndToEndTransformationDescription/rba_SysElem_e2eProfileCompatibilityPropsRef

Short Name	rba_SysElem_e2eProfileCompatibilityPropsRef
Long Name	e2eProfileCompatibilityPropsRef
Description	Reference to additional settings for the E2E state machine
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

9.3.1.262 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription/rba_SysElem_SOMEIPTTransformationDescription

Short Name	rba_SysElem_SOMEIPTTransformationDescription
Long Name	SOMEIPTTransformationDescription
Description	The SOMEIPTTransformationDescription is used to specify SOME/IP transformer specific attributes.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.263 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription/rba_SysElem_SOMEIPTransformationDescription/rba_SysElem_Alignment

Short Name	rba_SysElem_Alignment
Long Name	Alignment
Description	Specifies the alignment of dynamic data in the serialized data stream. The alignment shall be specified in Bits.
Multiplicity	1..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.264 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription/rba_SysElem_SOMEIPTransformationDescription/rba_SysElem_ByteOrder

Short Name	rba_SysElem_ByteOrder
Long Name	ByteOrder
Description	Defines which byte order shall be serialized by the SOME/IP transformer.
Multiplicity	1..1
Default Value	-
Enumeration Literals	mostSignificantByteFirst, mostSignificantByteLast, opaque
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.265 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription/rba_SysElem_SOMEIPTransformationDescription/rba_SysElem_InterfaceVersion

Short Name	rba_SysElem_InterfaceVersion
Long Name	InterfaceVersion
Description	The interface version the SOME/IP transformer shall use.
Multiplicity	1..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

9.3.1.266 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription/rba_SysElem_UserDefinedTransformationDescription

Short Name	rba_SysElem_UserDefinedTransformationDescription
Long Name	UserDefinedTransformationDescription
Description	The UserDefinedTransformationDescription is used to specify details and documentation for custom transformers.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

9.3.1.267 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationDescription/rba_SysElem_UserDefinedTransformationDescription/rba_SysElem_Description

Short Name	rba_SysElem_Description
Long Name	Desc
Description	This represents a general but brief (one paragraph) description what the object in question is about. It is only one paragraph! Desc is intended to be collected into overview tables. This property helps a human reader to identify the object in question
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

9.3.1.268 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_TransformationTechnologyForeignRef

Short Name	rba_SysElem_TransformationTechnologyForeignRef
Long Name	TransformationTechnology Foreign Reference
Description	Reference to this TransformationTechnology in the SystemExtract.
Multiplicity	1..1
Default Value	-
Type	ECUC-FOREIGN-REFERENCE-DEF

9.3.1.269 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology

gy/rba_SysElem_TransformerClass

Short Name	rba_SysElem_TransformerClass
Long Name	TransformerClass
Description	Specifies to which transformer class this transformer belongs.
Multiplicity	1..1
Default Value	-
Enumeration Literals	serializer, safety, security, custom
Type	ECUC-ENUMERATION-PARAM-DEF

9.3.1.270 rba_SysElem/rba_SysElem_SystemModel/rba_SysElem_TransformationTechnologies/rba_SysElem_TransformationTechnology/rba_SysElem_Version

Short Name	rba_SysElem_Version
Long Name	Version
Description	Version of the implemented protocol.
Multiplicity	1..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

9.4 Limitations

Version v12.7.0 has following limitations.

- No Support of Signals which are considered in the Aspect of VLANs.

9.5 Exclusions

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- There are no known exclusions in this version.

9.6 Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

- There are no known deviations in this version.

9.7 API Definition

List of supported API functions

9.8 Configuration Advice



NOTE

Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the "Configuration Generation" section of the RTA-CAR User Guide.

9.8.1 Configuration of System Extract for ConfGen

In order to automatically generate the configuration for Rba_SysElem (and rba_Xfrm), the following configuration must be provided in the System Description:

- DataTransformationSet with DataTransformation and TransformationTechnology configuration for the Transformers.
- For COMBased TransformationTechnology configuration:
 - Protocol attribute must be exactly "COMBased"
 - TransformerClass must be "SERIALIZER"
 - BufferProperties child configuration must be provided with relevant attributes.
- For SOME/IP TransformationTechnology configuration:
 - Protocol attribute must be exactly "SOMEIP"
 - TransformerClass must be "SERIALIZER"
 - BufferProperties child configuration must be provided with relevant attributes.
- For E2E TransformationTechnology configuration:
 - Protocol attribute must be exactly "E2E"
 - TransformerClass must be "SAFETY"
 - BufferProperties child configuration must be provided with relevant attributes.
 - EndToEndTransformationDescription must be configured with:
 - Category and ProfileName should be configured as either "PROFILE_01", "PROFILE_02", "PROFILE_04", "PROFILE_05", "PROFILE_06", "PROFILE_07".
 - Remaining attributes should be configured based on required operation of the E2E protection mechanisms.
- For each ISignalGroup that uses the DataTransformation:
 - DataTransformationRefConditional should be configured which refers to relevant DataTransformation
 - EndToEndTransformationISignalProps and EndToEndTransformationISignalPropsConditional must be configured when E2E Transformation is used.
- ImplementationDataTypes and corresponding PortInterfaces must be

created for receiving/sending data based on the SignalGroup data elements. The ImplementationDataTypeElements should not include the header added by the transformers in front of the data.

- The ImplementationDataTypeElements should refer to an ImplementationDataType, instead of BaseType. The category of this ImplementationDataType should be either: STRUCTURE, UNION, ARRAY or TYPE_REFERENCE
- Before running ConfGen, for the Signals/SignalGroups using TransformationTechnology, SystemDataMappings from Port DataElements to SystemSignalGroups and DataRecordElements/DataArrayElements to SystemGroupSignals must be provided in order to generate completed Rba_SysElem configuration.
- After ConfGen, before Generating RTE, these SystemDataMappings should be removed.

9.9 Integration Advice

9.9.1 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the ENTER and EXIT resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The Rba_SysElem module does not use any exclusive OS resources.

9.10 Hardware Requirements

Information relating to the hardware resources required by this module is not currently available.

10 rba_Xfrm

Module name	rba_Xfrm
Package	RTA-COM
AUTOSAR Module ID	N/A
AUTOSAR Version	22-11

10.1 Introduction

The rba_Xfrm module is an RTA-BSW specific extension to the AUTOSAR specification of transformers. See the Supported Features section for more details.

Module Configuration

This module is not currently supported by the RTA-BSW Configuration Generator (ConfGen) and must be manually configured.

10.2 Supported Features

10.2.1 General

The rba_Xfrm module provides the following features:

- The `TransformerTypes.h` header for the `SomelpXf`, `ComXf` and `E2EXf` Transformers, which includes the `StdTypes` and the `Xfrm Paramdef` for general Transformer Values.
- A memcpy API called `rba_Xfrm_MemCopy` and a memset API called `rba_Xfrm_MemSet` to provide process-conform Source Code for the Transformers (see the Extensions section for more details).
- `Xfrm_EcucParamDef.arxml` for general Transformers' Values (determines which Signal/SignalGroup is used with which Transformer Mechanism).

Based on the Importer, Transformer-related `EcucValues` will be imported from the `SystemExtract` and will be stored in either `rba_SysElem` or `rba_Xfrm`.

10.3 Extensions

10.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

- No known extension parameters are defined in this version.

10.3.2 Additional Services

The following API are available in version v12.7.0 in addition to those defined in AR v22-11

10.3.2.1 rba_Xfrm_MemCopy

Syntax	<code>rba_Xfrm_MemCopy(void* xDest_pv, const void* xSrc_pcv, uint32 numBytes_u32)</code>
Parameters In	<code>xDest_pv</code> Same as C90-memcpy. <code>xSrc_pcv</code> Same as C90-memcpy. <code>numBytes_u32</code> Same as C90-memcpy.
Description	Implements Memcopy with the same parameters as C90-memcpy. To save copy structures, it copies 8 bit, 16 bit, 32 bit and 64 bit data based on the source and destination alignment.

10.3.2.2 rba_Xfrm_MemSet

Syntax	<code>rba_Xfrm_MemSet(void* xDest_pv, sint32 xPattern_u32, uint32 numBytes_u32)</code>
Parameters In	<code>xDest_pv</code> Same as C90-memcpy. <code>xPattern_u32</code> Same as C90-memcpy. <code>numBytes_u32</code> Same as C90-memcpy.
Description	Implements Memset with the same parameters as C90-memcpy. It writes 32 bit data if possible, so it is safe to initialize structures which require consistent 32 bit data.

10.4 Limitations

Version v12.7.0 has following limitations.

- RTA-BSW does not currently support the automatic detection and selection of the **rba_Xfrm** module in the CodeGen Launch Configuration window. You must make a manual check to ensure that it is selected whenever the `rba_SysElem` module has been configured.

10.5 Exclusions

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- There are no known exclusions in this version.

10.6 Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

- There are no known deviations in this version.

10.7 API Definition

List of supported API functions

10.8 Configuration Advice

No configuration is required for the rba_Xfrm module, but each Transformer module must have an rba_Xfrm module configuration.

10.9 Integration Advice

This module is part of the Transformers Package, and it should be integrated along with the Transformers, the respective Adapters and the rba_SysElem EcuC Paramdef. The EcucValues will be filled by the Importer and during the build of the respective Transformers.

The following three parameters need to be configured under the rba_Xfrm container:

rba_Xfrm64BIT_MachineSupport

rba_Xfrm_LibMemcpy

rba_Xfrm_LibMemset

The tpl file has been removed; hence *rba_Xffrm_cfg.h* must be generated.

10.9.1 Init Functions

No specific handling necessary.

10.9.2 Delnit Functions

No specific handling necessary.

10.9.3 Scheduled Functions

There are no scheduled functions.

10.9.4 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the ENTER and EXIT resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The Rba_Xfrm module does not use any exclusive OS resources.

10.10 Hardware Requirements

Information relating to the hardware resources required by this module is not currently available.

11 rba Comstack Complex Device Driver

Module name	Rba_ComStackCdd
Package	RTA-COM
Module ID	N/A
AUTOSAR Version	22-11

11.1 Introduction

A complex device driver (CDD) is a software entity that is not standardized by AUTOSAR, and which can access or be accessed via AUTOSAR interfaces and/or the APIs of Basic Software modules.

A CDD is a specific module that is located in the Complex Drivers Layer of the Basic Software and which interacts with standard BSW modules or the RTE.

A CDD can also serve as a migration mechanism to introduce existing or new concepts into the AUTOSAR Software Architecture.

The project-specific implementation of a CDD will be dependent on the intended application, and on the micro-controller and ECU.

The `rba_ComStackCdd` module is the RTA-BSW implementation to support AUTOSAR CDDs.

Module Configuration

This module is not currently supported by the RTA-BSW Configuration Generator (ConfGen) and must be manually configured.

11.2 Supported Features

The `rba_ComStackCdd` module supports the following AUTOSAR-defined features for CDDs.

Interfacing to a CDD from modules of the layered software architecture

The CDD offers interface(s) that can be generically configured by the accessing AUTOSAR module. A typical example is the PDU Router: a CDD may implement the interface module of a new bus system. This is already taken care of within the configuration of the PDU Router.

Interfacing from a CDD to modules of the layered software architecture

This is only allowed if the respective modules of the layered software architecture offer the interfaces and are prepared to be accessed by a CDD. Usually this means that the CDD will take care of re-entrancy of interfaces. For non-re-entrant interfaces, only one caller can access the interface. For conditionally re-entrant interfaces, several callers may access the interface concurrently if they use different IDs

Interfacing with the Communication Stack (PduR and <Bus>Interfaces modules)

- **Interfacing with PduR:** The PduR is the exclusive bus and protocol independent access point to the communication stack for IPDU. The CDD can use

standard APIs of the PduR module to access IPDU. When the CDD interacts with the PduR, one container per CDD is configured within the PduR

- **Interfacing with <Bus>Interfaces modules:** The <Bus>Interfaces modules are the exclusive bus specific access point to the communication stack. The CDD can use standard APIs of the <Bus>Interfaces modules to access IPDU. When CDD interacts with <Bus>Interface, it uses the access functions defined for <Bus>Interface and <Bus>Interface callbacks configured according to CDD needs. The <Bus>Interface shall be configured to include the CDD_<MODULENAME>_Cbk.h header file. Refer to <BUS>Interface specifications and user manual for details.

11.3 Extensions

11.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

- No known extension parameters are defined in this version.

11.3.2 Additional Services

The following API are available in version v12.7.0 in addition to those defined in AR v22-11

11.4 Limitations

The following features are not supported for version v12.7.0

- The Postbuild Selectable and Postbuild Loadable configuration variants are not supported.
- Manual configuration of the handle IDs for Pdus is not supported.
- A single CDD component cannot support both If and Tp interfaces.
- The Cdd_RxIndication API can be called by both PduR and ComIf modules, so all the cddPduRUpperLayerRxPdu and cddComIfUpperLayerRxPdu are combined during Id assignment. This allows the Pdu handle Ids to be segregated (and be identified separately from ComIf Ids) when Cdd_RxIndication is called. This will only occur if a CDD is configured with both cddPduRUpperLayerRxPdu and cddComIfUpperLayerRxPdu.
- Contrary to AUTOSAR, CDD currently generates only one header file for both Symbolic name and Call back function declarations.
- CddSoAdUpperLayerRxPdu and CddSoAdUpperLayerTxPdu should not be configured in PduR. CddSoAdUpperLayerHandleId is an optional parameter for CddSoAdUpperLayerTxPdu, but CDD is always forwarding the handleId for this.

11.5 Exclusions

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- Cdd/CddComStackContribution/CddComMLowerLayerContribution/CddComMLowerLayerChannel/CddComMLowerLayerChannelRef
- Cdd/CddComStackContribution/CddGenericNmLowerLayerContribution
- Cdd/CddComStackContribution/CddGenericNmLowerLayerContribution/CddGenericNmLowerLayerChannel/CddGenericNmComMNetworkHandleRef
- Cdd/CddComStackContribution/CddJ1939RmContribution
- Cdd/CddComStackContribution/CddJ1939RmContribution/CddJ1939RmAckIndication
- Cdd/CddComStackContribution/CddJ1939RmContribution/CddJ1939RmRequestIndication
- Cdd/CddComStackContribution/CddJ1939RmContribution/CddJ1939RmRequestTimeoutIndication
- Cdd/CddComStackContribution/CddSoAdUpperLayerContribution/CddSoAdUpperLayerRxPdu/CddSoAdUpperLayerPduRef
- Cdd/CddComStackContribution/CddSoAdUpperLayerContribution/CddSoAdUpperLayerTxPdu/CddSoAdUpperLayerPduRef
- Cdd/CddEcucPartitionInteraction
- Cdd/CddEcucPartitionInteraction/CddEcucPartitionRef
- Cdd/CddEcucPartitionInteraction/CddPartitionStoppedFunctionName
- Cdd/CddGeneral
- Cdd/CddGeneral/CddInstancelId
- Cdd/CddGeneral/CddModuleId
- Cdd/CddGlobalTimeContribution
- Cdd/CddGlobalTimeContribution/CddGlobalTimeDomain
- Cdd/CddGlobalTimeContribution/CddGlobalTimeDomain/CddGlobalTimeDomainId
- Cdd/CddGlobalTimeContribution/CddGlobalTimeDomain/CddGlobalTimeMaster
- Cdd/CddGlobalTimeContribution/CddGlobalTimeDomain/CddGlobalTimeMaster/CddGlobalTimeMasterPdu
- Cdd/CddGlobalTimeContribution/CddGlobalTimeDomain/CddGlobalTimeMaster/CddGlobalTimeMasterPdu/CddGlobalTimeMasterPduRef
- Cdd/CddGlobalTimeContribution/CddGlobalTimeDomain/CddGlobalTimeMaster/CddGlobalTimeTxPeriod
- Cdd/CddGlobalTimeContribution/CddGlobalTimeDomain/CddGlobal-

TimeSlaveCddGlobalTimeSlave

- Cdd/CddGlobalTimeContribution/CddGlobalTimeDomain/CddGlobalTimeSlave/CddGlobalTimeSlavePdu
- Cdd/CddGlobalTimeContribution/CddGlobalTimeDomain/CddGlobalTimeSlave/CddGlobalTimeSlavePdu/CddGlobalTimeSlavePduRef
- Cdd/CddGlobalTimeContribution/CddGlobalTimeDomain/CddSynSynchronizedTimeBaseRef

11.6 Deviations

- There are no known deviations in this version.

11.7 API Definition

Table 11.2.rba_ComStackCdd Supported API functions

<CddIf>_Transmit	Requests transmission of an I-PDU.
<CddIf>_RxIndication	Indicates a received I-PDU from a lower layer communication interface.
<CddIf>_TxConfirmation	Confirmation from the lower layer communication interface of the successful transmission of the I-PDU.
<CddIf>_TriggerTransmit	The CDD has to copy the data of the I-PDU into the buffer provided by the lower layer.
<CddTp>_CopyRxData	Called when transport protocol module has data to copy to the receiving module.
<CddTp>_TpRxIndication	Called by the transport protocol module after an I-PDU has been received successfully or when an error occurred.
<CddTp>_StartOfReception	Called by the transport protocol module at the start of receiving an I-PDU.
<CddTp>_CopyTxData	Called by the transport protocol module to query the transmit data of an I-PDU segment.
<CddTp>_TpTxConfirmation	Called by the transport protocol module to indicate if the I-PDU transmission was successful or not.

11.8 Configuration Advice



NOTE

Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the "Configuration Generation" section of the RTA-CAR User Guide.

There is currently no specific configuration advice for rba_ComStackCdd.

11.9 Integration Advice

11.9.1 Init Functions

Dependent on the implementation of the CDD.

11.9.2 Delnit Functions

Dependent on the implementation of the CDD.

11.9.3 Scheduled Functions

Dependent on the implementation of the CDD.

11.9.4 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the ENTER and EXIT resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The Rba_ComStackCdd module does not use any exclusive OS resources.

11.10 Hardware Requirements

Information relating to the hardware resources required by this module is not currently available.

12 rba_XCoreCdd

Module name	Rba_XCoreCdd
Package	RTA-COM
Module ID	N/A
AUTOSAR Version	22-11

12.1 Introduction

The rba_XCoreCdd module is a non-AUTOSAR module designed as a complex device driver (CDD) of PduR. It is compliant with the AUTOSAR-defined interface for PduR CDDs.

The main purpose of rba_XCoreCdd is to transport Tx and Rx PDUs between cores in a multi-core environment in which memory areas are partitioned and therefore more secure. The PDUs are packed in a data structure - a "Pipe packet" - that contains a header and the actual PDU data before being transmitted/received via a Pipe object.

The rba_XCoreCdd module is intended to be used wherever high data rates and CPU core load balancing are required.

Module Configuration

This module is not currently supported by the RTA-BSW Configuration Generator (ConfGen) and must be manually configured.

12.2 Supported Features

The rba_XCoreCdd module has the following features:

12.2.1 Separation of the "Data Producer" and "Data Consumer" Tasks

This happens regardless of whether the tasks are hosted on the same core, or on different cores. By decoupling the Producer from the Consumer, the data processing can often be implemented without locks or other protection mechanisms, which can improve overall system performance.

12.2.2 Localization of Data

This occurs according to where the CPU processing is more intensive. The core that processes the PDUs (e.g. COM or signal conversion layer) is much faster if the PDUs are read from a local, in-sync memory area. rba_XCoreCdd enables higher CPU performance by copying the data to the best memory area for a given PDU.

12.2.3 Multi-ASIL Partition Support

This means that Tx and Rx PDUs can be transferred from any ASIL upper/lower-layer component to any other ASIL upper/lower-layer component in the BSW:

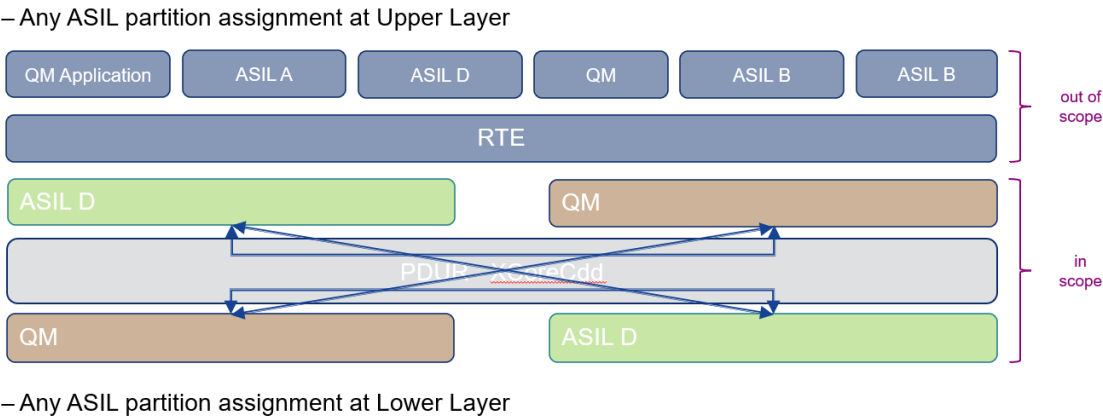


Figure 12.1. Multi-ASIL support

It is enabled by the Pipe memory allocation: specifically, objects *managing Pipe storage and FIFO updates* that separate Producer and Consumer data. In one object, only the Data Producer has WRITE access; in the other, the Data Producer only has READ access, while the Data Consumer has WRITE access.

This enhances *freedom from interference* i.e. the Producer and Consumer (running in the MainFunctions of their respective partitions) can *only* modify their own data and avoid critical issues such as data corruption, race conditions and fault propagation (from Producer to Consumer).

e.g.:

- The Data Producer is a lower ASIL function responsible for generating or producing data (e.g. sensor information/raw data from less critical systems).
- The Data Consumer is a higher ASIL function (e.g. a controller making safety-critical decisions based on the produced data).

NOTE

This separation applies to memory that holds data for *Pipe storage management*. The memory that holds the actual system data sent from Producer to Consumer is already separate.

12.3 Extensions

12.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

12.3.1.1 rba_XCoreCdd

Short Name	rba_XCoreCdd
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Description	Module to define XCoreCdd relevant configuration parameters
Multiplicity	0..1
Default Value	-
Type	ECUC-MODULE-DEF

12.3.1.2 rba_XCoreCdd/rba_XCoreCddGeneral

Short Name	rba_XCoreCddGeneral
Long Name	General
Description	Configuration of rba_XCoreCdd relevant code generation parameters
Multiplicity	1..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

12.3.1.3 rba_XCoreCdd/rba_XCoreCddGeneral/rba_XCoreCddDebug

Short Name	rba_XCoreCddDebug
Long Name	Debug
Description	DEBUG_INSTRUMENTATION_ACTIVE ON or OFF.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

12.3.1.4 rba_XCoreCdd/rba_XCoreCddGeneral/rba_XCoreCddDebugMeasurement

Short Name	rba_XCoreCddDebugMeasurement
Long Name	Debug Measurement
Description	DEBUG_TIMING_MEASUREMENT_ACTIVE ON or OFF.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

12.3.1.5 rba_XCoreCdd/rba_XCoreCddGeneral/rba_XCoreCddDevErrorDetect

Short Name	rba_XCoreCddDevErrorDetect
Long Name	Development Error Detection

Description	Enables/disables the detection and notification of development errors. If this parameter is enabled, then extended error checking is performed during runtime; any errors detected by these checks are reported to the Dem module. This feature should be switched off in series software.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

12.3.1.6 rba_XCoreCdd/rba_XCoreCddPipe

Short Name	rba_XCoreCddPipe
Long Name	Pipe
Description	Configuration of rba_XCoreCdd relevant code generation parameters
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

12.3.1.7 rba_XCoreCdd/rba_XCoreCddPipe/rba_XCoreCddLimitPdusforPipe

Short Name	rba_XCoreCddLimitPdusforPipe
Long Name	Limit Pdus to be processed from main function
Description	Limit Pdus to be processed from main function (This parameter is introduced to control data flow at bus level)
Multiplicity	0..1
Value Range	1..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

12.3.1.8 rba_XCoreCdd/rba_XCoreCddPipe/rba_XCoreCddPipeAllowBypassByTaskId

Short Name	rba_XCoreCddPipeAllowBypassByTaskId
Long Name	Is Bypass by task Id specific to PIPE required

Description	This parameter controls the bypassing of the PIPE specific calls and directly calling intended lower layer API. TRUE - Enables bypassing based on Task Id. No Pipe specific calls are invoked. Also locking of this Pipe is ensured independent of value configured rba_XcoreCddPipeSpecificLock, till rba_XCoreCdd_PipeRead API is called for the first time on this Pipe. FALSE - Disables bypassing based on taskId. Pipe Specific calls are invoked.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

12.3.1.9 rba_XCoreCdd/rba_XCoreCddPipe/rba_XCoreCddPipeDestCore

Short Name	rba_XCoreCddPipeDestCore
Long Name	Pipe Destination Core
Description	Destination Core ID of this pipe
Multiplicity	0..1
Value Range	0..255
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

12.3.1.10 rba_XCoreCdd/rba_XCoreCddPipe/rba_XCoreCddPipeDestPartitionRef

Short Name	rba_XCoreCddPipeDestPartitionRef
Long Name	Pipe Destination Partition Reference
Description	Reference to the Destination Partition of this pipe
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

12.3.1.11 rba_XCoreCdd/rba_XCoreCddPipe/rba_XCoreCddPipeDestTask

Short Name	rba_XCoreCddPipeDestTask
Long Name	Pipe Task Name
Description	Destination OS task in which the Pipe is processed
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

12.3.1.12 rba_XCoreCdd/rba_XCoreCddPipe/rba_XCoreCddPipeId

Short Name	rba_XCoreCddPipeId
Long Name	Pipe Id
Description	ID of this pipe
Multiplicity	1..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

12.3.1.13 rba_XCoreCdd/rba_XCoreCddPipe/rba_XCoreCddPipePdu

Short Name	rba_XCoreCddPipePdu
Long Name	Pipe Pdu
Description	Configuration of rba_XCoreCdd relevant code generation parameters
Multiplicity	1..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

12.3.1.14 rba_XCoreCdd/rba_XCoreCddPipe/rba_XCoreCddPipePdu/
rba_XCoreCddPipeDestPduRef

Short Name	rba_XCoreCddPipeDestPduRef
Long Name	Pipe Destination Pdu Ref
Description	Reference to the XCoreCdd upper Layer PDU
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

12.3.1.15 rba_XCoreCdd/rba_XCoreCddPipe/rba_XCoreCddPipePdu/
rba_XCoreCddPipeSrcPduRef

Short Name	rba_XCoreCddPipeSrcPduRef
Long Name	Pipe Source Pdu Ref
Description	Reference to the XCoreCdd lower Layer PDU
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

12.3.1.16 rba_XCoreCdd/rba_XCoreCddPipe/rba_XCoreCddPipePeriod

Short Name	rba_XCoreCddPipePeriod
Long Name	Pipe Period [s]
Description	Period in seconds, the function rba_XCoreCdd_PipeRead() shall be called. Parameter is needed to generate rba_XCoreCdd_PipeMainfunction_xxx() function.
Multiplicity	0..1
Value Range	0..3600
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

12.3.1.17 rba_XCoreCdd/rba_XCoreCddPipe/rba_XCoreCddPipeProcessingMode

Short Name	rba_XCoreCddPipeProcessingMode
Long Name	Pipe Processing Mode
Description	Polling, INTERRUPT, TASK
Multiplicity	1..1
Default Value	-
Enumeration Literals	POLLING, SWI, TASK
Type	ECUC-ENUMERATION-PARAM-DEF

12.3.1.18 rba_XCoreCdd/rba_XCoreCddPipe/rba_XCoreCddPipeProcessingType

Short Name	rba_XCoreCddPipeProcessingType
Long Name	Pipe Processing Type
Description	Rx, Tx, TxConfirmation...
Multiplicity	1..1
Default Value	-
Enumeration Literals	RX, TX, TXCONF
Type	ECUC-ENUMERATION-PARAM-DEF

12.3.1.19 rba_XCoreCdd/rba_XCoreCddPipe/rba_XCoreCddPipeRamSize

Short Name	rba_XCoreCddPipeRamSize
Long Name	Pipe RAM Size
Description	RAM_SIZE of this pipe in bytes. Minimum value 8 bytes is for the default invalid packet header at fifo end.

Multiplicity	0..1
Value Range	8..65535
Default Value	2048
Type	ECUC-INTEGER-PARAM-DEF

12.3.1.20 rba_XCoreCdd/rba_XCoreCddPipe/rba_XCoreCddPipeSpecificLock

Short Name	rba_XCoreCddPipeSpecificLock
Long Name	Is lock specific to PIPE required
Description	This parameter controls the Disabling and Restoring of local core interrupts when operating on a specific PIPE. TRUE ensures that disabling and restoring of local core IRQ's are taken care when operating on PIPE whereas FALSE ensures that local core IRQ's are not disabled and restored.
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

12.3.1.21 rba_XCoreCdd/rba_XCoreCddPipe/rba_XCoreCddPipeSrcCore

Short Name	rba_XCoreCddPipeSrcCore
Long Name	Pipe Source Core
Description	Source Core ID of this pipe
Multiplicity	0..1
Value Range	0..255
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

12.3.1.22 rba_XCoreCdd/rba_XCoreCddPipe/rba_XCoreCddPipeSrcPartitionRef

Short Name	rba_XCoreCddPipeSrcPartitionRef
Long Name	Pipe Source Partition Reference
Description	Reference to the Source Partition of this pipe
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

12.3.1.23 rba_XCoreCdd/rba_XCoreCddPipe/rba_XCoreCddPipeVariablePk-

tSizerba_XCoreCddPipeVariablePktSize

Short Name	rba_XCoreCddPipeVariablePktSize
Long Name	Pipe Variable Pkt Size
Description	This paramter controls what algorithm is used for the PIPE data. Variable Packet size is the only type of allocation used so far. This parameter is kept for future usage.
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

12.3.2 Additional Services

The following API are available in version v12.7.0 in addition to those defined in AR v22-11

- There are no additional API defined in this version.

12.4 Limitations

- Currently, for parameter `rba_XCoreCddPipeProcessingMode`, only polling mode is tested and verified. Validation added to avoid Task Interrupt.
- Currently, the memory barrier is not supported. The expectation, therefore, is that the Pipe must only be incremented *after* the completion of the write action using *memcpy* API, the counter responsible for allowing the Consumer read. For the supported devices (TC27XX Testboard, RH850P1X ECU) execution of the *memcpy* action and the counter increment are in sequence, so memory is not a constraint.
- The sequence of Pipe write and Pipe read need to be ensured; the *cache flush* mechanism is not yet supported.
- *Compiler abstraction support* is deprecated; henceforth, the component has to be only used for compilers $\geq$ 32-bit controllers.
- From AR42.8.4.0, a new DET reporting (`RBA_XCORECDD_E_WRONGTASK`) is supported. This will be reported when PDUs belonging to different tasks try to write into the same Pipe and no lock is enabled. Currently, this is only tested for PDUs coming to Tasks. Testing of calls to the API `Os_GetTaskID` in an interrupt context needs to be analyzed and tested.
- Ensure that `ecuCRbMetadata` is set to TRUE if this component is enabled in the Project.
- The `VAR_NON_CACHED_RAM` and `VAR_NON_CACHED_SHARED_RAM` macros will be mapped to normal RAM sections in a controller and the same can be overwritten at a Project level.
- If `LowerLayer_Transmit` returns `E_NOT_OK`, then `Up_TxConfirmation` will be called with `E_NOT_OK`, without checking whether or not the *TxConf* pipe is configured for the PDU.

- The debug variable *rba_XCoreCdd_PipeDbgInfoType_tst* has been moved to the *rba_XCoreCdd_PipeRuntimeInfoType_tst* structure. This debug structure will be separated into its own structure.
- Multi-ASIL partition support for rba_XCoreCdd cannot be tested in the BSWSIM environment.

12.5 Exclusions

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- There are no known exclusions in this version.

12.6 Deviations

- There are no known deviations in this version.

12.7 API Definition

List of supported functions:

rba_XCoreCdd_Init	Initializes the module.
rba_XCoreCdd_Transmit	PduR requests transmission of an I-PDU.
rba_XCoreCdd_RxIndication	Indication of a received I-PDU from a lower-layer communication interface via PduR.
rba_XCoreCdd_TxConfirmation	The lower-layer communication interface confirms the transmission of the I-PDU via PduR.
rba_XCoreCdd_PipeRead	Performs PDU read activities for the indicated PipeId.
rba_XCoreCdd_TriggerTransmit	The lower-layer communication interface callback for data transmission.
rba_XCoreCdd_CopyTxData	The lower-layer transport protocol API used during transmission of the TP PDU via PduR.
rba_XCoreCdd_TpTxConfirmation	The lower-layer transport protocol confirms the transmission of the I-PDU via PduR.
rba_XCoreCdd_StartOfReception	The lower-layer transport protocol initiates the reception of the TP PDU via PduR.
rba_XCoreCdd_CopyRxData	The lower-layer transport protocol provides the data pointer of the PDU to be received via PduR.
rba_XCoreCdd_TpRxIndication	The lower-layer confirms the completion of the transport protocol reception PDU via PduR.
rba_XCoreCdd_PipeCleanUp	Freezes Pipe activities. Typically called by a callback triggered by the Bus State Manager.
rba_XCoreCdd_CancelTransmit	Cancels transmission of a given PDU. Called by the PduR.
rba_XCoreCdd_CancelReceive	Cancels reception of a given PDU. Called by the PduR.

rba_XCoreCdd_PipeMainfunction	For each periodic task, this is a schedulable interface which is periodically called by the scheduler task.
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12.8 Configuration Advice

12.8.1 Sample Use Case

The following configuration advice is based on a sample Use Case of a multi-core system comprising two ECUs (CoreA and CoreB).

12.8.1.1 ECUC

Create the following PDUs for Transmit Message:

- CoreA_Com2PduR
- CoreA_PduR2XCoreCdd
- CoreB_XCoreCdd2PduR
- CoreB_PduR2CanIf

Create the following PDUs for Receive Message:

- CoreB_CanIf2PduR
- CoreB_PduR2XCoreCdd
- CoreA_XCoreCdd2PduR
- CoreA_PduR2Com

12.8.1.2 PduR

Create the following PduRRoutingPaths for Transmit Message:

- CoreA_Com2XCoreCdd
- CoreB_XCoreCdd2CanIf

Create the following PduRRoutingPaths for Receive Message:

- CoreB_CanIf2XCoreCdd
- CoreA_XCoreCdd2Com

12.8.1.3 rba_ComStackCdd

Create the following for Transmit Message:

- CddPduRLowerLayerTxPdu referenced to CoreA_PduR2XCoreCdd
- CddPduRUpperLayerTxPdu referenced to CoreB_XCoreCdd2PduR

Create the following for Receive Message:

- CddPduRLowerLayerRxPdu: refer to CoreA_XCoreCdd2PduR

- CddPduRUpperLayerRxPdu: refer to CoreB_PduR2XCoreCdd

12.8.1.4 rba_XCoreCdd

Under rba_XCoreCddPipe, with rba_XCoreCddPipeProcessingType configured as Tx, create a rba_XCoreCddPipePdu with the following:

- rba_XCoreCddPipeSrcPduRef referenced to CoreA_PduR2XCoreCdd
- rba_XCoreCddPipeDestPduRef referenced to CoreB_XCoreCdd2PduR

Under rba_XCoreCddPipe, with rba_XCoreCddPipeProcessingType configured as Rx, create a rba_XCoreCddPipePdu with the following:

- rba_XCoreCddPipeSrcPduRef: CoreB_PduR2XCoreCdd
- rba_XCoreCddPipeDestPduRef: CoreA_XCoreCdd2PduR

12.8.1.5 CanIf

- For Transmit Message, configure CanIfTxPduCfg/CanIfTxPduRef to refer to the CoreB_PduR2CanIf PDU
- For Receive Message, configure CanIfRxPduCfg/CanIfRxPduRef to refer to the CoreB_CanIf2PduR PDU

12.8.1.6 Com

- For Transmit Message, configure ComPduIdRef to refer to the CoreA_-Com2PduR PDU
- For Receive Message, configure ComPduIdRef to refer to the CoreA_PduR2Com PDU

12.8.2 General Advice

PduR/PduRBSwModules needs to be configured with the *shortName* of "rba_X-CoreCdd", and with the following child parameter settings:

- PduRCommunicationInterface: TRUE
- PduRLowerModule: TRUE
- PduRTxConfirmation: TRUE
- PduRUpperModule: TRUE

rba_ComStackCdd must be configured with the *shortName* "rba_XCoreCdd".

The rba_XCoreCddPipeProcessingMode parameter provides three different configurable options (POLLING, INTERRUPT and TASK), but for the current implementation only the POLLING mode has been fully tested and verified, so INTERRUPT and TASK should not be used.

12.9 Integration Advice

12.9.1 Init Functions

`rba_XCoreCdd` is initialized via `rba_XCoreCdd_Init`.

12.9.2 Delnit Functions

No specific handling necessary.

12.9.3 Scheduled Functions

In the `rba_XCoreCdd_Cfg_BSWMD.arxml` file generated for each configured `rba_XCoreCddPipe`, there will be a corresponding `BswTimingEvent` with the following naming rule, in which:

- #x is the Mainfunction period configured in the relevant `rba_XCoreCddPipe/rba_XCoreCddPipePeriod` parameter.
- #y is the associated `coreId` configured in `rba_XCoreCddPipe/rba_XCoreCddPipeDestCore`.

Naming rule:

`EV_SE_rba_XCoreCdd_PipeMainfunction_#xms_Core#y`

These events should be mapped to the relevant `OsTask` on the correct core.

12.9.4 Shared Memory Allocation

Because the Pipe runtime object `rba_XCoreCdd_PipeRuntimeInfo_ast` has to be accessed from different cores, the User should ensure that it is allocated in system-shared RAM. This can be done by adding the `#pragma section` directive for `RBA_XCORECDD_START_SYSTEM_SHARED_NON_CACHED_RAM` in `rba_XCoreCdd_Cfg_MemMap.h`.



NOTE

`rba_XCoreCdd_PipeRuntimeInfo_ast` is a shared memory structure for Pipe storage management that is used for debug/status data. Its shared status does not impact multi-ASIL partitioning.

12.9.5 EcuCRbMetadata

If `rba_XCoreCdd` is included in a Project, it must be ensured that the `EcuCRbMetadata` configuration parameter in the `EcuC` module is set to `TRUE`.

12.9.6 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the `ENTER` and `EXIT` resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in

the OS with the correct task/ISR allocation.

The following is a list of all exclusive areas that this module might use regardless of the user configuration. It is the job of the integrator to ensure that the implementation of these exclusive areas is correctly defined. This implementation can be done using the RTE exclusive area configuration wizard or by manual implementation of the corresponding BSW API calls to ensure exclusivity.

The Rba_XCoreCdd module uses the following OS resources:

- RBA_XCORECDD_ENTER
- RBA_XCORECDD_NONEST_DISABLE_CORE_LOCAL_IRQ

12.10 Hardware Requirements

Information relating to the hardware resources required by this module is not currently available.

13 Glossary

ASIL

Automotive Safety Integrity Level

BswM

AUTOSAR BSW Mode Manager module

Can

Controller Area Network Driver

CanIf

AUTOSAR CAN Interface module

CanNM

AUTOSAR CAN Network Management module

CanSM

AUTOSAR CAN State Machine module

CanTp

AUTOSAR CAN Transport Layer module

CBV

Control Bit Vector

CDD

Complex Device Driver

Com

AUTOSAR Communication module

ComM

AUTOSAR Communication Manager module

ComXf

AUTOSAR COM Based Transformer module

CWU

Car Wake Up

DCM

AUTOSAR Diagnostic Communication Manager module

E2EXf

AUTOSAR End to End Library Transformer module

EcuM

AUTOSAR ECU State Manager module

EIRA

External and Internal Request Array

ERA

External Request Array

Fr FlexRay Driver

FrIf AUTOSAR FlexRay Interface module

FrNm

AUTOSAR FlexRay Network Management module

FrTp

AUTOSAR FlexRay Transport Protocol module

I-PDU

Interaction Layer Protocol Data Unit

IPduM

AUTOSAR PDU Multiplexer module

J1939Nm

AUTOSAR J1939 Network Management module

J1939Tp

AUTOSAR J1939 Transport Protocol module

LdCom

AUTOSAR Large Data Com module

Lin Lin Driver**LinIf**

AUTOSAR Lin Interface module

LinNm

AUTOSAR Lin Network Management module

LinSM

AUTOSAR Lin State Manager module

LinTp

AUTOSAR Lin Transport Protocol module

Nm AUTOSAR Network Management Interface module**NM** Network Management**NMS**

Network Management State

Pdu

Protocol Data Unit

PduR

AUTOSAR PDU Router module

PNC

Partial Network Cluster

rba_ComStackCdd

RTA-BSW Module, provides support for AUTOSAR CDDs

rba_ComXfAdp

RTA-BSW Module, COM Based Transformer Adapter

rba_SysElem

RTA-BSW Module, holds information from the AUTOSAR system level regarding DataTransformations (SystemSignals, ISignals, DataMappings, DataTypes,...) in the form of ECU configuration values for use in BSW code-generation

rba_XCoreCdd

RTA-BSW Module, transports Tx and Rx PDUs between cores in a multi-core environment

rba_Xfrm

RTA-BSW Module, provides extension to the AUTOSAR specification of transformers

RPG

Routing Path Group

Rx Receipt/Reception

SOME/IP

Scalable Service-Oriented Middleware Over IP. An automotive middleware solution which can be used to send and receive control messages on a vehicle network

SomelpTp

AUTOSAR Scalable service-Oriented MiddlewarE over IP (SOME/IP) Transport Protocol module

SomelpXf

AUTOSAR SOME/IP Transformer module

TMS

Transmission Mode Selector

TP Transport Protocol

Transformer

A transformer employs a data transformation mechanism to linearize and transform data

Tx Transmit/Transmission

UdpNm

AUTOSAR UDP Network Management module